

Amino acid Metabolism and Urea cycle

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Amino acid metabolism

- **Amino acids** are the **final class of biomolecules** that, through their **oxidative degradation**, make a significant contribution to the **generation of metabolic energy**.
- The fraction of metabolic energy obtained from amino acids, whether they are derived from **dietary protein or from tissue protein**, varies greatly with the type of organism and with metabolic conditions.
- **Carnivores** can obtain (immediately following a meal) **up to 90% of their energy requirements** from amino acid oxidation, whereas **herbivores** may fill only a small fraction of their energy needs by this route.

Amino acid metabolism

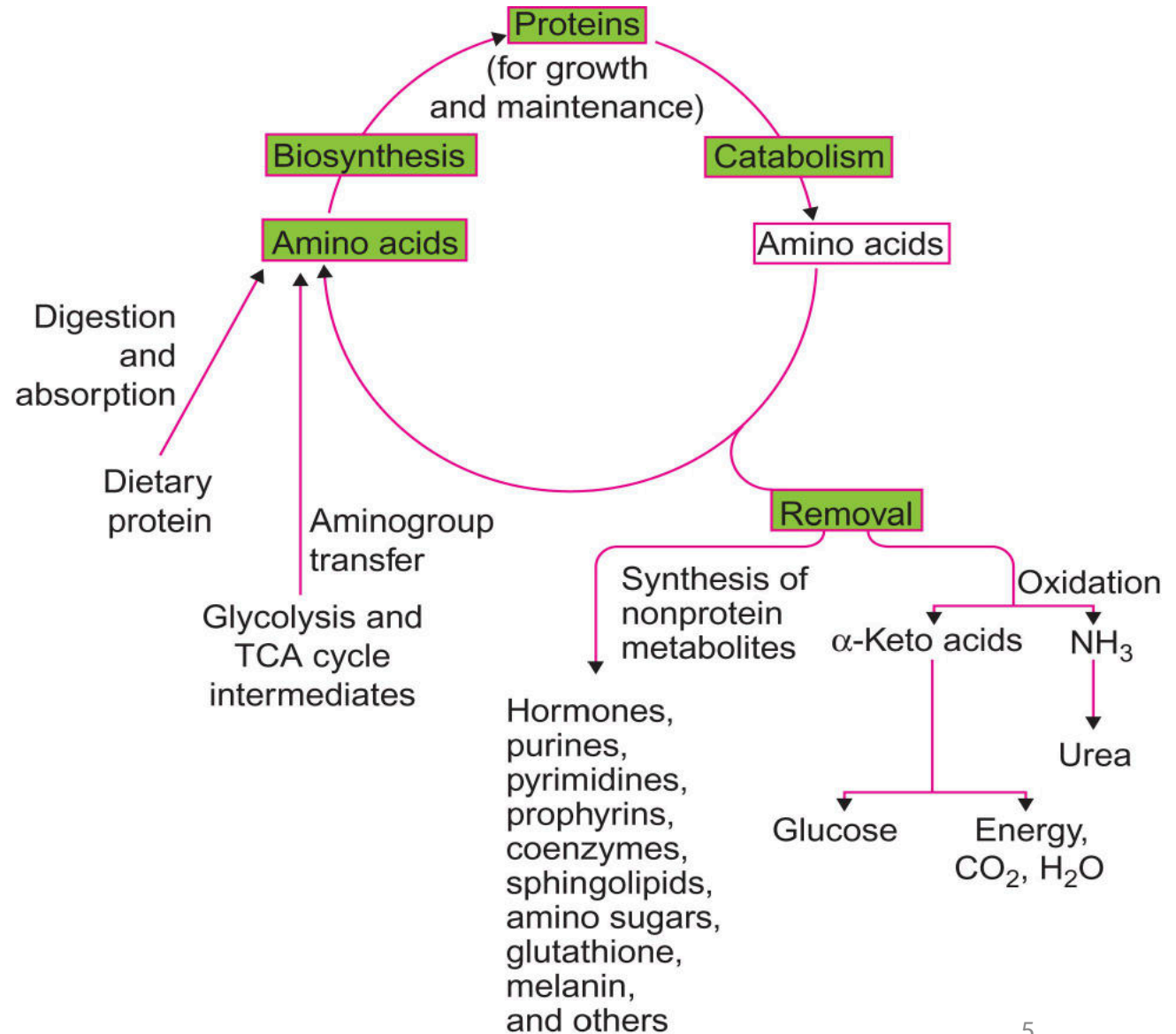
- Most microorganisms can scavenge amino acids from their environment and use them as fuel when required by metabolic conditions.
- Plants, however, rarely if ever oxidize amino acids to provide energy; the carbohydrate produced from CO₂ and H₂O in photosynthesis is generally their sole energy source.

Amino acid metabolism

- **In animals, amino acids undergo oxidative degradation in three different metabolic circumstances:**
 1. During the **normal synthesis and degradation of cellular proteins** some amino acids that are released from protein breakdown and are **not needed** for new protein synthesis undergo oxidative degradation.
 2. **During intake of protein rich diet** and the ingested amino acids **exceed the body's needs for protein synthesis**, the surplus is catabolized; **amino acids cannot be stored**.
 3. **During starvation or in uncontrolled diabetes mellitus**, when carbohydrates are either **unavailable or not properly utilized**, cellular proteins are used as fuel.

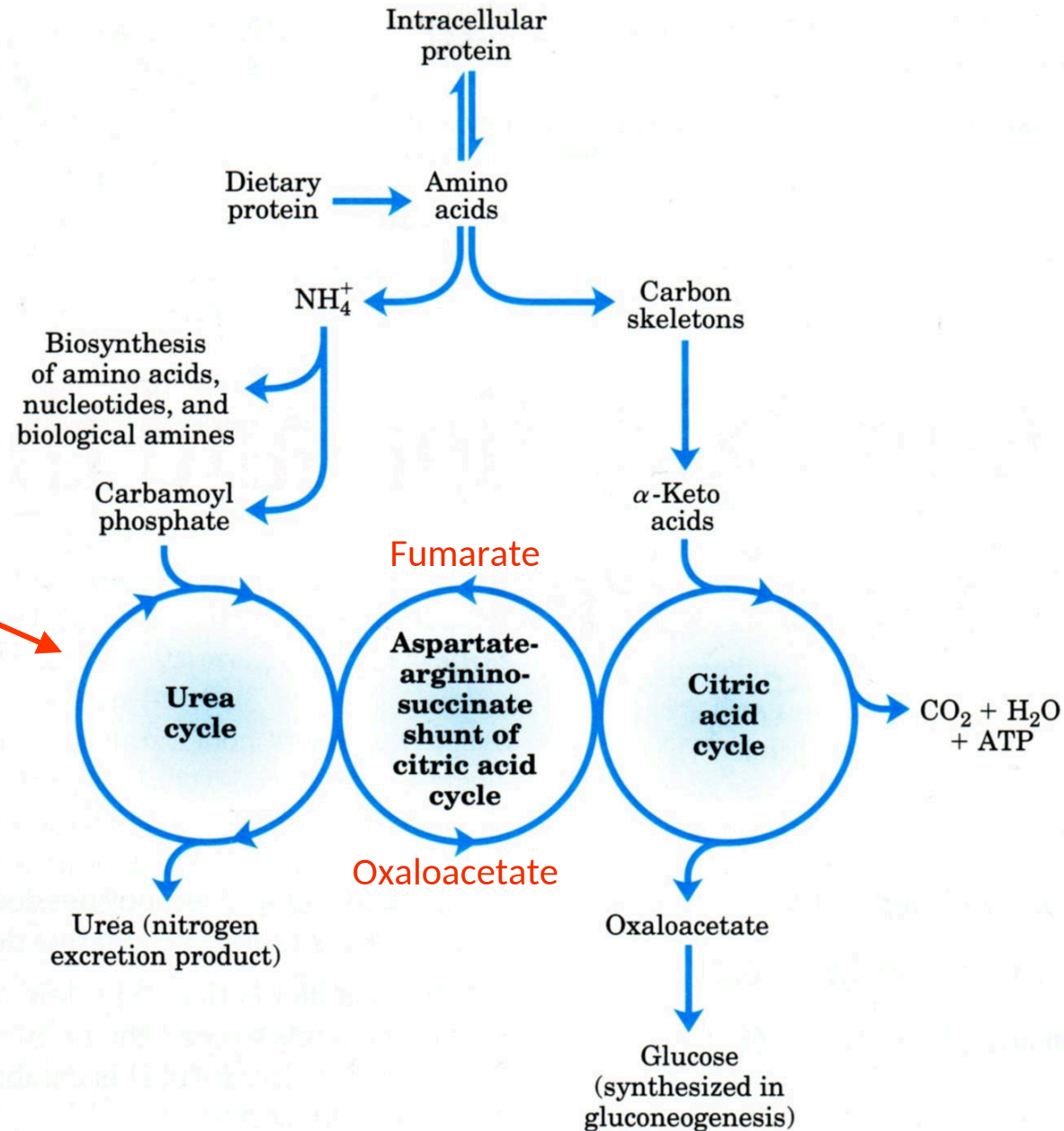
Overview of amino acid metabolism

- Proteins constantly undergo **turnover**.
- Amino acids are also used to synthesize some non-protein metabolites.
- No protein stores, so essential amino acids must come from diet.



Two choices

1. Reuse
2. Urea cycle



Metabolic fates of amino groups

- **Transfer of Amino Groups to Glutamate:**
- The α -amino groups of the 20 L-amino acids, commonly found in proteins, are removed during the oxidative degradation of the amino acids.
- If not reused for the synthesis of new amino acids, these amino groups are channeled into a single excretory product.
- Many aquatic organisms simply release ammonia as NH_4^+ into the surrounding medium.
- Most terrestrial vertebrates first convert ammonia into either urea (e.g., humans, other mammals and adult amphibians) or uric acid (e.g., reptiles, birds).

Metabolic fates of amino groups

- The first step in the catabolism of most of the L-amino acids is the removal of the **α -amino group (i.e., transamination) by a group of enzymes called aminotransferases (= transaminases).**
- In these reactions, the α -amino group is transferred to the α -carbon atom of α -ketoglutarate, leaving behind the corresponding α -keto acid analogue of the amino acid.
- The effect of transamination reactions is to collect the amino groups from many different amino acids in the form of only one, namely L-glutamate.
- Cells contain several different aminotransferases, many of which are specific for α ketoglutarate as the amino group acceptor.

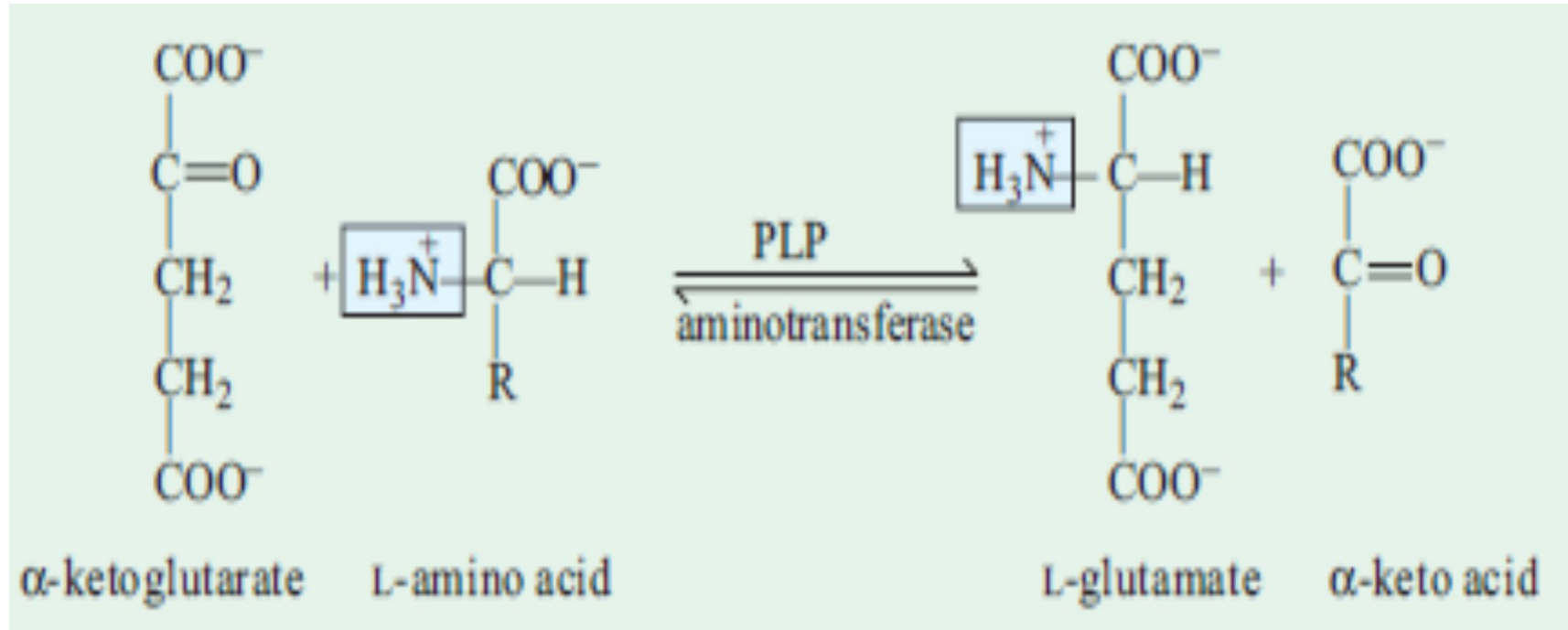


Fig. 26–4. The transamination (or the aminotransferase) reaction

Metabolic Fates of Amino Groups

- **Nitrogen, N_2** , is abundant in the atmosphere but is **too inert for use** in most biochemical processes.
- Because only a few microorganisms can convert N_2 to biologically useful forms such as NH_3 , amino groups are carefully conserved in biological systems.

Metabolic Fates of Amino Groups

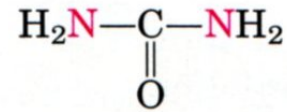
- Amino acids derived from **dietary protein** are the **source of most amino groups**.
- Most amino acids are **metabolized in the liver**.
- Some of the **ammonia** generated in this process is **recycled and used in a variety of biosynthetic pathways**; the excess is either excreted directly or **converted to urea or uric acid for excretion**, depending on the organism.
- **Excess ammonia generated in other (extrahepatic) tissues** travels to **the liver** for conversion to the excretory form.

Excretory forms of nitrogen



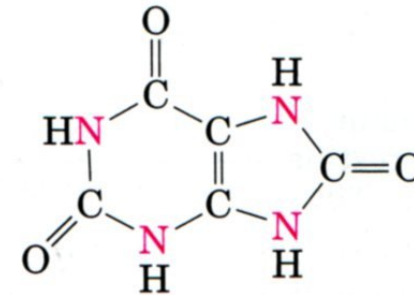
Ammonia (as
ammonium ion)

Ammonotelic animals:
most aquatic vertebrates,
such as bony fishes and
the larvae of amphibia



Urea

Ureotelic animals:
many terrestrial
vertebrates; also sharks



Uric acid

Uricotelic animals:
birds, reptiles

- a) Excess NH_4^+ is excreted as ammonia (microbes, aquatic vertebrates or larvae of amphibia),
- b) Urea (many terrestrial vertebrates)
- c) or uric acid (birds and terrestrial reptiles)

Metabolic Fates of Amino Groups

- **Four amino acids** play central roles in nitrogen metabolism.
- **glutamate, glutamine, alanine, and aspartate.**
- The **special place of these four amino acids** is due to that these particular amino acids are the **ones most easily converted into citric acid cycle intermediates: glutamate and glutamine to -ketoglutarate, alanine to pyruvate, and aspartate to oxaloacetate.**
- **Glutamate and glutamine are especially important**, acting as a kind of general collection point for amino groups.
- In the **cytosol of hepatocytes**, amino groups from most amino acids are transferred to -ketoglutarate to form glutamate, which enters mitochondria and gives up its amino group to form NH_4^+ .

Metabolic Fates of Amino Groups

- **Excess ammonia** generated in most other tissues is converted to the amide nitrogen of glutamine, which passes to the liver, then into liver mitochondria.
- Glutamine or glutamate or both are present in higher concentrations than other amino acids in most tissues.
- **In skeletal muscle, excess amino groups are generally transferred to pyruvate to form alanine**, another important molecule in the transport of amino groups to the liver.

Ammonia has to be eliminated

- **Ammonia is toxic**, especially for the **CNS**, because it reacts with α -**keto glutarate to form glutamate**, thus making it **limiting for the TCA cycle** ➡ **decrease in the ATP level**.
- **Liver damage** or metabolic disorders associated with elevated ammonia can lead to tremor, slurred speech, blurred vision, coma, and death
- **Normal conc. of ammonia in blood: 19- 87 ug /dl**

Nitrogen removal from amino acids

Step 1: Remove amino group

Step 2: Take amino group to liver for nitrogen excretion

Step 3: Entry into mitochondria

Step 4: Prepare nitrogen to enter urea cycle

Step 5: Urea cycle

Nitrogen carriers

1. Glutamate

- Transfers one amino group WITHIN cells:
- **Aminotransferase** → makes glutamate from α -ketoglutarate
 - Aminotransferase (PLP) → α -ketoglutarate → glutamate
- **Glutamate dehydrogenase** → opposite
 - Glutamate dehydrogenase (no PLP) → glutamate → α -ketoglutarate (in liver)

2. Glutamine

Transfers two amino group BETWEEN cells → releases its amino group in the liver

- **Glutamine synthase** → glutamate → glutamine
- **Glutaminase** → glutamine → glutamate (in liver)

3. Alanine

Transfers amino group from tissue (muscle) into the liver

Step 1: Removal of amino group

- Amino groups can be removed by **transamination**
- In liver cytosol, amino groups are passed to α -KG, forming **glutamate**.
- **Transaminases** (aka aminotransferases) require pyridoxal phosphate cofactor.

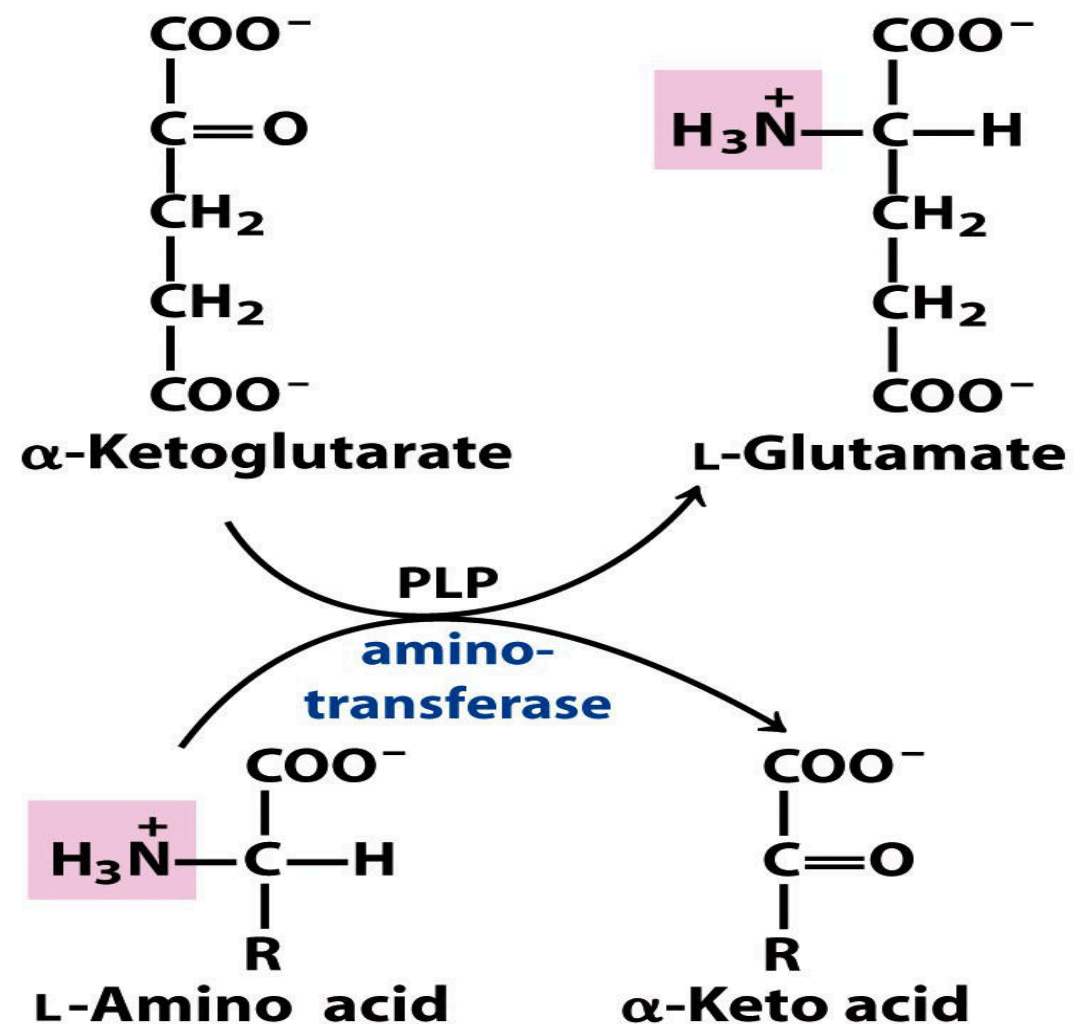
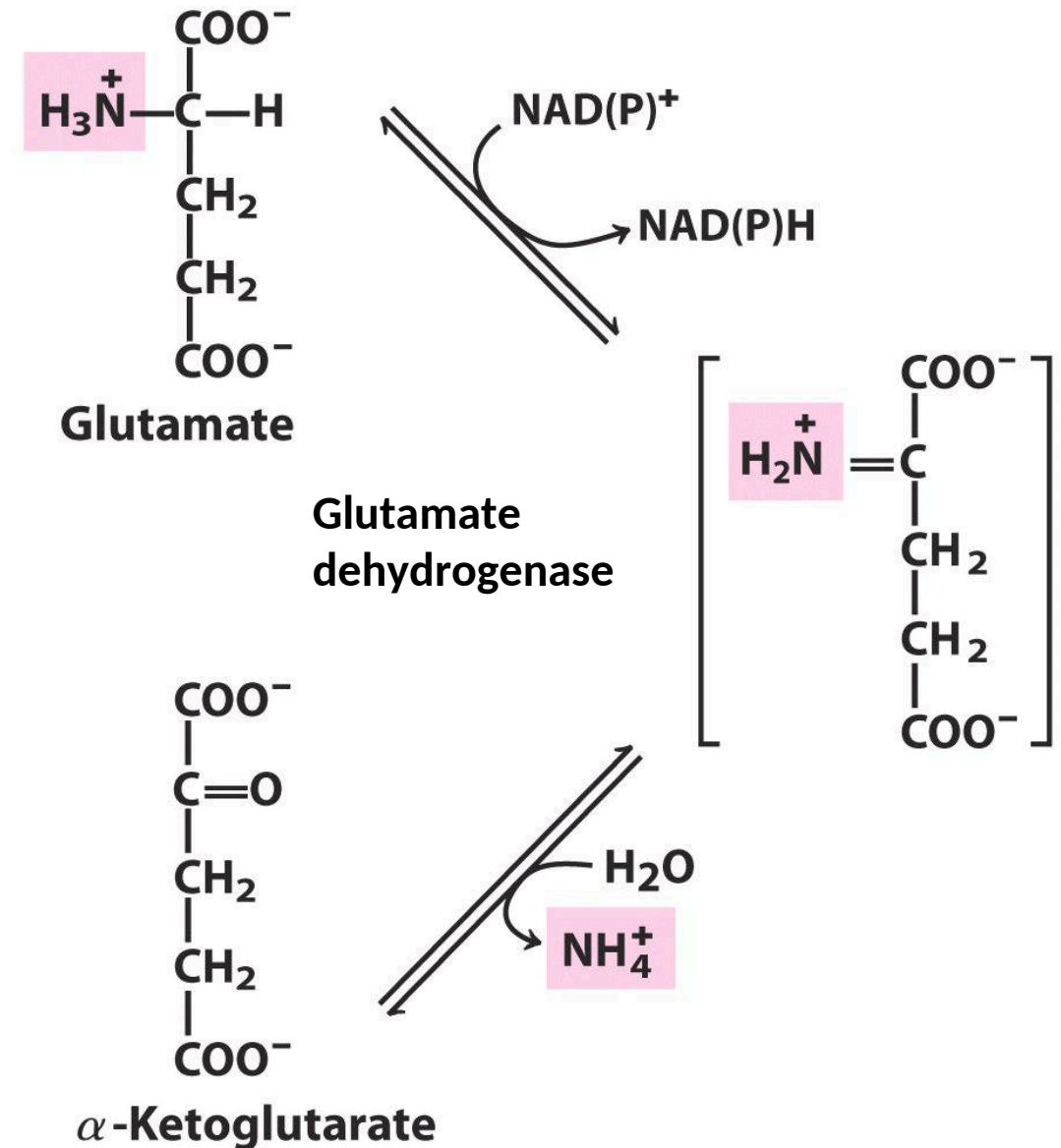


Figure 18-4
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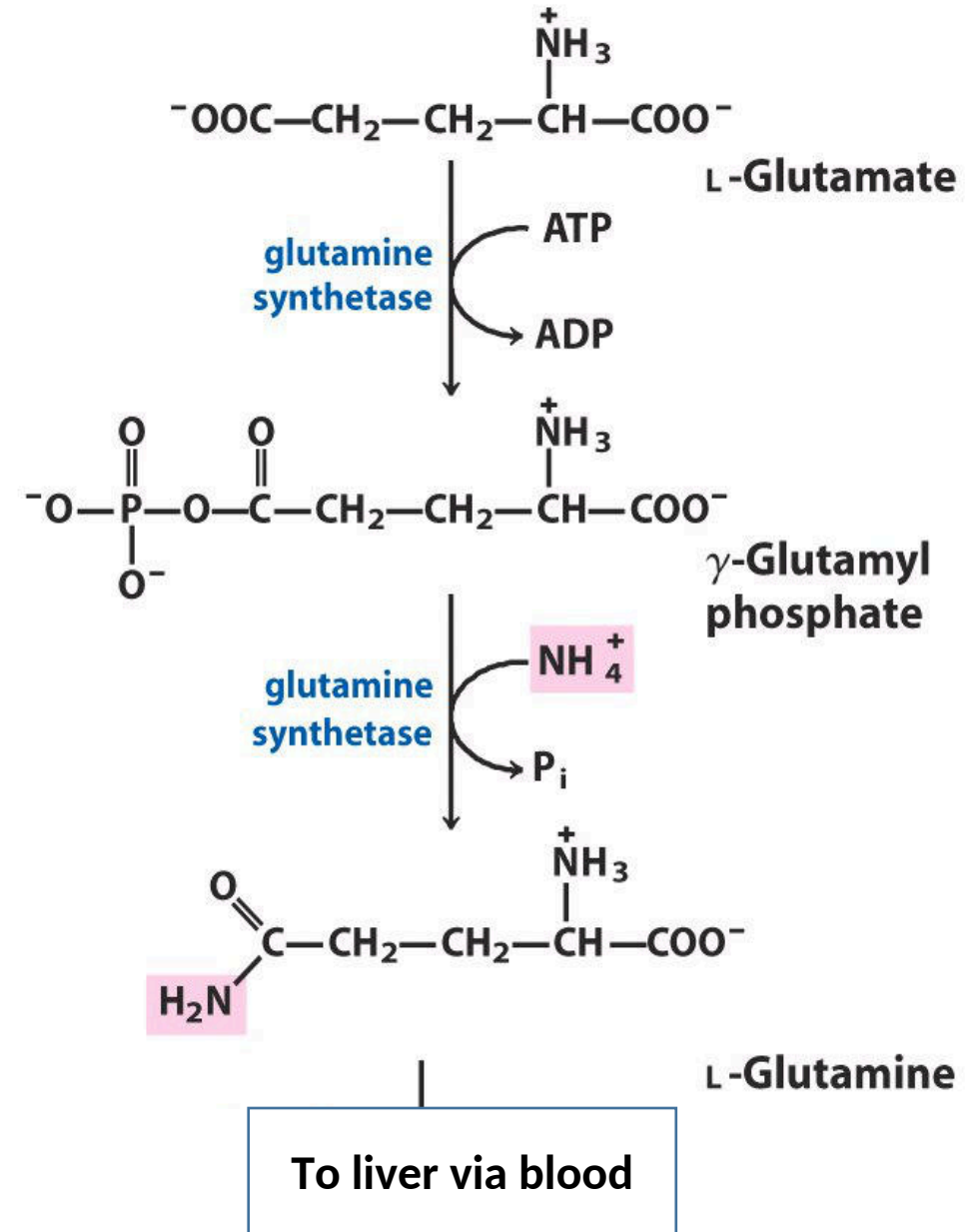
Step 2: Take amino group to liver for nitrogen excretion(oxidative deamination)

- Glutamate in the liver cytosol enters the mitochondrial matrix, where its amino group is removed by **glutamate dehydrogenase**.
- The **amino groups** from many of the **α-amino acids** are collected in the liver in the form of the **amino group of L-glutamate molecules**.
- The GDH of mammalian liver has the unusual capacity to use either NAD^+ or NADP^+ as cofactor



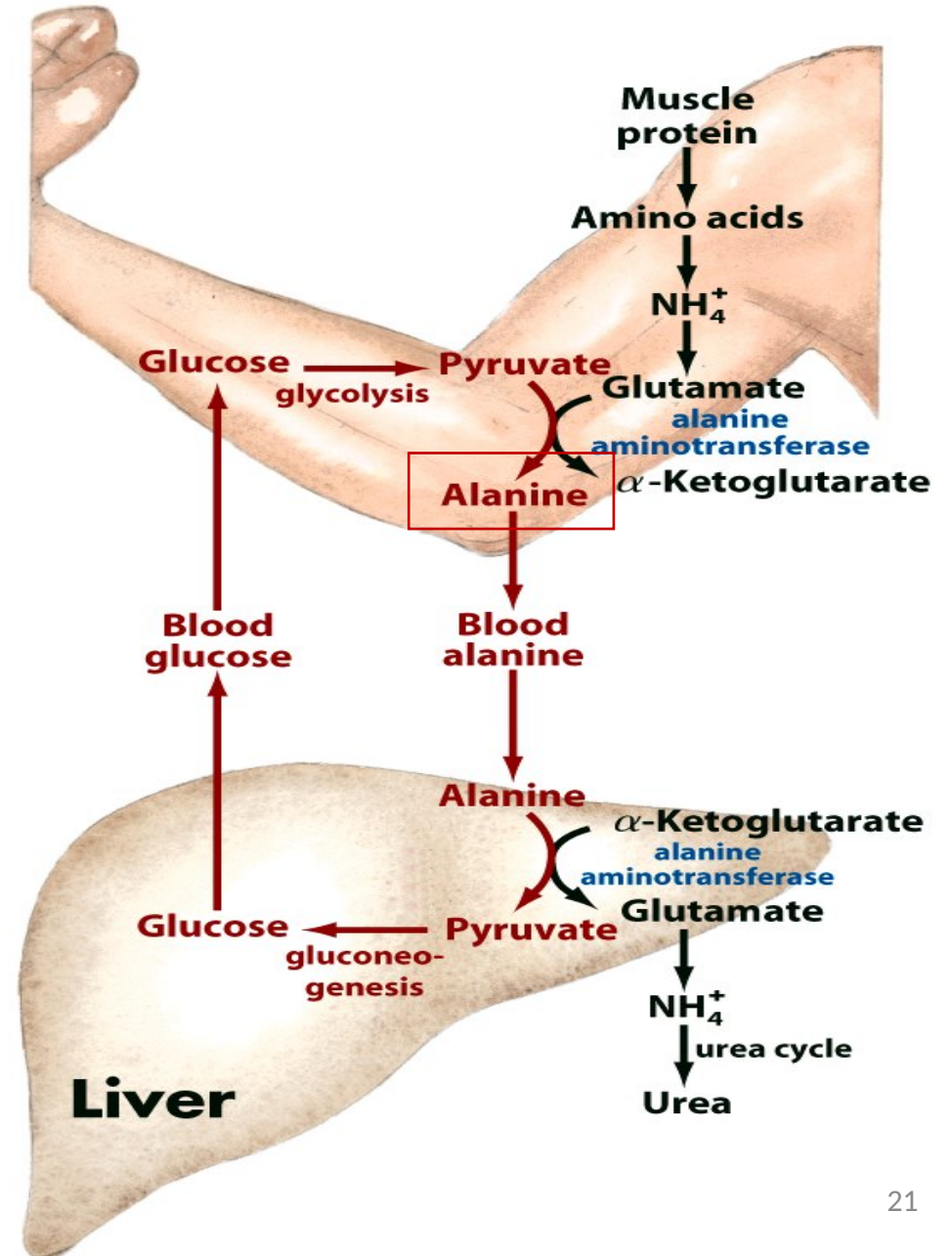
Transport of amino groups as glutamine

- Peripheral tissues may send their **amino groups as glutamine** through the bloodstream to the liver for processing.



Transport of amino groups as alanine (glucose-alanine cycle)

- In concert with the **Cori cycle**, skeletal muscle may send **pyruvate through bloodstream as alanine** (the glucose-alanine cycle).
- Operates when muscle proteins are undergoing catabolism.



Why Urea?

- **Non toxic**
- **Water soluble**
- **Combines two waste products into one molecule:**
 - CO_2
 - NH_3

Production of Urea from Ammonia: The Urea Cycle

- A moderately active man consuming about 300 g of carbohydrate, 100 g of fat and 100 g of protein daily must excrete about 16.5 g of nitrogen daily.
- 95% is eliminated by the kidneys and the remaining 5% in the faeces.
- The major pathway of N₂ excretion in humans is as urea which is synthesized in the liver, released into the blood, and cleared by the kidney.
- In humans eating an occidental diet, urea constitutes 80-90% of the nitrogen excreted.
- The urea cycle spans two cellular compartments (It begins inside the mitochondria of liver cells (= hepatocytes), but 3 of the steps occur in the cytosol.
- Whatever its source, the NH_4^+ generated in liver mitochondria is immediately used, together with HCO_3^- produced by mitochondrial respiration, to form carbamoyl phosphate in the matrix.
- This ATP-dependent reaction is catalyzed by carbamoyl phosphate synthetase I, a regulatory enzyme present in liver mitochondria of all ureotelic organisms including man.
- In bacteria, glutamine rather than ammonia serves as a substrate for carbamoyl phosphate synthesis.

The Urea Cycle

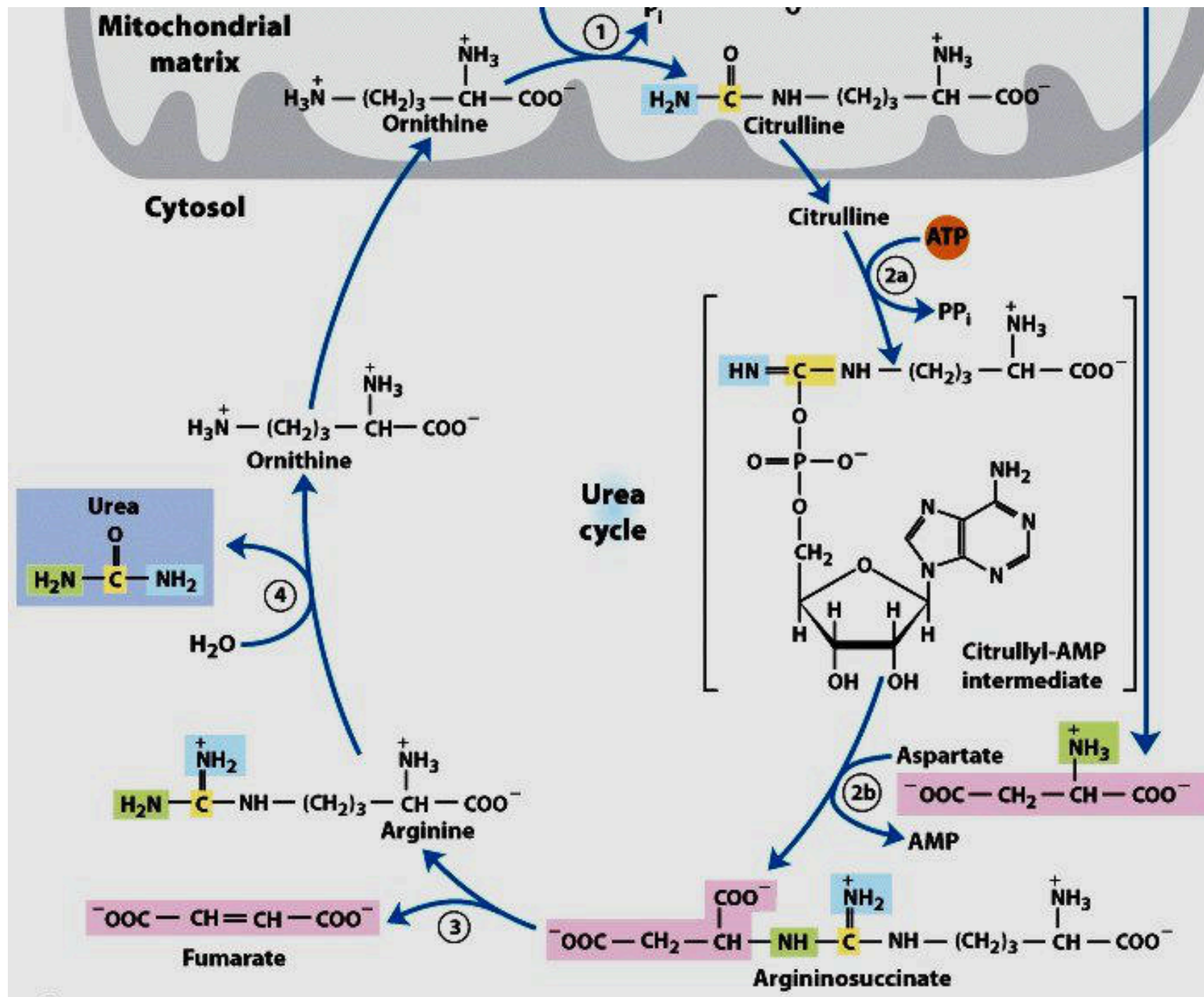
- The carbamoyl phosphate now enters the urea cycle, which itself consists of 4 enzymatic steps. These are:
- **Step 1: Synthesis of Citrulline:** Carbamoyl phosphate has a high transfer potential because of its anhydride bond. It, therefore, donates its carbamoyl group to ornithine to form citrulline and releases inorganic phosphate.
- The reaction is catalyzed by **L-ornithine transcarbamoylase** of liver mitochondria.
- The citrulline is released from the mitochondrion into the cytosol.
- **Step 2: Synthesis of argininosuccinate:** The second amino group is introduced from aspartate (produced in the mitochondria by transamination and transported to the cytosol) by a condensation reaction between the amino group of aspartate and the ureido (= carbonyl) group of citrulline to form argininosuccinate.
- The reaction is catalyzed by **argininosuccinate synthetase** of the cytosol.
- It requires ATP which cleaves into AMP and pyrophosphate and proceeds through a citrullyl-AMP intermediate.

The Urea Cycle

- **Step 3: Cleavage of argininosuccinate to arginine and fumarate:** Argininosuccinate is then reversibly cleaved by **argininosuccinate lyase** (= argininosuccinase), a cold-labile enzyme of mammalian liver and kidney tissues, to form free arginine and fumarate, which enters the pool of citric acid cycle intermediates.
- These two reactions, which transfer the amino group of aspartate to form arginine, preserve the carbon skeleton of aspartate in the form of fumarate.
- **Step 4: Cleavage of arginine to ornithine and urea:** The arginine so produced is cleaved by the cytosolic enzyme **arginase**, present in the livers of all ureotelic organisms, to yield urea and ornithine.
- Smaller quantities of **arginase** also occur in **renal tissue, brain, mammary gland, testes and skin.**
- Ornithine is, thus, regenerated and can be transported into the mitochondrion to initiate another round of the urea cycle.

Urea cycle –(Sequence of reactions)

- **Carbamoyl phosphate** formation in mitochondria is a prerequisite for the urea cycle
 - (**Carbamoyl phosphate synthetase**)
- **Citrulline** formation from carbamoyl phosphate and ornithine
 - (**Ornithine transcarbamoylase**)
- **Aspartate** provides the **additional nitrogen** to form **argininosuccinate** in cytosol
 - (**Argininosuccinate synthase**)
- **Arginine** and **fumarate** formation
 - (**Argininosuccinate lyase**)
- Hydrolysis of arginine to **urea** and ornithine
 - (**Arginase**)



The Cost of Urea Synthesis

2 ATPs are used in carbamoyl-phosphate synthesis.

1 ATP is used in arginosuccinate synthesis.

The overall equation for the urea cycle is:



However, fumarate was produced which feeds into the citric acid cycle and is converted to oxaloacetate, producing a molecule of NADH (2.5 ATP)

Regulation of urea cycle

The activity of urea cycle is regulated at two levels:

- **Dietary intake of proteins** → much urea (amino acids are used for fuel)
- **Prolonged starvation** → break down of muscle proteins → much urea also
- The rate of **synthesis of four urea cycle enzymes** and **carbamoyl phosphate synthetase I (CPS-I)** in the liver is regulated by changes in demand for urea cycle activity.
- Enzymes are synthesized at higher rates in animals during:
 - starvation
 - in very high protein diet
- Enzymes are synthesized at lower rates in
 - well-fed animals with carbohydrate and fat diet
 - animals with protein-free diets

Ammonia toxicity

- **Ammonia encephalopathy** : Increased **concentration of ammonia in the blood** and other biological fluids → ammonia diffuses into cells, across blood/brain barrier → **increased synthesis of glutamate from α -ketoglutarate, increased synthesis of glutamine**
- **α -ketoglutarate is depleted from CNS → inhibition of TCA cycle and production of ATP**
- Neurotransmitters – glutamate (excitatory neurotr.) and GABA (inhibitory neurotr.), may contribute to the CNS effects ?

Deficiencies of urea cycle enzymes

- The synthesis of urea in the liver is the major pathway of the removal of NH_4^+ .
- A blockage of carbamoyl phosphate synthesis or any of the 4 steps of the urea cycle has serious consequences because there is no alternative pathway for the synthesis of urea.
- They all lead to an elevated level of NH_4^+ in the blood (hyperammonemia) and encephalopathy.
- Some of these genetic defects become evident a day or two after birth, when the afflicted infant becomes lethargic and vomits periodically.
- Coma and irreversible brain damage may ensue.

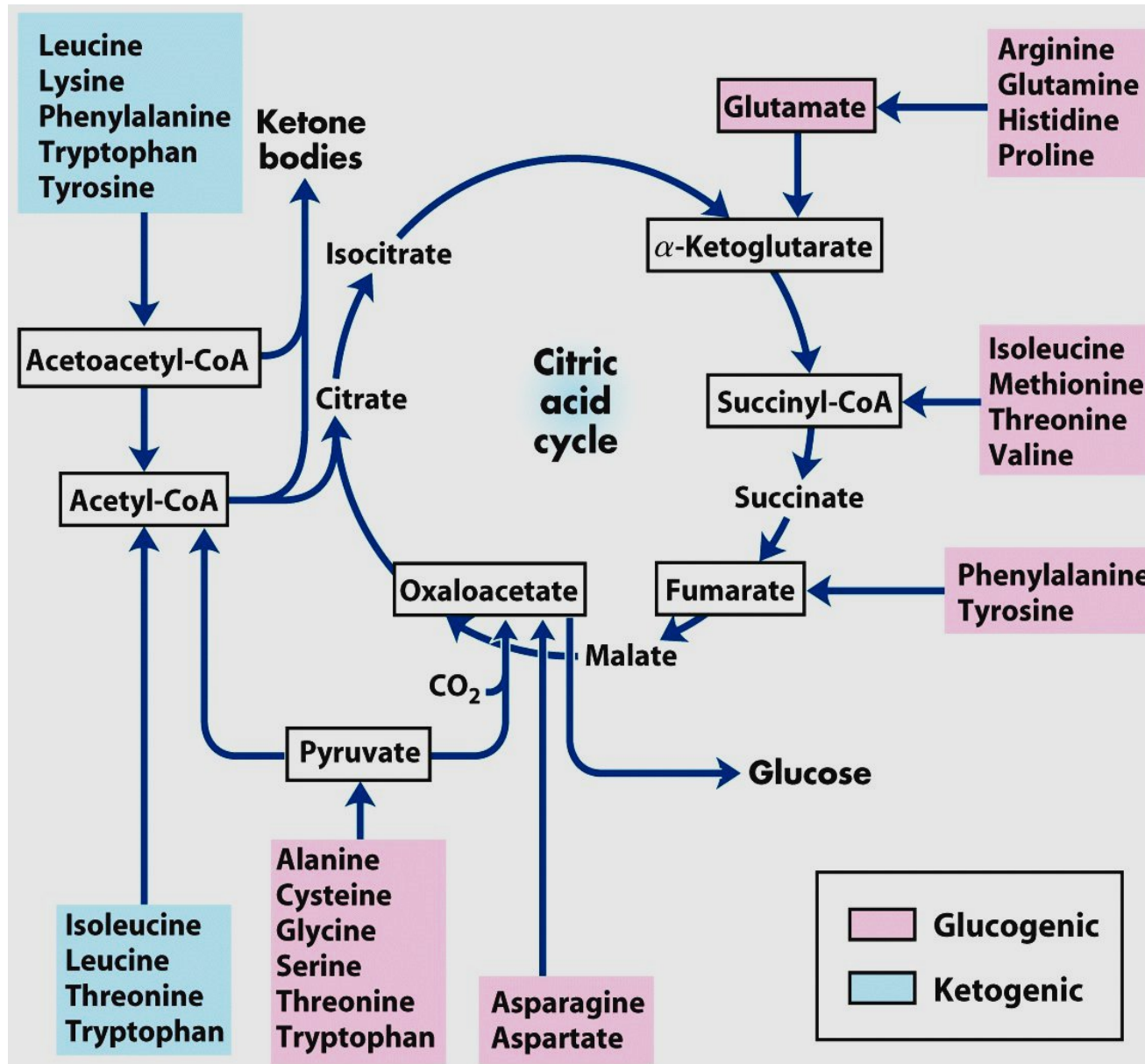
TABLE 18-2 Some Human Genetic Disorders Affecting Amino Acid Catabolism

<i>Medical condition</i>	<i>Approximate incidence (per 100,000 births)</i>	<i>Defective process</i>	<i>Defective enzyme</i>	<i>Symptoms and effects</i>
Albinism	<3	Melanin synthesis from tyrosine	Tyrosine 3-monooxygenase (tyrosinase)	Lack of pigmentation: white hair, pink skin
Alkaptonuria	<0.4	Tyrosine degradation	Homogentisate 1,2-dioxygenase	Dark pigment in urine; late-developing arthritis
Argininemia	<0.5	Urea synthesis	Arginase	Mental retardation
Argininosuccinic acidemia	<1.5	Urea synthesis	Argininosuccinase	Vomiting; convulsions
Carbamoyl phosphate synthetase I deficiency	<0.5	Urea synthesis	Carbamoyl phosphate synthetase I	Lethargy; convulsions; early death
Homocystinuria	<0.5	Methionine degradation	Cystathionine β -synthase	Faulty bone development; mental retardation
Maple syrup urine disease (branched-chain ketoaciduria)	<0.4	Isoleucine, leucine, and valine degradation	Branched-chain α -keto acid dehydrogenase complex	Vomiting; convulsions; mental retardation; early death
Methylmalonic acidemia	<0.5	Conversion of propionyl-CoA to succinyl-CoA	Methylmalonyl-CoA mutase	Vomiting; convulsions; mental retardation; early death
Phenylketonuria	<8	Conversion of phenylalanine to tyrosine	Phenylalanine hydroxylase	Neonatal vomiting; mental retardation

Fates of carbon skeleton

- Based on their catabolic products, amino acids are classified as **glucogenic or ketogenic**.
- Those amino acids that generate **precursors of glucose, e.g., pyruvate or citric acid cycle intermediate (i.e., α -ketoglutarate, succinyl-CoA, fumarate or oxaloacetate)**, are **referred to as glucogenic**. Ala, Arg, Asn, Asp, Cys, Gln, Glu, Gly, His, Met, Pro, Ser, Thr and Val belong to this category.
- Those amino acids that are degraded to **acetyl-CoA or acetoacetyl-CoA** are **termed as ketogenic because they give rise to ketone bodies**.
- Their ability to form ketones is particularly evident in untreated diabetes mellitus, in which large amounts of ketones are produced by the liver, not only from fatty acids but from the ketogenic amino acids.
- Leucine (Leu) exclusively belongs to this category.
- The remaining 5 amino acids (Ile, Lys, Phe, Trp and Tyr) are both glucogenic and ketogenic.
- Some of their carbon atoms emerge in acetyl-CoA or acetoacetyl-CoA, whereas others appear in potential precursors of glucose.

Fates of carbon skeleton



Lipid metabolism and Beta oxidation

Noor Ullah

Fatty acid oxidation

- The oxidation of **long chain fatty acids to acetyl-CoA** is a central energy yielding pathway in many organisms and tissues.
- **In mammalian heart and liver**, for example, it provides as much as **80% of the energetic needs under all physiological circumstances**.
- **The electrons removed from fatty acids during oxidation pass through the respiratory chain, driving ATP synthesis.**
- The acetyl-CoA produced from the fatty acids may be completely oxidized to **CO₂** in the citric acid cycle, resulting in further energy conservation.

Fatty acid oxidation

- In some species and in some tissues, the acetyl-CoA has alternative fates.
- In liver, **acetyl-CoA may be converted to ketone bodies**—water-soluble fuels exported to the brain and other tissues when glucose is not available.
- In higher plants, acetyl-CoA serves primarily as a biosynthetic precursor, only secondarily as fuel.
- Although the biological role of fatty acid oxidation differs from organism to organism, the mechanism is essentially the same.

Fatty acid activation (conversion of a fatty acid to a fatty acyl–CoA)

- The conversion is catalyzed by **fatty acyl–CoA synthetase** and **inorganic pyrophosphatase**.
- Fatty acid activation by formation of the fatty acyl–CoA derivative occurs in two steps. The overall reaction is highly exergonic.

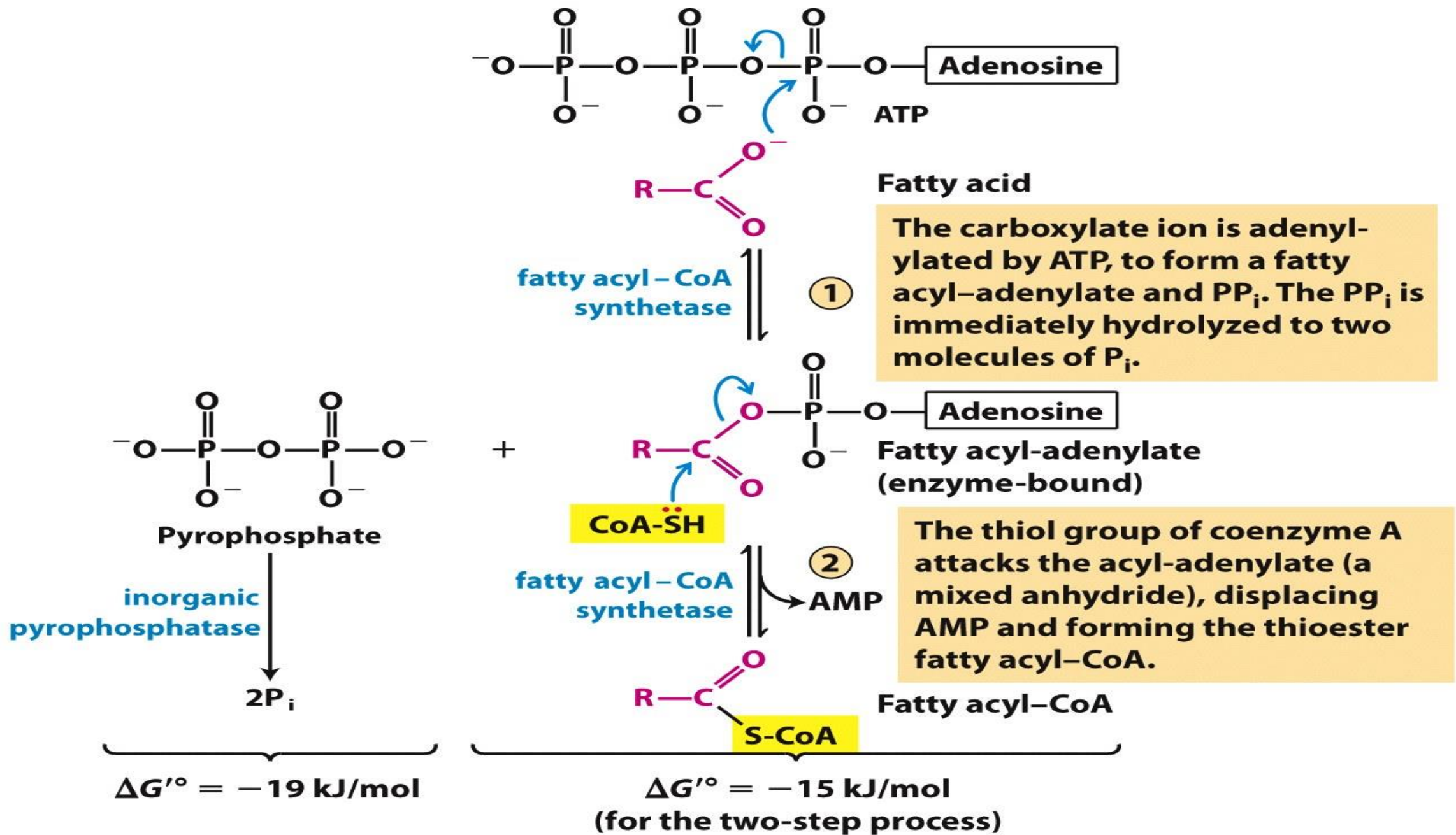


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Fatty Acids Are Activated and Transported into Mitochondria

- The enzymes of fatty acid oxidation in animal cells are located in the mitochondrial matrix.
- The fatty acids with **chain lengths of 12 or fewer carbons enter mitochondria without the help of membrane transporters.**
- **Those with 14 or more carbons- cannot pass directly through the mitochondrial membranes—they must require the carnitine shuttle- the three-step enzymatic process for transferring fatty acids into the mitochondrion—esterification to CoA, trans-esterification to carnitine followed by transport, and transesterification back to CoA—links two separate pools of coenzyme A and of fatty acyl–CoA, one in the cytosol, the other in mitochondria.**

Fatty Acids Are Activated and Transported into Mitochondria

- These pools have different functions. **Coenzyme A** in the mitochondrial matrix is largely used in oxidative degradation of pyruvate, fatty acids, and some amino acids, whereas cytosolic coenzyme A is used in the biosynthesis of fatty acids.
- Fatty acyl–CoA in the cytosolic pool can be used for membrane lipid synthesis or can be moved into the mitochondrial matrix for oxidation and ATP production.
- Conversion to the carnitine ester **commits the fatty acyl moiety to the oxidative fate.**
- **The carnitine-mediated entry process is the rate-limiting** step for oxidation of fatty acids in mitochondria and is a **regulation point** .
- Once inside the mitochondrion, the fatty acyl–CoA is acted upon by a set of enzymes in the matrix.

Fatty acid entry into mitochondria via the acyl-carnitine/carnitine transporter

- After **fatty acyl – carnitine** is formed at the outer membrane or in the intermembrane space, it moves into the matrix by **facilitated diffusion** through the transporter in the inner membrane.
- In the matrix, the acyl group is transferred to mitochondrial coenzyme A, freeing carnitine to return to the intermembrane space through the same transporter.
- **Acyltransferase I is inhibited by malonyl-CoA**, the first intermediate in fatty acid synthesis.
- This inhibition prevents the simultaneous synthesis and degradation of fatty acids.

Fatty acid are transported into mitochondria

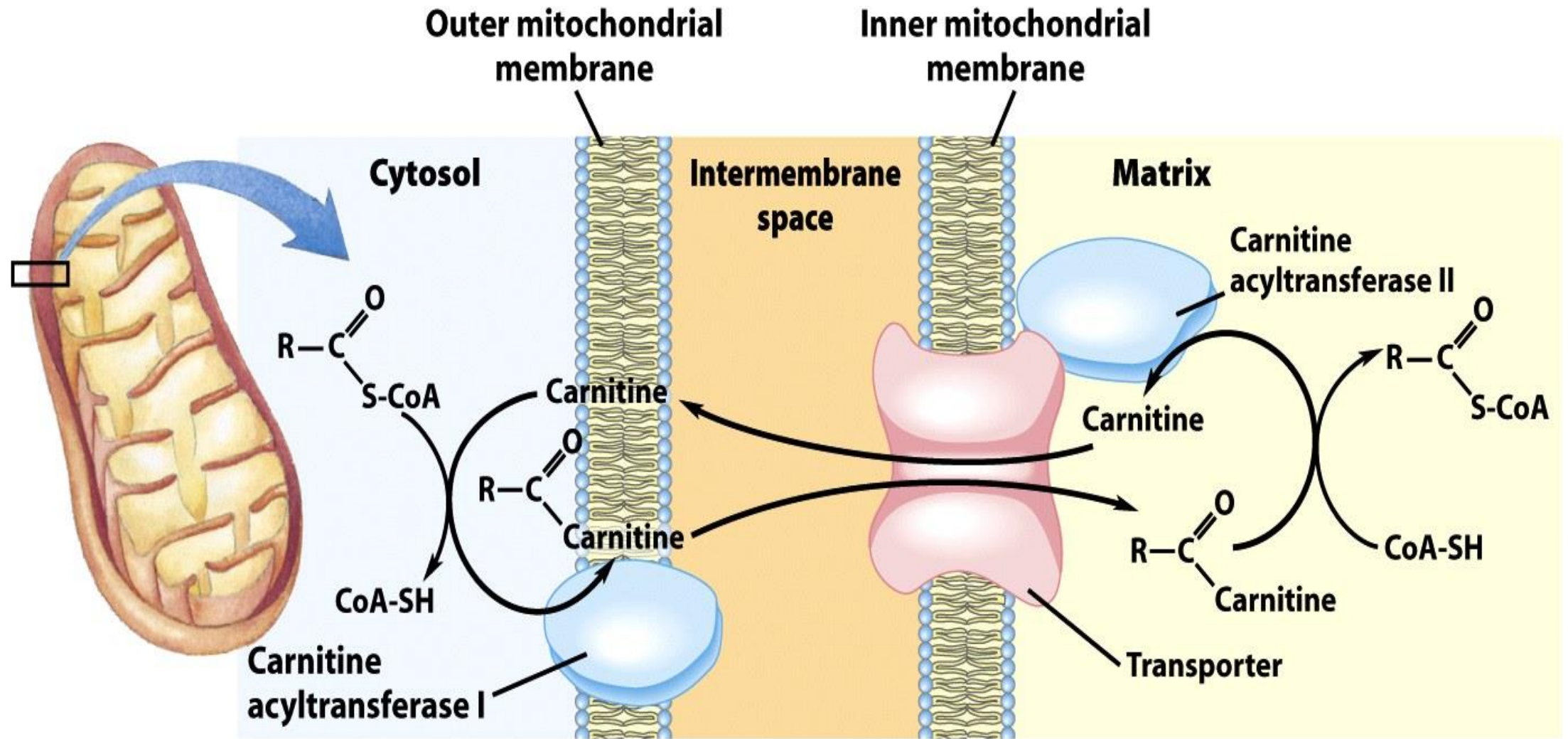


Figure 17-6

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Fatty acid oxidation

- Mitochondrial oxidation of fatty acids takes place in **three stages**:
- In the first stage— **β -Oxidation**—fatty acids undergo oxidative removal of successive **two-carbon units in the form of acetyl-CoA**, starting from the **carboxyl end of the fatty acyl chain**.
- For example, the **16-carbon palmitic acid** undergoes **seven passes** through the oxidative sequence, in each pass **losing two carbons as acetyl-CoA**.
- At the end of seven cycles the last two carbons of palmitate remain as acetyl-CoA.
- The overall result is the conversion of the **16-carbon chain of palmitate to eight two-carbon acetyl groups of acetyl-CoA molecules**.
- Formation of each acetyl-CoA requires removal of **four hydrogen atoms (two pairs of electrons and four H^+)** from the fatty acyl moiety by **dehydrogenases**.

Fatty acid oxidation

- In the second stage of fatty acid oxidation, the **acetyl groups of acetyl-CoA** are **oxidized to CO₂** in the **citric acid cycle**, which also takes place in the mitochondrial matrix.
- **In the third stage** electrons derived from the oxidations of stages 1 and 2 pass to O₂ via the mitochondrial respiratory chain, providing the energy for ATP synthesis by oxidative phosphorylation.
- The first two stages of fatty acid oxidation produce the **reduced electron carriers NADH and FADH₂**, which in the third stage donate electrons to the mitochondrial respiratory chain, through which the electrons pass to oxygen with the concomitant phosphorylation of ADP to ATP.
- The energy released by fatty acid oxidation is thus conserved as ATP.

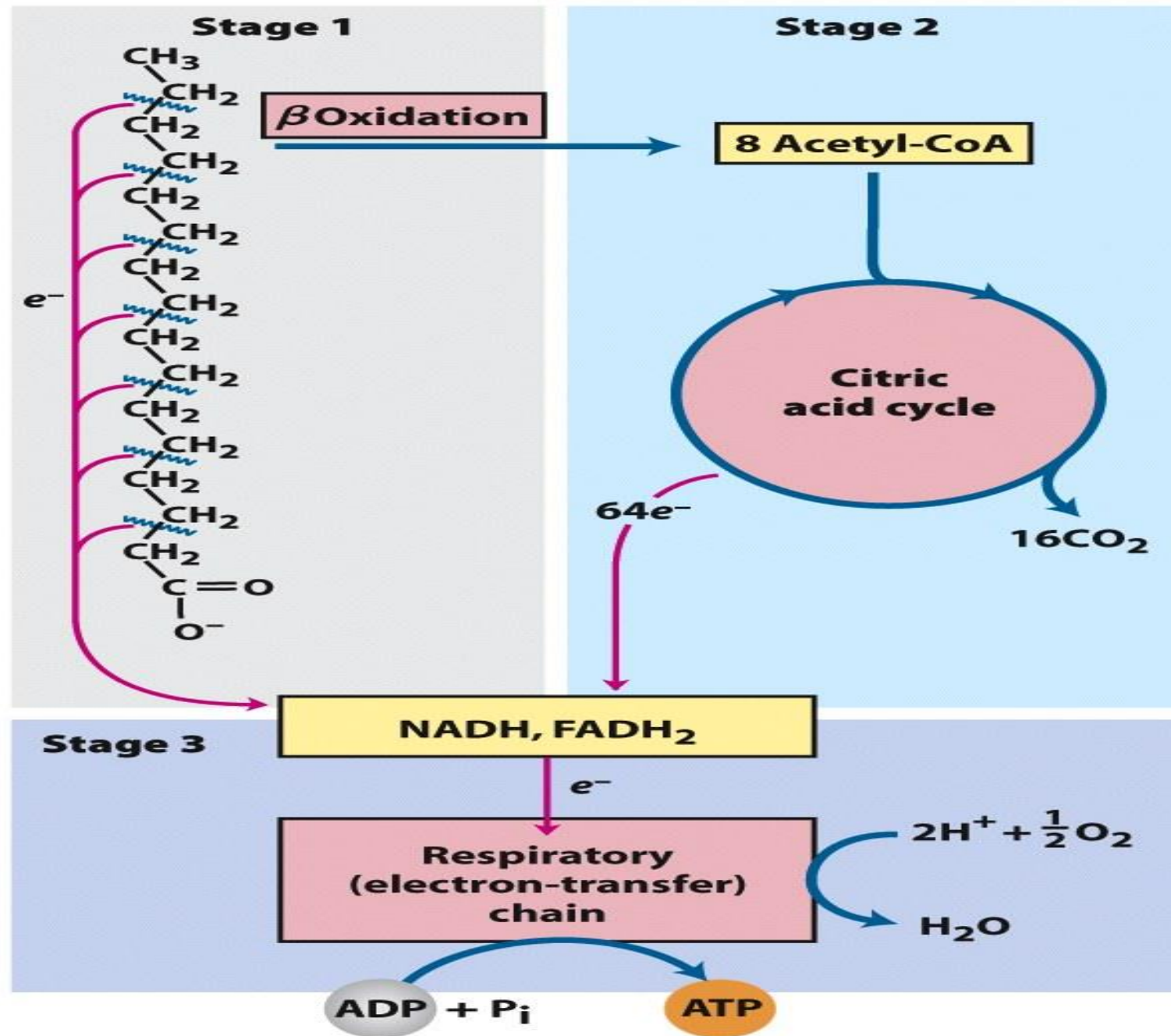


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The β -Oxidation pathway

- In each pass through this **four-step sequence**, **one acetyl residue (shaded in pink)** is removed in the form of **acetyl-CoA** from the carboxyl end of the fatty acyl chain—in this example palmitate (C_{16}), which enters as **palmitoyl-CoA**.
- Six more passes through the pathway yield seven more molecules of acetyl-CoA, the seventh arising from the last two carbon atoms of the 16-carbon chain.
- Eight molecules of acetyl-CoA are formed in all.

The β -Oxidation pathway

1. Acyl-CoA dehydrogenase: dehydrogenation of fatty-acyl CoA produces a double bond between α and β carbon.

2. Enoyl-CoA hydratase: water is added to the double bond of the trans-enoyl-CoA, analogous to the fumarase reaction

3. β -Hydroxyacyl-CoA dehydrogenase: NADH formed and then transfer electron to NADH dehydrogenase in ETS. Analogous to the malate dehydrogenase

4. Thiolase: thiolysis reaction

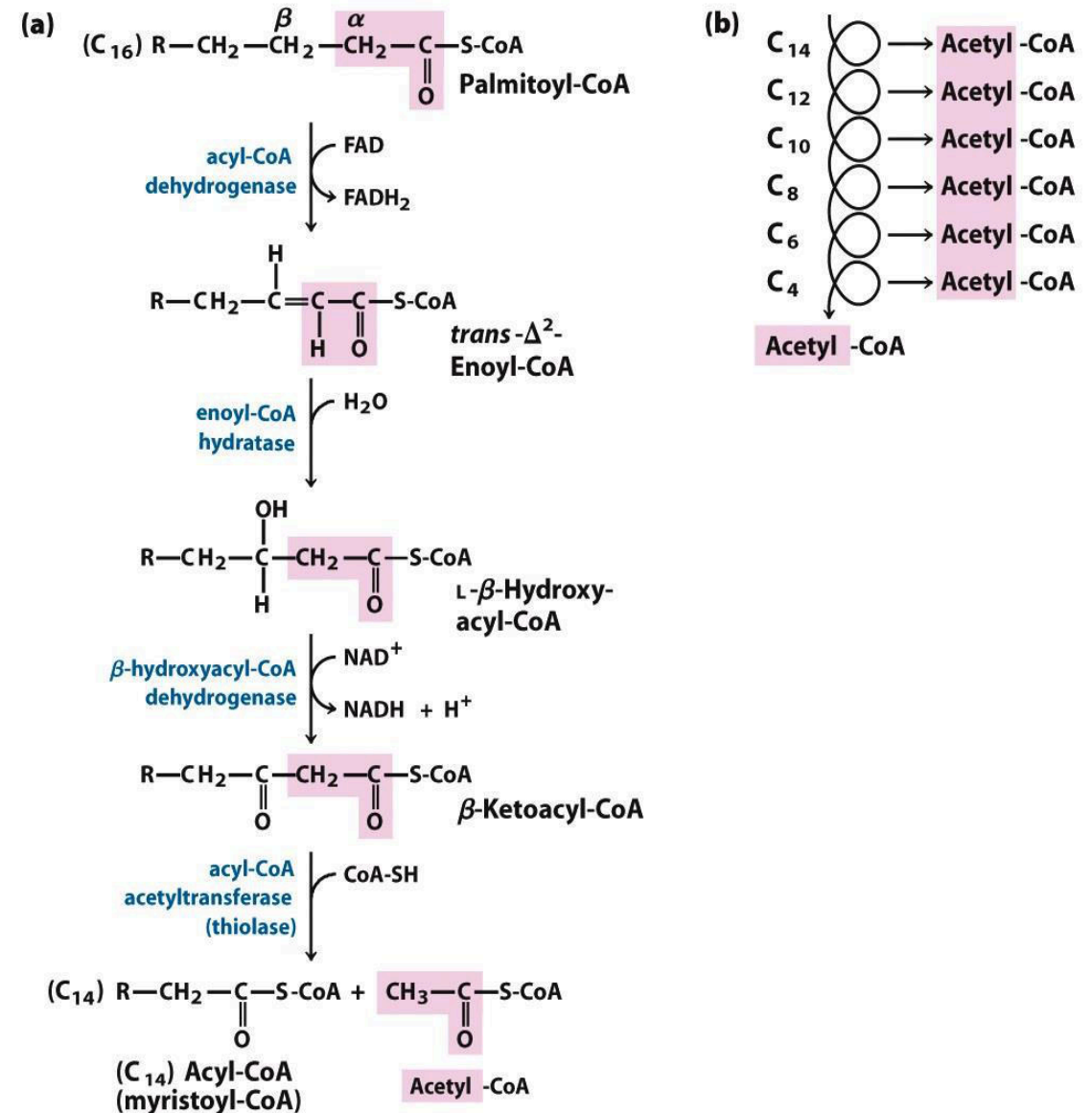


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TABLE 17-1 Yield of ATP during Oxidation of One Molecule of Palmitoyl-CoA to CO₂ and H₂O

Enzyme catalyzing the oxidation step	Number of NADH or FADH ₂ formed	Number of ATP ultimately formed*
Acyl-CoA dehydrogenase	7 FADH ₂	10.5
β -Hydroxyacyl-CoA dehydrogenase	7 NADH	17.5
Isocitrate dehydrogenase	8 NADH	20
α -Ketoglutarate dehydrogenase	8 NADH	20
Succinyl-CoA synthetase		8†
Succinate dehydrogenase	8 FADH ₂	12
Malate dehydrogenase	8 NADH	20
Total		108

*These calculations assume that mitochondrial oxidative phosphorylation produces 1.5 ATP per FADH₂ oxidized and 2.5 ATP per NADH oxidized.

Oxidation of unsaturated fatty acid (Monounsaturated)

- **Oxidation of a monounsaturated fatty acid.** Oleic acid, as **oleoyl-CoA (Δ^9)**, is the example used here.
- Oxidation requires an additional enzyme, **enoyl-CoA isomerase**, to reposition the double bond, converting the cis isomer to a trans isomer, a normal intermediate in β oxidation.

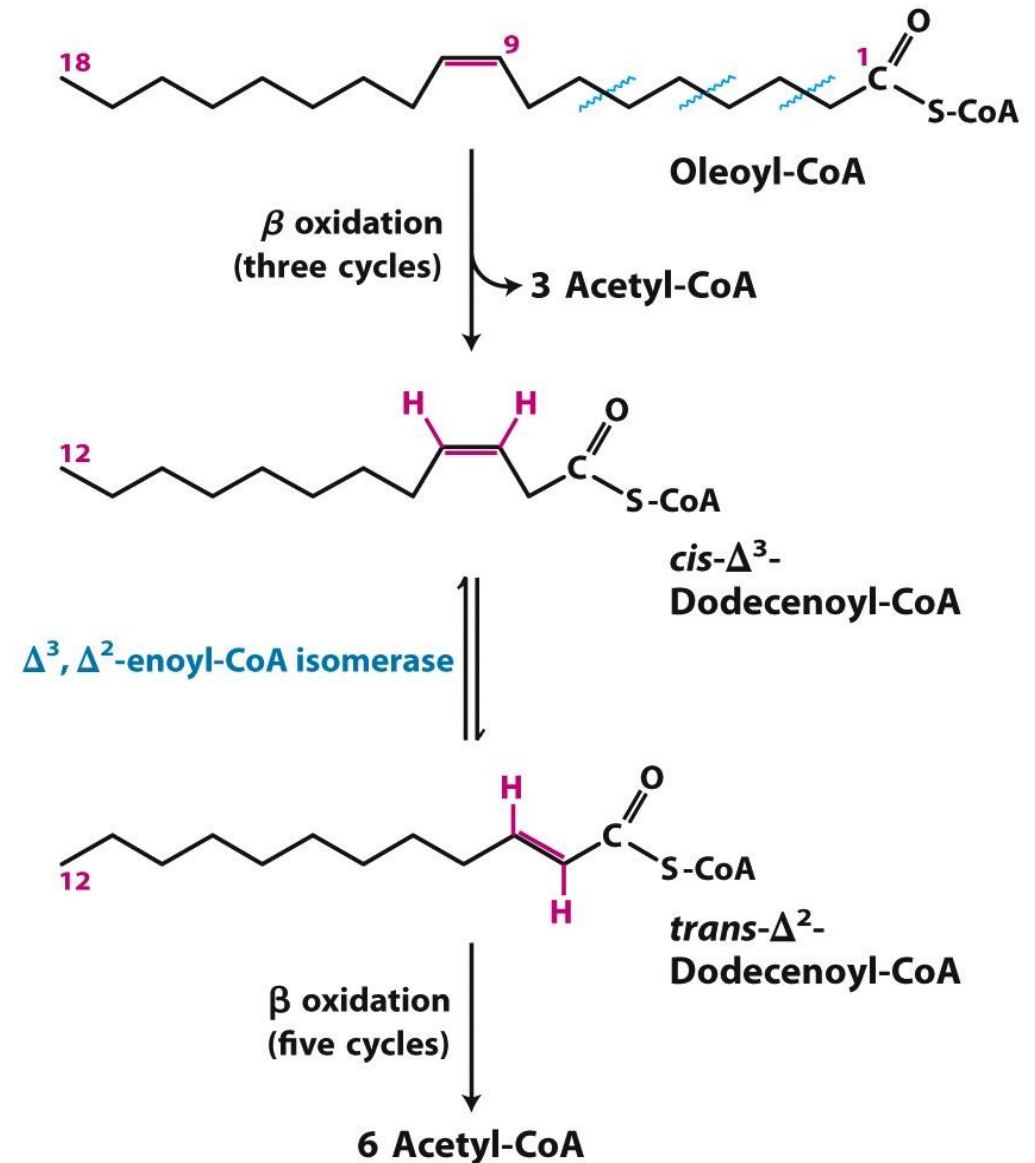


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Oxidation of unsaturated fatty acid (polyunsaturated)

- The example here is **linoleic acid**, as **linoleoyl-CoA ($\Delta^{9,12}$)**.
- Oxidation requires a second auxiliary enzyme in addition to **enoyl-CoA isomerase**: **NADPH-dependent 2,4-dienoyl-CoA reductase**.
- The combined action of these two enzymes converts a ***trans*- Δ^2 ,*cis*- Δ^4 -dienoyl-CoA** intermediate to the ***trans*- Δ^2 -enoyl-CoA** substrate necessary for β oxidation.

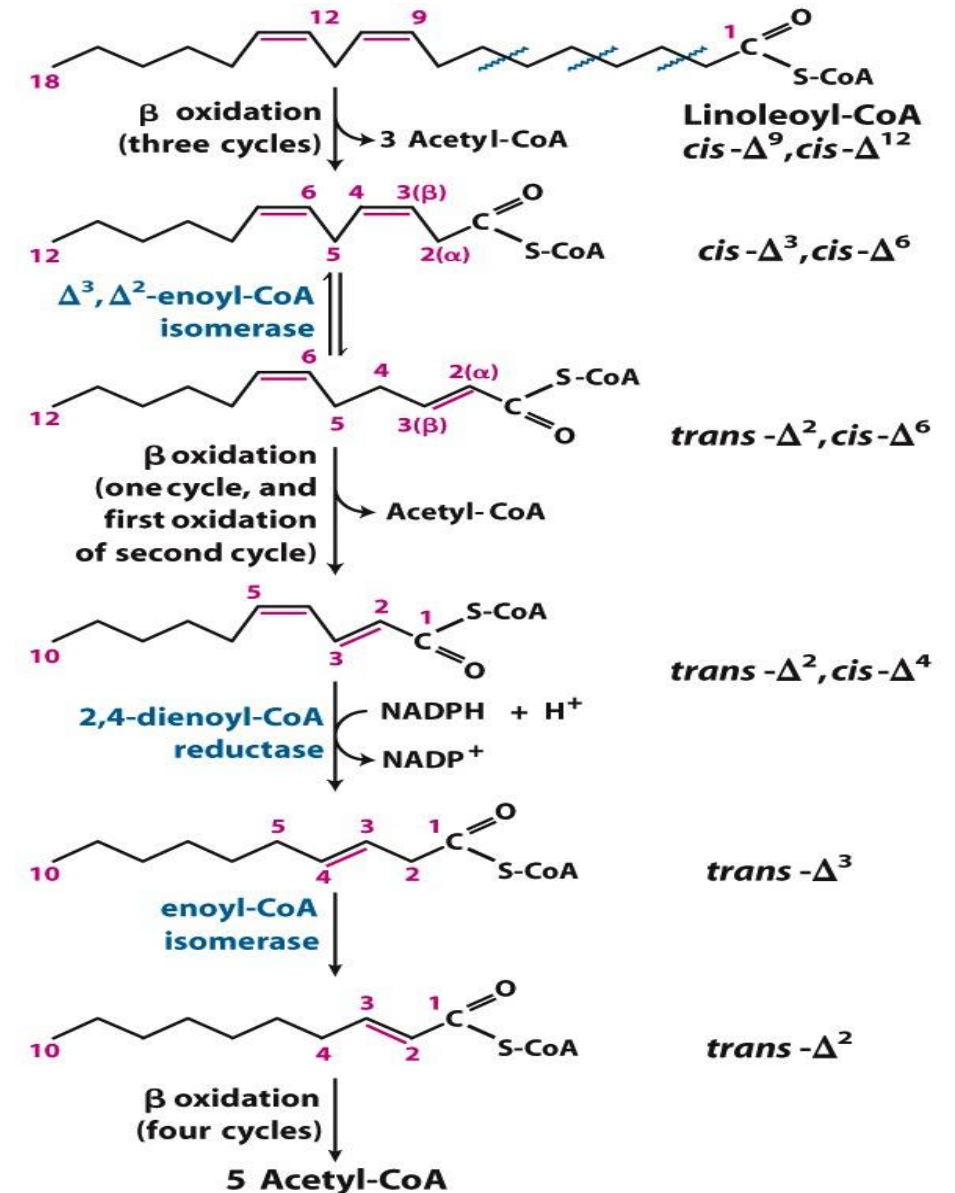


Figure 17-10

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Omega-oxidation of fatty acids- endoplasmic reticulum

- This alternative to **β oxidation** begins with oxidation of the carbon most distant from the **α carbon—the ω (omega) carbon.**
- The substrate is usually a **medium-chain fatty acid**; shown here is **lauric acid (laurate)**.
- This pathway is generally not the major route for oxidative catabolism of fatty acids.

Omega-oxidation of fatty acids- endoplasmic reticulum

1. There is another pathway in some species, including vertebrates, that involves oxidation of **the omega carbon**.
2. The enzymes unique to omega oxidation are located in the **ER of liver and kidney**
3. The preferred substrates are **fatty acids of 10 to 12 carbon**.
4. The first step introduces a **hydroxyl group on to omega carbon by mixed function oxidase** involved in cytochrome p450 and NADPH
5. Produce a carboxyl group at each end.
6. In each pass through the β -oxidation pathway, the **double-ended fatty acid yields dicarboxylic acid such as succinic acid, which can enter citric acid cycle**.

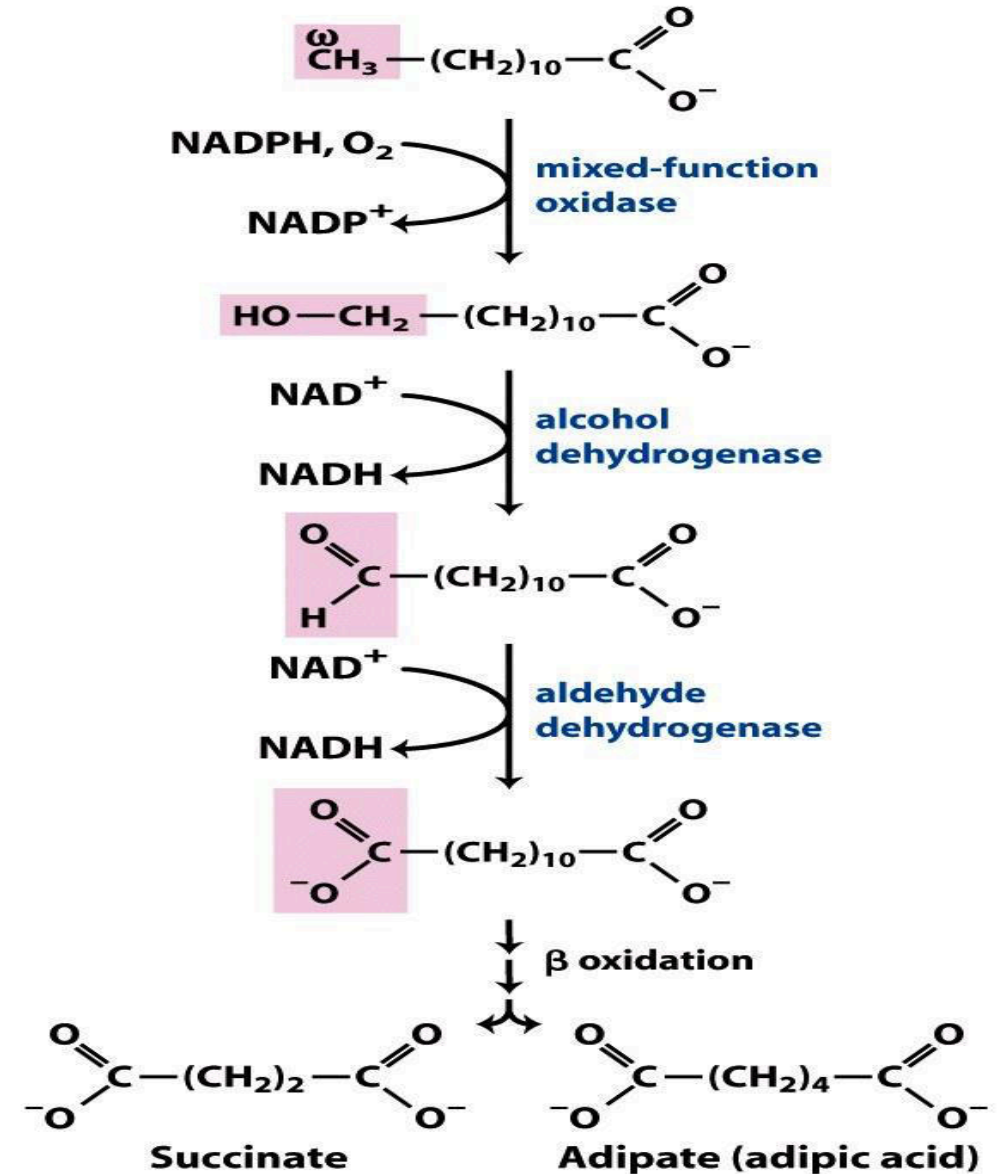


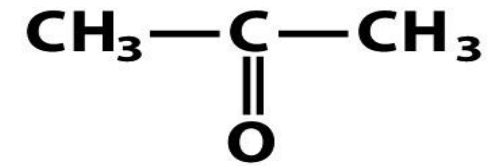
Figure 17-16
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Fatty acid oxidation is tightly regulated

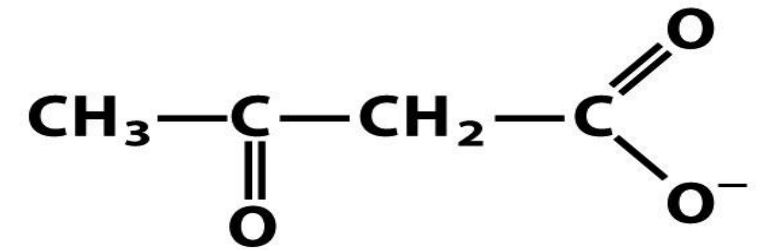
1. The **three step process (carnitine shuttle)** is **rate-limiting** for fatty acid oxidation and is an important point of regulation
2. **Malonyl-CoA**, the first intermediate in the cytosolic biosynthesis of fatty acid from acetyl-CoA, **inhibits carnitine acyltransferase I**.
3. **[NADH]/[NAD⁺] ratio is high, β -hydroxyl-CoA dehydrogenase is inhibited.**
4. High concentrations of **acetyl-CoA** **inhibit thiolase**.
5. **[AMP]** is high, AMPK (AMP-activated protein kinase) is activated and phosphorylates acetyl-CoA carboxylase, which catalyzes the formation of malonyl-CoA, thereby decreasing the malonyl-CoA concentration. → allowing β -oxidation
6. Transcription factors turn on the synthesis of proteins for lipid catabolism:
 - **PPAR α (peroxisome proliferator-activated receptors)** acts in muscle, adipose tissue, and liver to turn on a set of genes essential for fatty acid oxidation.
 - **Glucagon → [cAMP] → CREB (cAMP response element binding) →** turn on a set of genes essential for fatty acid.

Ketone bodies

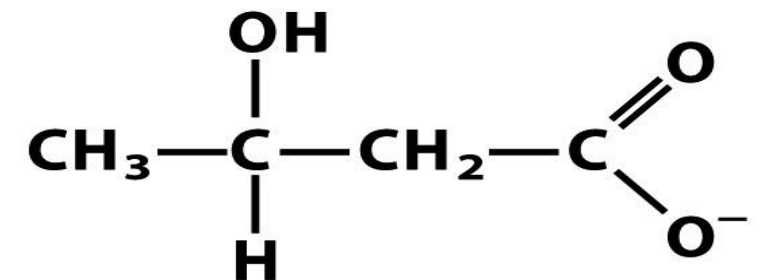
- In humans and most other mammals, acetyl-CoA formed in the liver during oxidation of fatty acids can either **enter the citric acid cycle** or undergo conversion to the “**ketone bodies**,”
- **acetone, acetoacetate, and D- β-hydroxybutyrate.**
- **Acetone**, produced in smaller quantities than the other ketone bodies, **is exhaled.**
- **Acetoacetate** and **D- β-hydroxybutyrate** are transported by the blood to extrahepatic tissues, where they are **converted to acetyl-CoA** and **oxidized in the citric acid cycle**, providing much of the energy required by tissues such as skeletal and heart muscle and the renal cortex.
- **The brain**, which preferentially **uses glucose as fuel**, can adapt to the use of acetoacetate or **D- β-hydroxybutyrate**, when glucose is unavailable.
- The production and export of ketone bodies from the liver to extrahepatic tissues allows continued oxidation of fatty acids in the liver when acetyl-CoA is not being oxidized in the citric acid cycle.



Acetone



Acetoacetate



D-β-Hydroxybutyrate

Ketone bodies are overproduced in diabetes and starvation

1. During starvation, **gluconeogenesis depletes citric acid cycle intermediates, diverting acetyl-CoA to ketone body production.**
2. Untreated DM → insulin down → cannot uptake glucose → malonyl-Co A fall → fatty acid oxidation increase → ketone body → acidosis (ketosis)
3. Individual on very low-calorie diets, using the fats stored in adipose tissue as their major energy source, also have increased level of ketone bodies in their blood and urine.

Ketone body formation and export from the liver

- Conditions that promote **gluconeogenesis** (untreated diabetes, severely reduced food intake) slow the citric acid cycle (by drawing off oxaloacetate) and enhance the conversion of acetyl-CoA to acetoacetate.
- The released coenzyme A allows continued β oxidation of fatty acids.

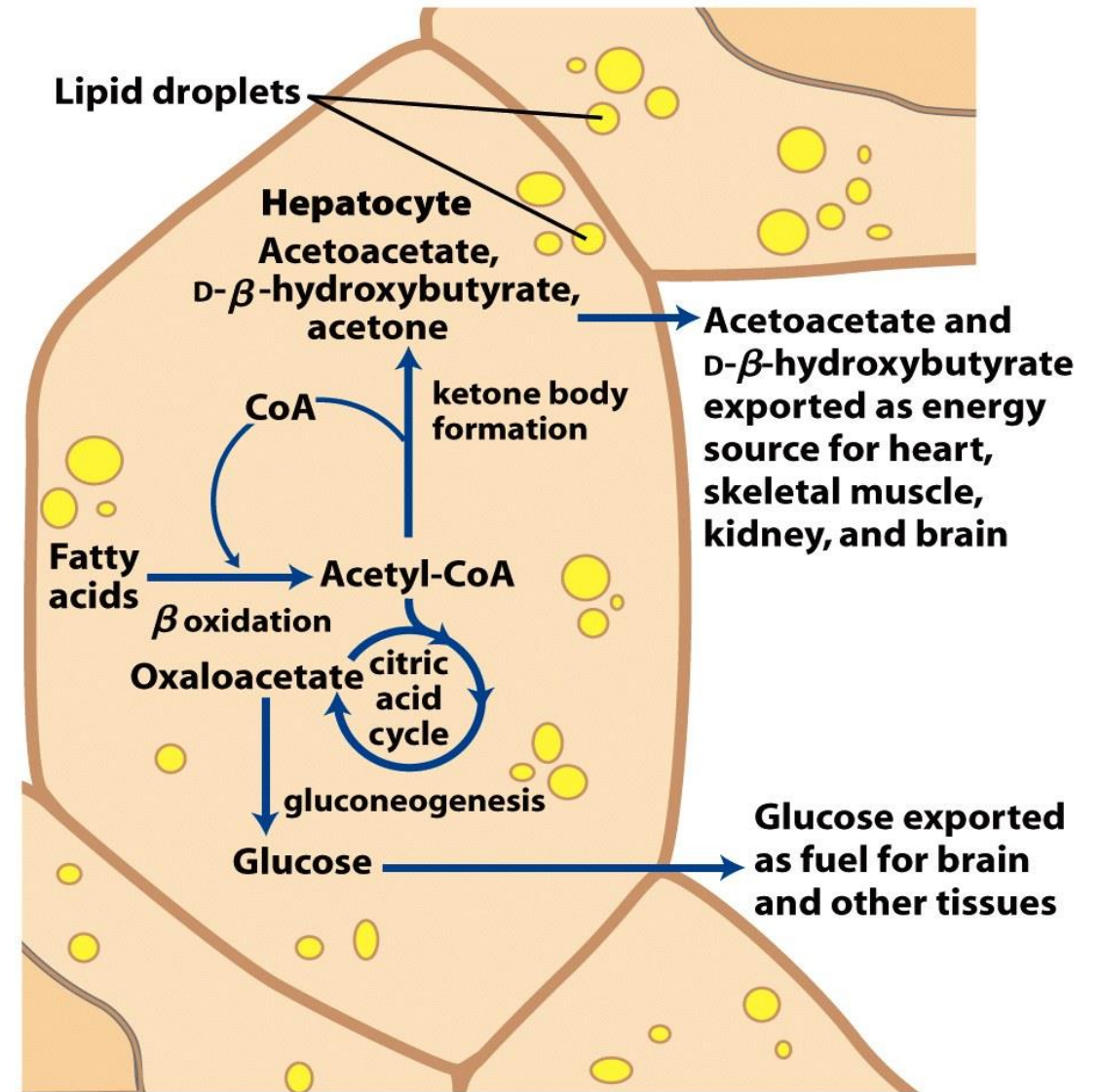
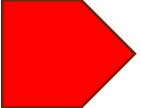


Figure 17-20
Lehninger Principles of Biochemistry, Fifth Edition
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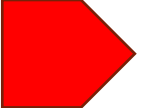
Carbohydrate Metabolism

By: Abdur Rehman

Lecturer IPMS KMU

Subject: Biochemistry (2nd semester)





Carbohydrates Metabolism

- The monosaccharide glucose is the central molecule in carbohydrate metabolism since all the major pathways of carbohydrate metabolism relate to it (Fig.13.1).
- The other monosaccharides important in carbohydrate metabolism are fructose, galactose and mannose
- Liver plays a key role in monitoring and stabilizing blood glucose levels.
 - Thus, liver may be appropriately considered as **glucostat monitor**



OXIDATIVE PATHWAYS

Glycolysis ←

Hexose mono-
phosphate shunt ←

Uronic acid
pathway ←

G
L
U
C
O
S
E

SYNTHETIC PATHWAYS

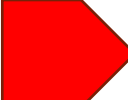
→ Other carbohydrates
(galactose, fructose)

→ Glycogenesis

→ Lipogenesis
(synthesis of fat)

→ Non-essential
amino acids

Fig. 13.1 : Overview of glucose metabolism.
(Note : For majority of the pathways, glucose participates as glucose 6-phosphate).



glycolysis

**Break down of
glucose**

**The oxidation of
glucose to
pyruvate and
lactate.**

Citric acid cycle (Kreb Cycl/TCA

**The oxidation of
acetylCoA to CO₂**

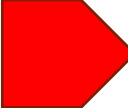
**Commono xidative
pathway for
carbohydrates, fats
or amino acids,
through acetyl CoA**

Gluconeogenesis

**The synthesis of
glucose from non-
carbohydrate
precursors**

**(e.g.amino acids,
glycerol etc.)**





Glycogenesis

The formation of glycogen
from glucose.

Glycogenolysis

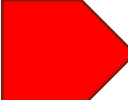
The breakdown of glycogen to
glucose

Hexose
monophosphate
shunt

alternative to
glycolysis and
TCA cycle

oxidation of glucose
directly to carbon
dioxide and water





Uronic acid pathway

Glucose is converted to glucuronic acid, pentoses and ascorbic acid in animal

an alternative oxidative pathway for glucose.

Galactose metabolism

conversion of galactose to glucose

synthesis of lactose.

Fructose metabolism

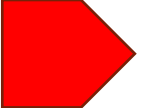
The oxidation of fructose to pyruvate

fructose and glucose metabolism.

Amino sugar metabolism

synthesis of amino sugars and mucopolyscharides



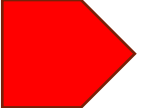


Entry of glucose into cells

- Glucose concentration is very low in the cells compared to plasma (for humans < 100 mg/dl)
- Two specific transport systems are recognized for the entry of glucose into the cells (No simple diffusion)

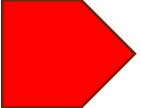
1. Insulin-independent transport system of glucose:

- This is a carrier mediated uptake of glucose not dependant on insulin
 - Mainly in hepatocytes, erythrocytes (red blood cells) and brain
- 



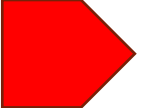
2.insulin-dependent transport system : Insulin increases the number and promotes the activity of GLUT-4 in skeletal muscle and adipose tissue

- In type 2 diabetes mellitus, insulin resistance is observed in these tissues due to the reduction in the quantity of GLUT-4 in insulin deficiency
 - In recent years, at least six glucose transporters (GLUT-1 to GLUT-5 and GLUT-7) in the cell membranes have been identified.
 - They exhibit tissue specificity. For instance, GLUT-1 is abundant in erythrocytes
- 



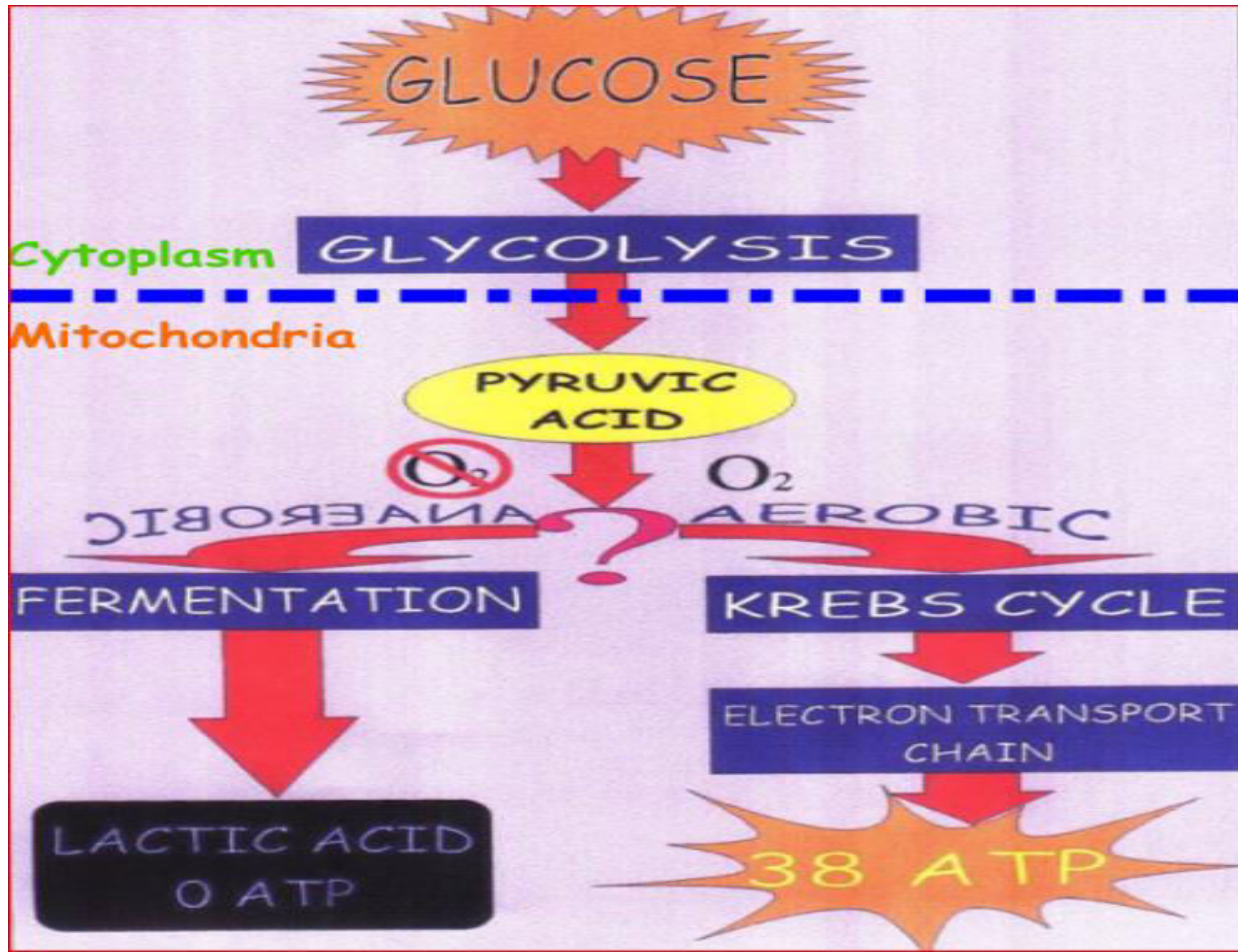
GLYCOLYSIS

- Glycolysis is derived from the Greek words (glycose—sweet or sugar; lysis—dissolution)
 - Glycolysis is defined as the sequence of reactions converting glucose (or glycogen) to pyruvate or lactate, with the production of ATP
 - This pathway is often referred to as **Embden-Meyerhof pathway (E.M. pathway)** in honor of the two biochemists who made a major contribution to the knowledge of glycolysis
- 



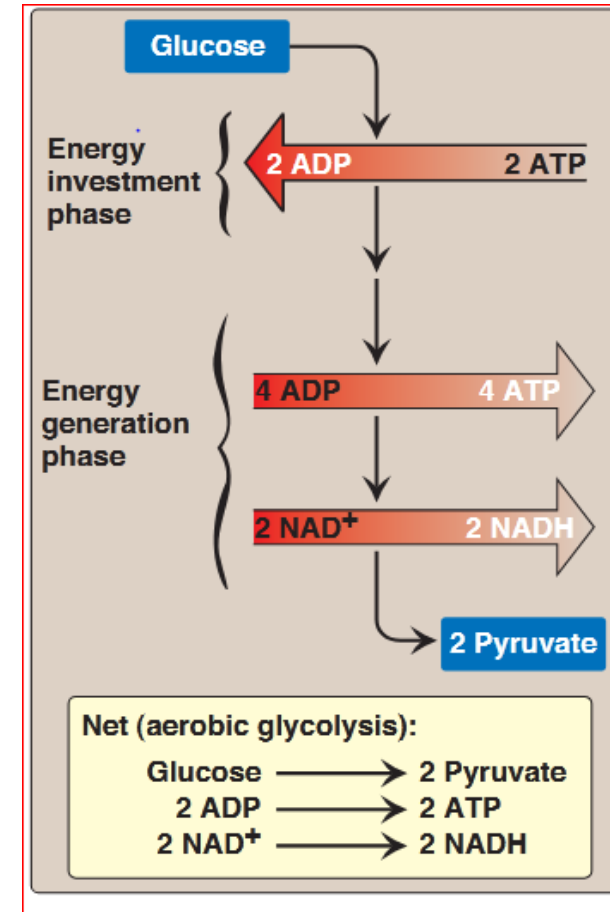
Salient Features of glycolysis

- Glycolysis takes place in all cells of body and enzyme present in cytosomal fraction of cell.
 - Glycolysis can be **aerobic** having end product of pyruvate and Co_2 and H_2O on further oxidation and can be anaerobic having end product lactate.
 - Tissue lacking mitochondria such as cornea, red blood cells etc, glycolysis is major pathway of ATP production
 - Glycolysis is very essential for brain which is dependent on glucose for energy.
 - Glycolysis is a central metabolic pathway with many of its intermediates providing branch point to other pathways such as synthesis of fats and amino acid
 - Reversal of glycolysis with few rearrangement steps result in
- 



Reactions of Glycolysis

The pathway can be divided into three distinct phases



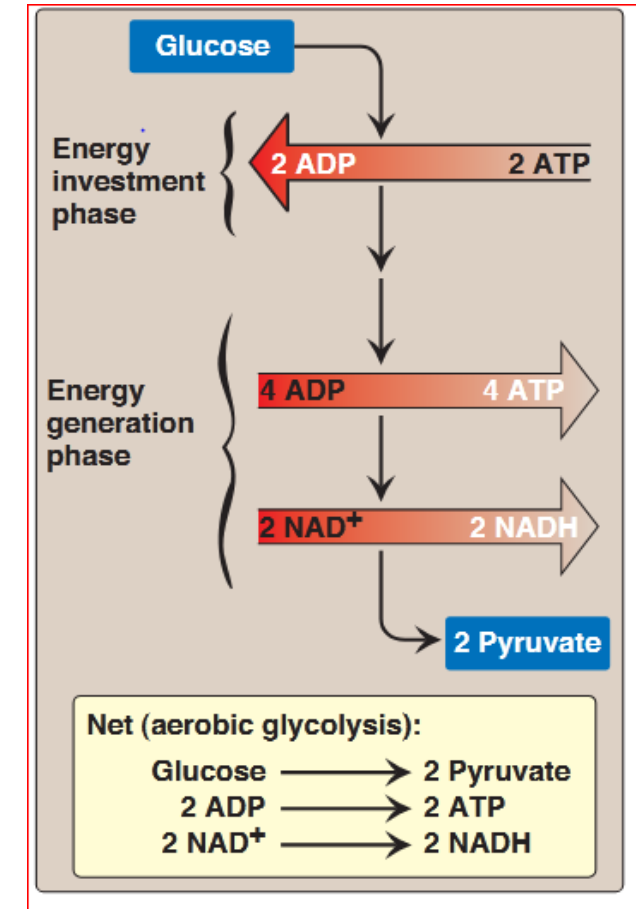
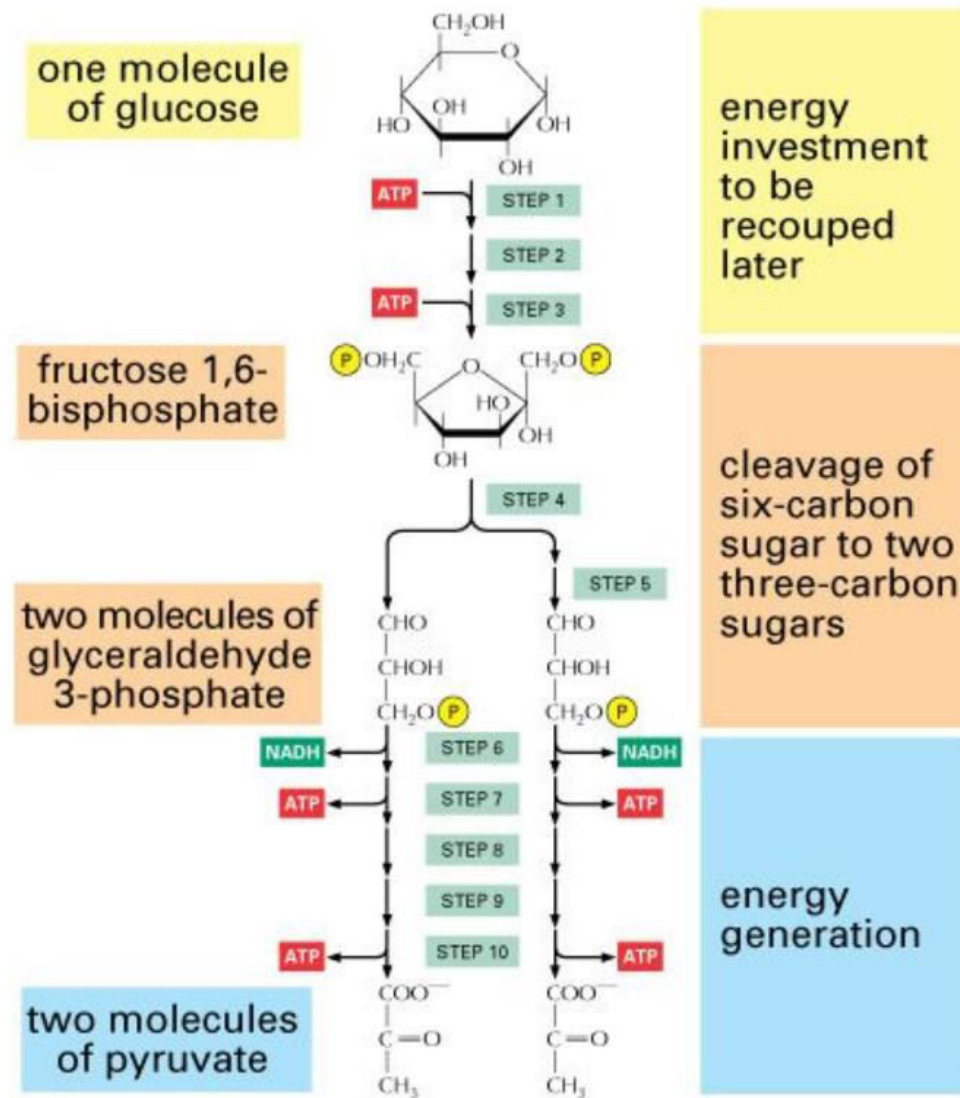
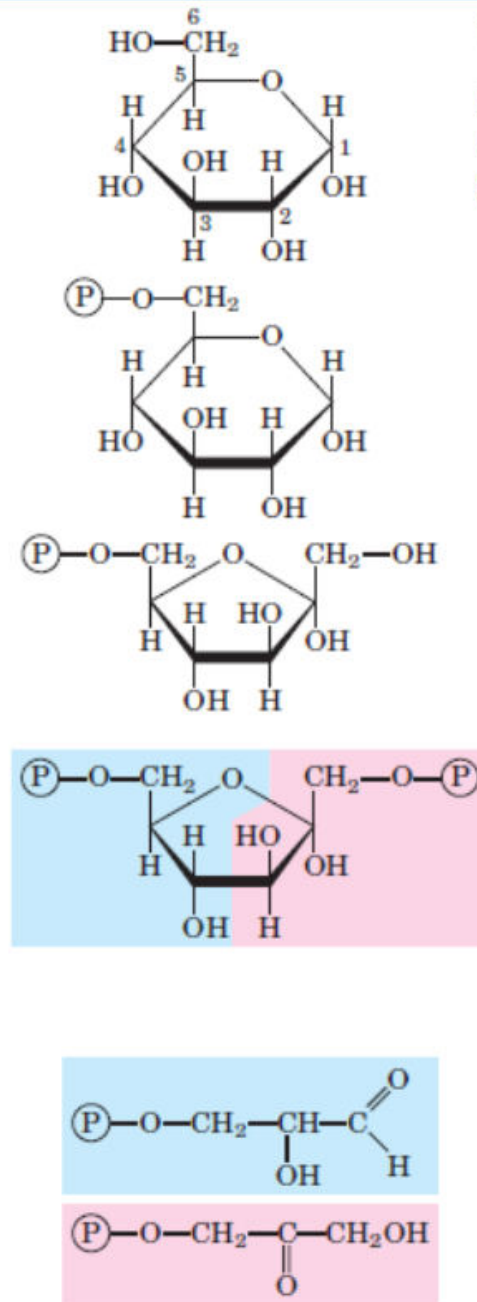
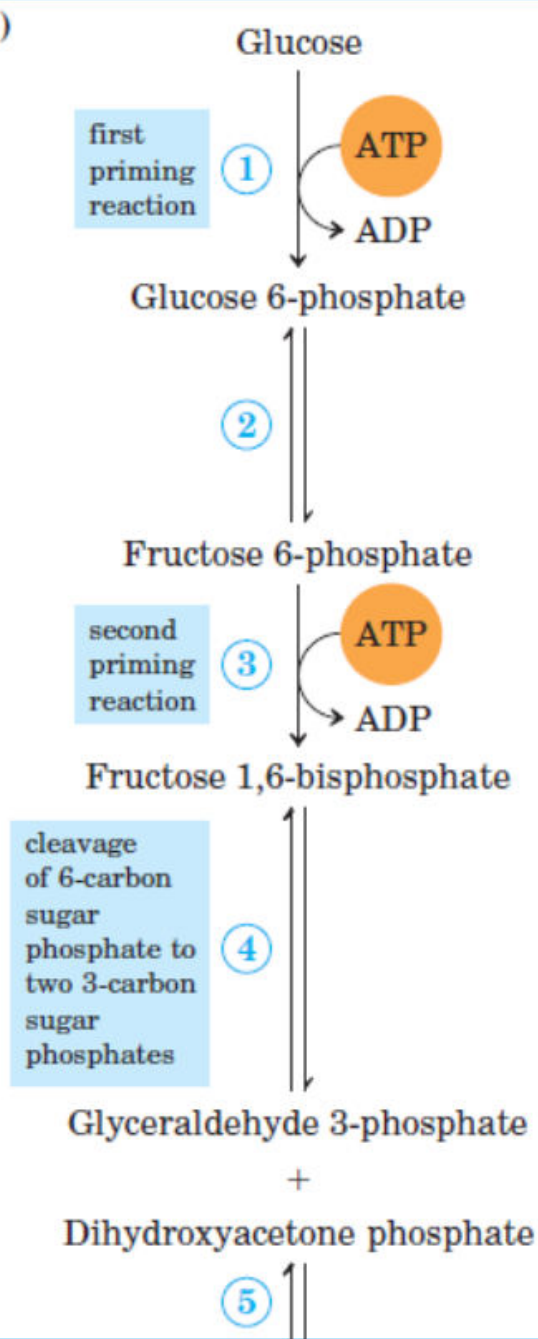


Figure 13-3 Essential Cell Biology, 2/e. (© 2004 Garland Science)

(a)

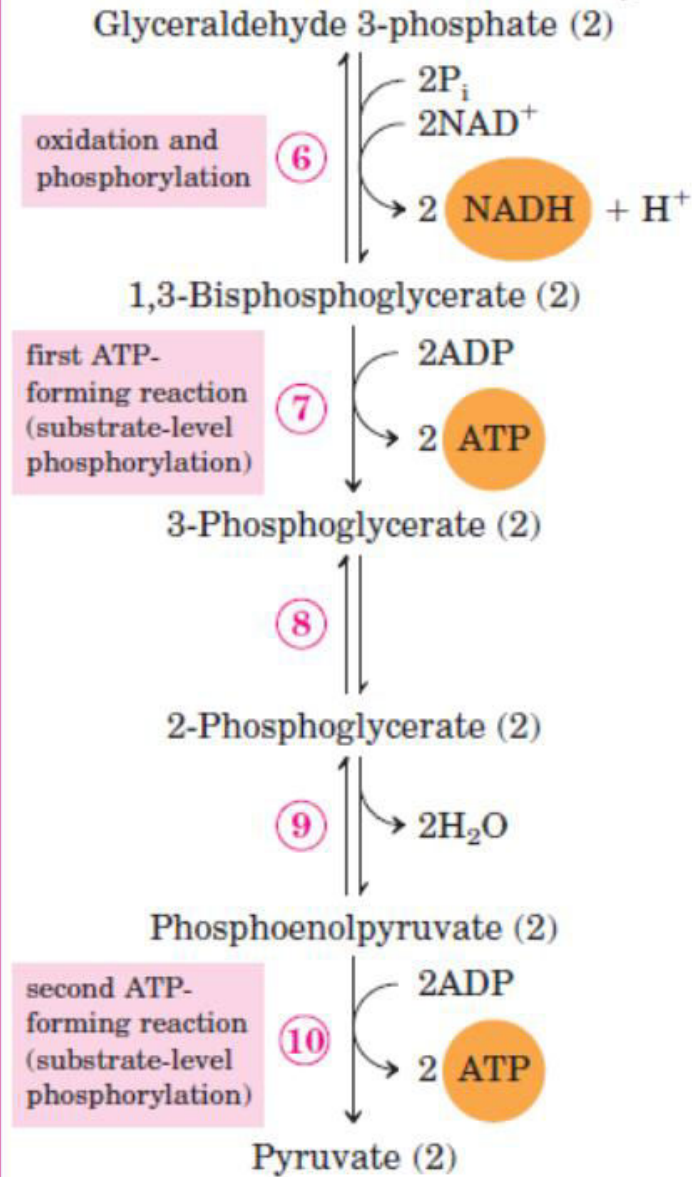


Preparatory phase

Phosphorylation of glucose and its conversion to glyceraldehyde 3-phosphate

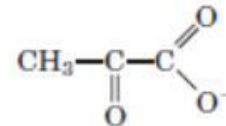
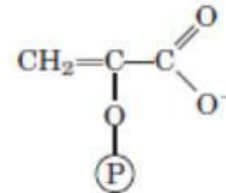
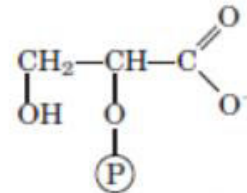
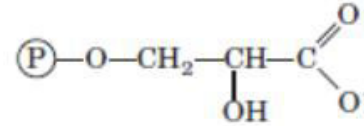
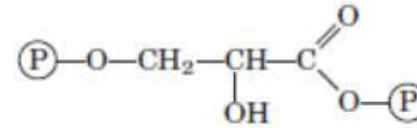
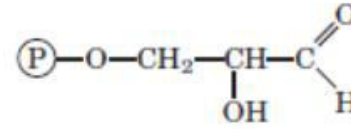
- ① Hexokinase
- ② Phosphohexose isomerase
- ③ Phosphofructokinase-1
- ④ Aldolase
- ⑤ Triose phosphate isomerase

(b)



Payoff phase

Oxidative conversion of glyceraldehyde 3-phosphate to pyruvate and the coupled formation of ATP and NADH



(6) Glyceraldehyde 3-phosphate dehydrogenase

(7) Phosphoglycerate kinase

(8) Phosphoglycerate mutase

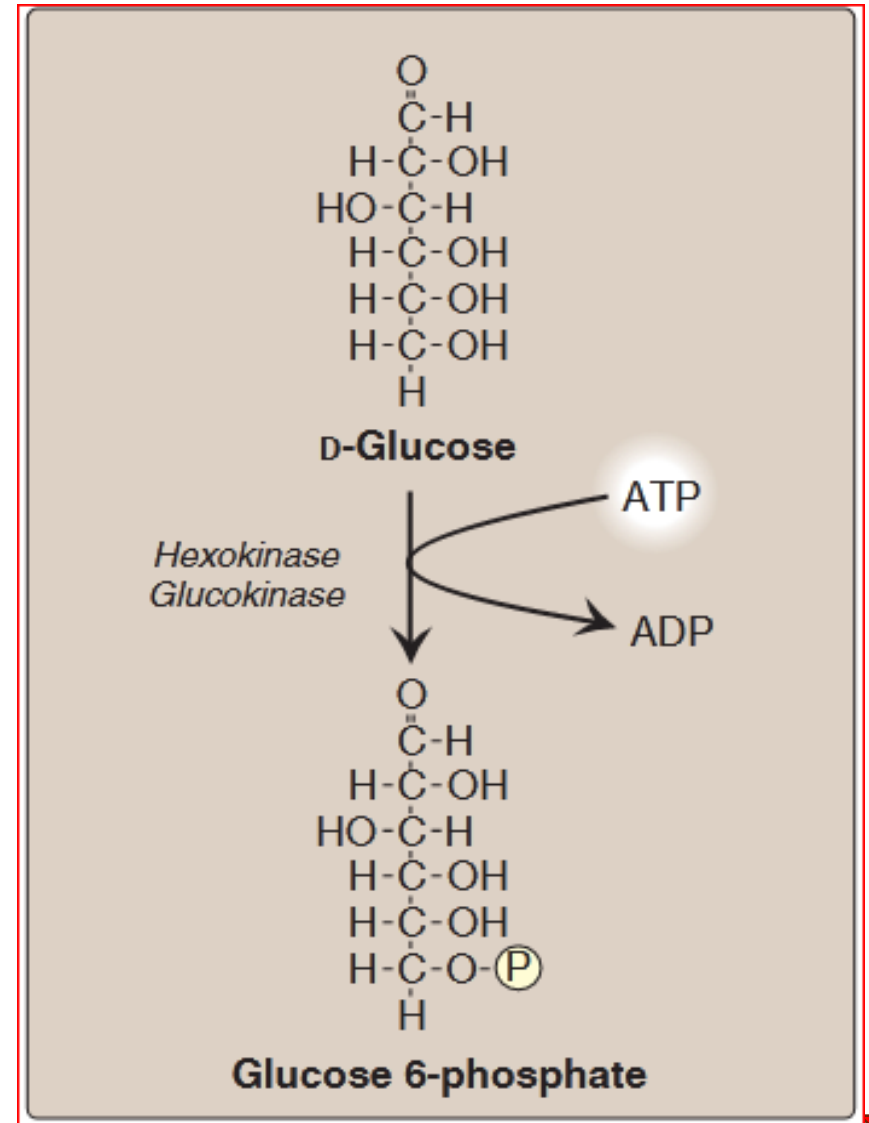
(9) Enolase

(10) Pyruvate kinase

A. Energy Investment Phase

1. Glucose Phosphorylated to Glucose 6 Phosphate:

- Glucose 6-phosphate is impermeable to the cell membrane.
- It is a central molecule with a variety of metabolic fates—glycolysis, glycogenesis, gluco-neogenesis and pentose phosphate pathway.
- This is an irreversible reaction, dependent on ATP and Mg^{2+}

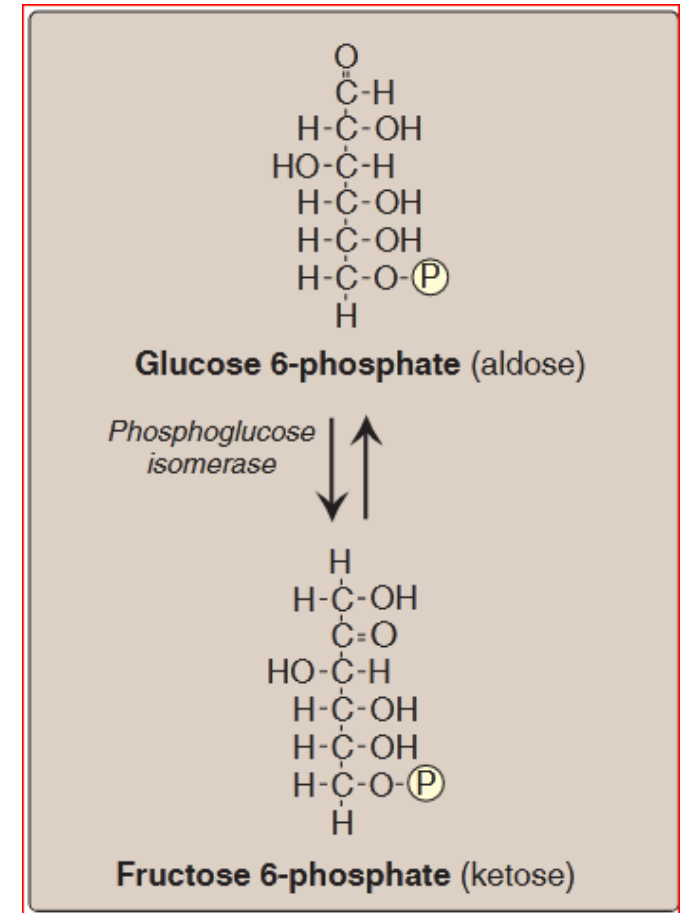


ROLE OF HEXOKINASE AND GLUCOKINASE

	Glucokinaase	Hexokinase
1. Site	Liver only	All tissue cells
2. Affinity to glucose	Low affinity (high K_m) i.e. it acts only in the presence of high blood glucose concentration.	High affinity (low K_m) i.e. it acts even in the presence of low blood glucose concentration.
3. Substrate	Glucose only	Glucose, galactose and fructose
4. Effect of insulin	Induces synthesis of glucokinase.	No effect
5. Effect of glucose-6-p	No effect	Allosterically inhibits hexokinase
6. Function	Acts in liver after meals. It removes glucose coming in portal circulation, converting it into glucose -6-phosphate.	It phosphorylates glucose inside the body cells. This makes glucose concentration more in blood than inside the cells. This leads to continuous supply of glucose for the tissues even in the presence of low blood glucose concentration.

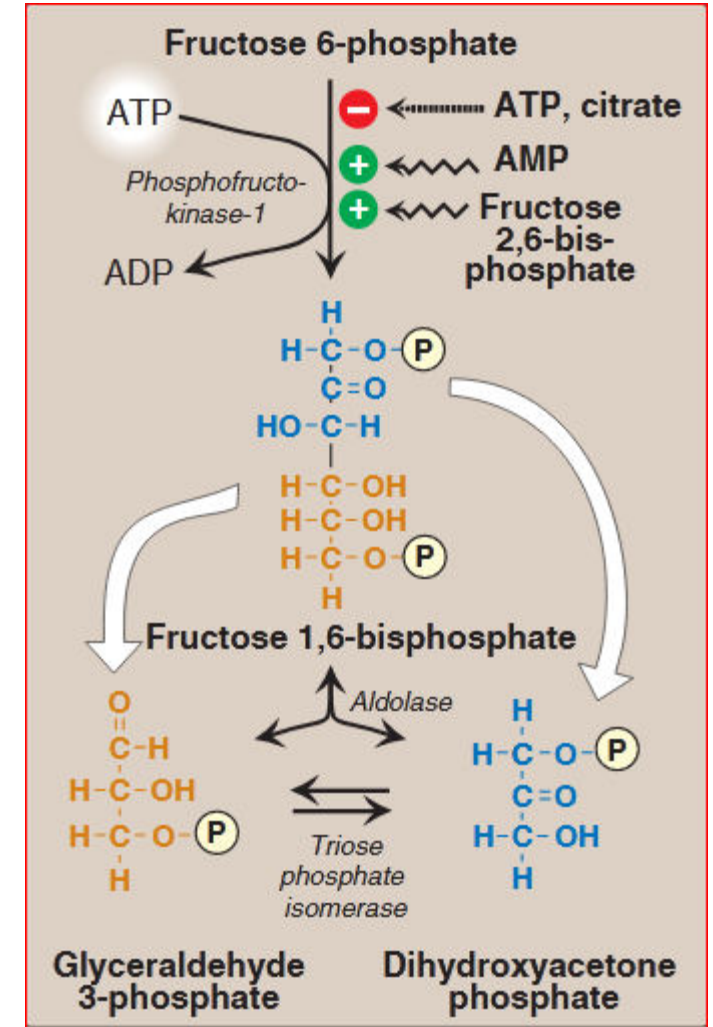
2. Isomerization of glucose 6-phosphate:

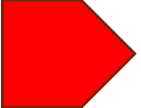
- The isomerization of glucose 6-phosphate to fructose 6-phosphate is catalyzed by phospho glucose isomerase
- The reaction is readily reversible and is not a rate-limiting or regulated step.



3. Phosphorylation of fructose 6-phosphate:

- Fructose 6-phosphate is phosphorylated to fructose 1,6-bisphosphate by phosphofructokinase (PFK).
- This is an irreversible and a regulatory step in glycolysis.





Regulation of Phosphofructo kinase (PFK1)

- PFK-1 is inhibited allosterically by elevated levels of ATP (already energy rich so no need)
 - Conversely, PFK-1 is activated allosterically by high concentrations of AMP (indicating energy depleting)
 - Fructose 2,6-bisphosphate is the most potent activator of PFK-1, it is able to activate the enzyme even when ATP levels are high.
 - During the well-fed state, decreased levels of glucagon and elevated levels of insulin, cause an increase in fructose 2,6-bis-phosphate and, thus, in the rate of glycolysis in the liver
 - During starvation elevated levels of glucagon and low levels of insulin result in low 2,6 bis-phosphate, so decrease glycolysis and increase gluconeogenesis
- 

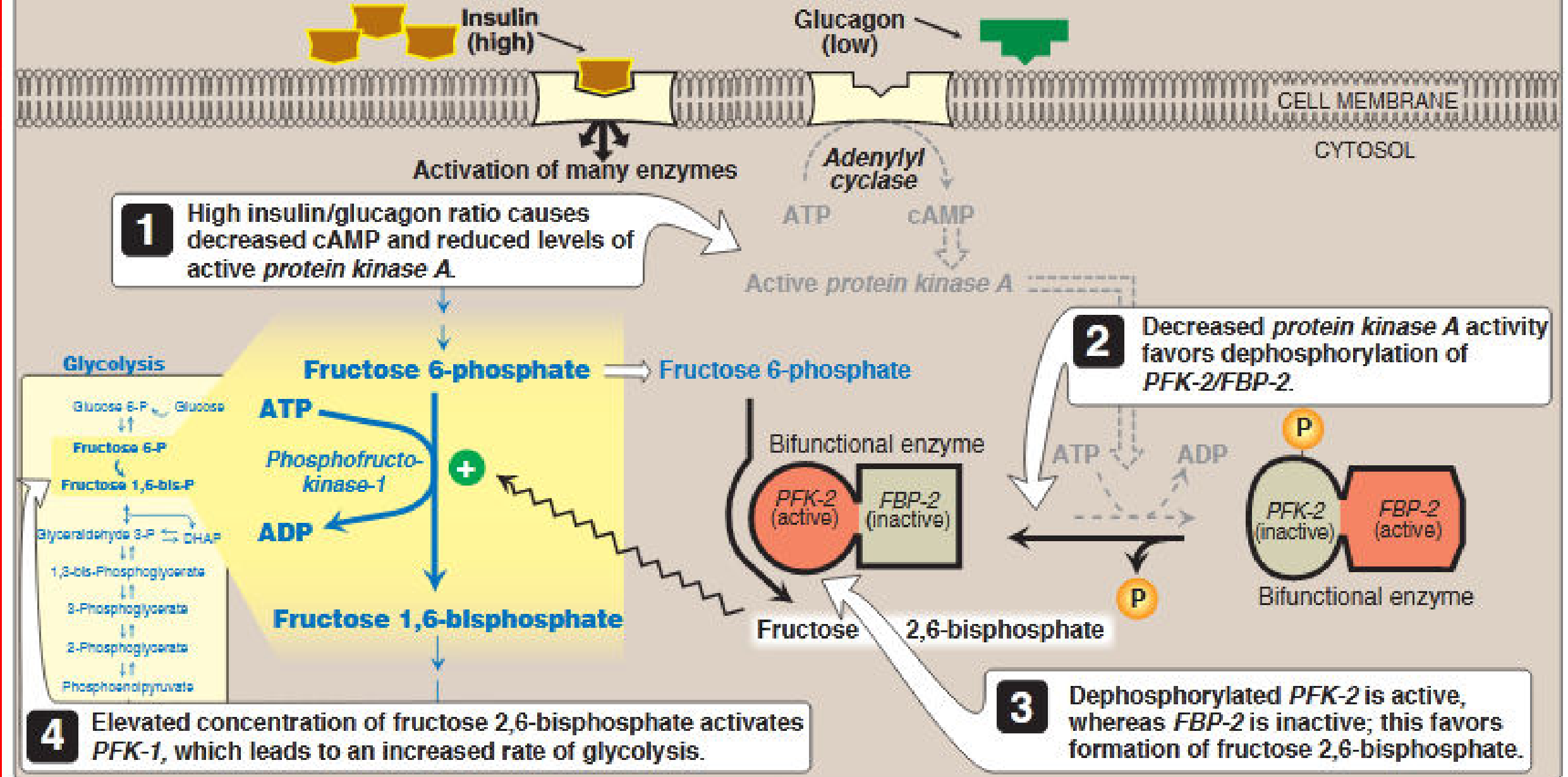


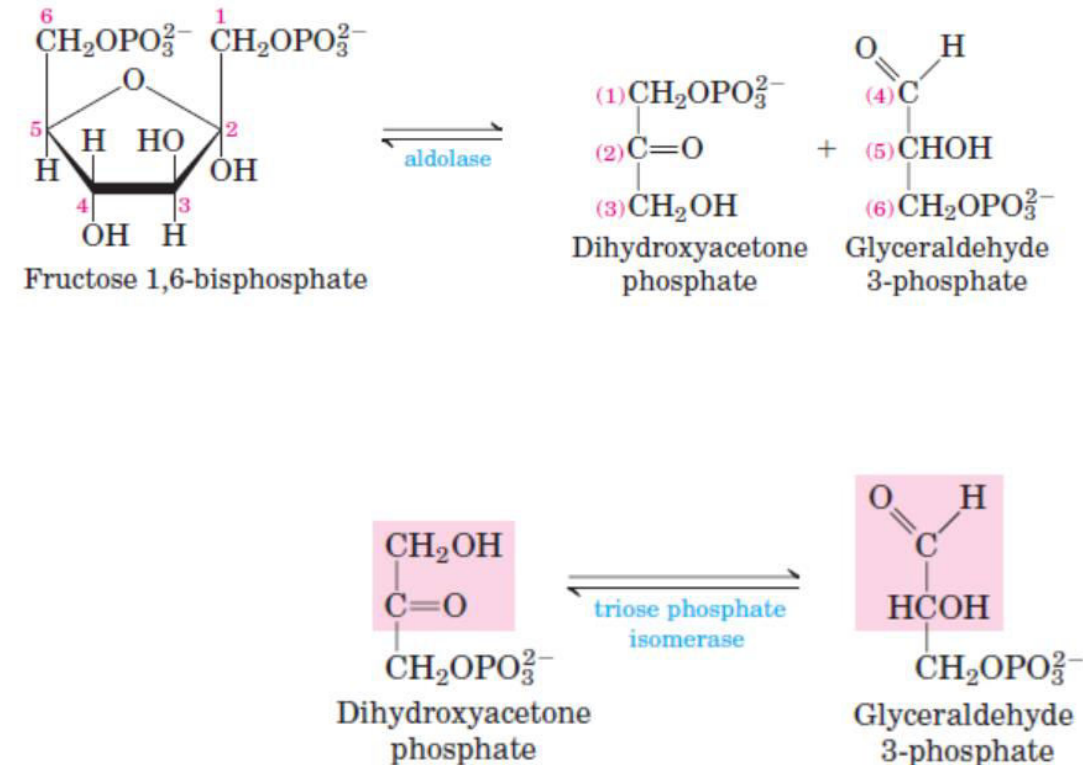
Figure 8.17

Effect of elevated insulin concentration on the intracellular concentration of fructose 2,6-bisphosphate in liver.

PFK-2 = phosphofructokinase-2 *FBP-2* = fructose biphosphatase-2

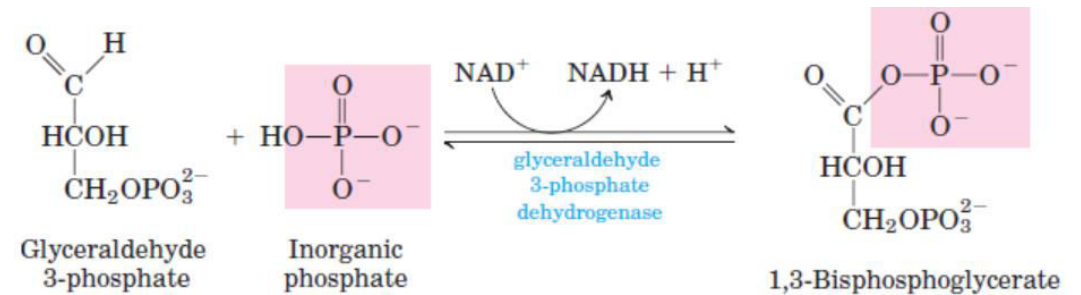
4. Cleavage of fructose 1,6-bisphosphate:

- Aldolase cleaves fructose 1,6-bisphosphate to dihydroxy acetonephosphate and glyceraldehyde 3-phosphate
- The reaction is reversible and not regulated.
- Triose phosphate isomerase interconverts dihydroxyacetone phosphate and glyceraldehyde 3-phosphate
- Dihydroxy -acetone phosphate must be isomerized to glyceraldehyde 3-phosphate for further metabolism by the glycolytic pathway.



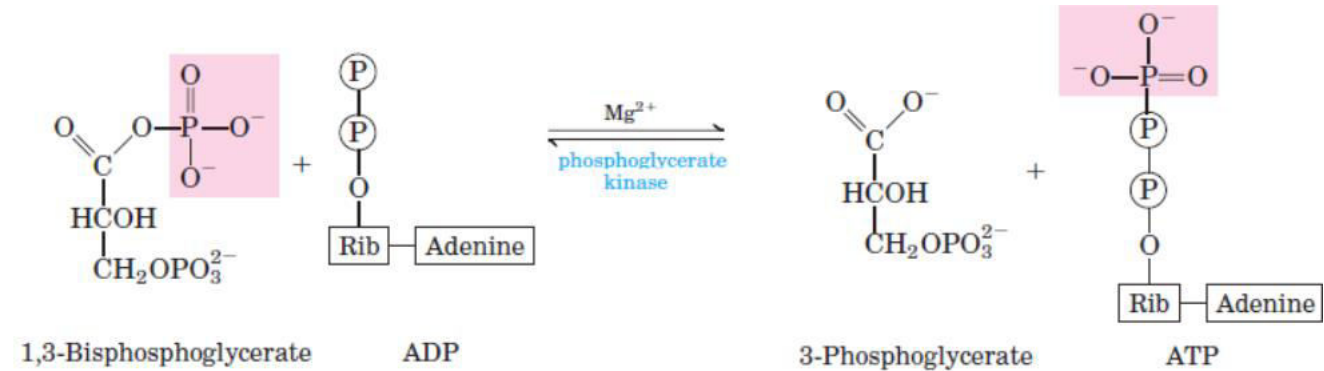
5. Dehydrogenation of Glyceraldehyde-3-Phosphate

- Glyceraldehyde-3-phosphate is oxidised to a high energy compound 1,3-bisphosphoglycerate by enzyme glyceraldehyde-3-phosphate dehydrogenase
- It is a NAD dependent reversible reaction which generates NADH

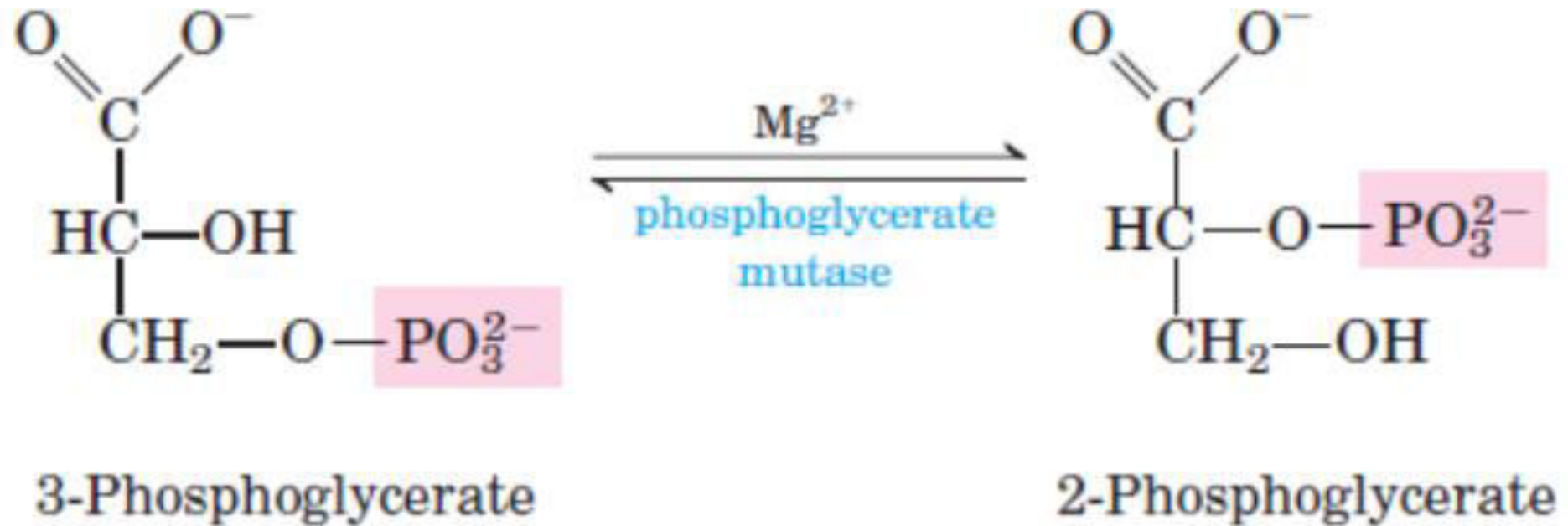


6. Conversion of 1,3-Bisphosphoglycerate to 3-Phosphoglycerate

- The enzyme **phosphoglycerate kinase** transfers the high-energy phosphoryl group from the carboxyl group of 1,3-bisphosphoglycerate to ADP, forming ATP and 3-phosphoglycerate
- This is only kinase reaction in glycolysis, which is reversible
- This step generates ATP at substrate level phosphorylation

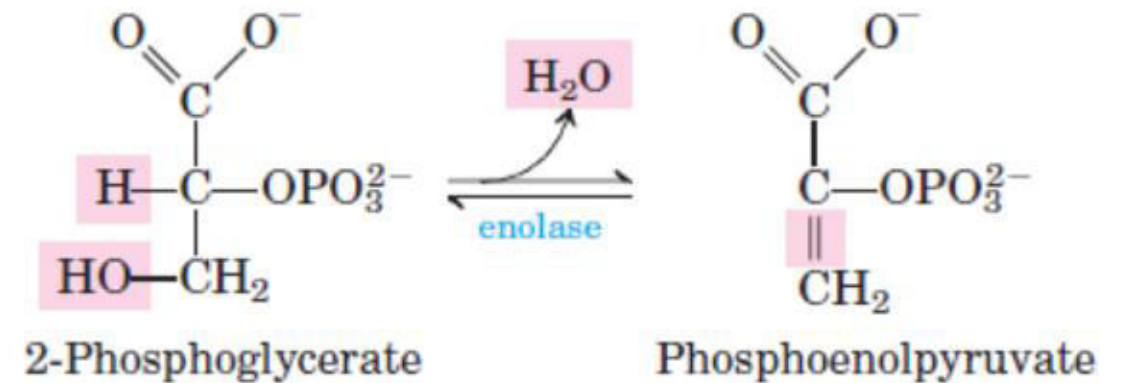


7-Inter-Molecular Shift of Phosphate Group:



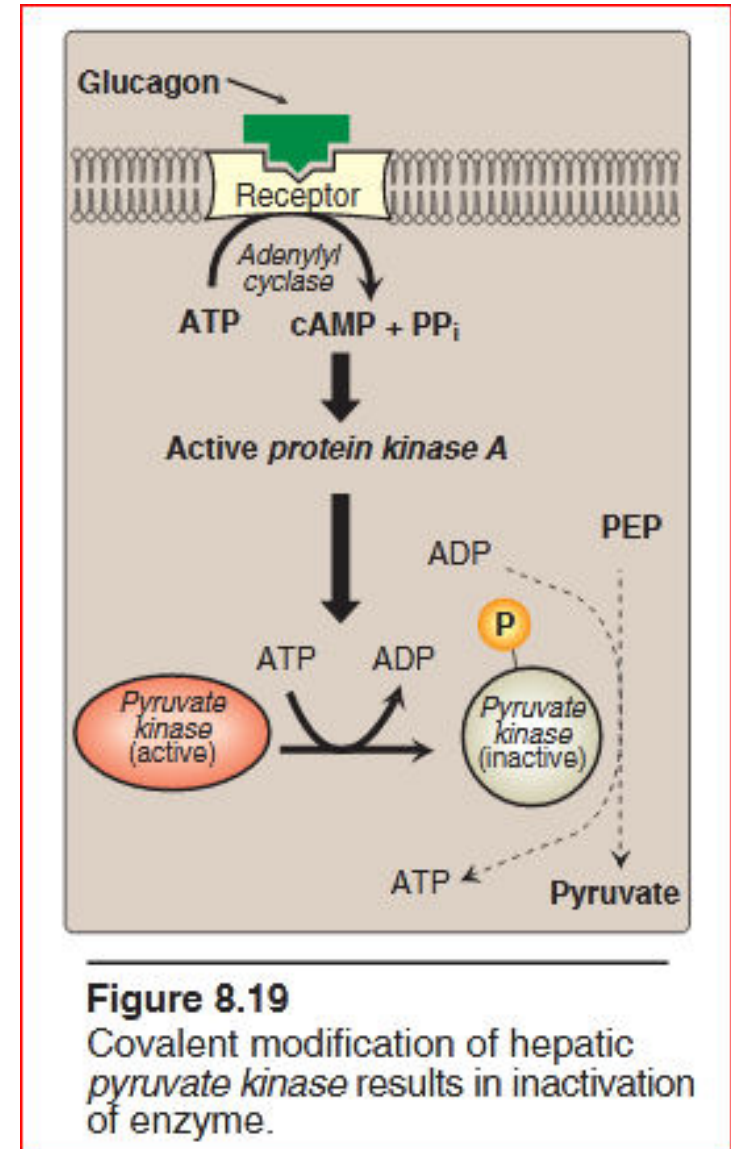
8-Dehydration of 2-Phosphoglycerate

- Enolase is dependent on Mn^{2+} and Mg^{2+}
- Fluoride inhibits enolase. This property can be used when it is required to prevent glycolysis in blood prior to the estimation of blood glucose



9-Fate of PEP and pyruvate

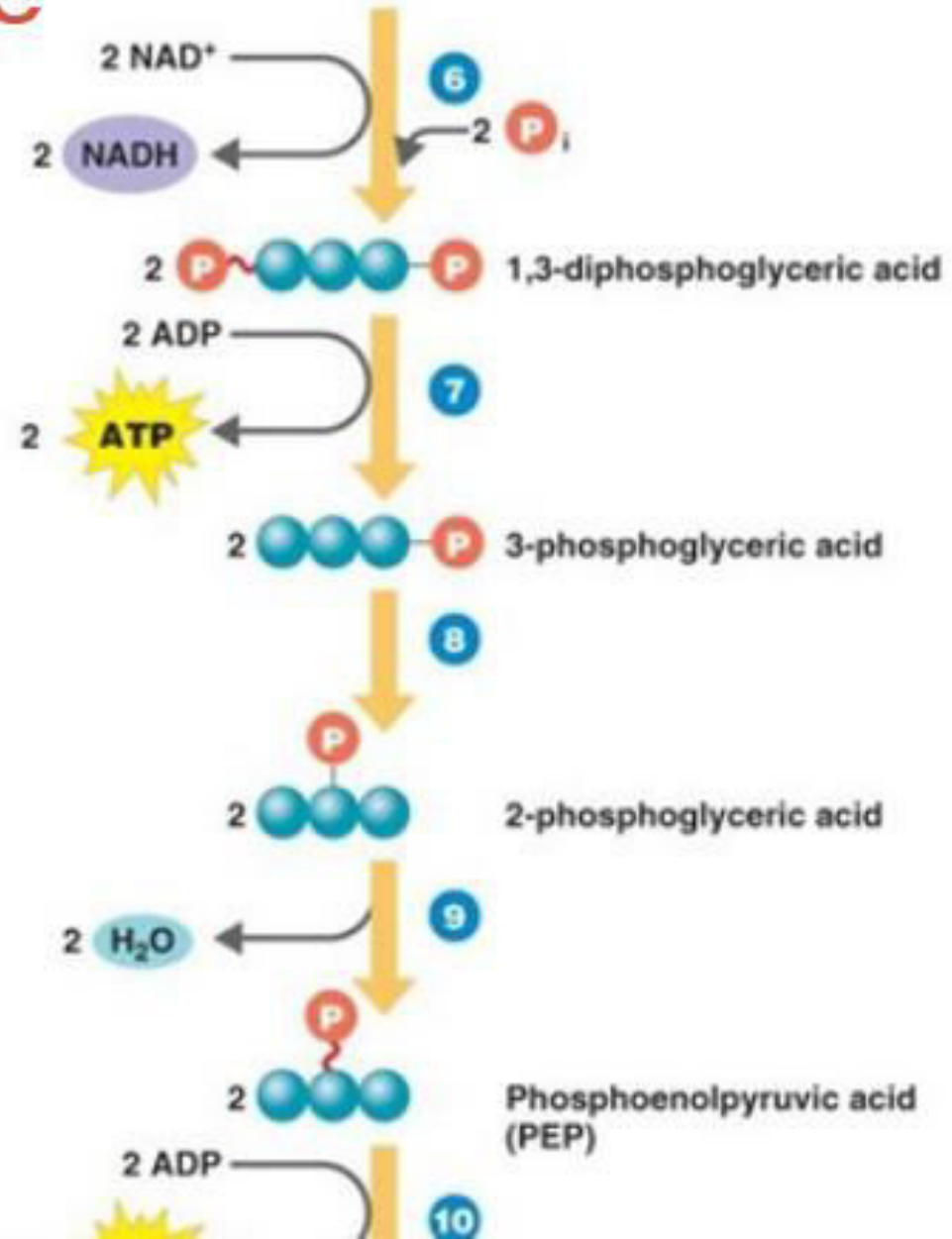
- Phosphorylation by a cAMP-dependent protein kinase leads to inactivation of pyruvate kinase in the liver (Figure 8.19).
- When blood glucose levels are low, elevated glucagon increases the intracellular level of cAMP, which causes the phosphorylation and inactivation of pyruvate kinase.
- Therefore, PEP is unable to continue in glycolysis, but instead enters the gluconeogenesis pathway.



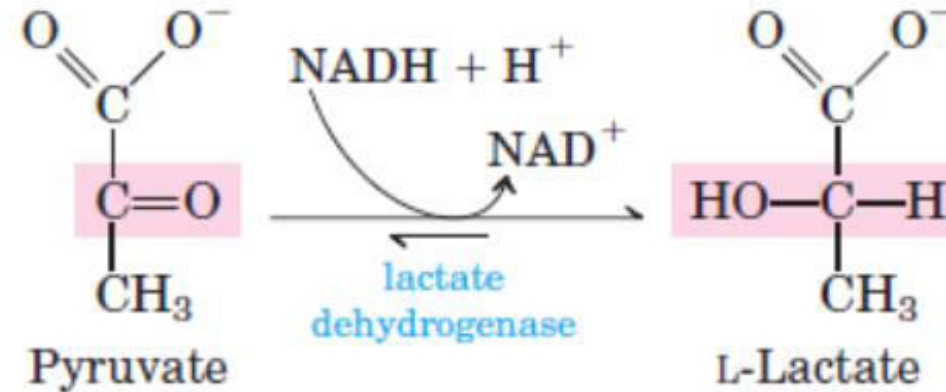
Energy-Generating Stage of Glycolysis

- 2 glucose-3-phosphate oxidized to 2 pyruvic acid
- 4 ATP produced
- 2 NADH produced

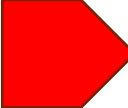
Energy-conserving stage



Additional Step in Anaerobic Condition



- When animal tissues cannot be supplied with sufficient oxygen to support aerobic oxidation of the pyruvate and NADH produced in glycolysis, NAD⁺ is regenerated from NADH by the reduction of pyruvate to lactate by **Lactate Dehydrogenase (LDH)**
- Tissues that function under hypoxic conditions eg. skeletal muscle, smooth muscle, erythrocytes produce lactate
- In erythrocytes, even under aerobic conditions, glycolysis terminates in lactate because of absence of mitochondria



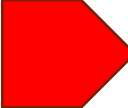
Energy Yield in Aerobic Glycolysis

Step No.	Enzyme	Source	No. of ATP
1	Hexokinase	–	-1
3	Phosphofructokinase	–	-1
6	Glyceraldehyde-3-phosphate dehydrogenase	NADH	$(+2.5) \times 2 = 5$
7	Phosphoglycerate Kinase	ATP	$(+1) \times 2 = 2$
10	Pyruvate Kinase	ATP	$(+1) \times 2 = 2$
	Net Energy Yield		9-2= 7 ATPs

If the cytosolic NADH uses malate-aspartate shuttle, 3 ATP are generated from each molecule of NADH

This is in contrast to glycerolphosphate shuttle that produces only 2 ATP





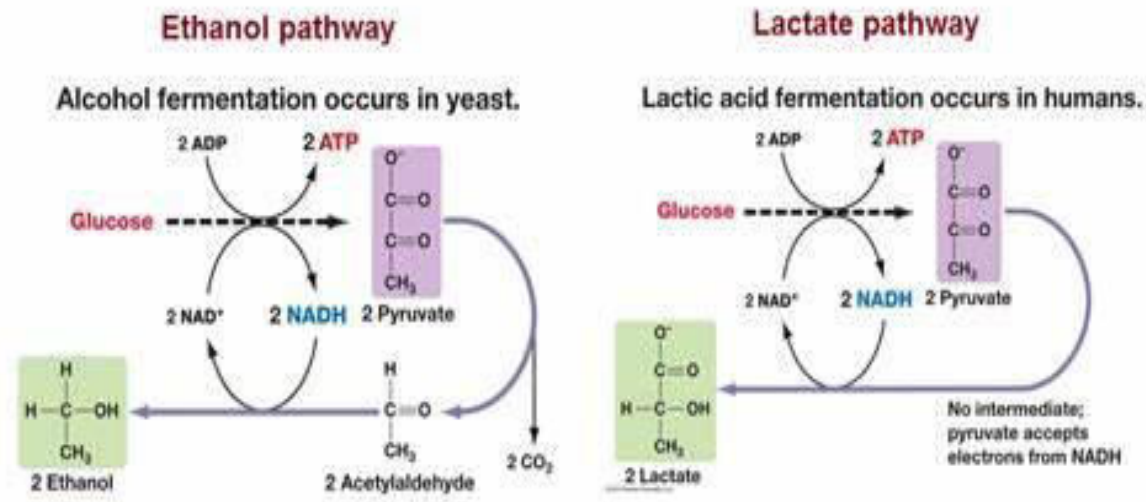
Energy Yield in Anaerobic Glycolysis

Step No.	Enzyme	Source	No. of ATP
1	Hexokinase	–	-1
3	Phosphofructokinase	–	-1
7	Phosphoglycerate Kinase	ATP	$(+1) \times 2 = 2$
10	Pyruvate Kinase	ATP	$(+1) \times 2 = 2$
	Net Energy Yield		$4 - 2 = 2$ ATPs



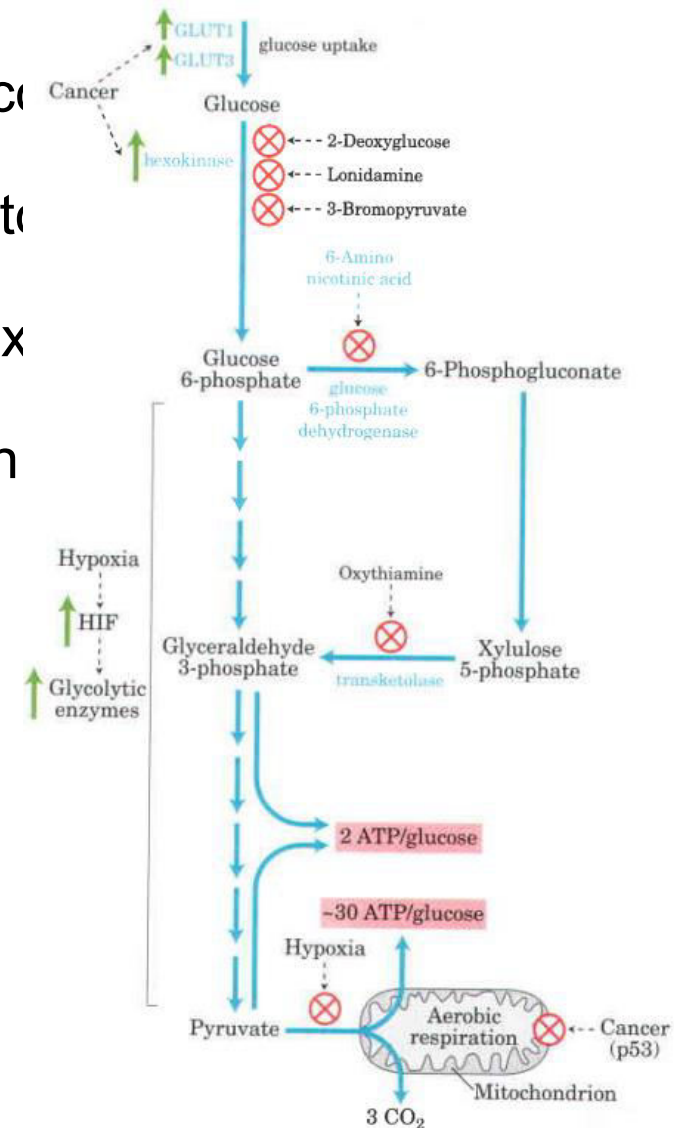
Fermentation

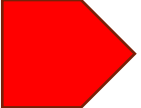
- Fermentation is a general term for the anaerobic degradation of glucose or other organic nutrients to obtain energy
- Occurs in anaerobic microorganisms such as brewer's or baker's yeast
- Decarboxylates pyruvate to acetaldehyde, which is reduced to ethanol
- Regenerates NAD^+ to continue glycolysis



Cancer and glycolysis

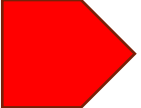
- Cancer cells display increased uptake of glucose and glycolysis
- As the tumors grow rapidly, the blood vessels are unable to supply adequate oxygen, and thus a condition of hypoxia exists
- The cancer cells get adapted to hypoxic condition result in activation hypoxia-inducible transcription factor (HIF)
- HIF increases the synthesis of glycolytic enzymes and the glucose transporters





Pasteur effect

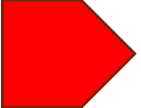
- The inhibition of glycolysis by oxygen (aerobic condition) is known as Pasteur effect
 - In the aerobic condition, the levels of glycolytic intermediates from fructose 1,6- biphosphate onwards decrease while the earlier intermediates accumulate
 - Clearly indicating Pasteur effect is due to the inhibition of the enzyme phosphofructokinase
 - The inhibitory effect of citrate and ATP (produced in the presence of oxygen) on phosphofructokinase
- 



Crabtree effect

- The phenomenon of inhibition of oxygen consumption by the addition of glucose to tissues having high aerobic glycolysis is known as Crabtree effect
- This is opposite to that of Pasteur effect
- Crabtree effect is due to increased competition of glycolysis for inorganic phosphate (P_i) and NAD^+ which limits their availability for phosphorylation and oxidation

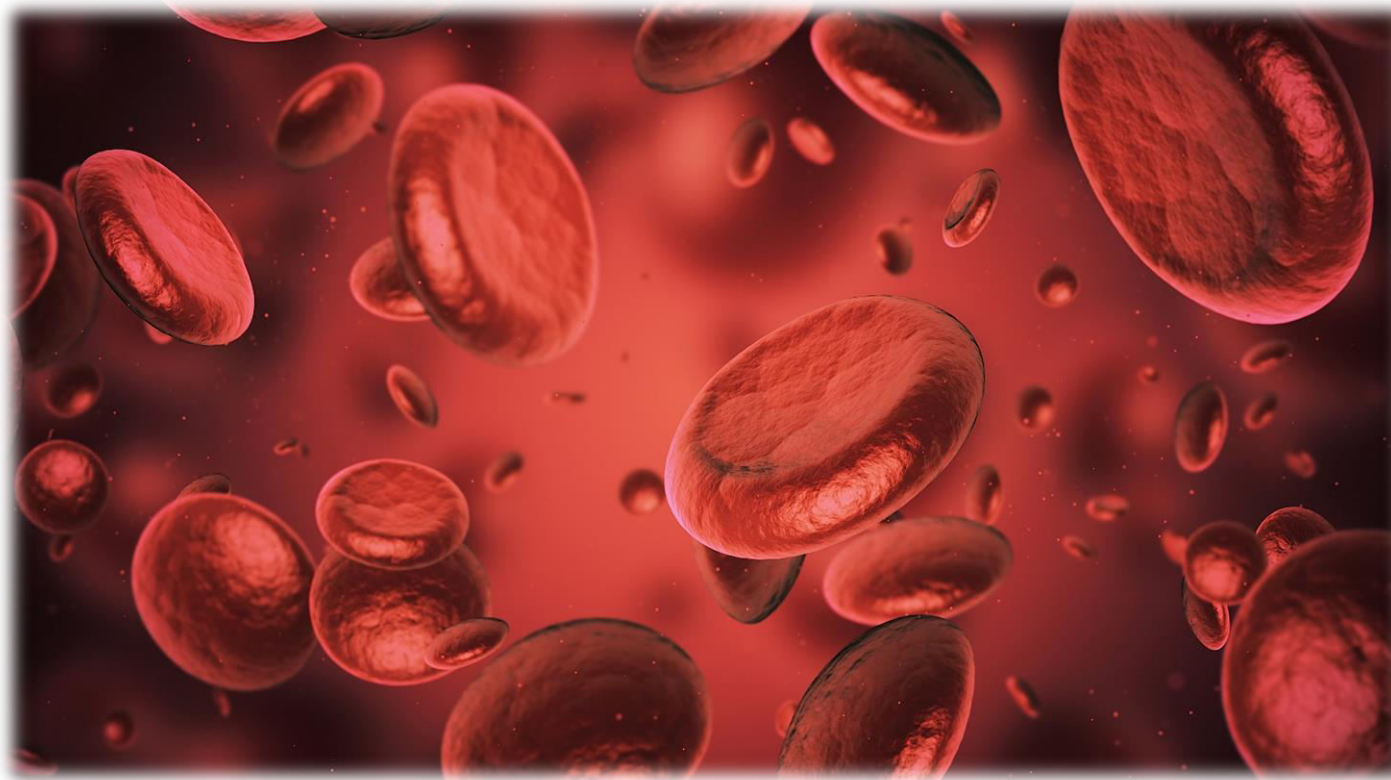


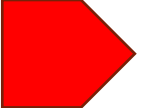


Glycolysis and dental caries

- It the destruction or decalcification of hard teeth due to organic acids released by bacterial infections
 - Anaerobic bacteria (e.g. Streptococcus mutans, Lactobacillus sps) main contributor
 - Grow on fermentable sugar & Initiate anaerobic glycolysis
 - Lactic acid production erode tooth enamel and dentin, and dissolve hydroxyapatite matrix lead to cavities
 - Fluoride in tooth inhibit enolase enzyme which inhibit glycolysis & prevent cavities
- 

GLYCOLYSIS IN RED BLOOD CELLS/ RAPAPORT-LEUBERING CYCLE

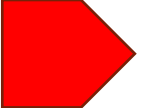




Unique Features of Glycolysis in RBCs

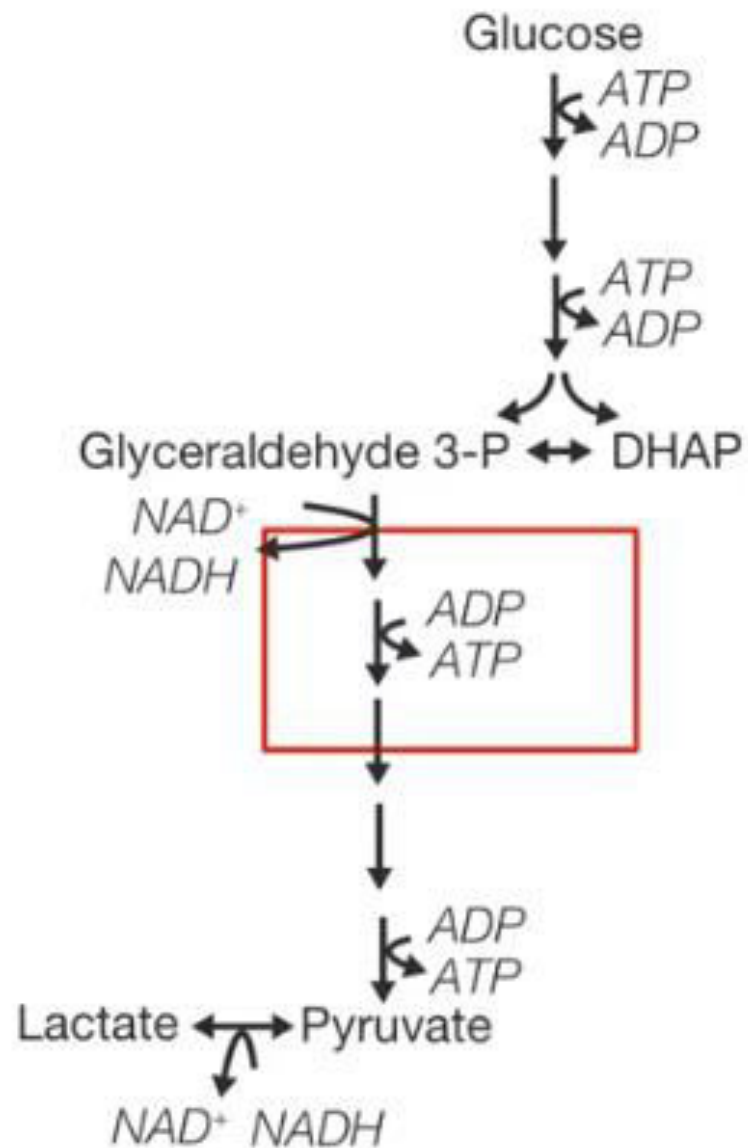
- RBCs are deficient mitochondria, so there is no respiratory chain, no citric acid cycle, and no oxidation of fatty acids or ketone bodies
- Glucose , main source of energy in RBCs
- Glycolysis The only source of ATP production which lead to lactic acid production (anaerobic glycolysis)
 - ATP utilize to keep biconcave shape of RBCs
 - To regulate ions concentration inside & outside RBCs e.g., $\text{Na}^+ - \text{K}^+$ -ATPase)



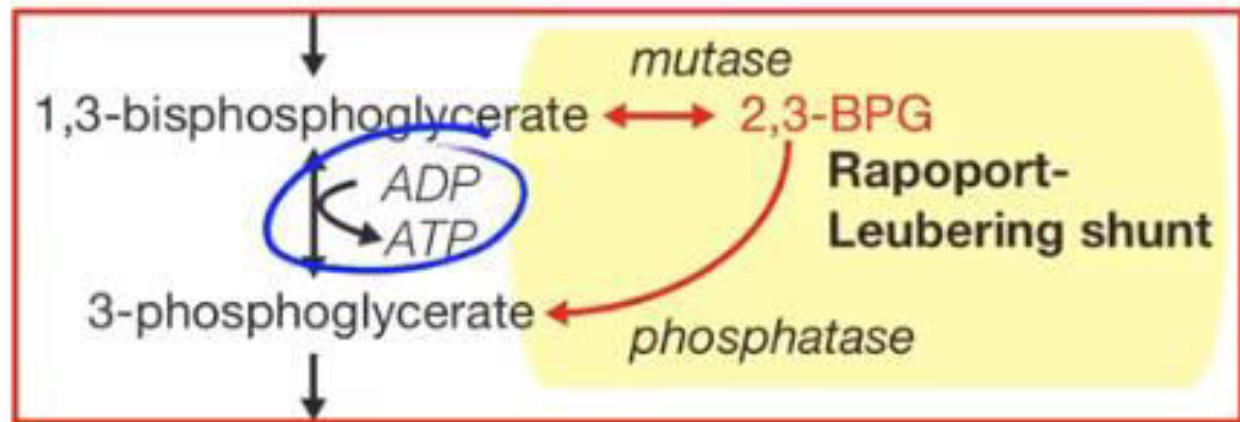


Glycolysis in RBCs

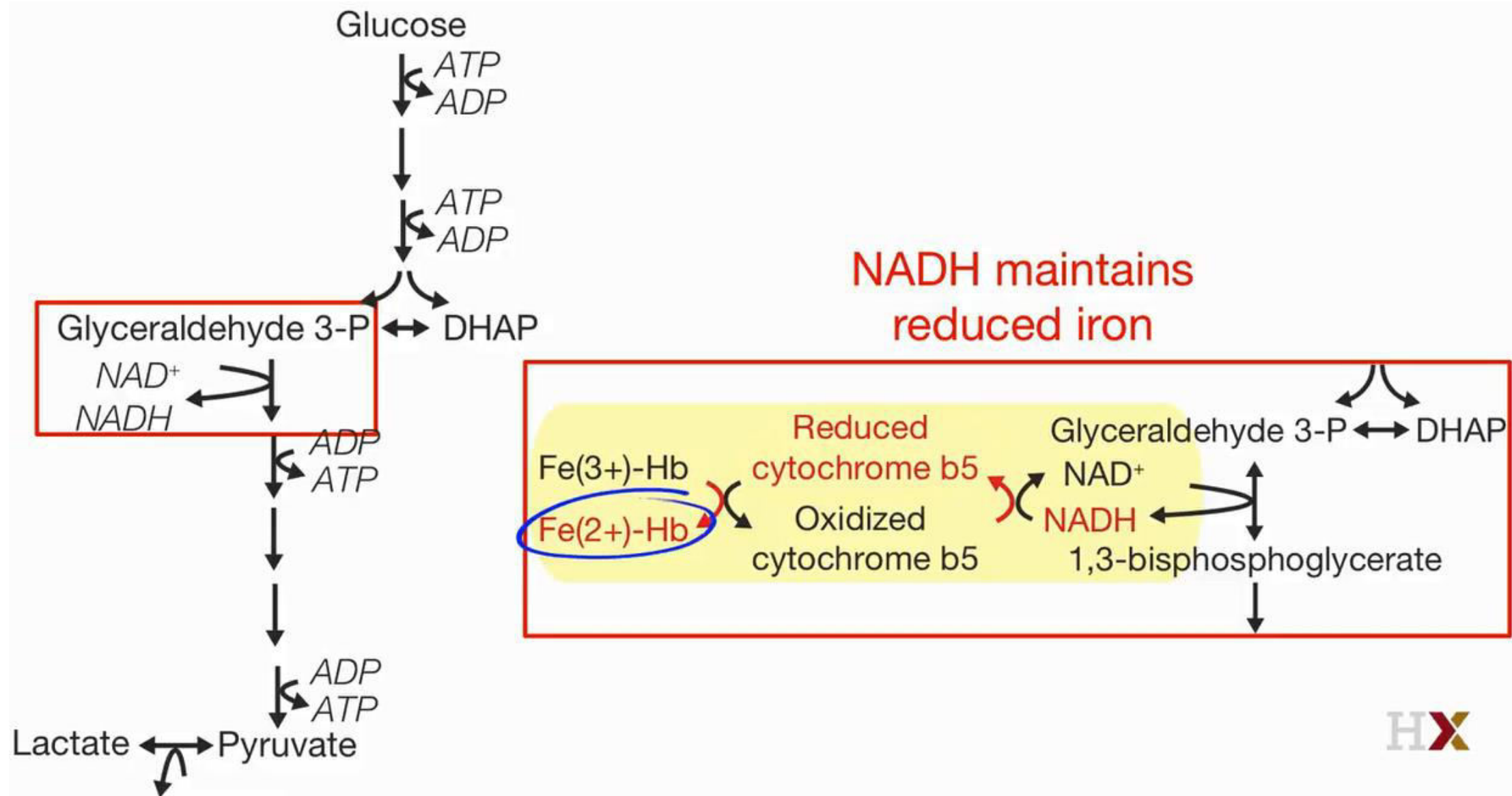
- Rapaport-Leubering cycle is a supplementary pathway to glycolysis
 - Rapaport-Leubering cycle is mainly concerned with the synthesis of 2,3-bisphosphoglycerate (2,3-BPG) in the RBC
 - **1,3-Bisphosphoglycerate (1,3-BPG)** produced in glycolysis is converted to **2,3-BPG** by the enzyme **2,3-bisphosphoglycerate mutase**
 - **2,3-BPG** is hydrolysed to **3-phosphoglycerate** by **bisphosphoglycerate phosphatase**
 - bisphosphoglycerate mutase is a bifunctional enzyme with mutase and phosphatase activities catalyzed by two different sites present on the same enzyme
 - 15-25% of the glucose that gets converted to lactate in erythrocytes goes via 2,3-BPG
- 

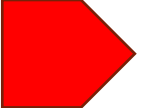


2,3-BPG production



Loss of 1 molecule of ATP per glyceraldehyde 3-phosphate (2 per glucose)



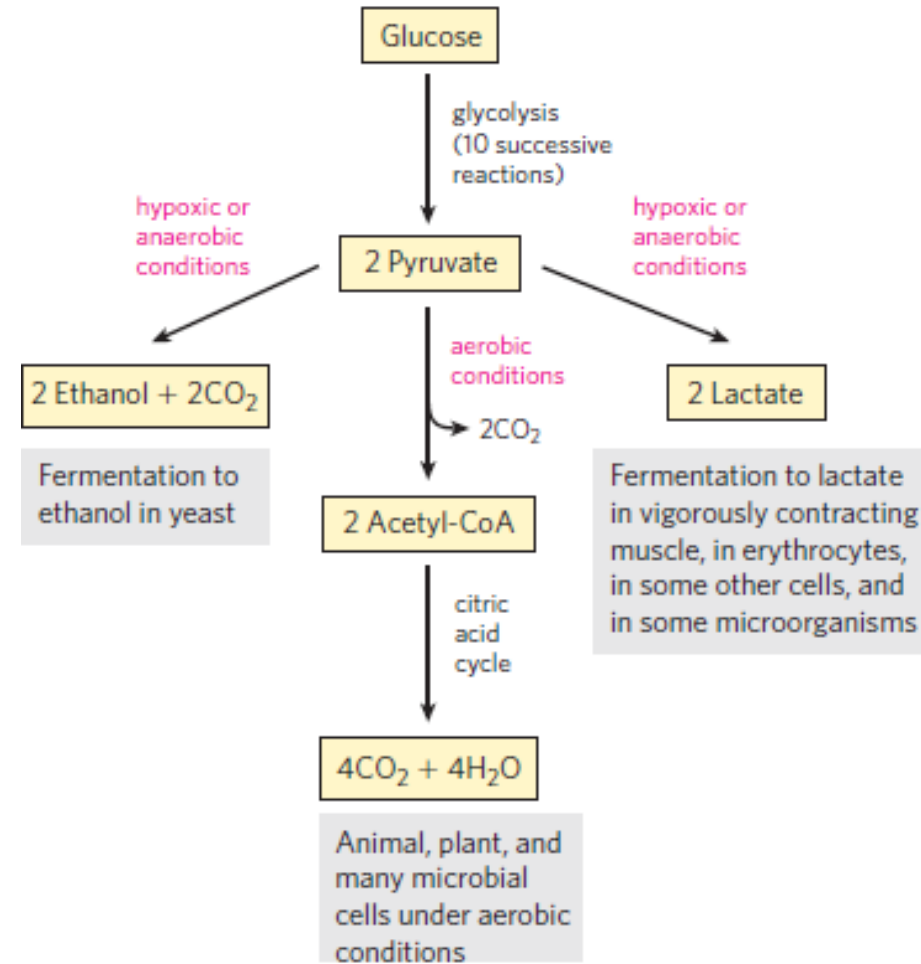


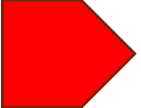
Why RBCs Need 2,3-BPG

- Production of 2,3-BPG via Rapaport-Leubering cycle is a shunt pathway of glycolysis to
 - Dissipate or waste the energy not needed by erythrocytes
 - 2,3-BPG combines with hemoglobin (Hb) and reduces Hb affinity with oxygen to help oxyhemoglobin unloads more oxygen to the tissues
 - Increase in erythrocyte 2,3-BPG is observed in (to enhance supply of oxygen to tissues)
 - hypoxic condition, high altitude, fetal tissues, anemic conditions etc
 - Deficiency of Hexokinase lead to low production of 2,3-BPG in RBCs hence high oxygen affinity
 - pyruvate kinase deficiency, lead to high production of 2,3-BPG in RBCs is high, resulting in low oxygen affinity



Fate of Pyruvate





Reference

- Biochemistry by Satyanarayana and Chakrapani
- [Lehninger Principles of Biochemistry, 5th Edition](#)



Clinical Enzymology

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Clinical Enzymology

- Clinical enzymology- the branch of science which **deals with the application of enzyme analysis in the diagnosis and treatment of disease.**

Enzymes

- Enzymes are distributed in **cytosol, nucleus, rough and smooth ER, Golgi bodies, mitochondria, lysosomes, plasma and organellar membranes.**
- Enzymes are mainly synthesized in **cytosol or ER** (stay, or target to other organelles and export to extracellular environment, or extracellular enzymes).
- A relatively small numbers are synthesized in the **mitochondria** and these enzymes stay within this space.

Enzymes

- Most enzymes are present in **cells at much higher concentrations than in plasma.**
- The **normal levels reflect the balance between the rate of synthesis and release into plasma during cell turnover, and the rate of clearance from the circulation.**

The enzyme level in plasma may be:

- ***Increased*** due to proliferation of cells, an increase in rate of cell turnover or damage or in enzyme synthesis, or to reduced clearance from plasma;
- ***Lower than normal***, due to reduced synthesis, congenital deficiency.

Enzymes

- Biological catalysis
- Very efficient –can increase reaction rates at the order of $\times 10$
- All are proteins- so liable to denaturation
- Specific to substrates
- Partly specific to tissues
- Assay by measure of rate of specific reaction catalyzed by that enzyme

Enzymes- isoforms

- **Isozymes** (different genes)- Catalyze same reactions but are formed from structurally different polypeptides.
- May have **different subcellular distribution**: (cytoplasm, mitochondria).
- Isozymes **allow the existence of similar enzymes with different characteristics, “customized” to specific tissue requirements or metabolic conditions.**

Enzymes- isoforms

- One example of the advantages of having isoenzymes for adjusting the metabolism to different conditions and/ or in different organs is the following:
- Glucokinase and Hexokinase are typical examples of isoenzymes. In fact, there are four Hexokinases: I, II, III and IV.
- Hexokinase I is present in **all mammalian tissues**, and **Hexokinase IV (Glucokinase)**, is found mainly in **liver, pancreas and brain**.

Enzymes- isoforms

- **Allozymes** (different alleles at same genetic locus)
- **Proenzyme** (zymogen) – inactive form of enzyme that becomes active after partial proteolysis example: pepsinogen 🤢🤮 pepsin
- **Isoform**– more general term, includes true isoenzymes and pseudoisoenzymes (posttranslational variations)- cell and tissue specific forms- e.g. liver- and bone-specific alkaline phosphatase (ALP) differ only in carbohydrate contents attached to the ALP proteins.

Measurement of serum enzymes

- Diagnostic enzymology
- Enzymes normally- **intracellular**. Very **LOW** concentration in blood
- Enzyme **leakage** in blood indicates cell damage (**cell -death, hypoxia, intracellular toxicity**)
- Quantitative measure of cell/tissue damage
- Fairly non invasive- possible to do repeated tests
- **Organ specificity**- but not absolute specificity in spite of same gene content.
- Most enzymes are present in most cells- differing amounts

Enzymes- Markers

- Presence of disease
- Organs involved
- Etiology /nature of disease- differential diagnosis
- Extent of disease- more damaged cells- more leaked enzymes in blood
- Time course of disease

Measurement of enzyme activity

- Since the amount of active enzymes present in the blood sample is very low, measuring their catalytic activities is a very sensitive and specific method for the measurement of the enzymes present in the serum sample.
- Conversion of substrate into product in the presence of enzyme under optimum condition is followed, then either substrate consumed or product formed is measured in a defined time interval.

Enzyme activity

- **Enzyme activity can be measured** as either an increase in product concentration, a decrease in substrate concentration, a decrease in coenzyme concentration, or an increase in concentration of altered coenzyme.
1. **Endpoint measurements** are performed after a reaction proceeds for a designated length of time, then is stopped.
 - Measurement is made of the amount of reaction that has occurred.
 2. **Kinetic measurements** are multiple measurements of absorbance change made at specific time intervals.
 3. An **international unit (U) of enzyme activity** is the amount of enzyme that catalyzes the reaction of 1 μmol of substrate per minute under specific conditions (e.g., temperature, pH).

Enzymes of clinical significance

A. **Phosphatases** promote the hydrolysis of phosphate esters.

1. **Acid phosphatase (ACP)** promotes the hydrolysis of orthophosphate esters, although the physiologic function of ACP is unknown.
 - Almost every body tissue contains ACP, but significant levels are found in RBCs and platelets. In adult men, 50% of ACP is found in the prostate gland. High levels of ACP are found in semen as well.
 - Falsely low values are produced by improper anticoagulant (e.g., fluoride, oxalate, heparin) use, whereas hemolysis produces false elevations.
 - The **major diagnostic significance** of ACP is its aid in detecting metastatic carcinoma.
 - Other types of cancer and bone disease exhibit increases of enzyme.

ALP

2. **Alkaline phosphatase (ALP)** hydrolyzes phosphate esters, but the function of this enzyme is relatively unknown.

- ALP exists in a wide variety of tissues, with significant amounts found in liver, bone, intestine, kidney, and placenta.
- Improper anticoagulant use produces decreased values of ALP, and a hemolyzed specimen demonstrates a false increase in ALP value.
- The **diagnostic significance** of ALP is its elevation during the third trimester of pregnancy and in persons with liver disease (hepatitis, cirrhosis), bone disease, hyperthyroidism, or diabetes mellitus.

Enzymes of clinical significance

- There are **four isoenzymes of ALP**: bone, liver, intestinal, and placental fractions.
- In disease states, the fractions are affected as follows:
- The **bone fraction** levels are elevated in persons with bone disease (osteomalacia, rickets and high in children.
- The **liver fraction** level is greatly elevated in persons with hepatobiliary obstruction.
- The **placental fraction level** becomes elevated between the 16th and 20th weeks of pregnancy.

Transaminases

B. Transaminases catalyze the transfer of amino groups and acids.

- This is an important reaction in the synthesis and degradation of amino acids.
- 1. **Aspartate transaminase (AST)** transfers an amino group between aspartate and keto acids (aspartate + α -ketoglutarate $\rightarrow$ oxaloacetate + glutamate).
- Pyridoxal phosphate (vitamin B6) is the coenzyme in this reaction.
- AST exists in cardiac tissue, liver, skeletal muscle, and RBCs.
- The **clinical significance** of AST elevations is that the elevations occur in patients with myocardial infarction, viral hepatitis, and skeletal muscle disease.

Transaminases

2. **Alanine transaminase (ALT)** catalyzes the transfer of an amino group from alanine to α -ketoglutarate with the formation of glutamate and pyruvate.
- **Distribution.** ALT is localized in the liver, with some in the heart, skeletal muscle, and RBCs.
- The **clinical significance** of ALT is in the evaluation of hepatocellular disorders.

CK

- **Creatine kinase (CK)** catalyzes a reaction responsible for the formation of ATP in tissues, especially contractile systems.
- When muscle contraction occurs, ATP is hydrolyzed to ADP to produce chemical energy for the contraction process.
- CK is involved in the storage of high-energy creatine phosphate (creatine + ATP $\rightarrow$ creatine phosphate + ADP).
- **Distribution.** CK is localized in skeletal muscle, brain, and cardiac muscle in addition to many other tissues.
- Hemolysis must be avoided when analyzing because of the presence of **adenylate kinase (AK)** in RBCs. AK induces false elevations in CK assays.
- **The clinical significance of CK** is in its elevation in muscle disorders.
- CK and one of its isoenzymes is a sensitive indicator of acute myocardial infarction and muscular dystrophy.

Enzymes of clinical significance

- **The structure of CK** allows for the formation of three isoenzymes.
- **CK-MM** is the major form found in striated and cardiac muscle and normal serum.
- Hypothyroidism and intramuscular injections result in CK-MM elevations.
- **CK-MB** is localized in cardiac muscle.
- Elevated levels are indicative of ischemic heart disease.
- If the serum level of CK-MB is >6% of total CK, it is a highly specific indicator of myocardial damage.
- Following **acute myocardial infarction**, CK-MB levels rise within 48 to 72 hours, peak at 12 to 24 hours, and return to normal within 48 to 72 hours.
- **CK-BB** is present in small quantities in normal serum and appears if extensive damage to the brain occurs, but is present in many other tissues.

TABLE 8-34**Rise and Fall Pattern of Creatine Kinase–Myocardial Bound and Troponins After Acute Myocardial Infarction**

Analyte	Rise (hr After AMI)	Peak (hr After AMI)	Back to Normal
CK-MB	4-6	12-24	72 hr after AMI
Troponin I	4-6	12-24	3-7 days
Troponin T	4-6	12-24	7-10 days

LD

- **Lactate dehydrogenase (LD)** catalyzes the interconversion of lactic and pyruvic acids.
- It is a hydrogen-transfer enzyme that uses the coenzyme NAD.
- **Distribution.** LD is widely distributed in all tissues, with high concentrations found in the heart and liver and low levels in RBCs, skeletal muscle, and kidneys
- **The diagnostic significance of LD** is in the diagnosis of cardiac, hepatic, skeletal muscle, and renal disease.

LD

- Highest LD levels are seen in persons with pernicious anemia, viral hepatitis, cirrhosis, and crush injuries
- There are, in normal serum, **five different isoforms of LD (LD1 to LD5)**, with each isoenzyme being composed of four protein subunits.
- **LD1** is localized predominantly in heart muscle and RBCs and constitutes approximately 25% of total LD.
- **LD2** localization is similar to LD1, but it also is found in the kidneys and comprises approximately 35% of total LD.

LD

- **LD3** is found in lung, lymphocytes, spleen, and pancreas. It constitutes approximately 22% of total LD.
- **LD4** is found mostly in liver and skeletal muscle and comprises approximately 10% of total LD.
- **LD5** is similar to LD4 in both localization and portion of total.
- After **myocardial infarction**, LD1 concentration is greater than LD2, which presents as a “flipped pattern” in electrophoresis.

Miscellaneous enzymes

- **Miscellaneous enzymes** that are clinically significant include γ -glutamyl-transferase (GGT), amylase, cholinesterase, pseudocholinesterase, and 5-nucleotidase.
- **GGT** transfers a γ -glutamyl residue to an amino acid (transpeptidation).
- This function is involved in peptide and protein synthesis.
- **Distribution.** Tissue sources of GGT include kidney, brain, prostate, pancreas, and liver. Urine contains significant amounts of GGT.
- The **clinical significance** of GGT is as follows.
- GGT levels are elevated in almost all hepatobiliary disorders or biliary tract obstructions as well as in patients taking enzyme-inducing drugs like warfarin, phenobarbital, and Dilantin.

GGT

- GGT levels are often examined with ALP levels. If both GGT levels and ALP values are high, some type of liver disorder is suspected. If GGT is normal and ALP is high, bone disease is likely present.
- Patients with acute pancreatitis exhibit increased GGT levels.
- In persons with diabetes, GGT is typically increased as triglyceride levels rise.
- Because alcohol has enzyme-inducing properties, the most useful application for GGT is in the **detection of alcoholism** and the monitoring of alcohol intake by patients during treatment.

Amylase

- **Amylase (AMS)** is a hydrolase that catalyzes the breakdown of starch and glycogen and produces products consisting of glucose, maltose, and dextrans.
- AMS is the smallest in size of all enzymes.
- **Distribution.** Tissue sources of AMS include the pancreas and salivary glands, as well as skeletal muscle.
- The **clinical significance** of AMS is its use in the diagnosis of acute pancreatitis.
- AMS levels rise 2 to 12 hours after onset of an attack and peak at 24 hours.
- Salivary gland lesions (e.g., mumps) also produce high levels of AMS.
- Patients with hyperlipidemia have low-to-normal AMS levels.

Lipase

- Increased in pancreatitis
- Remains elevated longer than amylase
- More specific for acute pancreatitis

Cholinesterase

- **Cholinesterase** is an enzyme that hydrolyzes the esters of choline.
- There are two types: acetylcholinesterase (AChE) and pseudocholinesterase (PChE).
- Destroys acetylcholine after nerve transmission
- Severe decrease results in serious neuromuscular effects, one of few enzymes in which decrease is clinically significant
- Decrease cholinesterase is seen in organophosphate poisoning and genetic susceptibility to certain anesthetic agents
- **Distribution.** AChE is found in RBCs, brain, and nerve cells.

G6PD

- **G6PD** is an oxidoreductase that catalyzes the oxidation of G6P to 6-phosphogluconate, which is an important first step in the pentose-phosphate shunt of glucose metabolism.
- **Distribution.** Tissue sources of G6PD include the adrenal glands, spleen, thymus, RBCs, and lymph nodes.
- In the RBC, G6PD functions to maintain NADPH in a reduced form to protect hemoglobin from oxidation and to prevent red cell hemolysis.
- **Diagnostic significance.** Clinically, deficiency of G6PD is an inherited sex-linked trait most common in African-Americans.
- Increases of G6PD are seen in persons with myocardial infarction and megaloblastic anemia.

5-nucleotidase (5-NT)

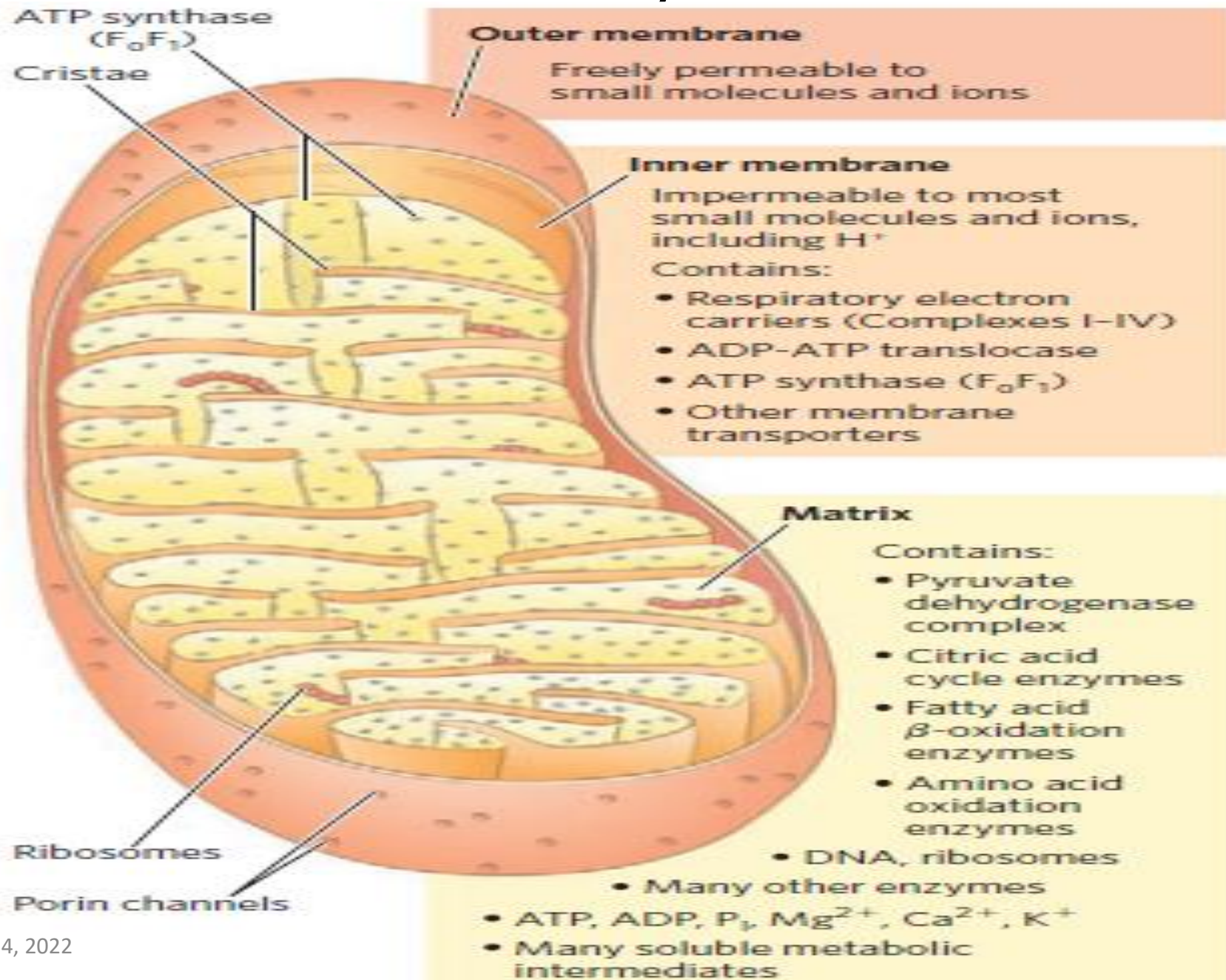
- **5-nucleotidase (5-NT)** originates in the liver or diseased bone.
- 5-NT becomes elevated in liver disease, as does ALP, but 5-NT does not increase in patients with bone disease.
- Serum levels of 5-NT are more sensitive to liver cancer.
- Biliary obstruction

ETC

Electron transport and oxidative phosphorylation

- **Oxidative phosphorylation** is the termination of **energy yielding metabolism in aerobic organisms**.
- All oxidative steps in the degradation of **carbohydrates, fats, and amino acids** converge at this final stage of cellular respiration, in which the **energy of oxidation drives the synthesis of ATP**.
- **In eukaryotes, oxidative phosphorylation occurs in mitochondria.**

Biochemical anatomy of mitochondria



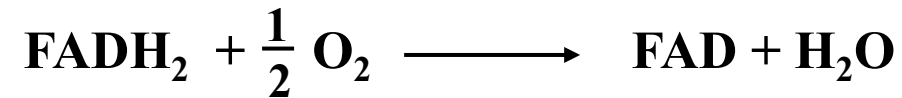
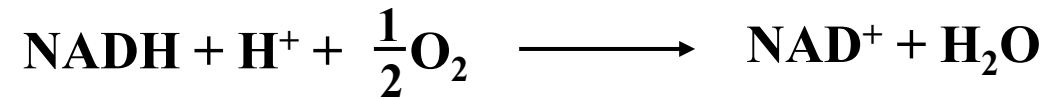
Electron transport and oxidative phosphorylation

- Each turn of TCA cycle generates **3NADH and 1 FADH₂**
- Electron transport and Oxidative Phosphorylation occurs in the mitochondria
- NADH and FADH₂ ultimately pass electrons to O₂ and produce H₂O.
- Energy from NADH can be used to drive synthesis of ATP several times.

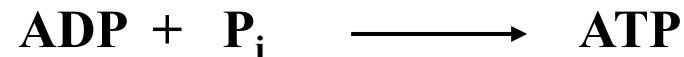
Oxidative Phosphorylation

Oxidative phosphorylation is the process by which the energy stored in NADH and FADH_2 is used to produce ATP.

A. Oxidation step: electron transport chain



B. Phosphorylation step



Electron carriers

- The **mitochondrial respiratory chain** consists of a series of **sequentially acting electron carriers**, most of which are **integral proteins** with prosthetic groups capable of **accepting and donating** either one or two electrons.
- **Three types of electron transfers** occur in oxidative phosphorylation.
 - (1) **direct transfer of electrons**, as in the reduction of **Fe^{+3} to Fe^{+2}** .
 - (2) transfer as a hydrogen atom (**$\text{H}^+ + \text{e}^-$**).
 - (3) transfer as a hydride ion (**$:\text{H}^-$**), which bears two electrons.
- The term **reducing equivalent** is used to designate a single electron equivalent transferred in an oxidation-reduction reaction.

Electron carriers

- In addition to **NAD** and **flavoproteins**, **three other types** of electron carrying molecules function in the **respiratory chain**.
- A **hydrophobic quinone (ubiquinone)** and two different types of **iron-containing proteins (cytochromes and iron-sulfur proteins)**.
- Ubiquinone (**also called coenzyme Q, or simply Q**) is a lipid soluble benzoquinone with a long isoprenoid side chain.
- The **cytochromes** are proteins with characteristic strong absorption of visible light, due to their **iron containing heme prosthetic groups**.
- Mitochondria contain **three classes of cytochromes**, designated a, b, and c, which are distinguished by differences in their light absorption spectra.

ETC

- ETC is made up of **4 different complexes** and contain different electron carriers.
- Complex I: NADH-CoQ reductase
- Complex II: Succinate-CoQ reductase
- Complex III: CoQ-cytochrome C reductase
- Complex IV: Cytochrome C oxidase.
- Each sequential carrier in the ETC has **increasing reduction potential (greater affinity for electrons)** and energy released during the transfer is used to transport H^+ across the inner mitochondrial membrane against an electrochemical gradient.

ETC

- Complex 1 uses NADH to reduce CoQ and then transfers to electron carriers of complex 3
- Complex II uses FADH₂ produced in the conversion of succinate to fumarate in the CA cycle and also reduces CoQ which then enters complex 3.

ETC

- Both complex 1 and 2 produce reduced CoQ which will transfer H^+ to carriers in complex 3. However that NADH generates more energy than $FADH_2$ since 6 H^+ are moved into the intermembrane space compared to 2 H^+ for complex 2.
- Energy released in the electron transfers in complexes 1,3,4 will be used to transport H^+ across the membrane and the electrons will eventually be transferred to O_2 to form water

ETC

- The Respiratory complexes are **proton pumps**. As electrons pass through complexes I, III, and IV, **hydrogen ions are pumped across the inner mitochondrial membrane into the intermembrane space**.
- The proton concentration in the intermembrane space increases relative to the mitochondrial matrix which generates a proton-motive force.
- ATP synthase complex (complex V): Hydrogen ions pass back into the matrix through V, this drives ATP synthesis.
 - $\text{NADH} \rightarrow 3\text{ATP}$
 - $\text{FADH}_2 \rightarrow 2\text{ATP}$
- ATP synthesized in the matrix is transported out of the matrix via an ATP/ADP translocase (an antiport) also coupled to proton motive force.

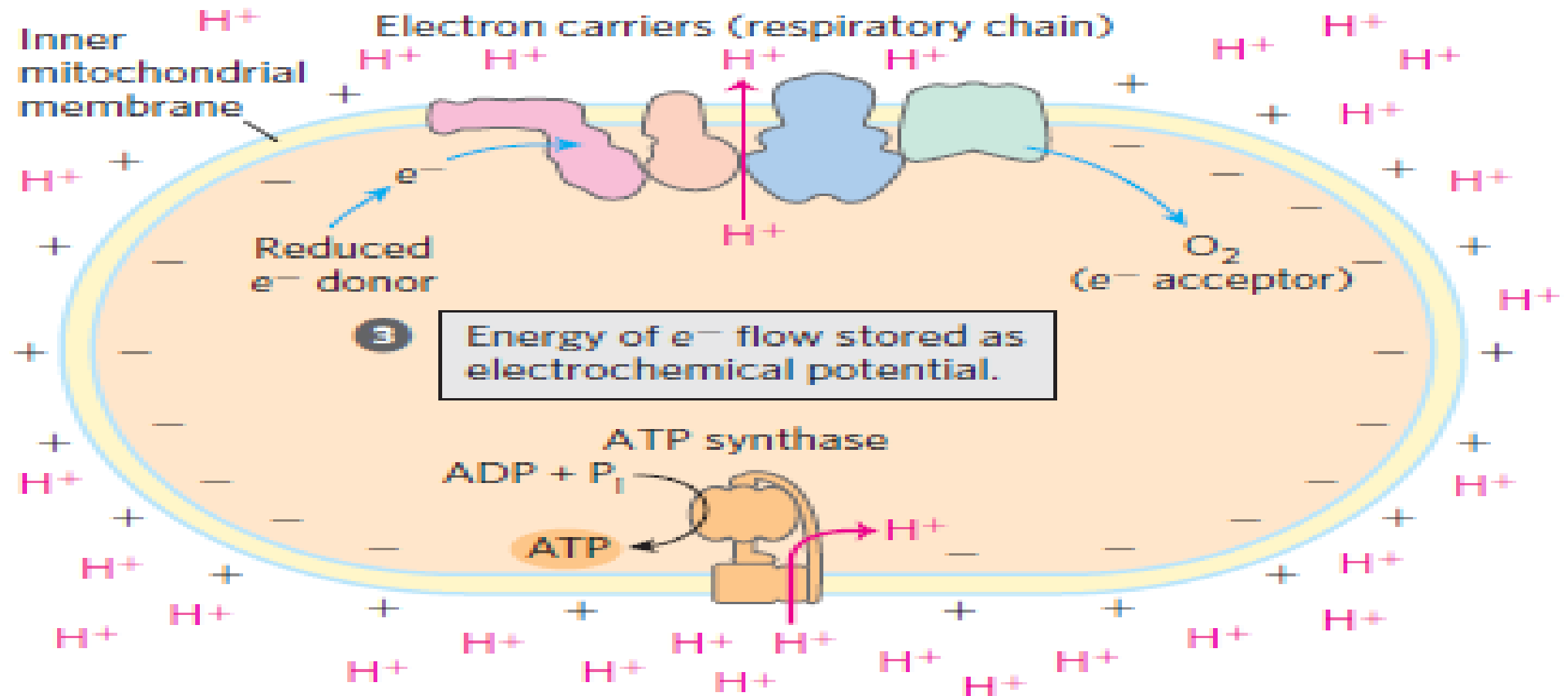
(a) Mitochondrion

1

Reduced substrate (fuel) donates e^- .

2

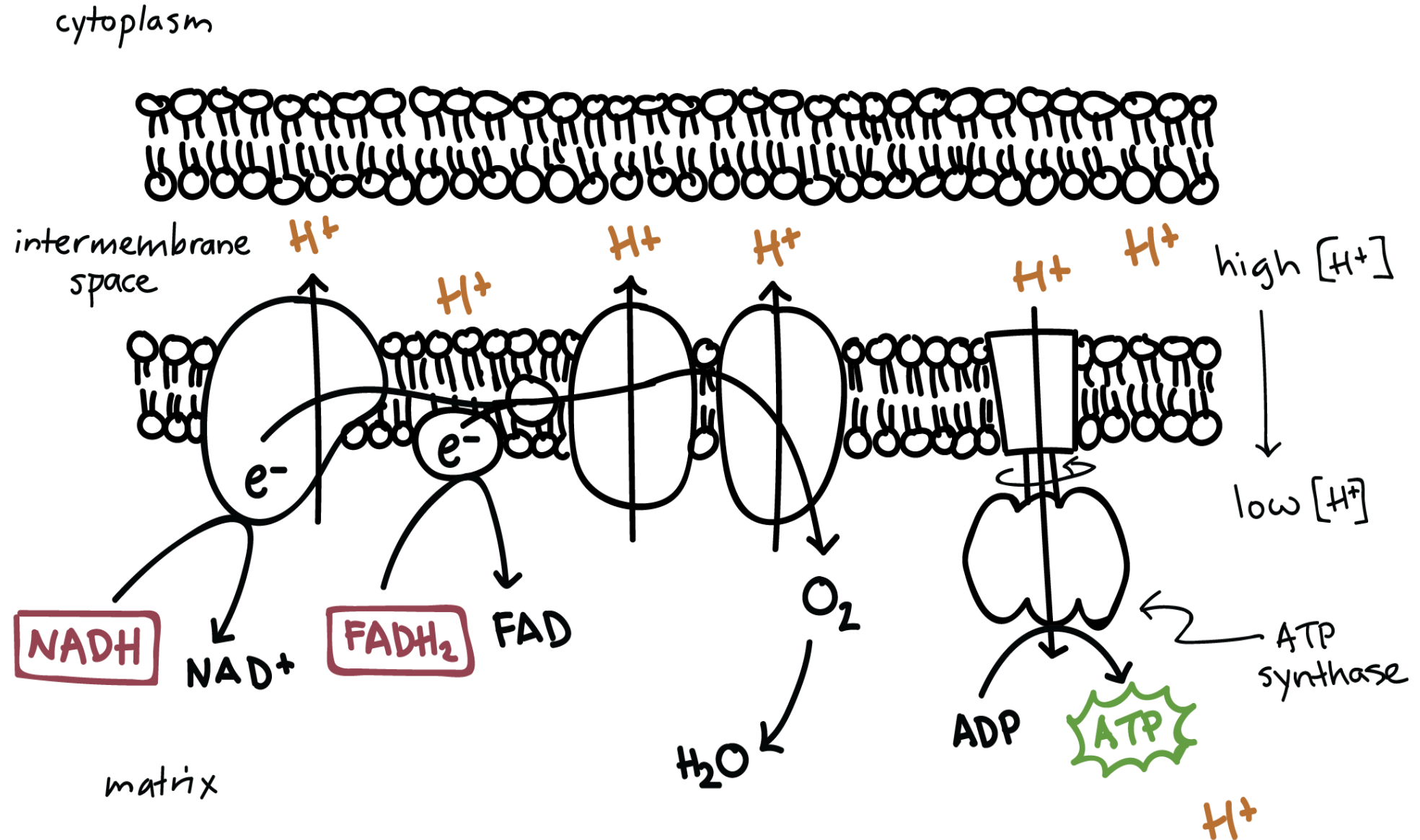
Electron carriers pump H^+ out as electrons flow to O_2 .



4

ATP synthase uses electrochemical potential to synthesize ATP.

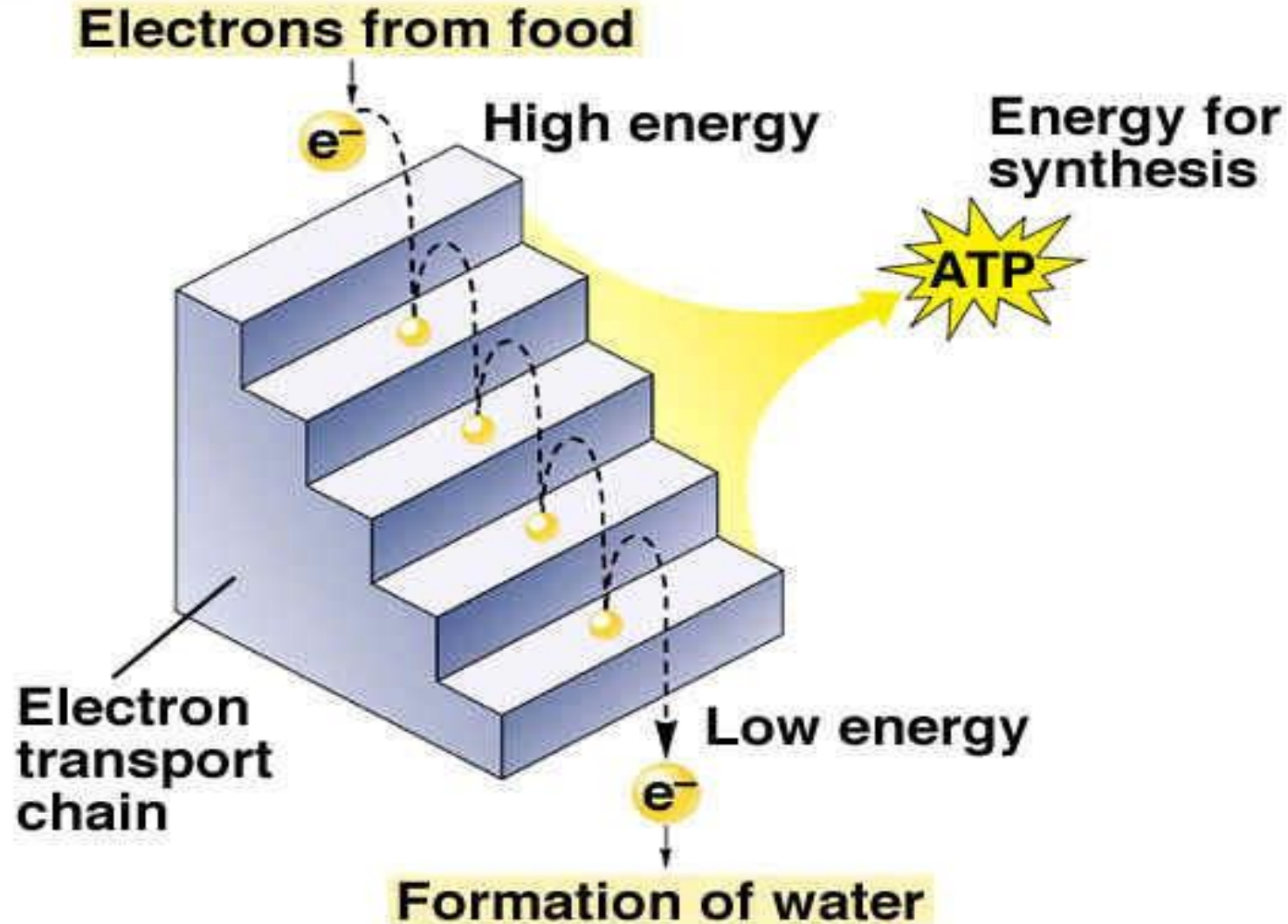
ETC



ETC

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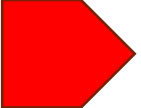
Electron Transport Schematic



Feeder Pathway For Glycolysis

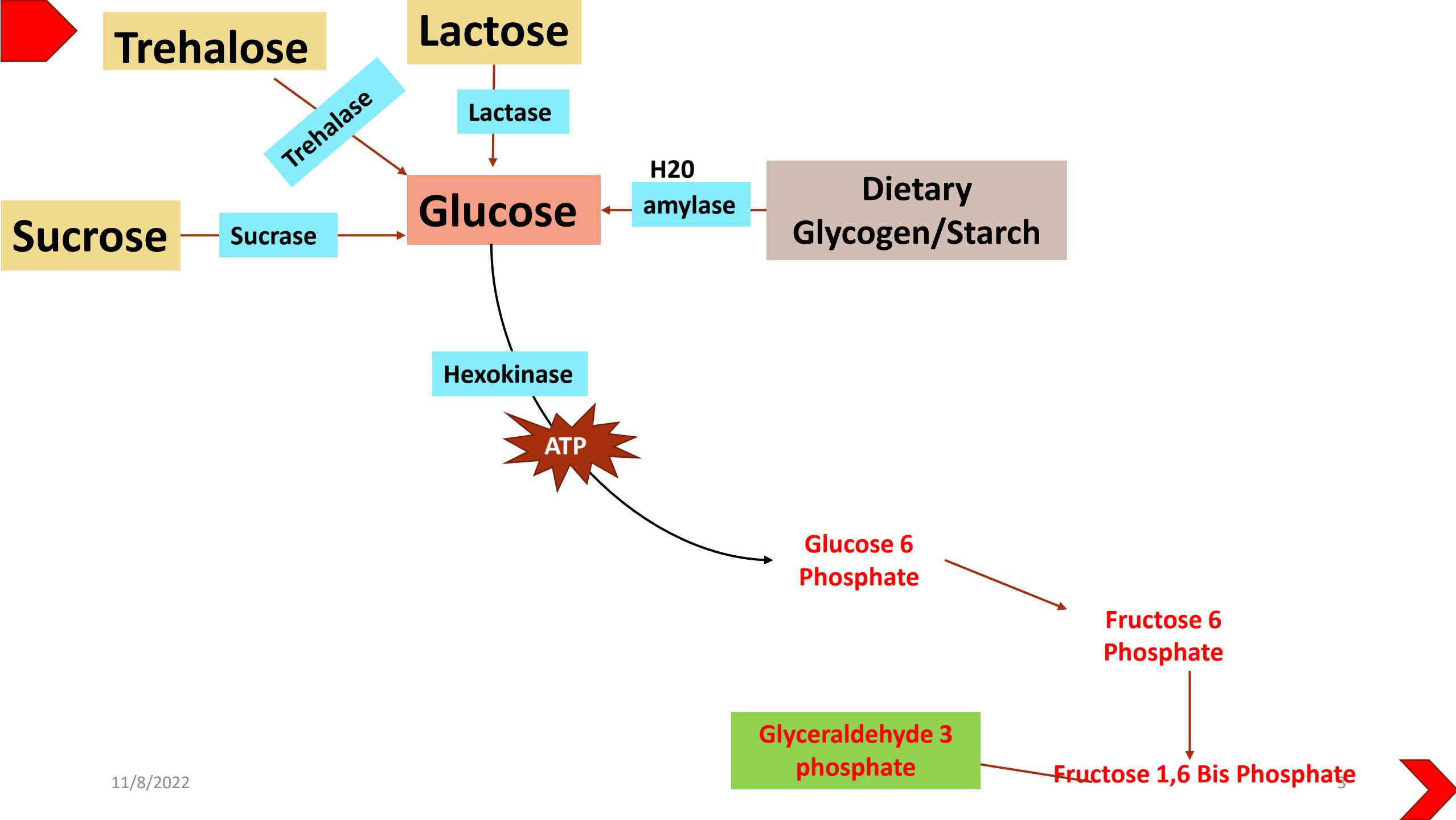
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- Lecturer Biochemistry

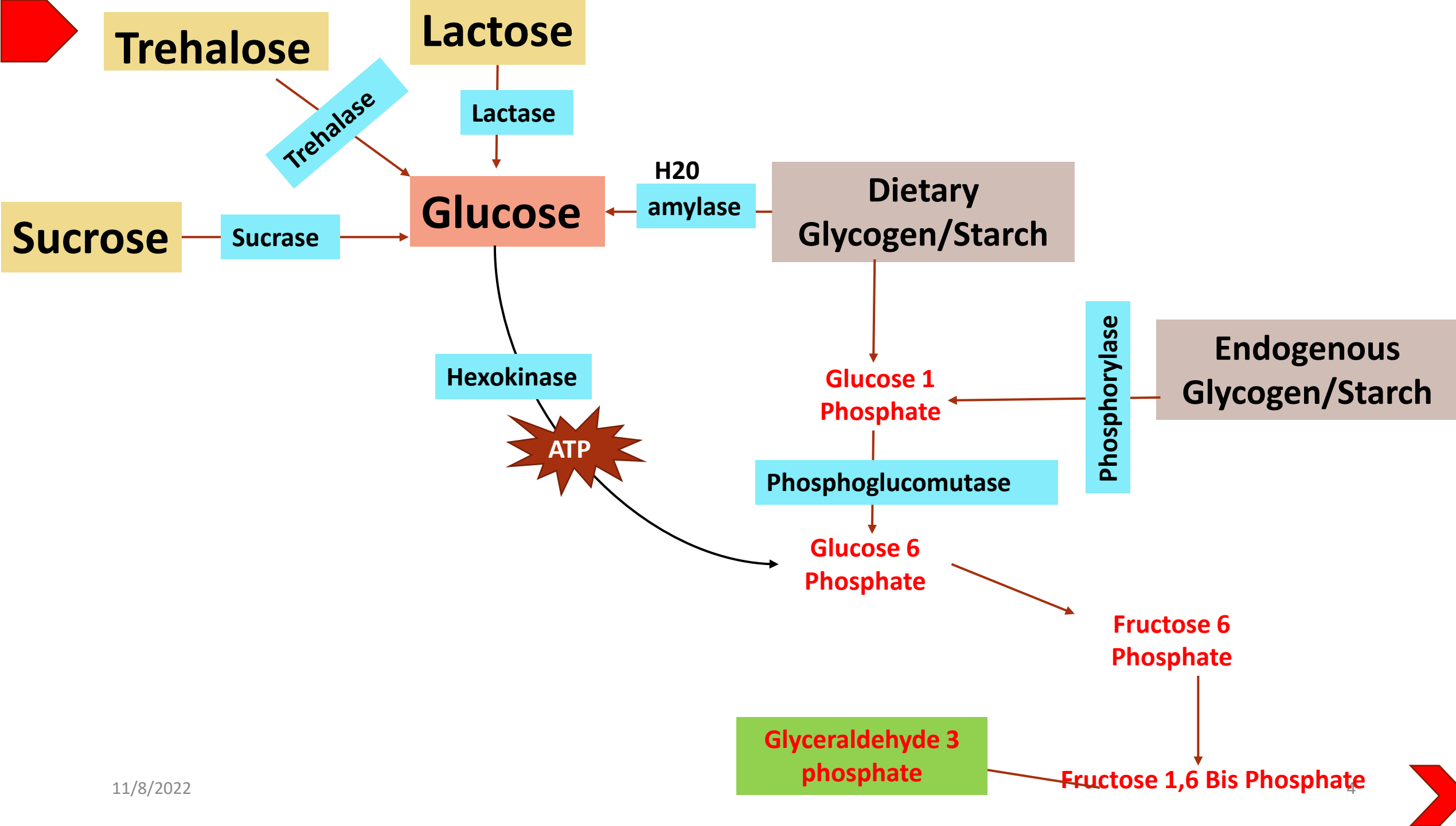


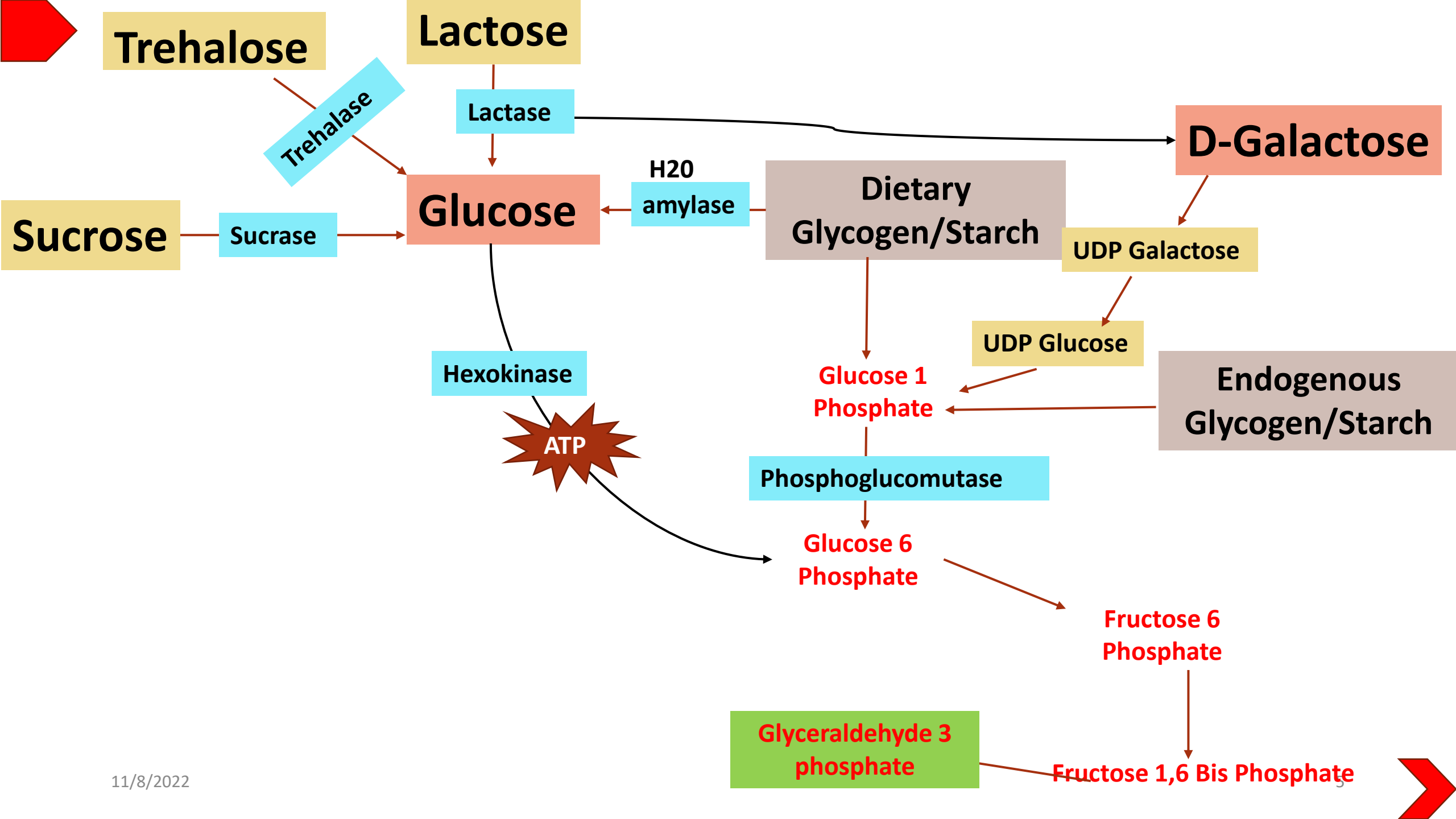


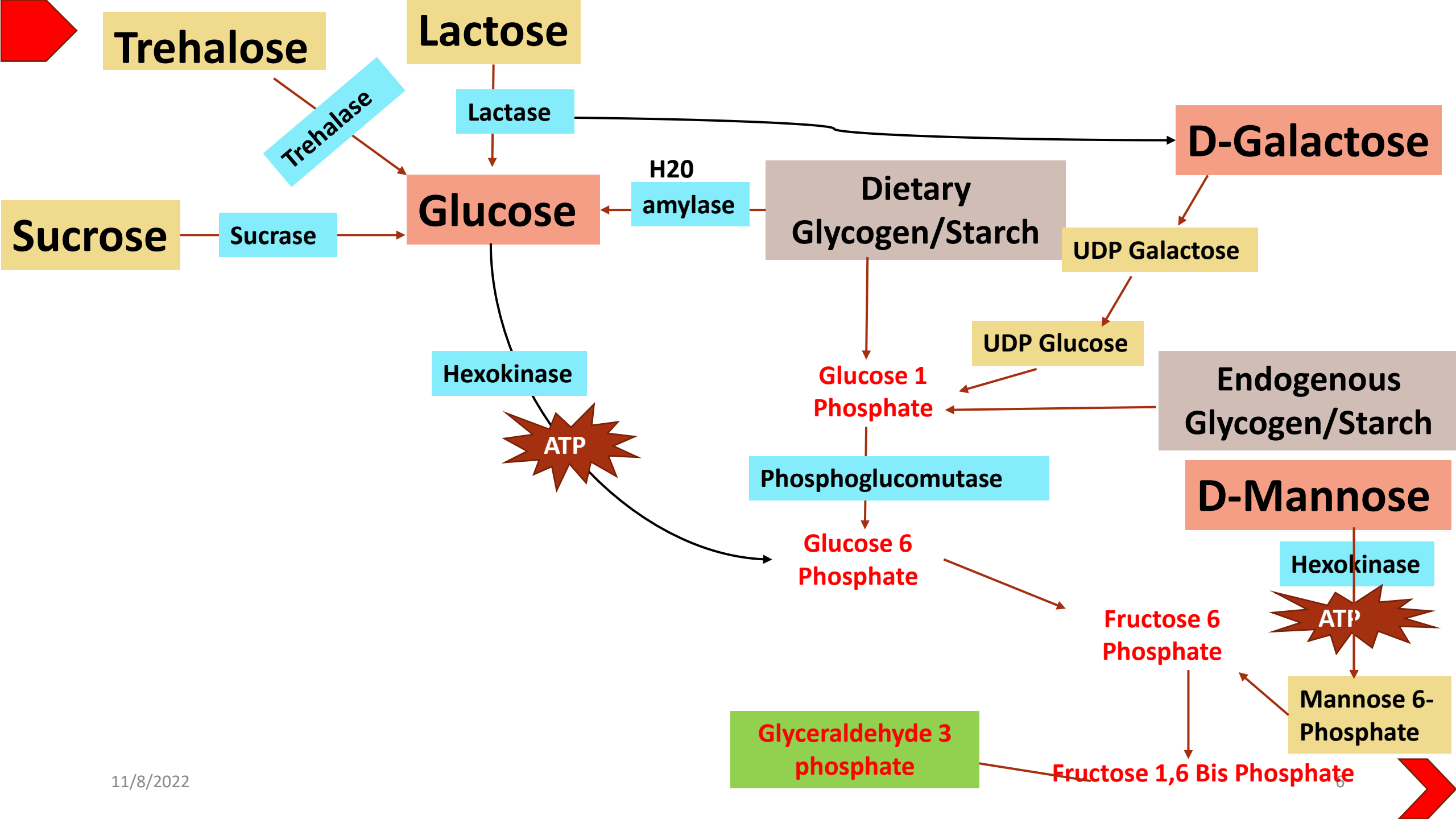
Feed Pathway for Glycolysis

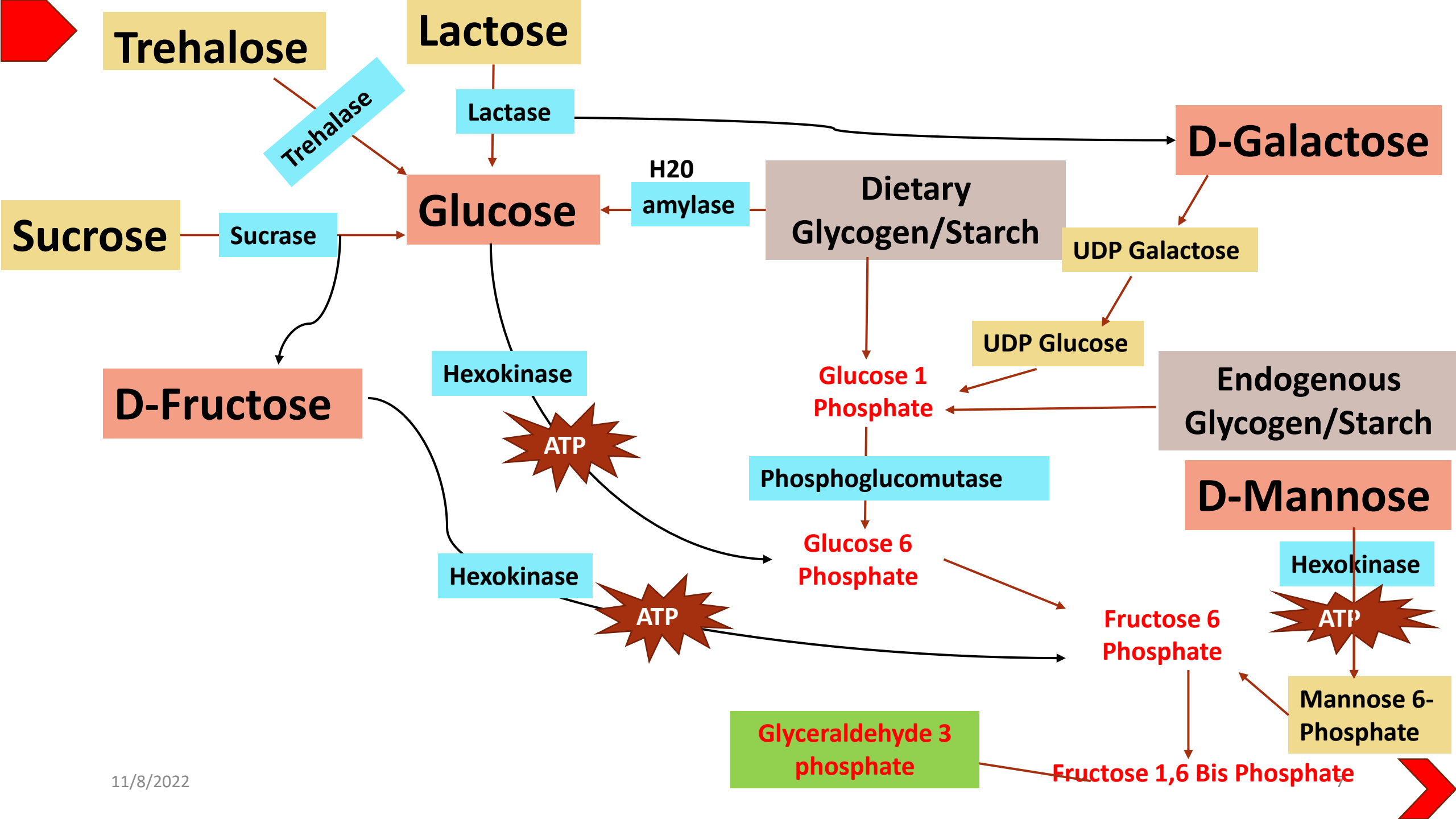
- Apart from glycolysis other carbohydrates meet their catabolic fate in glycolysis
 - After being transformed into one of the glycolytic intermediates
- The most significant of these carbohydrates are
 - Starch or Glycogen
 - Obtain from diet source OR
 - Endogenous (with in the cells)
 - Disaccharides e.g., maltose. Lactose, trehalose. and sucrose
 - Monosaccharides e.g., fructose, mannose, and galactose

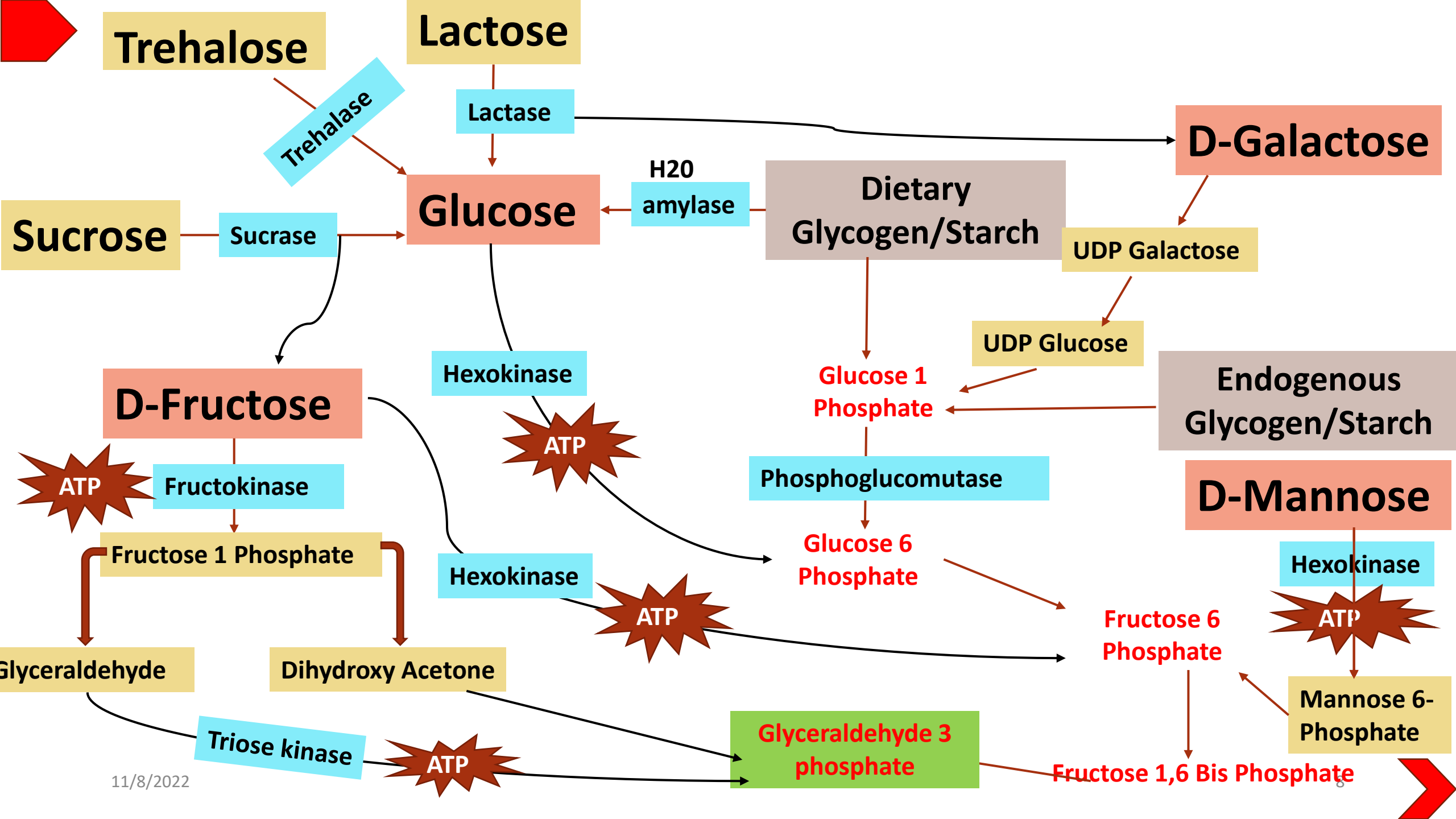


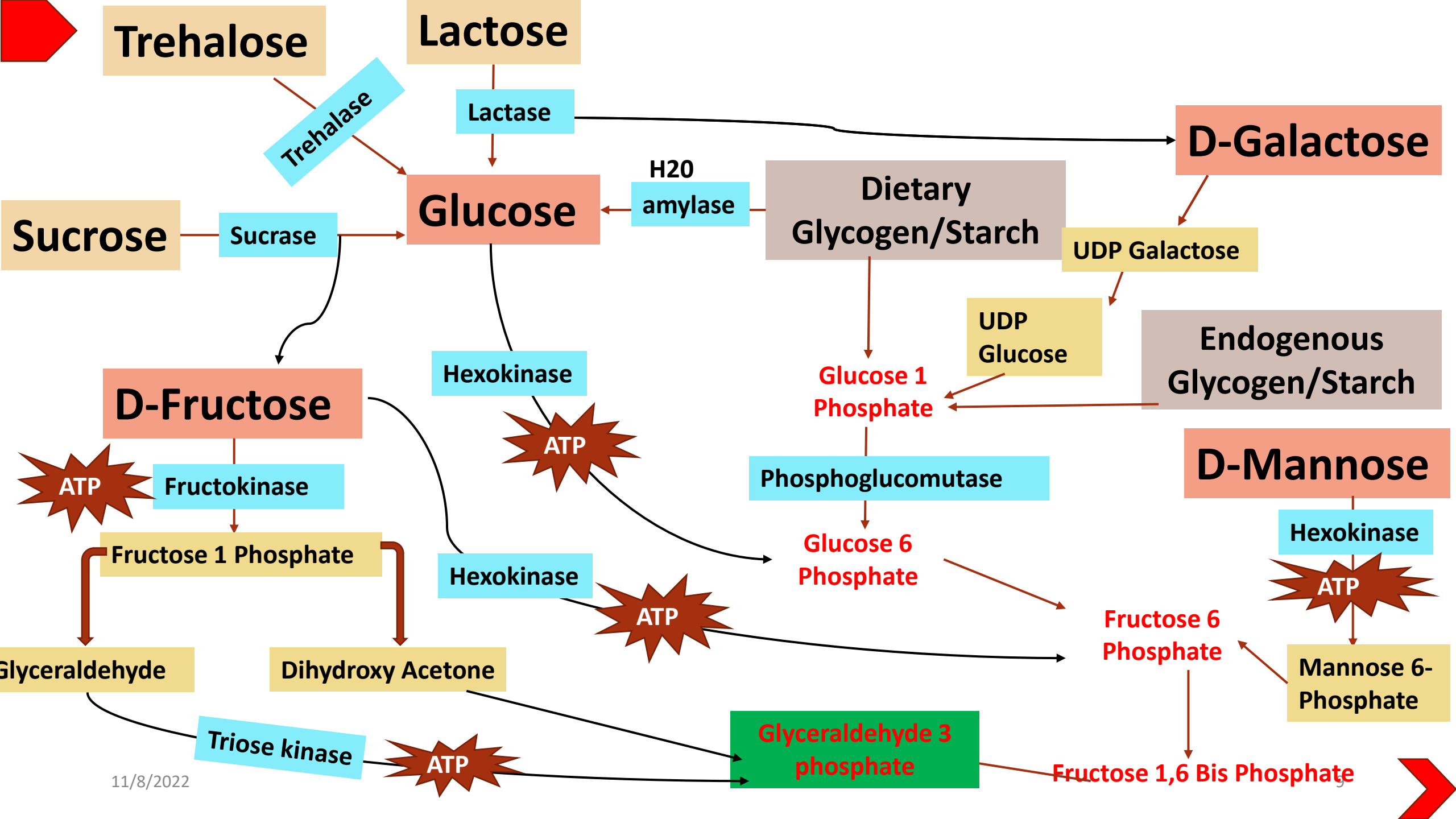


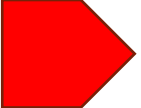






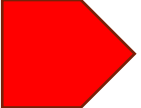






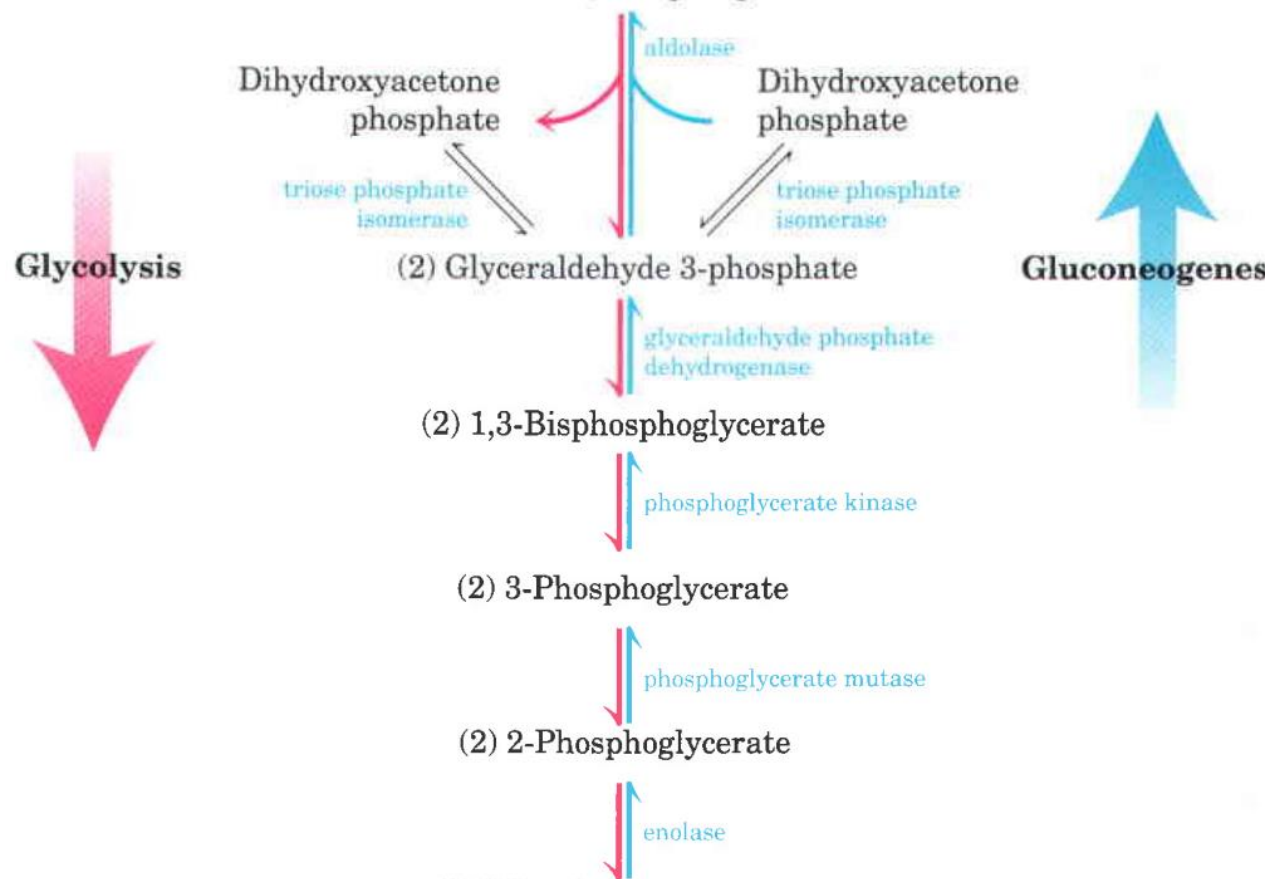
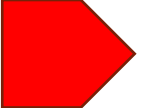
Gluconeogenesis

- The formation of new glucose molecules from non carbohydrate precursors, occurs primarily in the liver
 - Precursor molecules include lactate, pyruvate, glycerol, and certain -keto acids (molecules derived from amino acids)
- Under certain conditions (i.e., metabolic acidosis or starvation) the kidney can make small amounts of new glucose
- Between meals adequate blood glucose levels are maintained by the hydrolysis of liver glycogen
 - When liver glycogen is depleted (e.g., owing to prolonged fasting or vigorous exercise), the gluconeogenesis pathway provides the body with adequate glucose

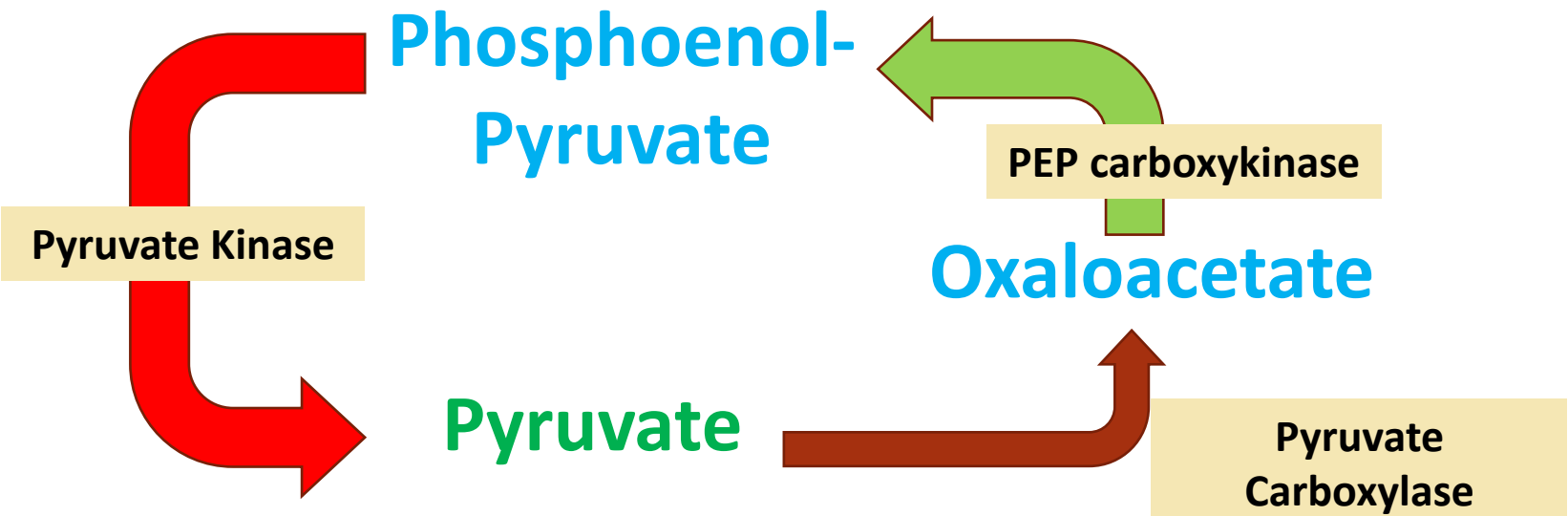


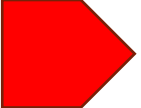
Gluconeogenesis Reaction

- The reaction sequence in gluconeogenesis is largely the **reverse of glycolysis with three rearrangement steps**
- The three Irreversible glycolytic reactions are catalyzed by
 - **Hexokinase**
 - **Phosphofructokinase-1,**
 - **pyruvate kinase**
 - In gluconeogenesis, alternate reactions catalyzed by different enzymes are used to bypass these obstacles. The reactions unique to gluconeogenesis are listed next



1st
Rearrangement
Step

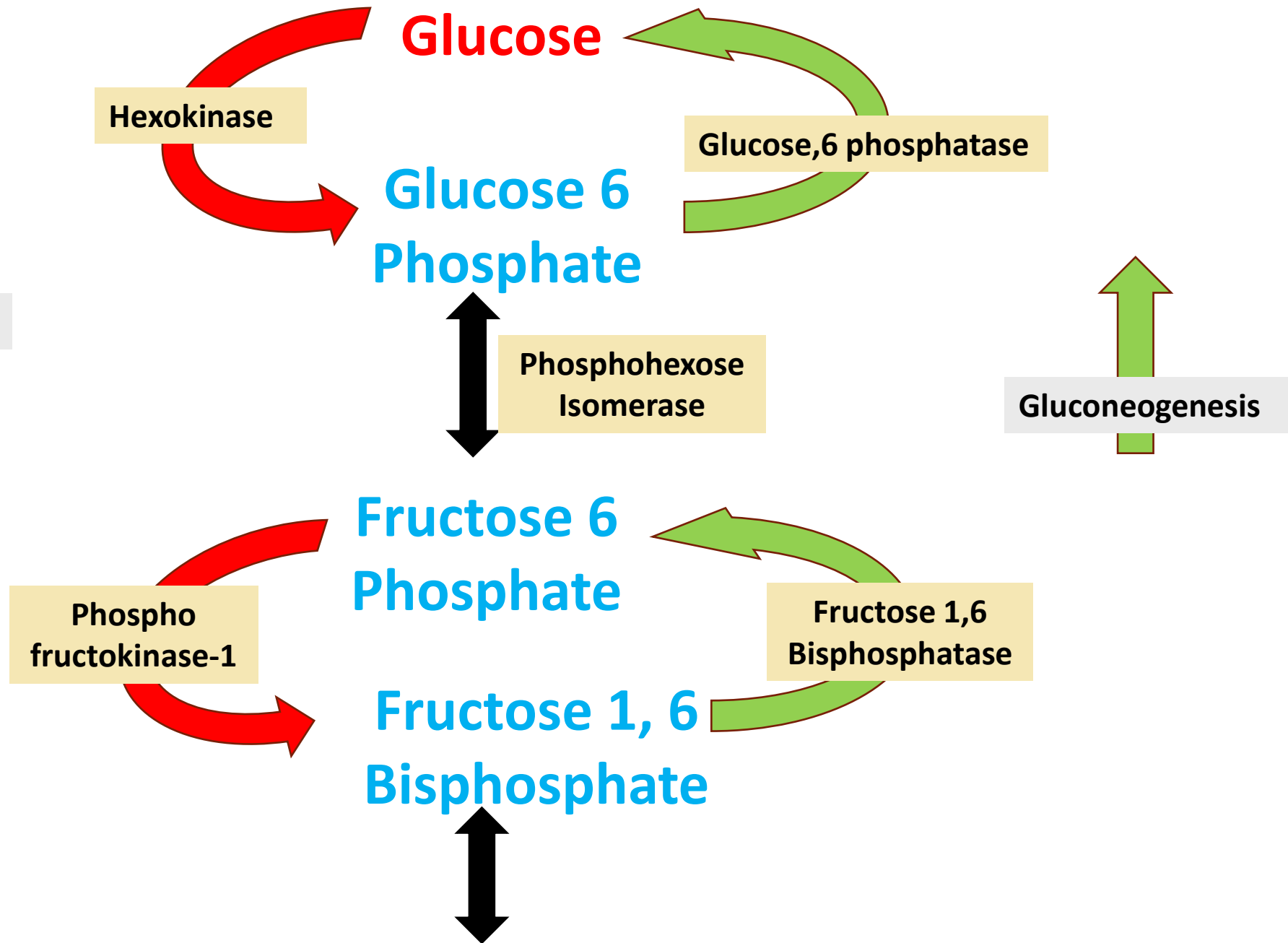


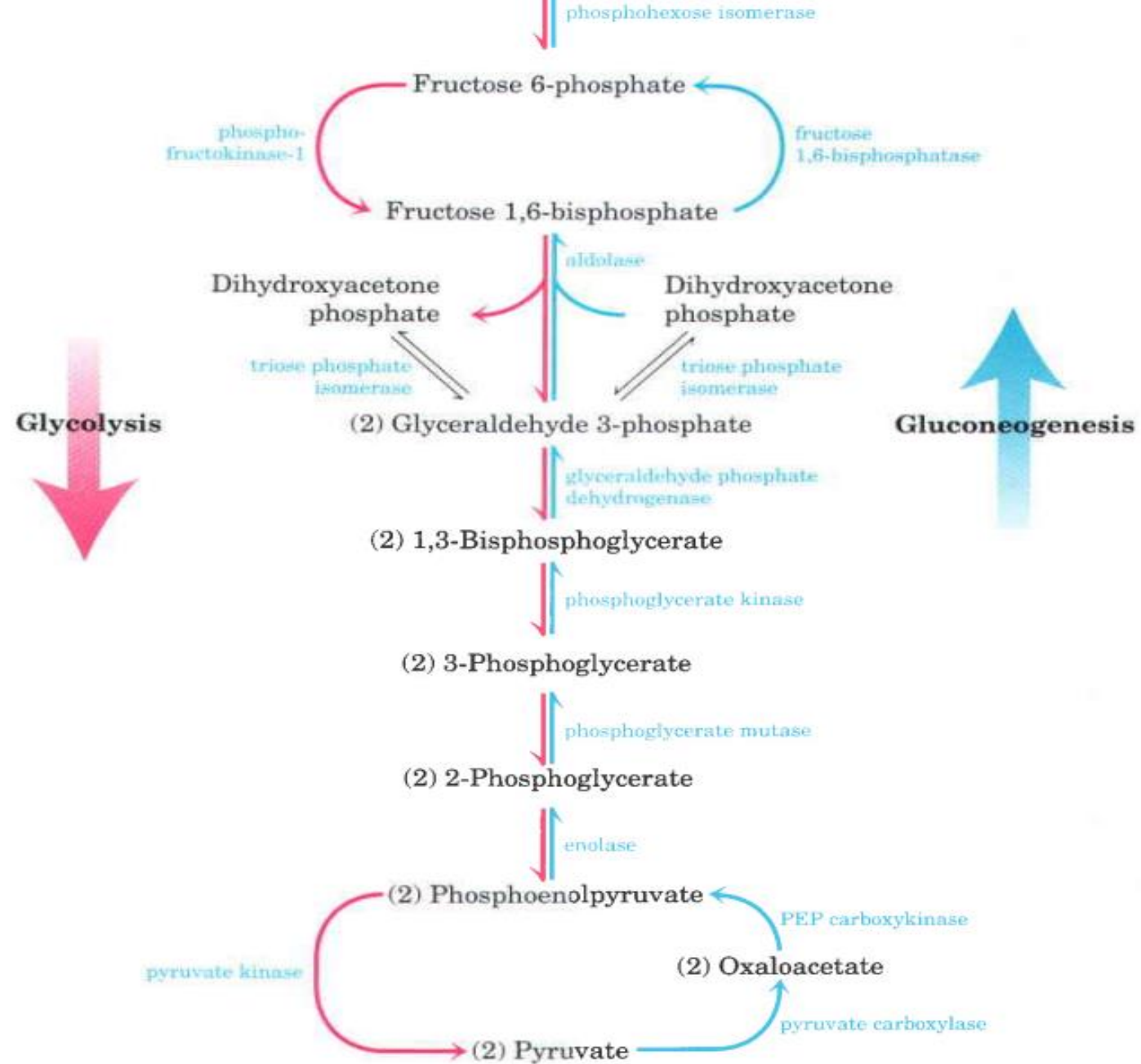


3rd
Rearrangement
Step

Glycolysis

2nd
Rearrangement
Step

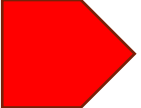




Feeder Pathway For Glycolysis

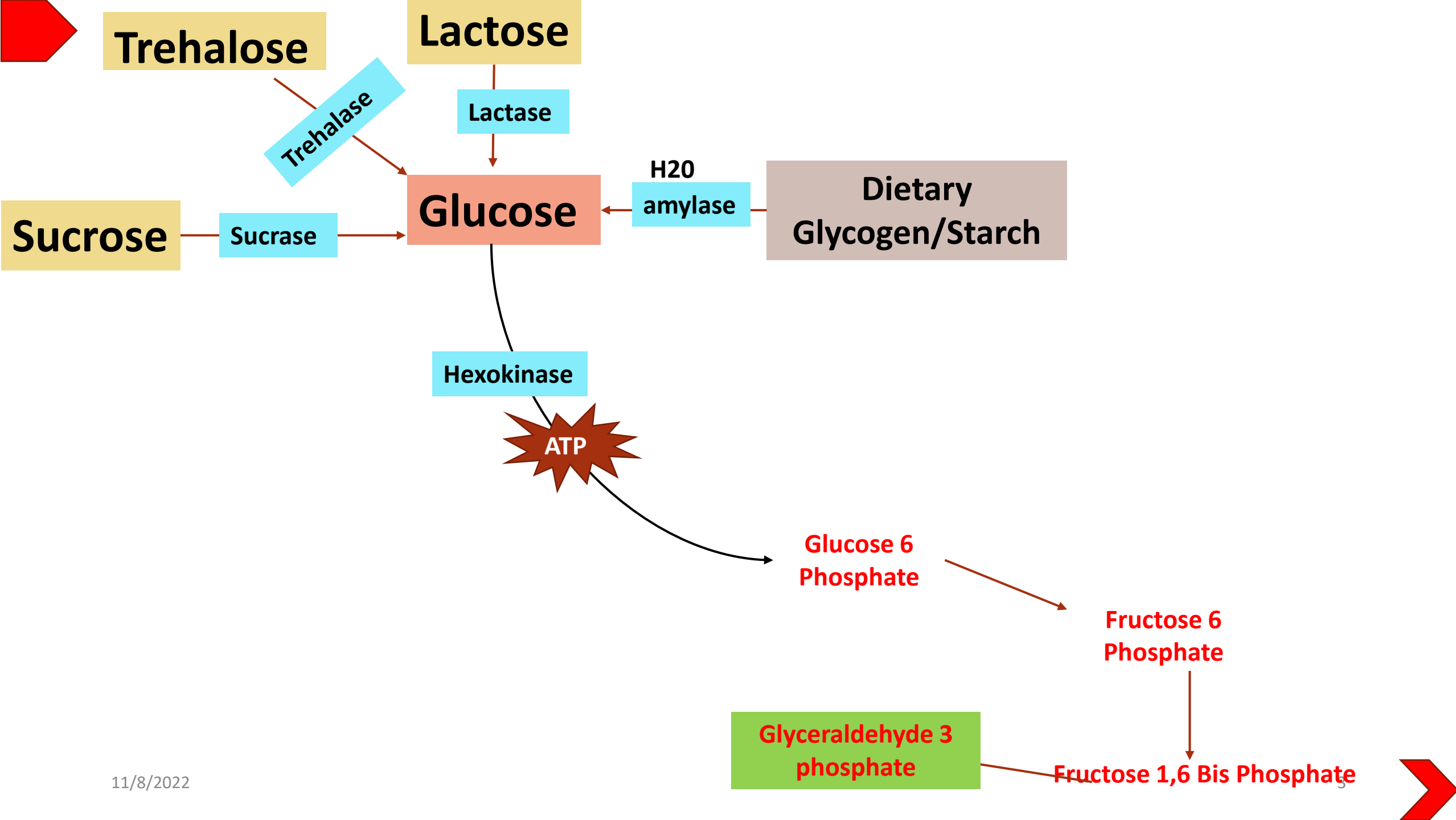
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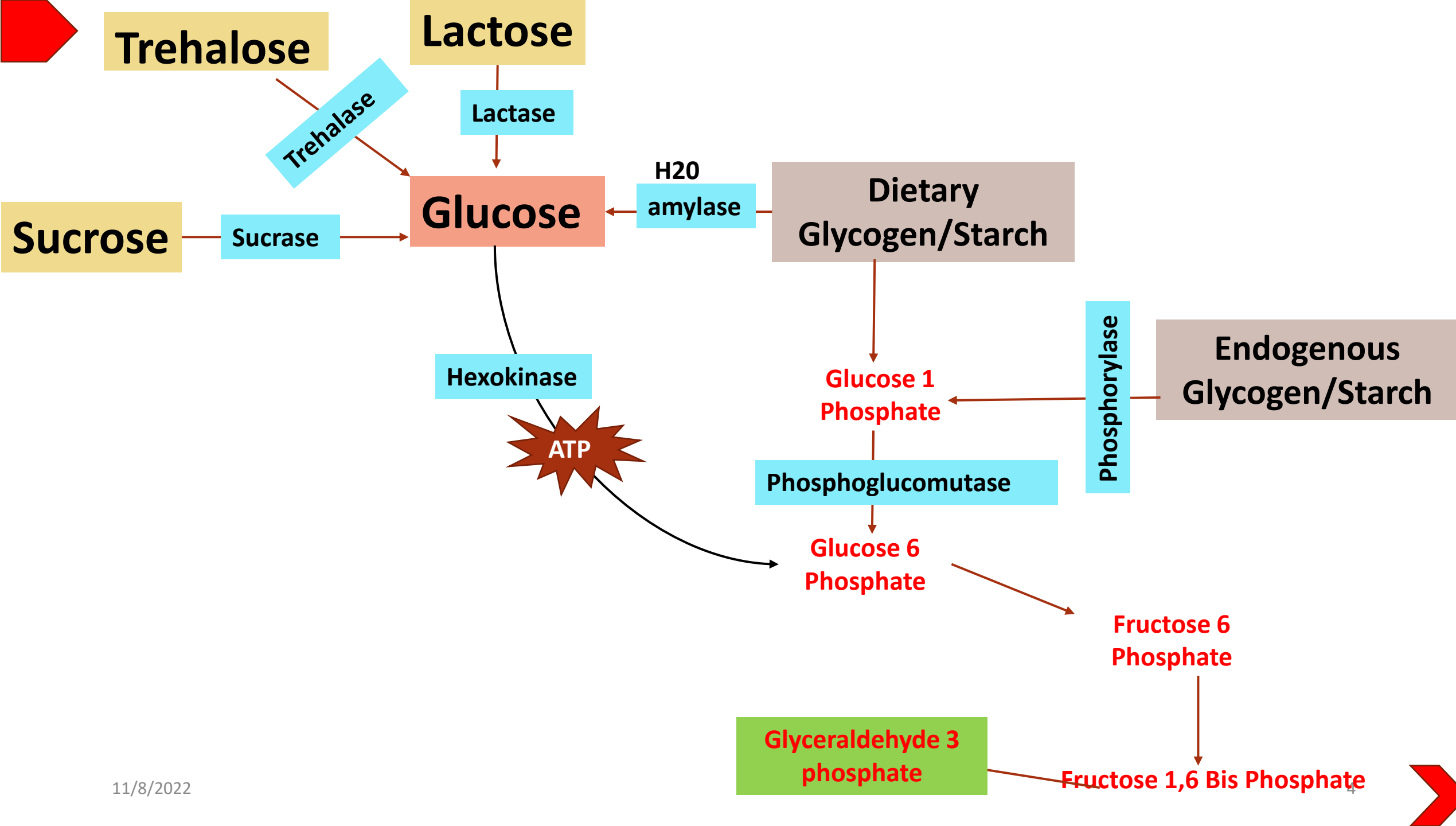


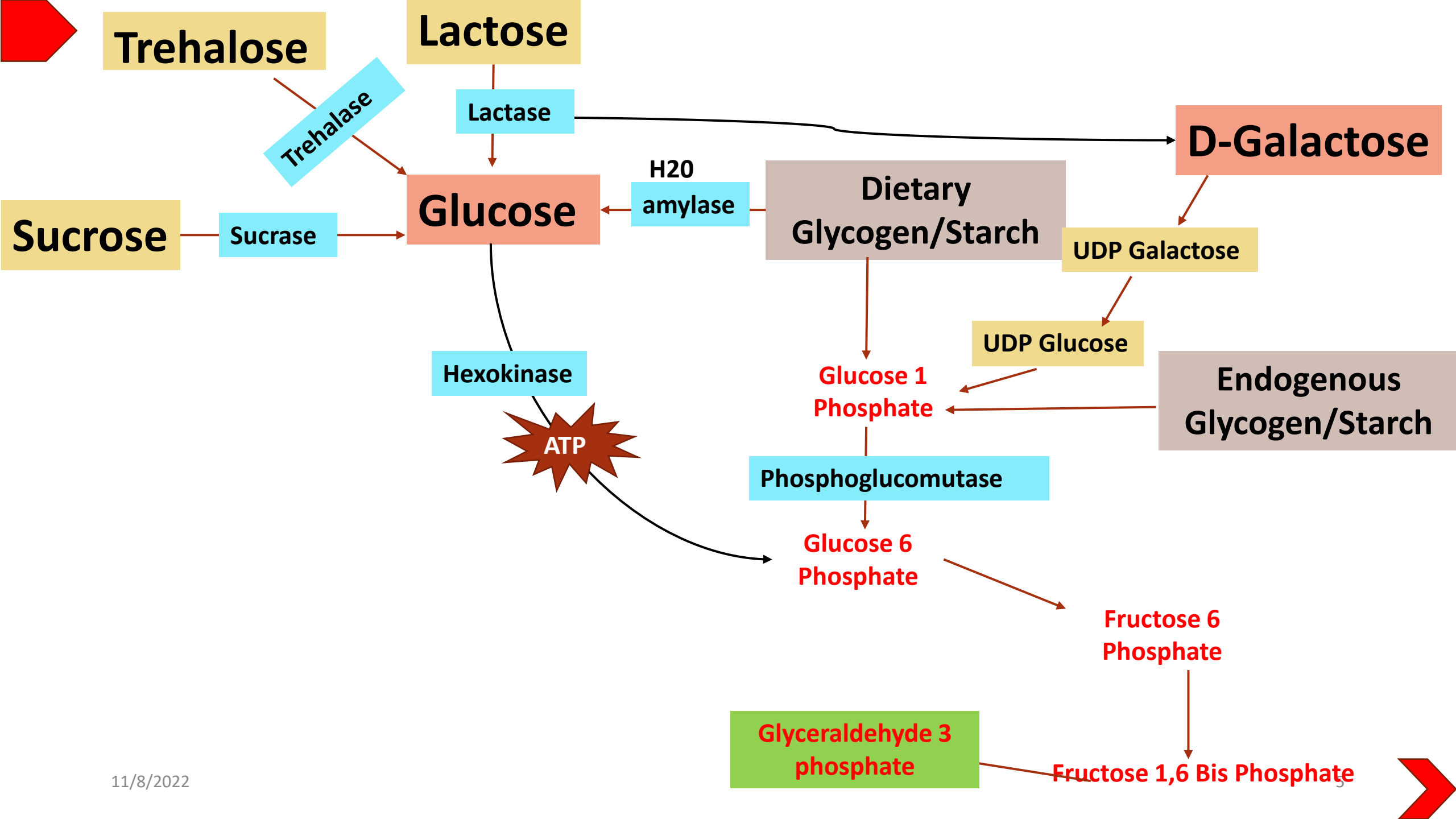


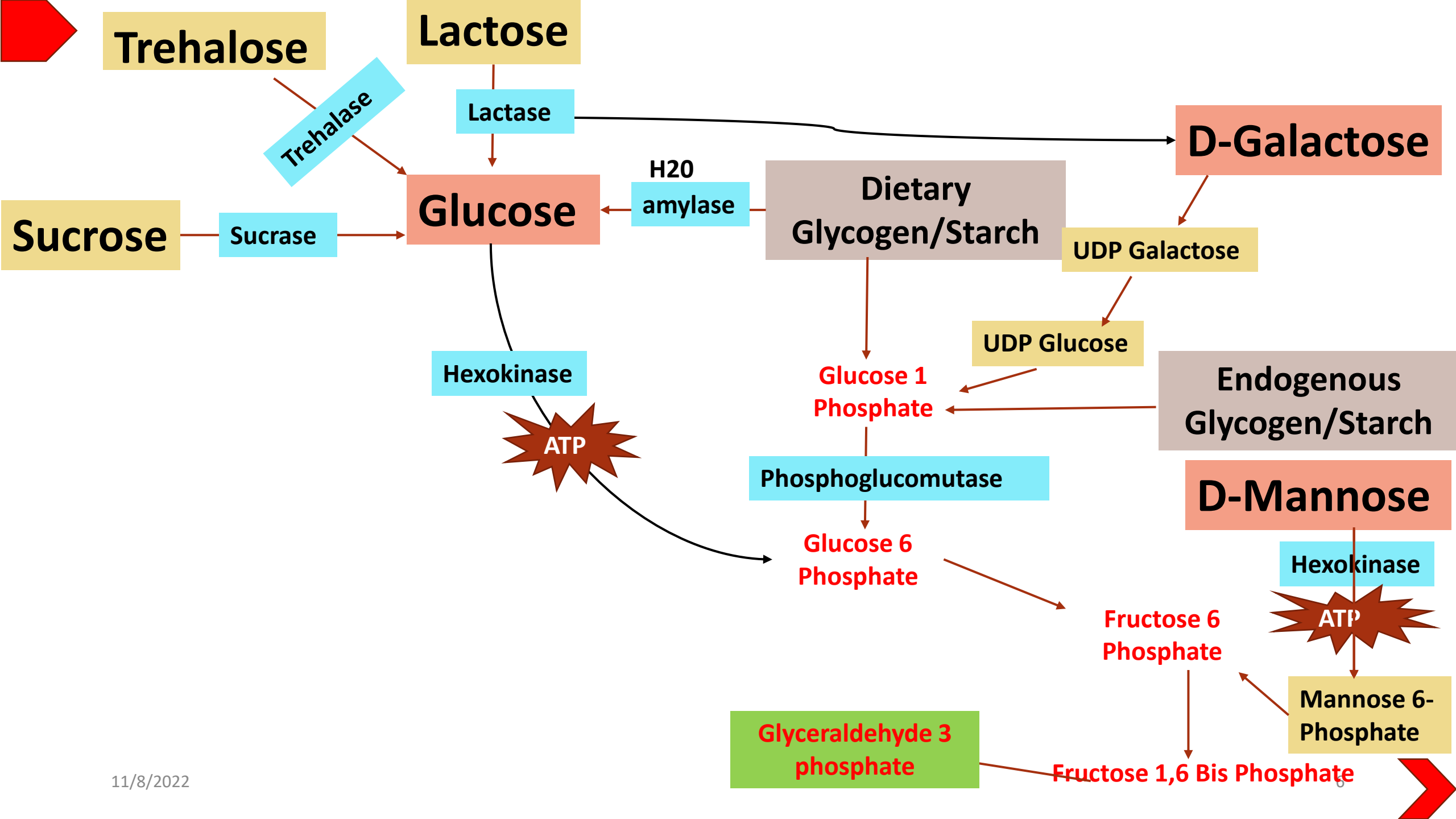
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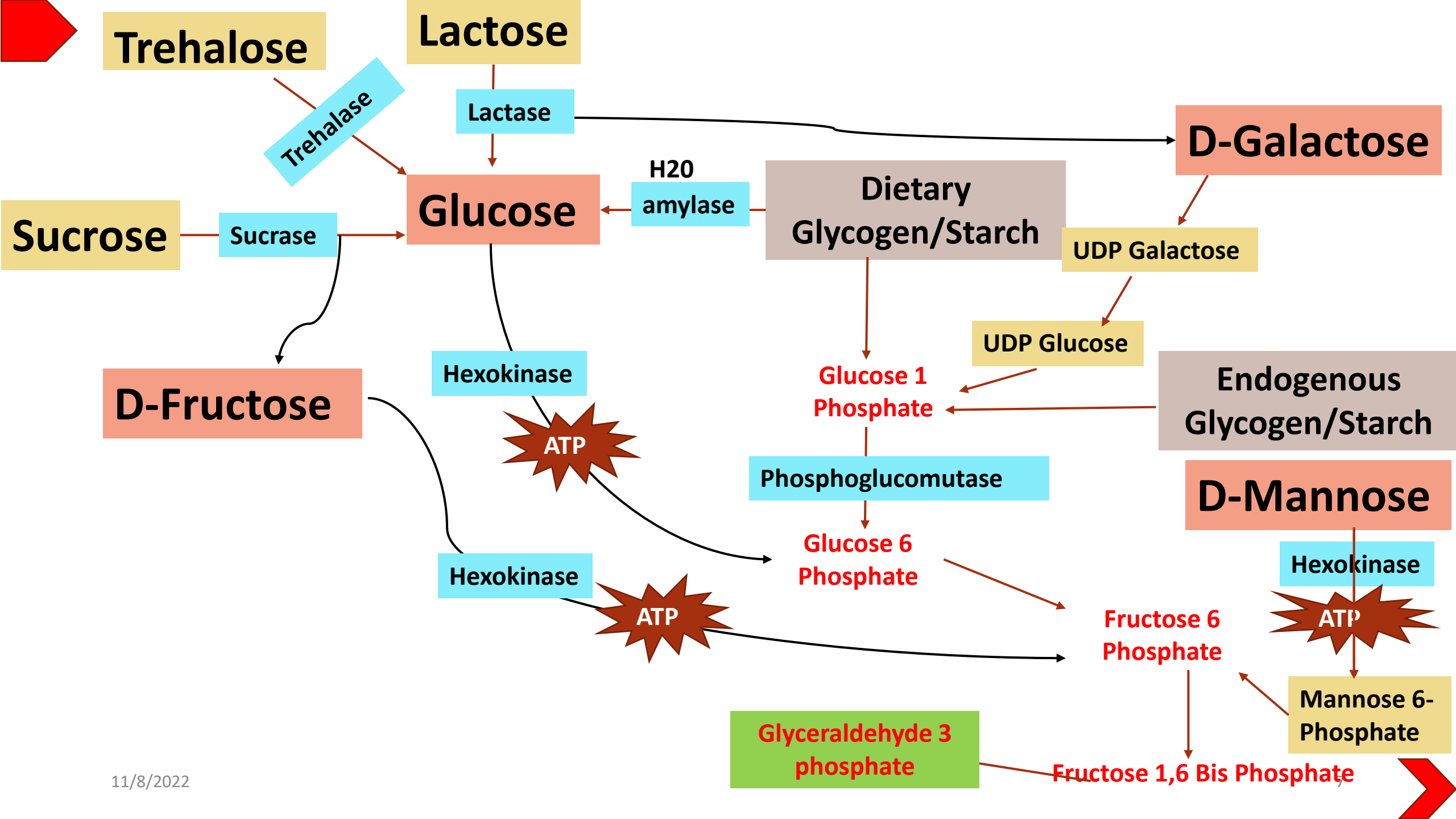
- Apart from glycolysis other carbohydrates meet their catabolic fate in glycolysis
 - After being transformed into one of the glycolytic intermediates
- The most significant of these carbohydrates are
 - Starch or Glycogen
 - Obtain from diet source OR
 - Endogenous (with in the cells)
 - Disaccharides e.g., maltose. Lactose, trehalose. and sucrose
 - Monosaccharides e.g., fructose, mannose, and galactose

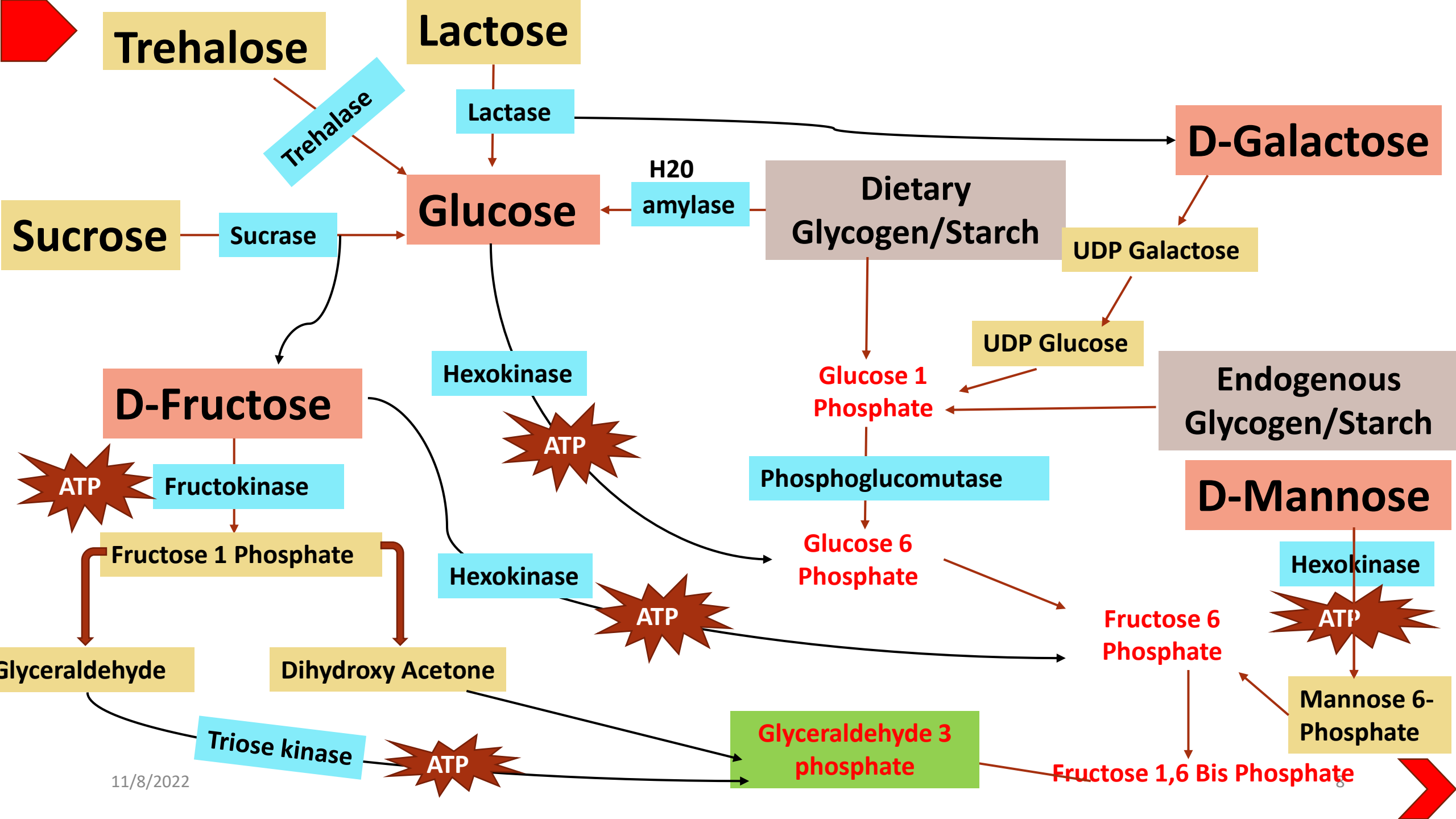


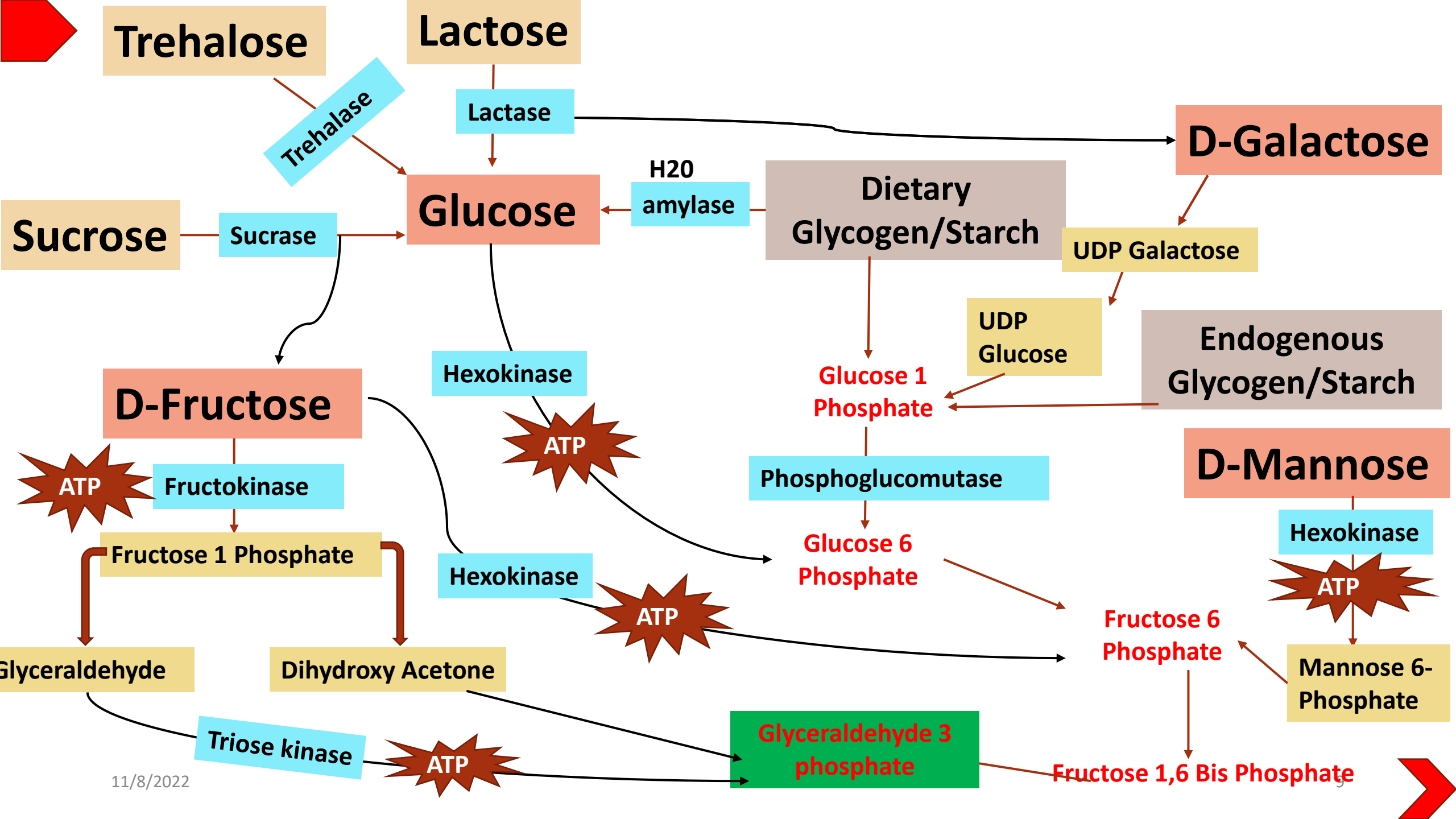


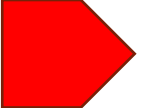






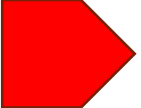






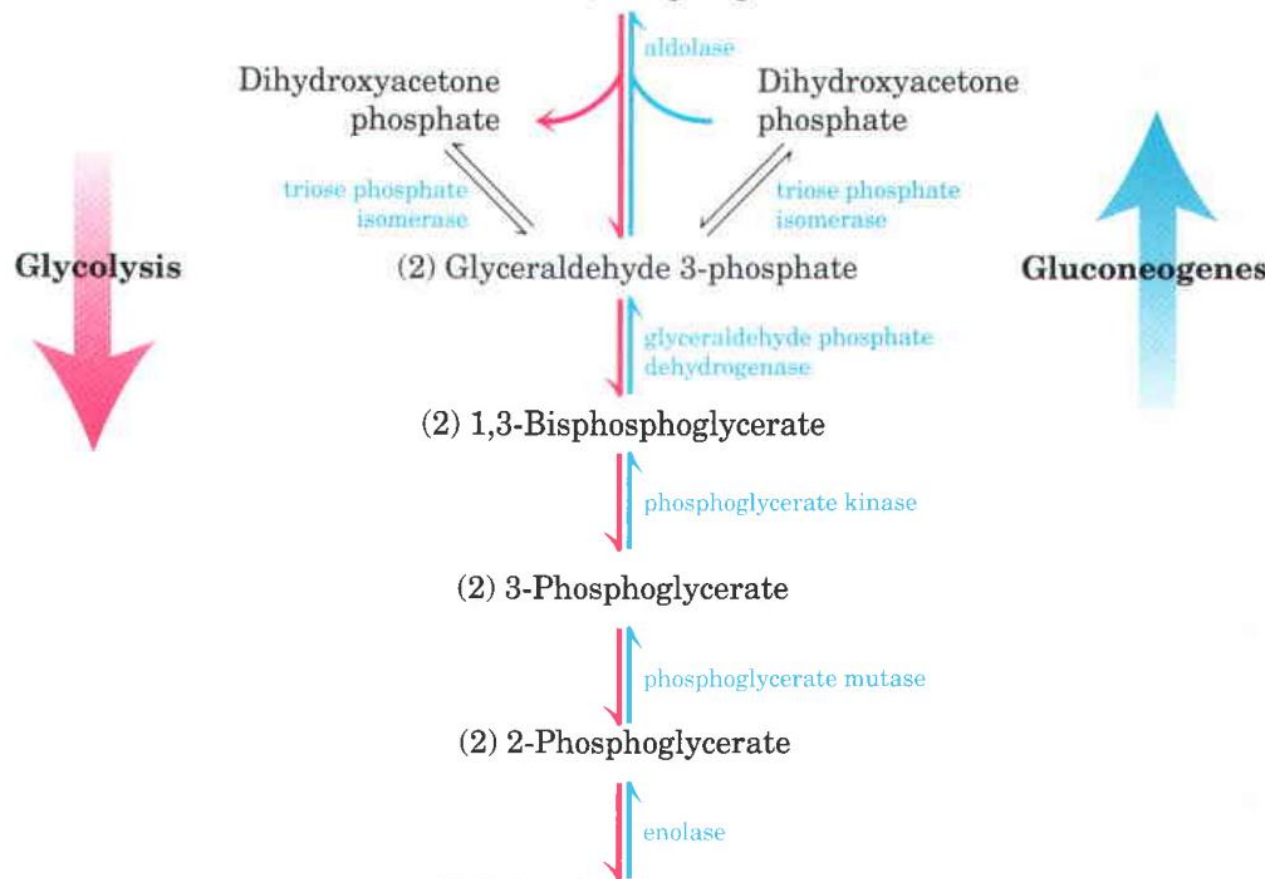
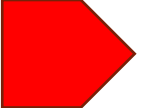
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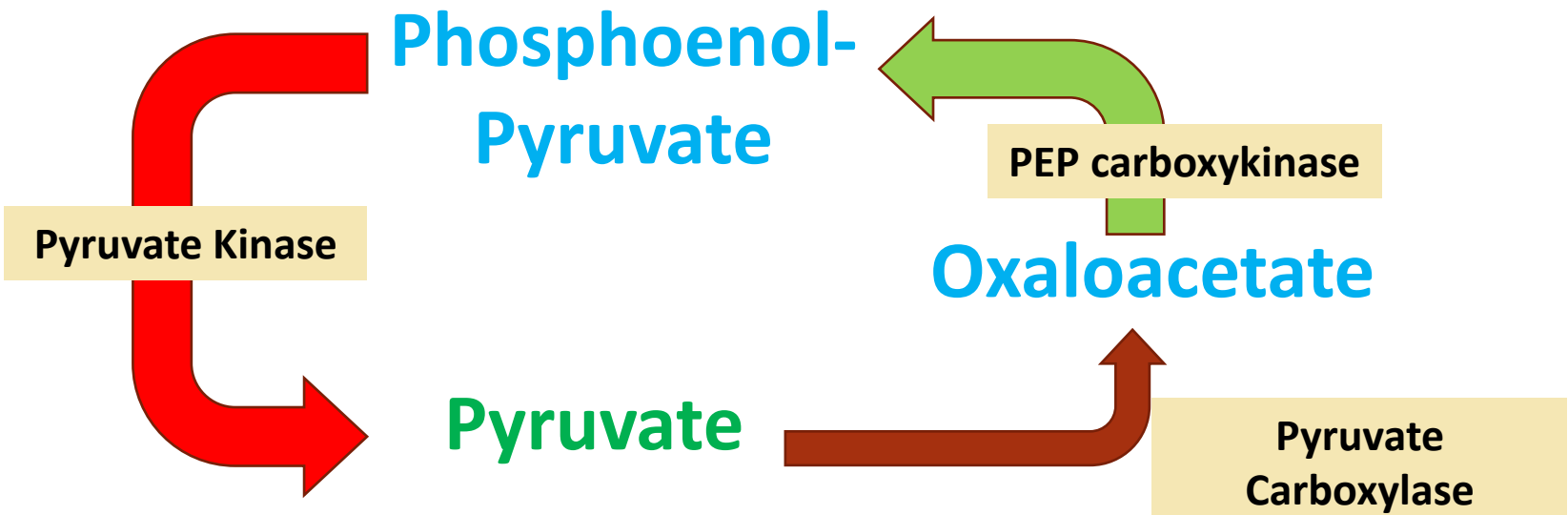


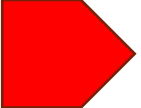
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 - **Phosphofructokinase-1,**
 - **pyruvate kinase**
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1st
Rearrangement
Step

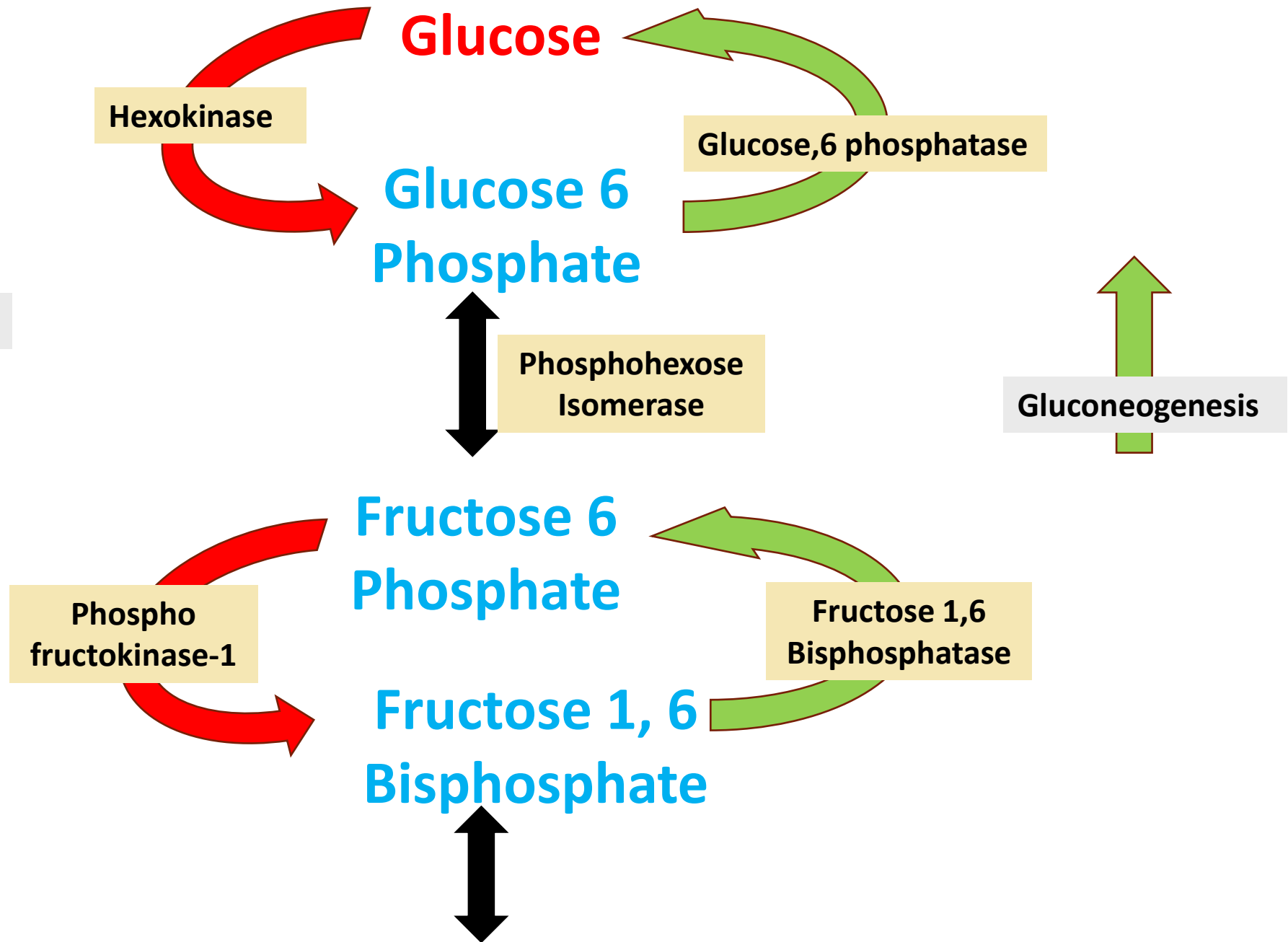


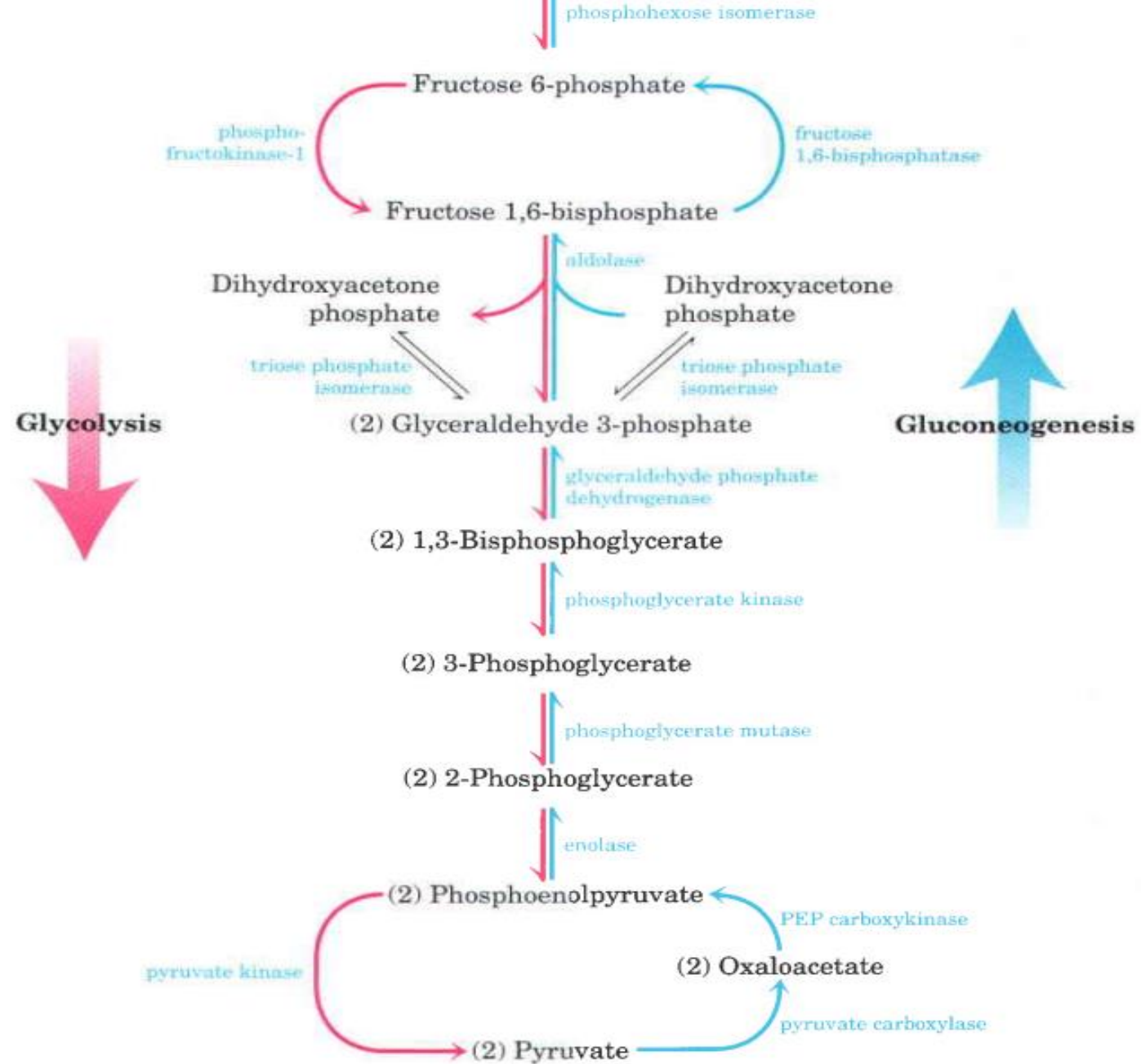


3rd
Rearrangement
Step

Glycolysis

2nd
Rearrangement
Step





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Gluconeogenesis

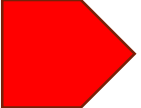


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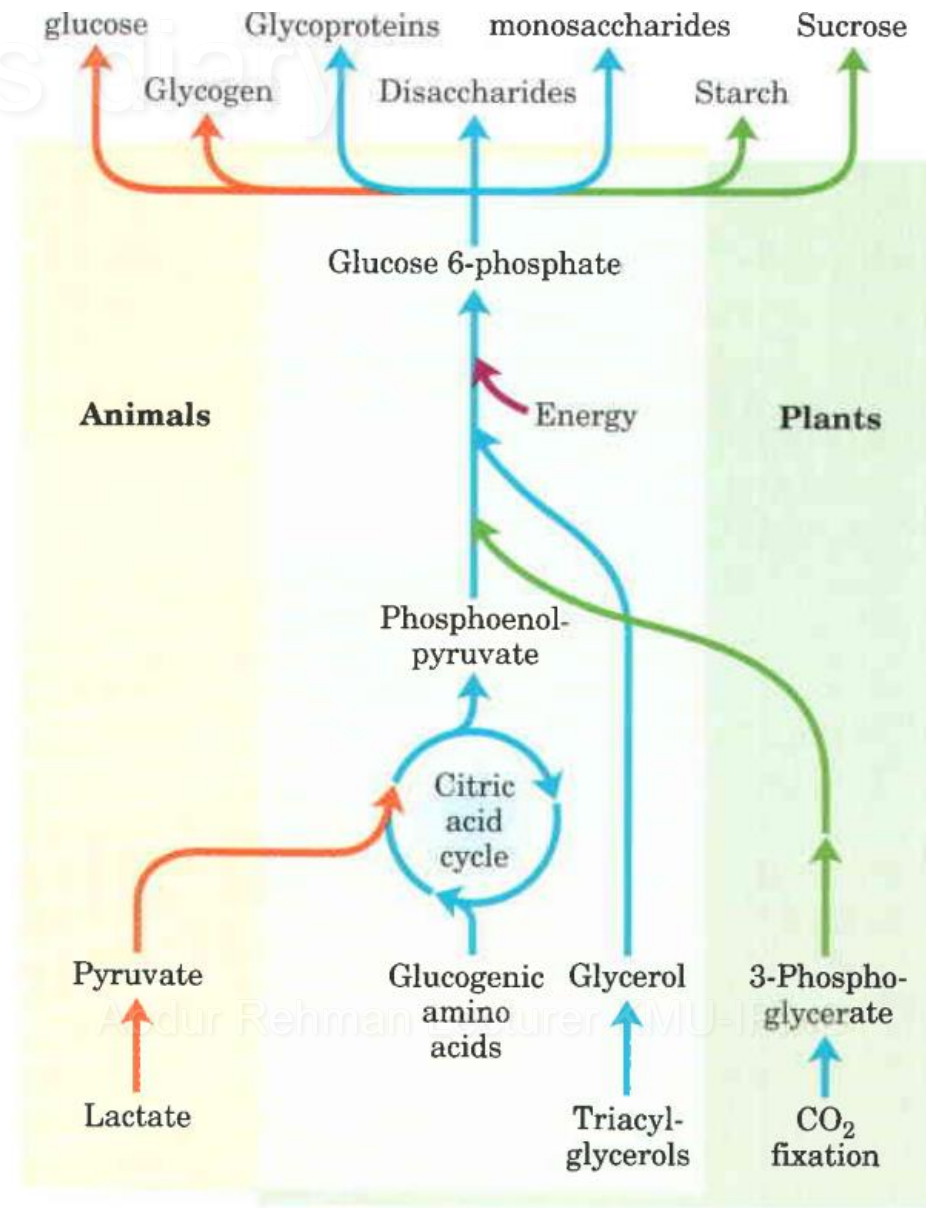
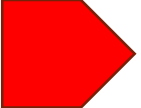


FIGURE 14-15 Carbohydrate synthesis from simple precursors. The



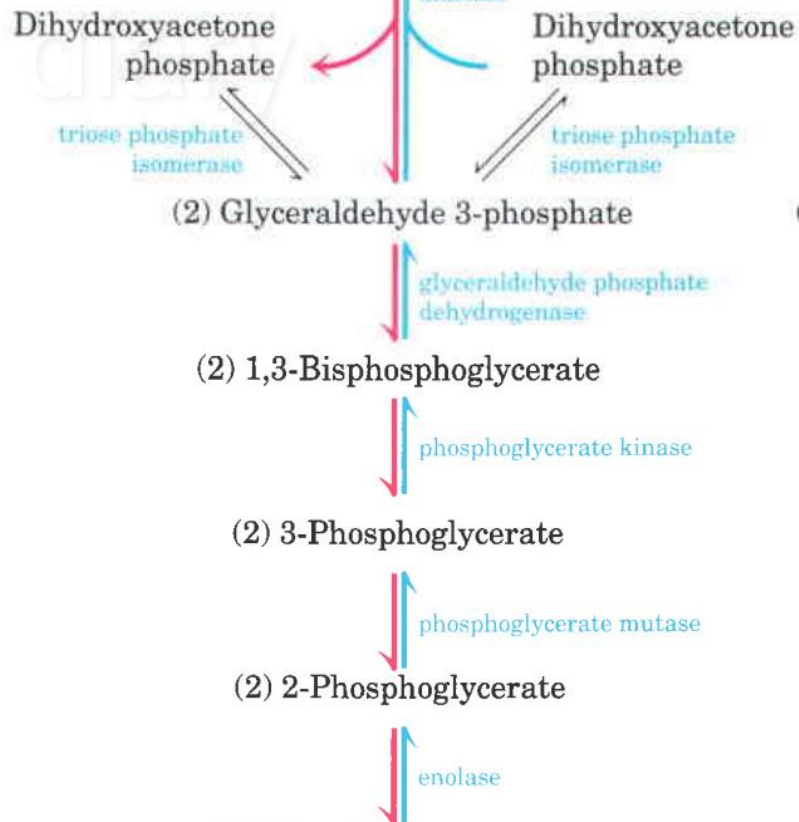
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Glycolysis

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Pyruvate Kinase

ATP

Phosphoenol-
Pyruvate

GTP

PEP carboxykinase

Oxaloacetate

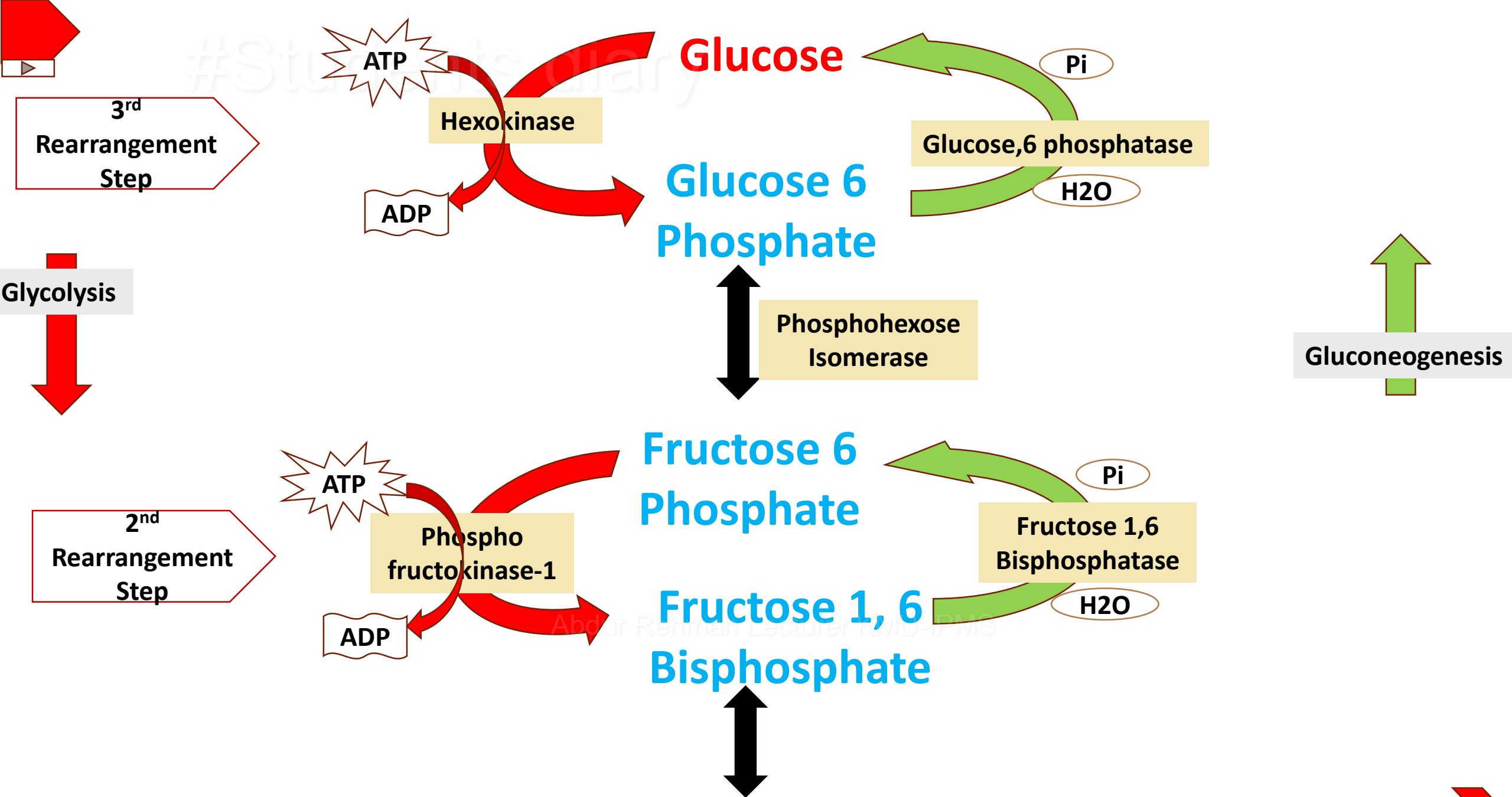
Pyruvate

ATP

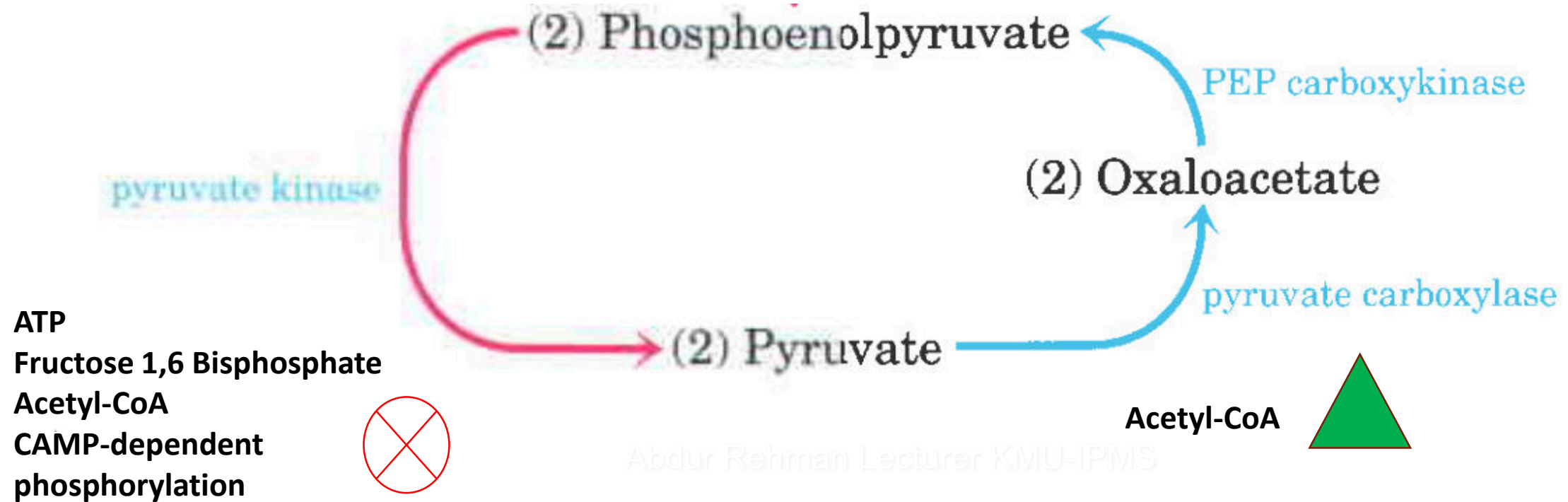
Pyruvate
Carboxylase

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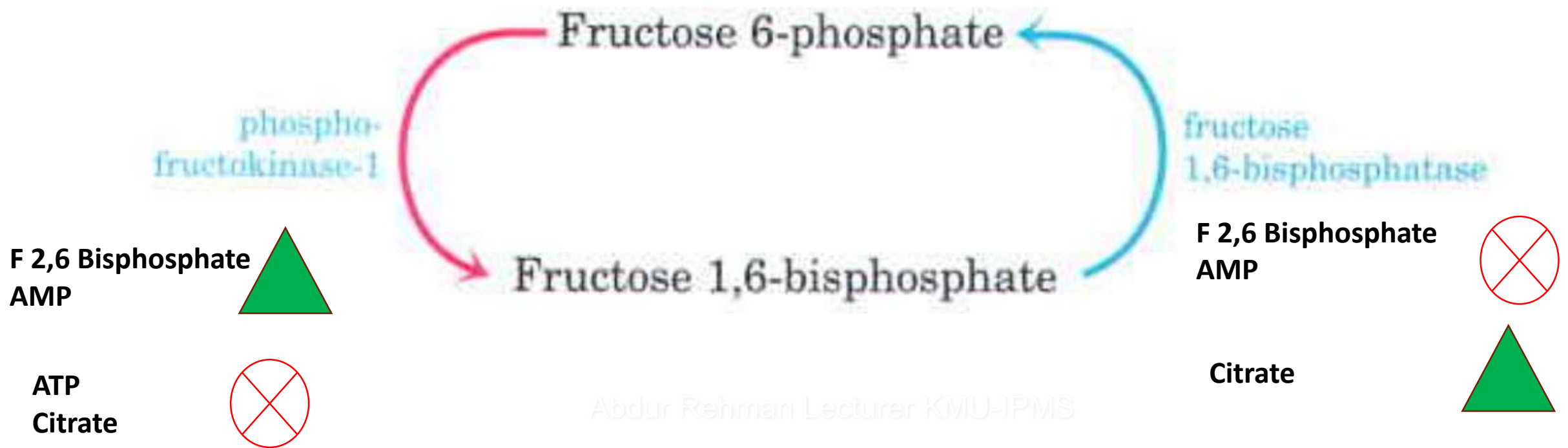
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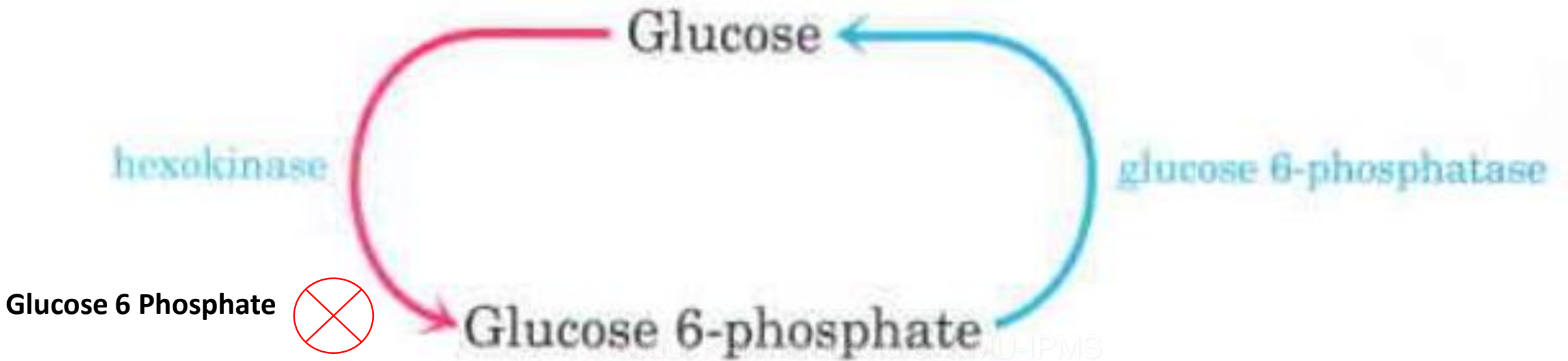
Regulation of Gluconeogenesis



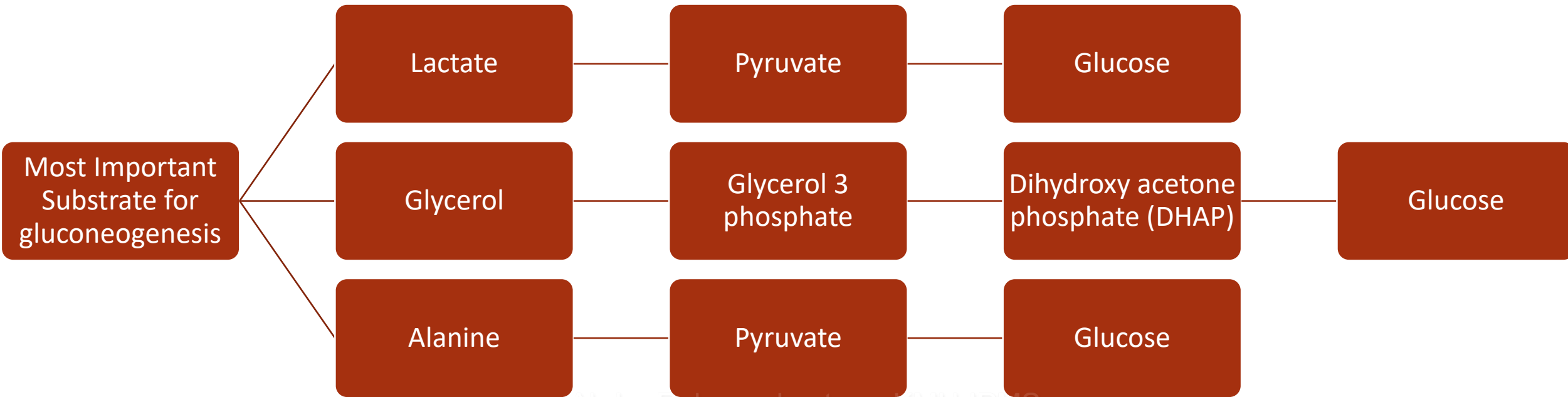
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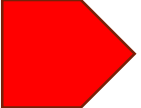
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Substrate of Gluconeogenesis



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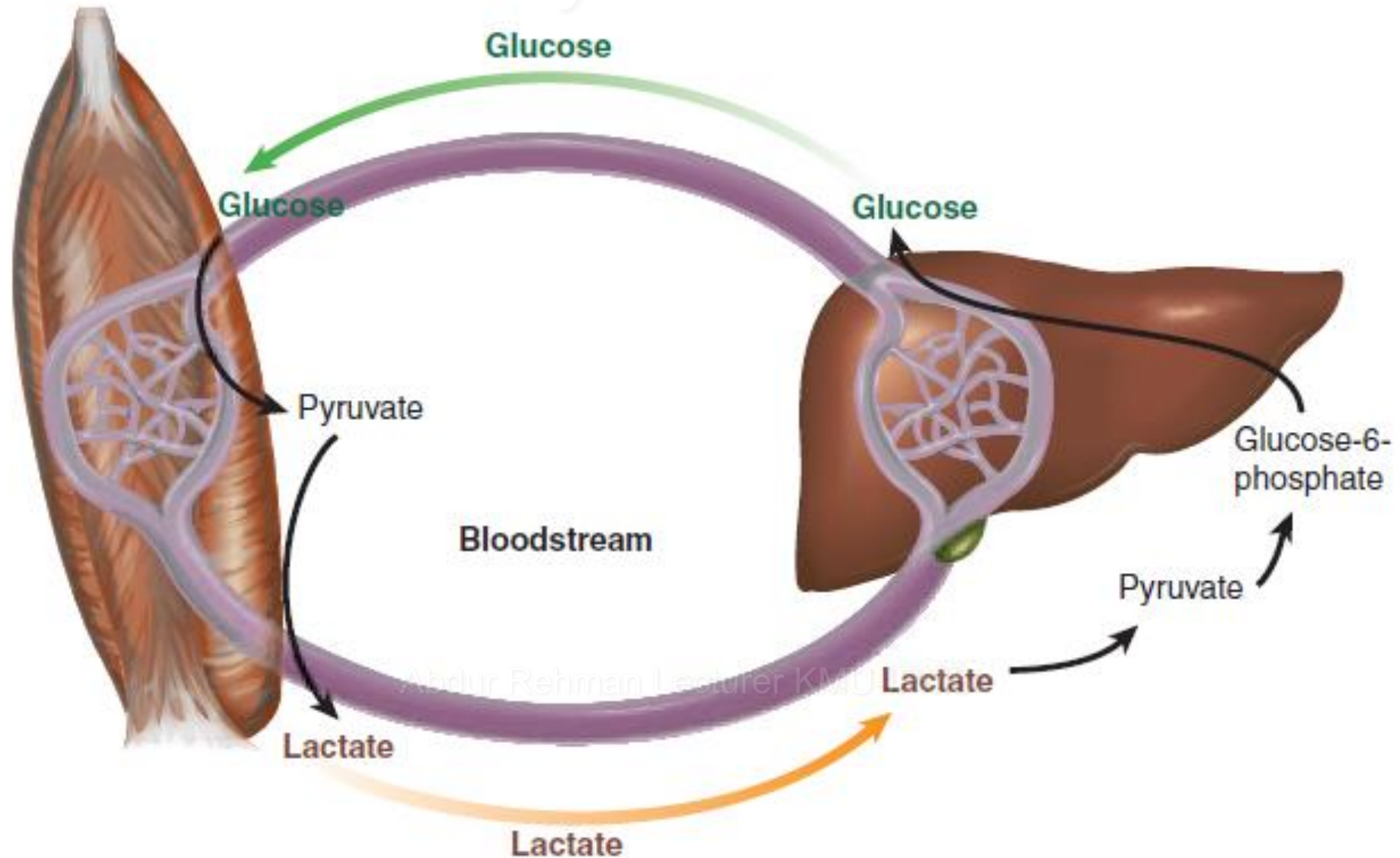


Cori cycle

- **Lactate** is released by **red blood cells** and **other cells that lack mitochondria or have low oxygen concentrations**
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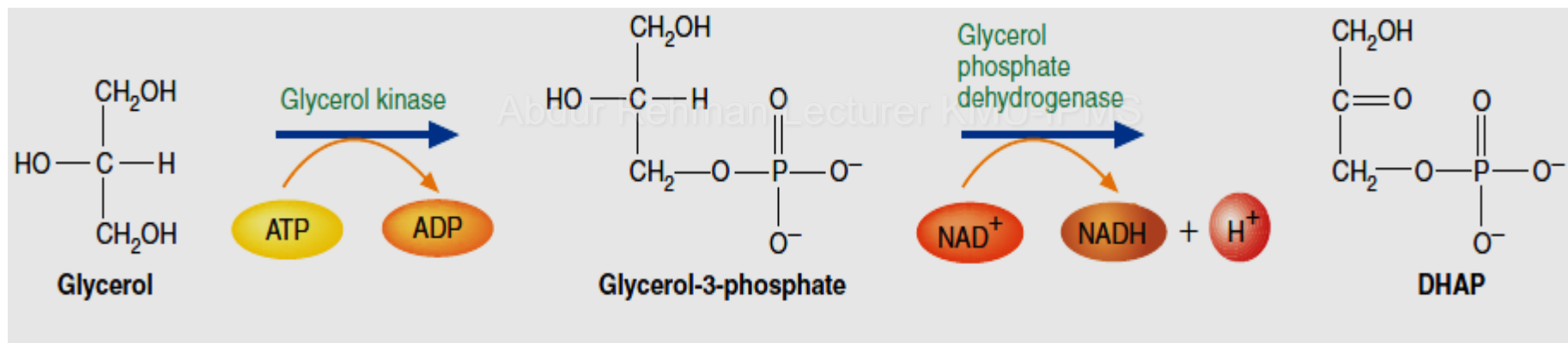
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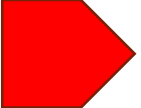
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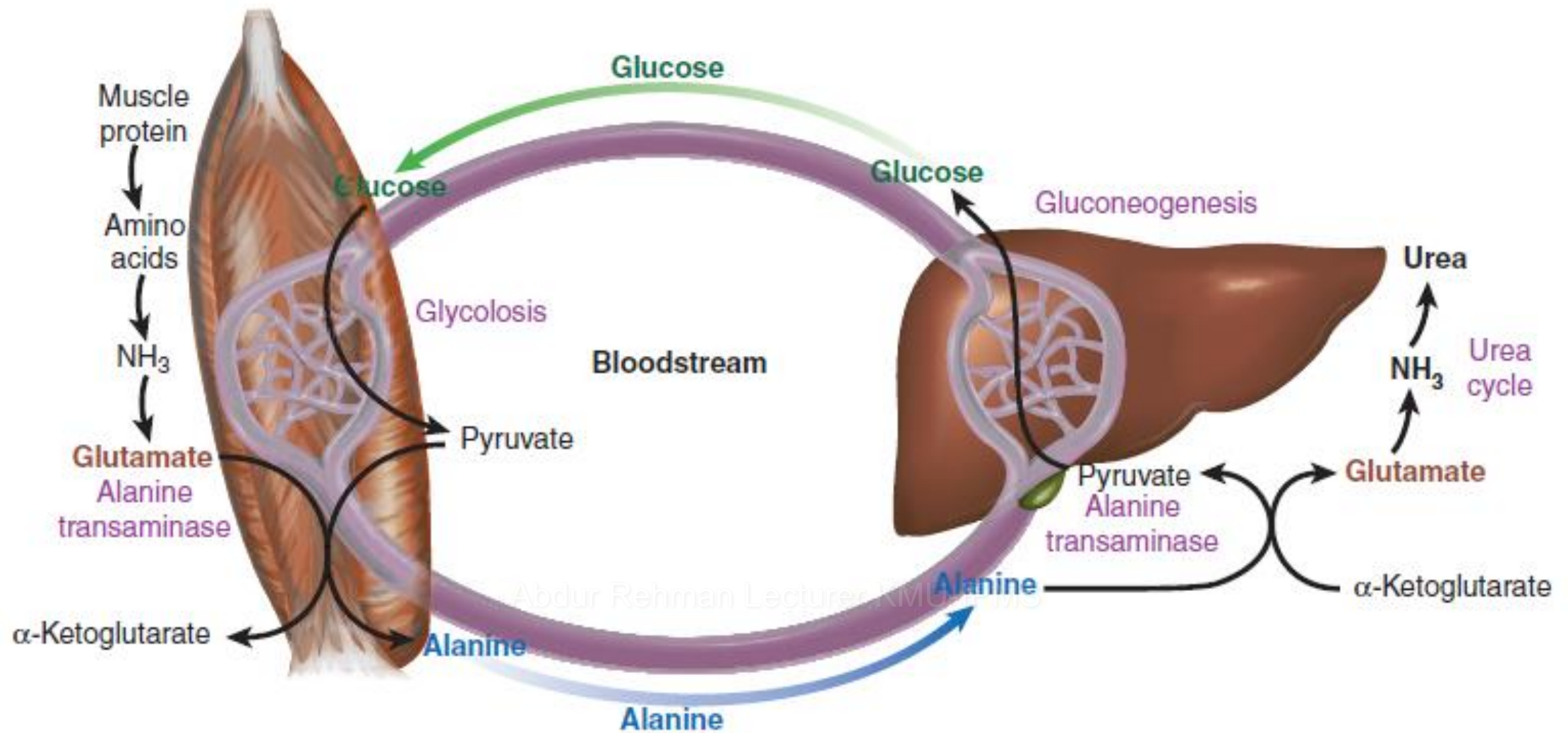


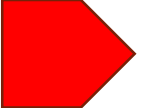


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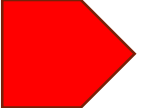




#Students diary **Reference**

- Biochemistry by Satyanarayana and Chakrapani
- Lehninger Principles of Biochemistry, 5th Edition

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Gluconeogenesis



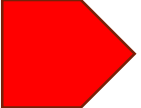
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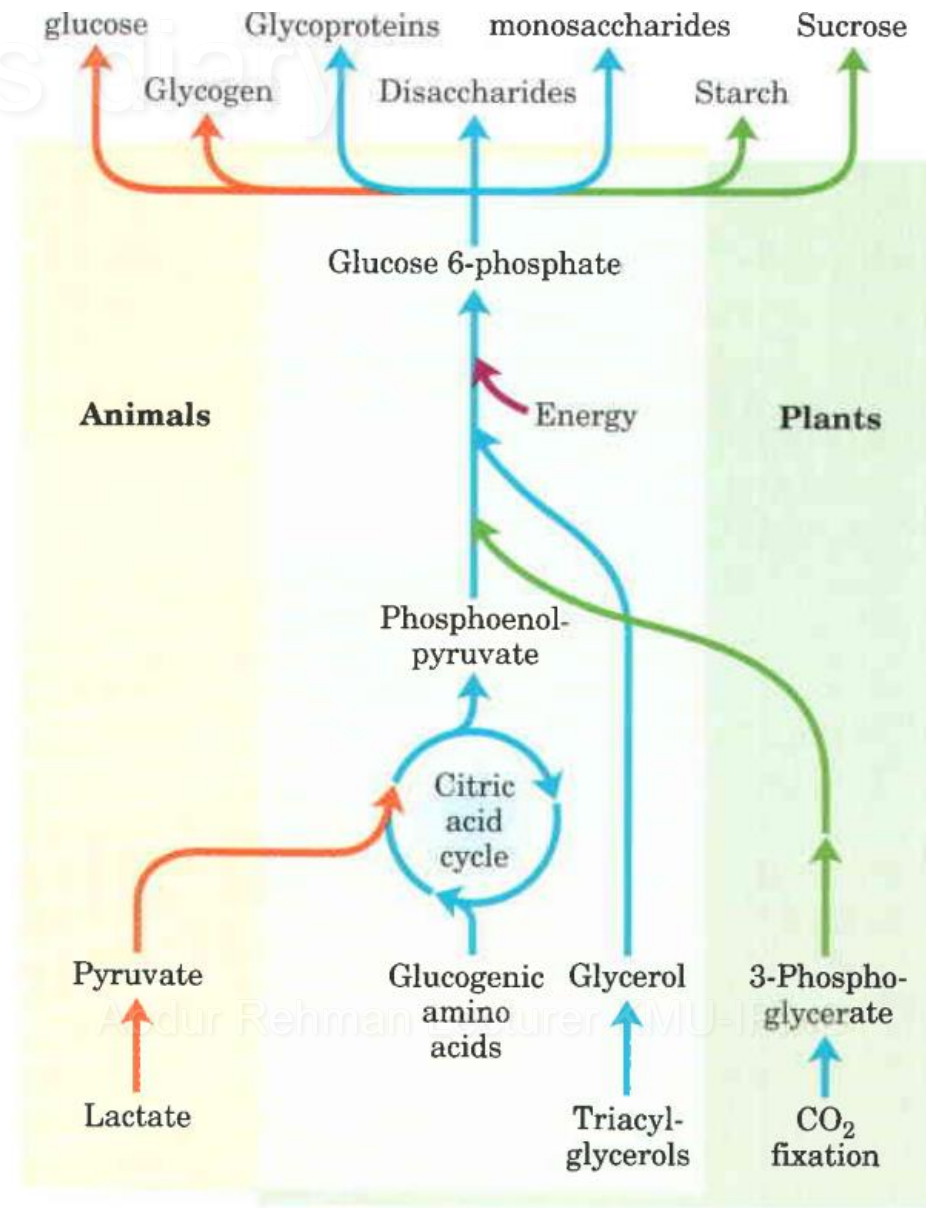
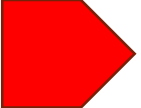


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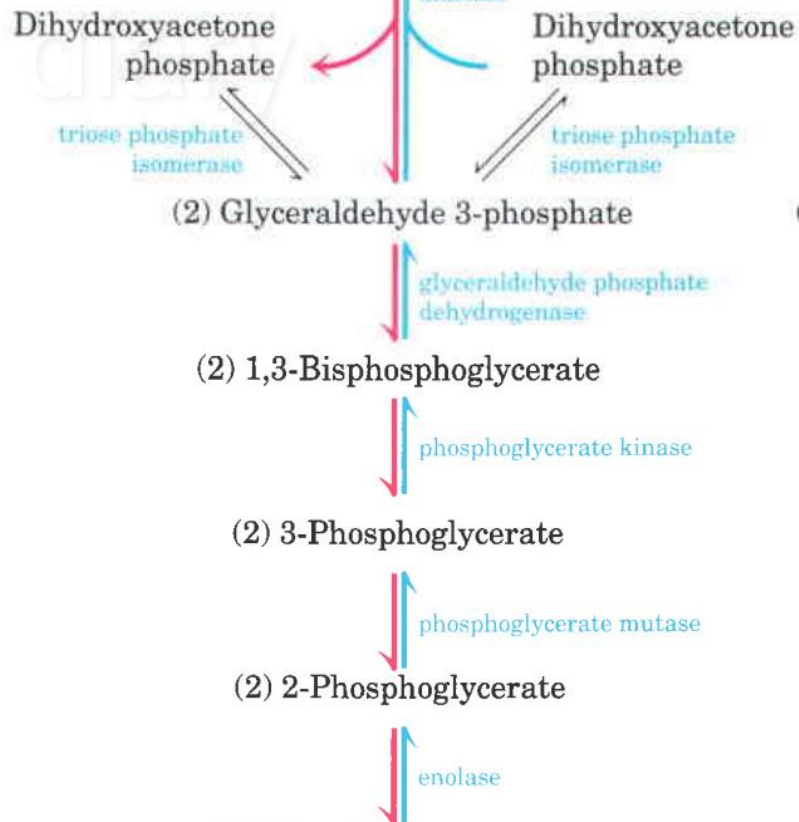
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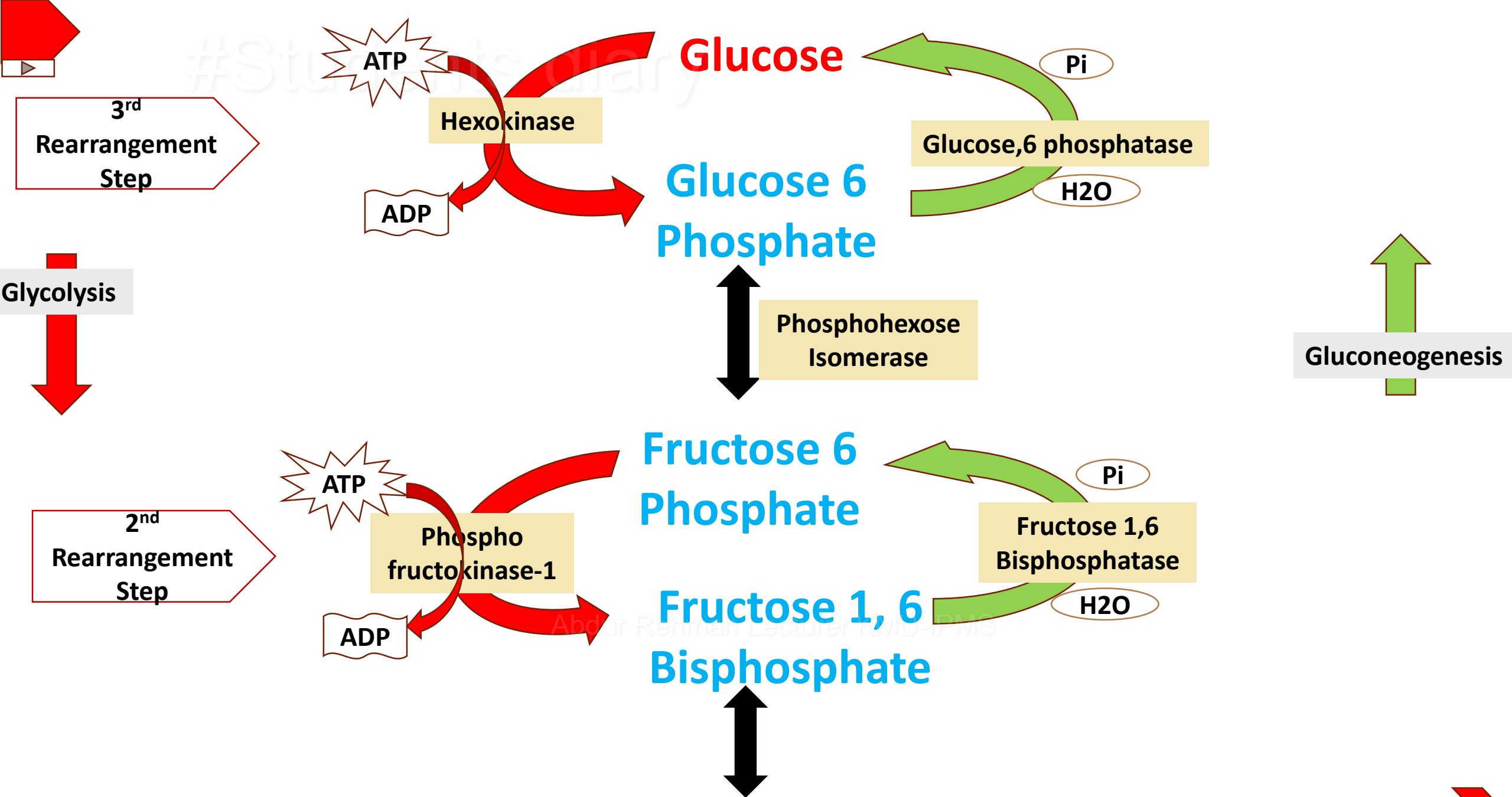
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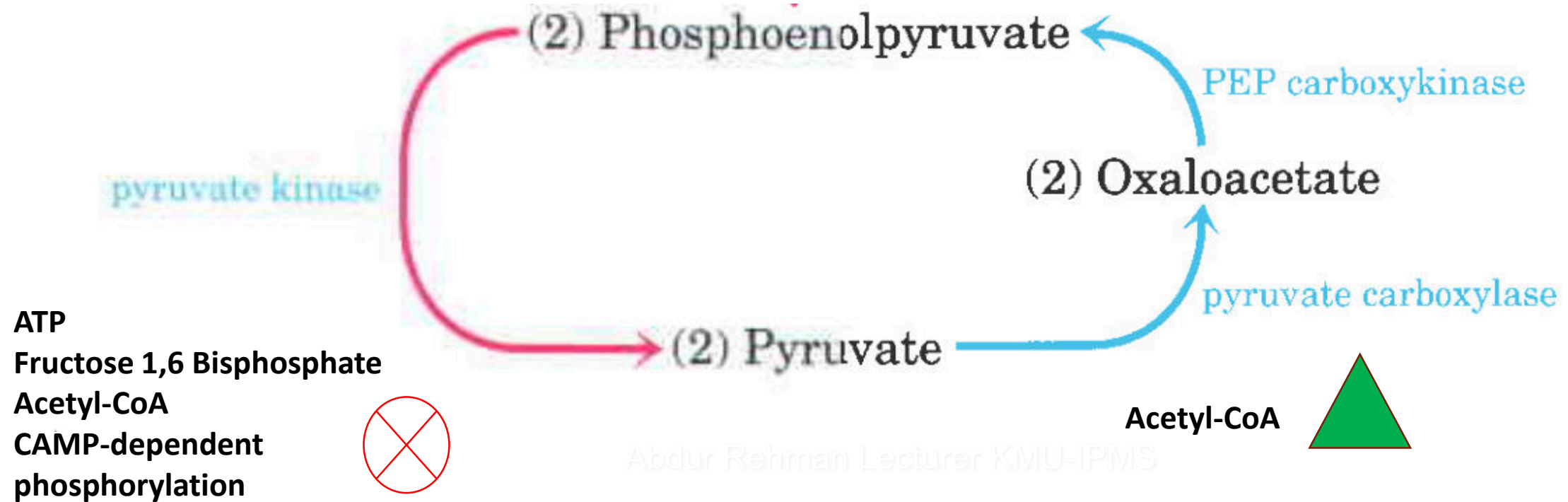
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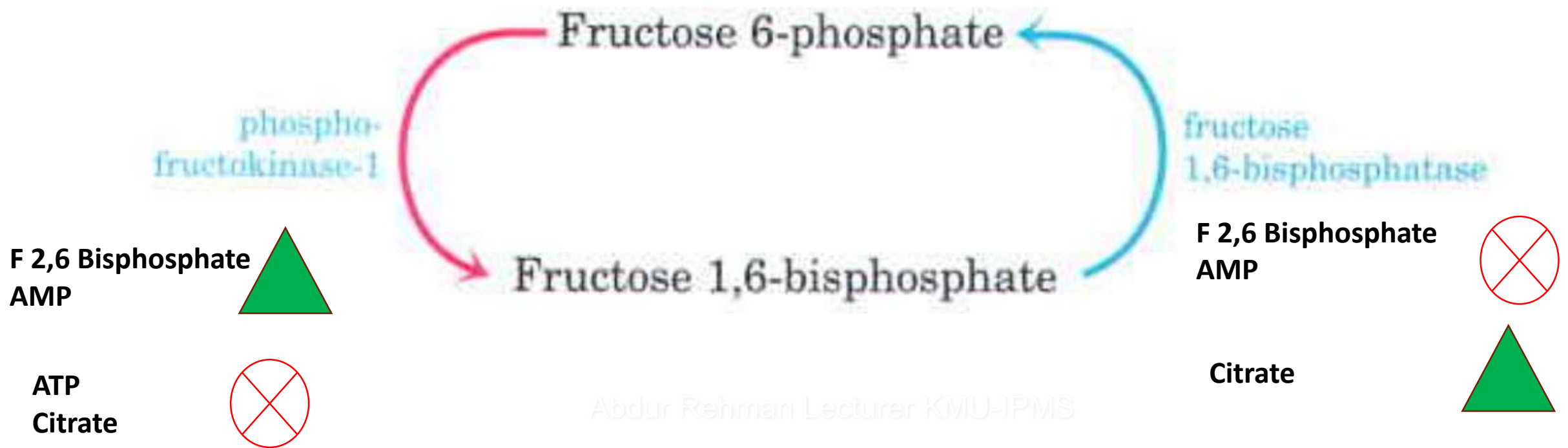
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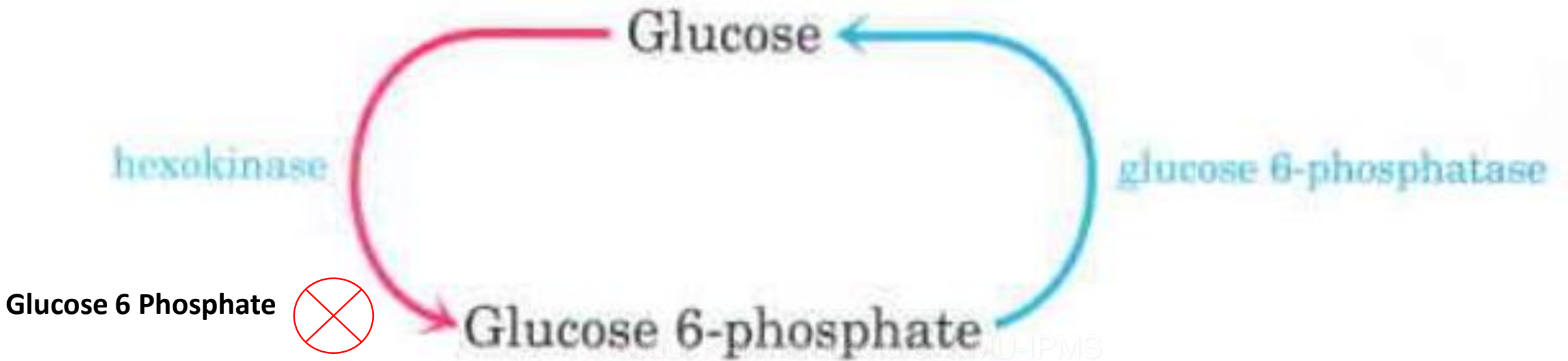
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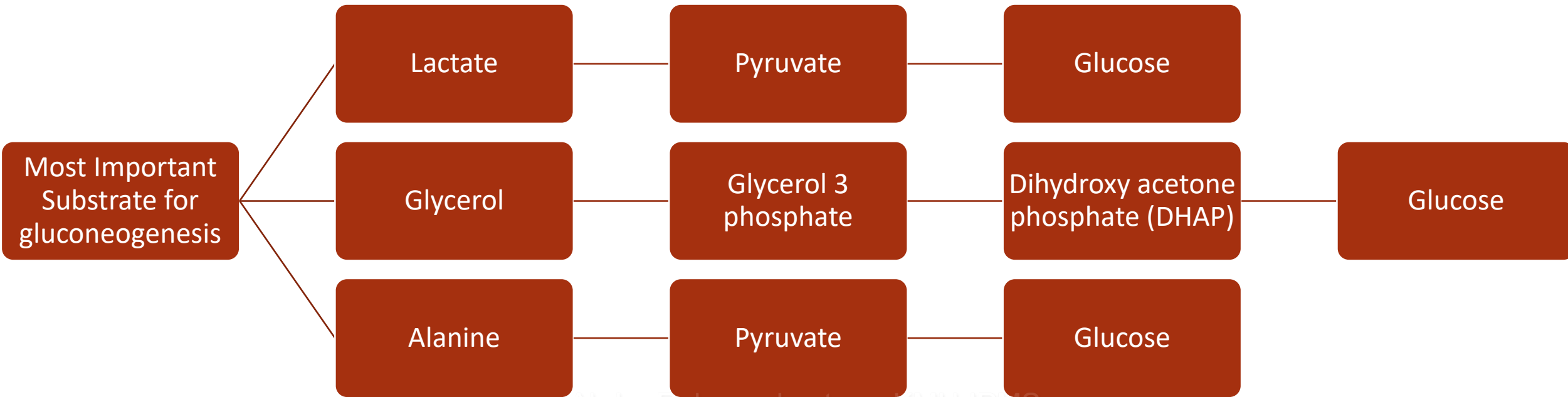
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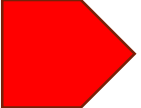
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Substrate of Gluconeogenesis



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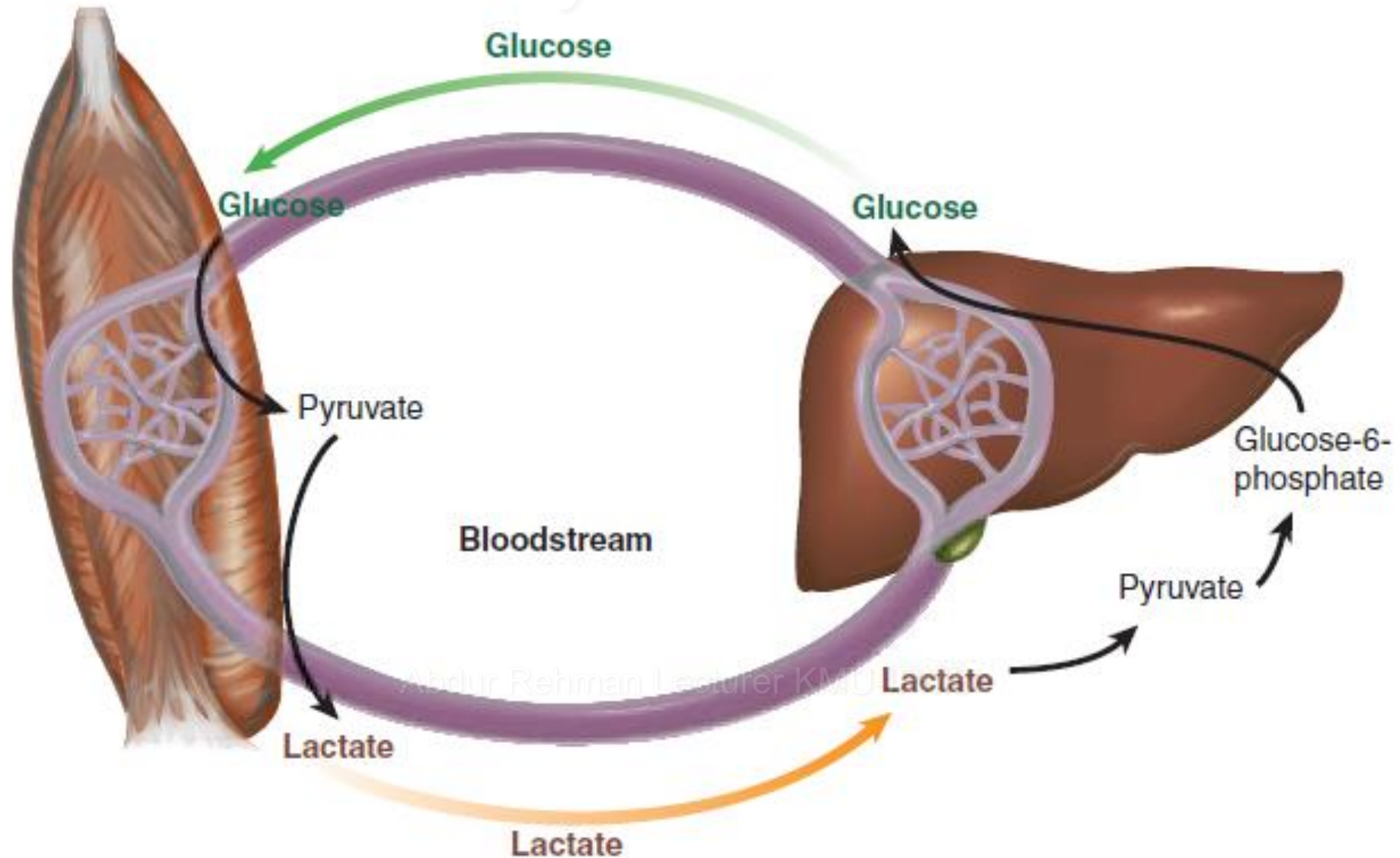


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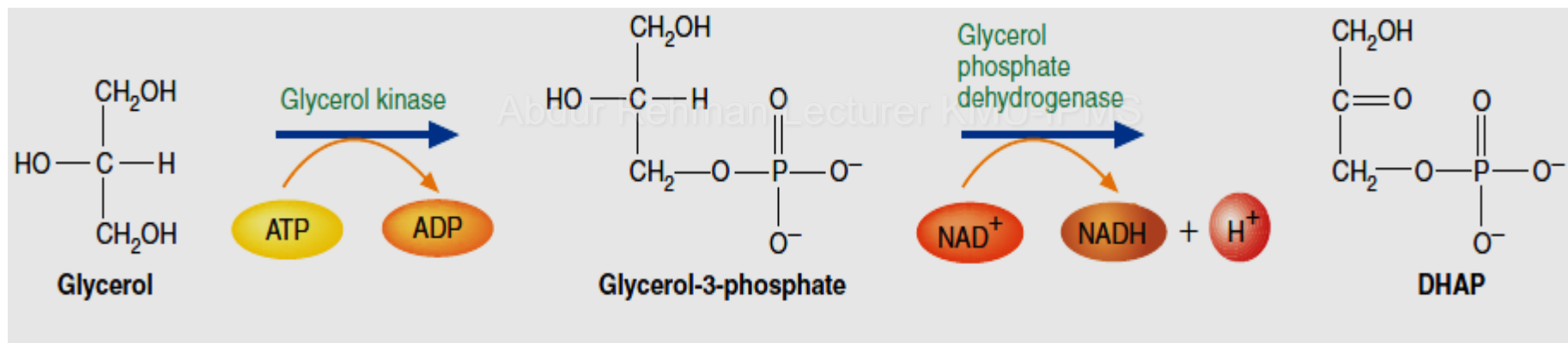
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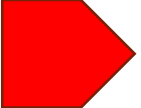
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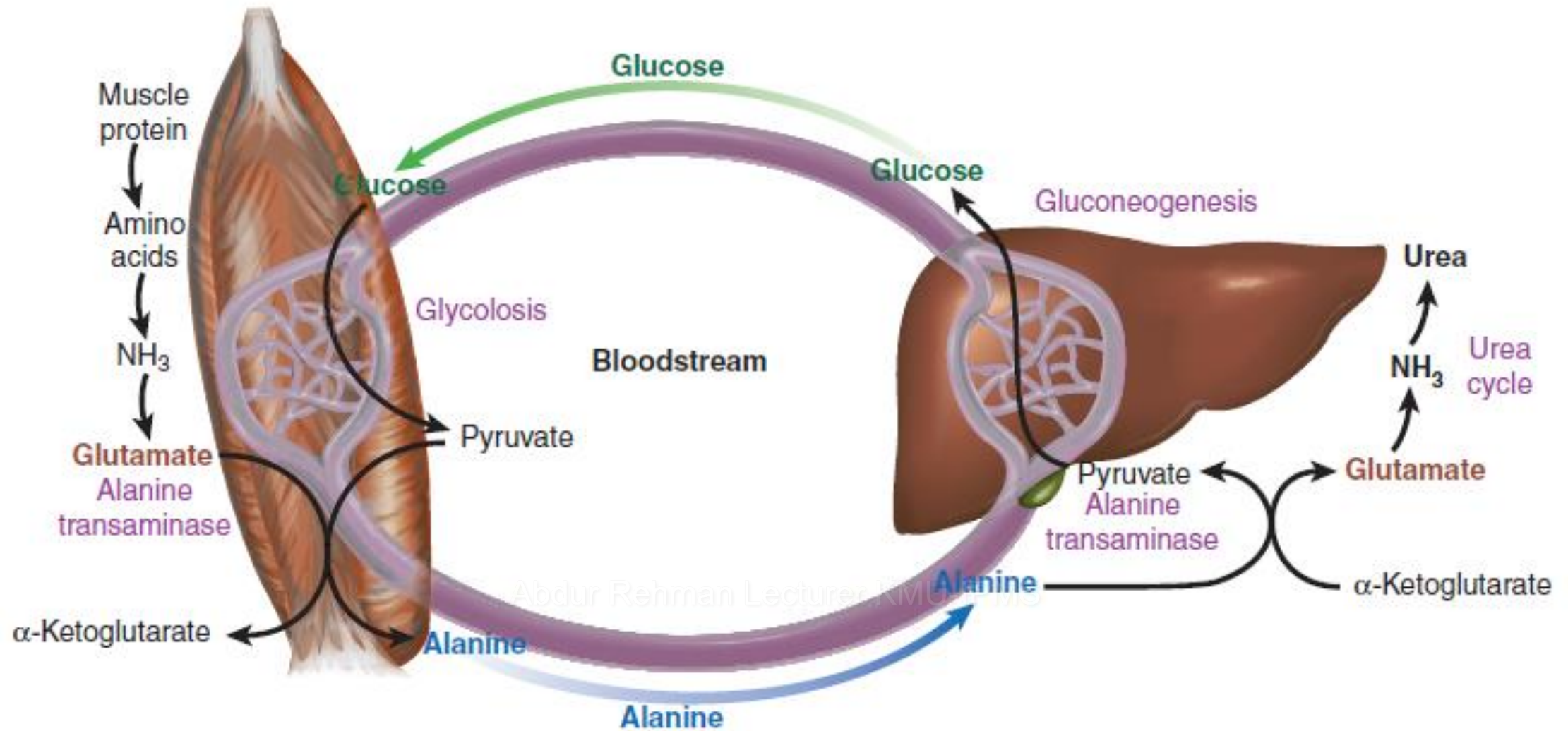


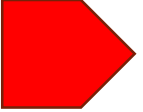


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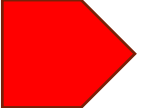
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Reference

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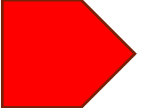


Glycogen Metabolism

By; Abdur Rehman

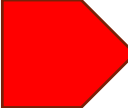
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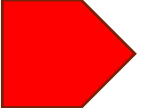
Glycogen Metabolism

- Glycogen is the storage form of glucose in animals, as is starch in plants
 - Liver 6-8%
 - Muscle 1-2% (glycogen reserves three times higher in muscle due to more muscle mass)
- Glycogen is stored as granules in the cytosol where all the machinery for synthesis & degradation are present
- Liver glycogen maintain the blood glucose level between the meals & time of demand
- Muscle glycogen act as fuel reserve provide ATP for muscular activity



Why store glycogen as a fuel reserve?

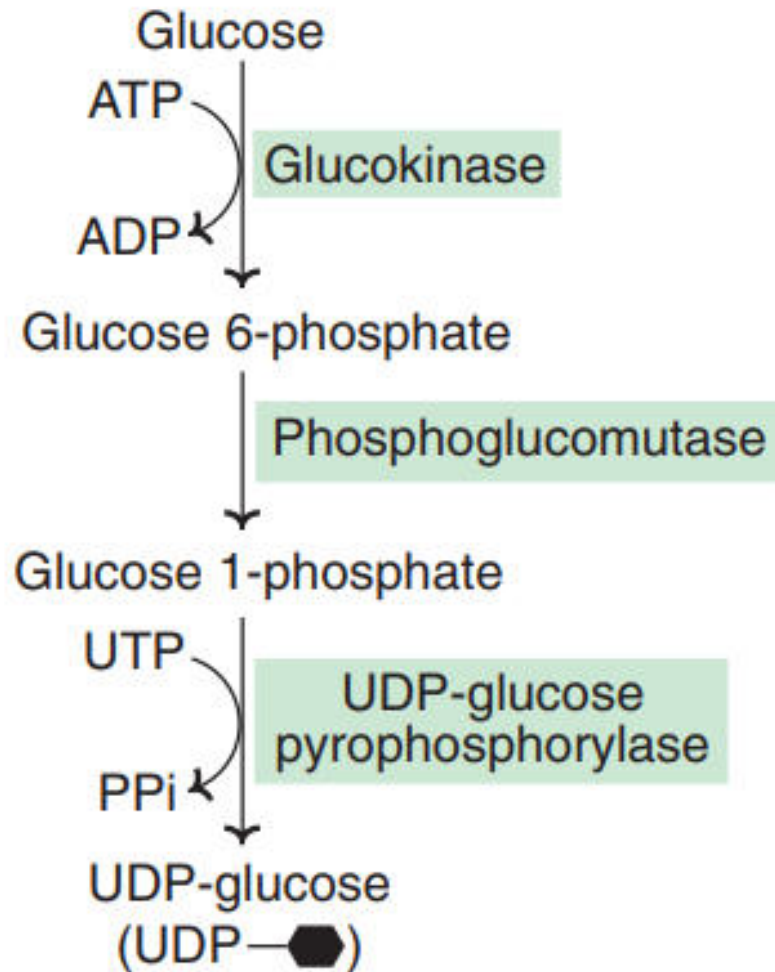
- Glycogen can be rapidly mobilized
- Glycogen can generate energy in the absence of oxygen
- Brain depends on continuous glucose supply (which mostly comes from glycogen)



Glycogenesis

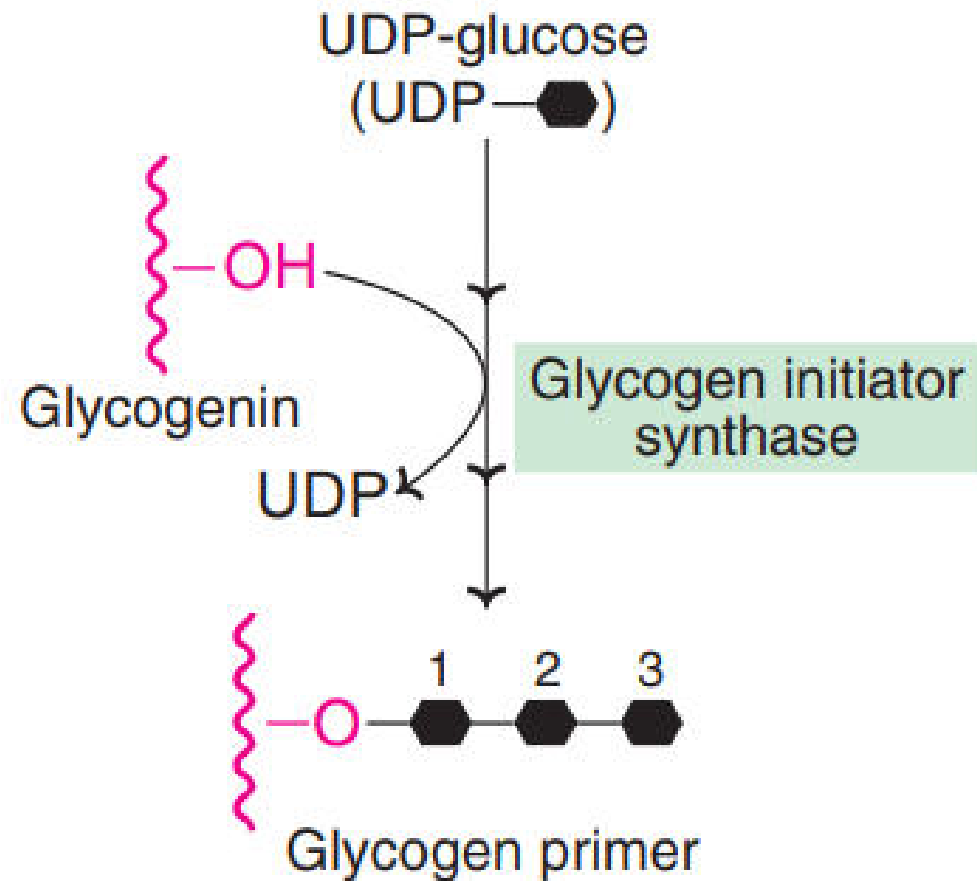
- Glycogen synthesis occurs in liver after a meal, when blood glucose levels are high
- Glycogenesis occurs by four major steps
 - 1. Synthesis of UDP-glucose**
 - 2. Requirement of primer to initiate glycogenesis**
 - 3. Glycogen synthesis by glycogen synthase**
 - 4. Formation of branches in glycogen**

1-Synthesis of UDP-glucose



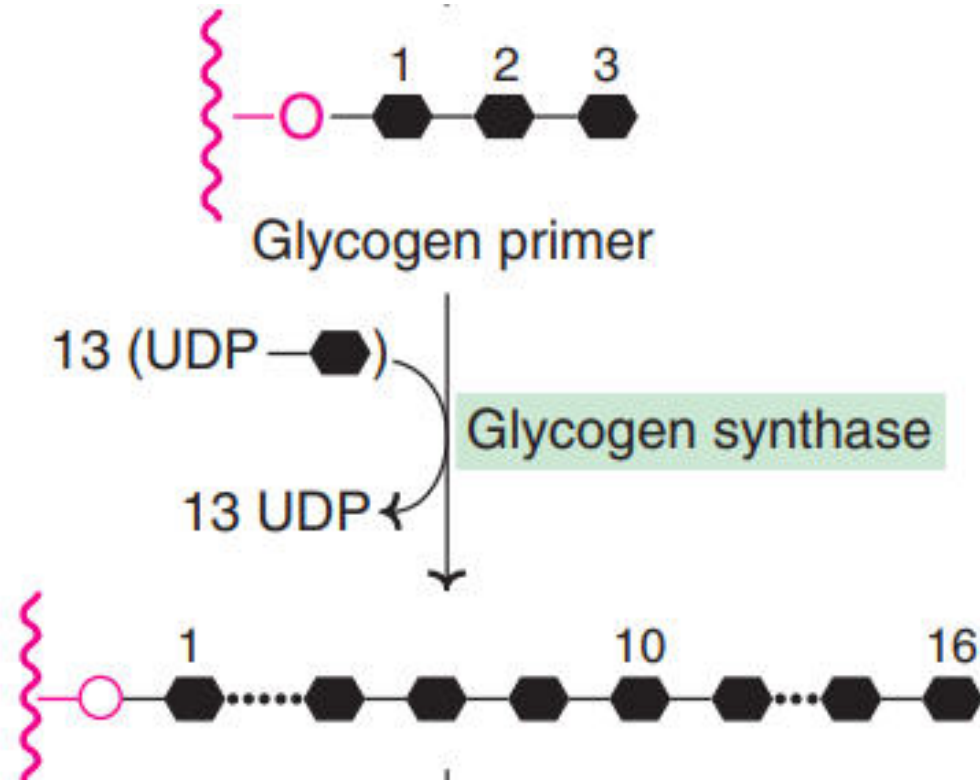
- Hexokinase (in muscle) and glucokinase (in liver) convert glucose to glucose 6-phosphate
- Phosphoglucomutase catalyses the conversion of glucose 6-phosphate to glucose 1-phosphate
- Uridine diphosphate glucose (UDPG) is synthesized from glucose 1-phosphate and UTP by UDP-glucose pyrophosphorylase

2. Requirement of primer to initiate glycogenesis



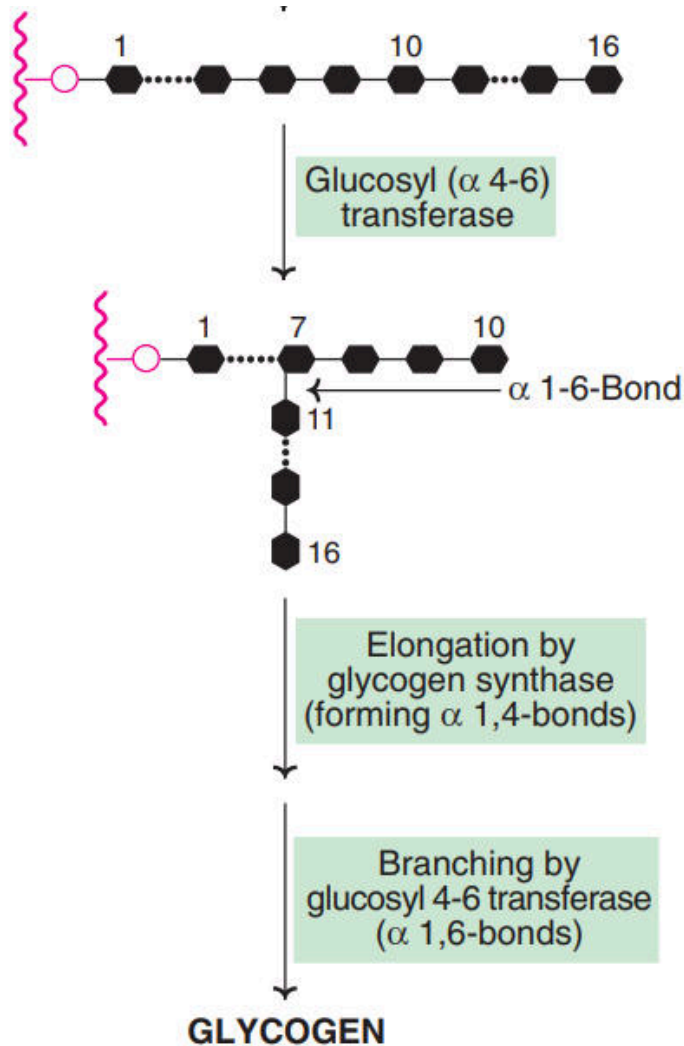
- pre-existing glycogen must act as a 'primer' to initiate glycogen synthesis
- In the absence of glycogen primer, a specific protein—namely 'glycogenin'—can accept glucose from UDPG
- The hydroxyl group of the amino acid tyrosine of glycogenin is the site at which the initial glucose unit is attached
- The enzyme glycogen initiator synthase transfers the first molecule of glucose to glycogenin

3. Glycogen synthesis by glycogen synthase

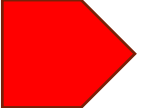


- Glycogen synthase is responsible for the formation of 1,4-glycosidic linkages
- This enzyme transfers the glucose from UDP-glucose to the non-reducing end of glycogen to form D1,4 linkages

4. Formation of branches in glycogen



- Glycogen synthase can catalyse the synthesis of a linear unbranched molecule with 1,4 D glycosidic linkages
- The formation of branches is brought about by the action of a branching enzyme, namely glucosyl D-4-6 transferase
- 2 ATP use in addition of glucose molecule

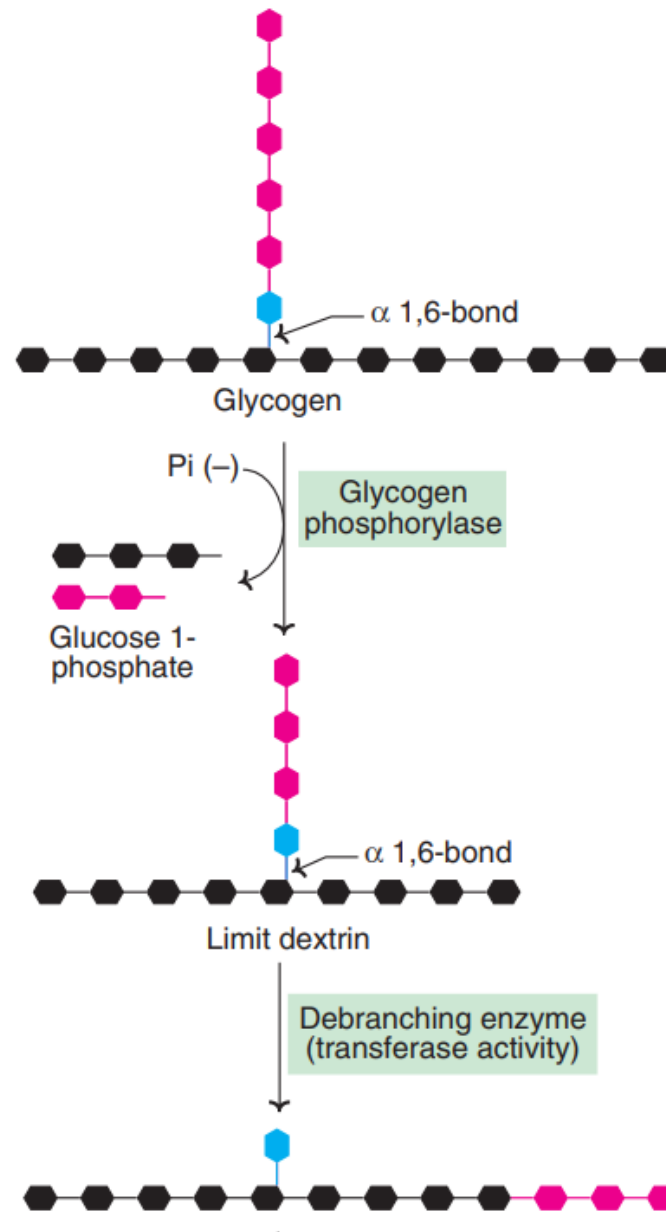


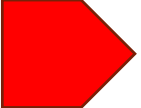
Glycogenolysis

- The enzymatic breakdown of stored (not dietary) glycogen
- Glycogen degradation take place in two process

1. Glucose Removal from non reducing end of glycogen:

- **Glycogen phosphorylase** uses **inorganic phosphate (Pi)** to cleave the alpha (1,4) linkages on the outer branches of glycogen to yield glucose-1-phosphate
- Glycogen phosphorylase stops when it comes within four glucose residues of a branch point
 - A glycogen molecule that has been degraded to its branch points is called a limit dextrin





Glycogenolysis

2. Hydrolysis of the $\alpha(1,6)$ glycosidic bonds at branch points of glycogen:

- **Amylo- $\alpha(1,6)$ -glucosidase (debranching enzyme)** begins the removal of $\alpha(1,6)$ branch points by transferring the outer three of the four glucose residues attached to the branch point to a nearby non reducing end
- It then removes the single glucose residue attached at each branch point
 - The product of this latter reaction is free glucose
 - The fate of glucose 6-phosphate depends on the tissue

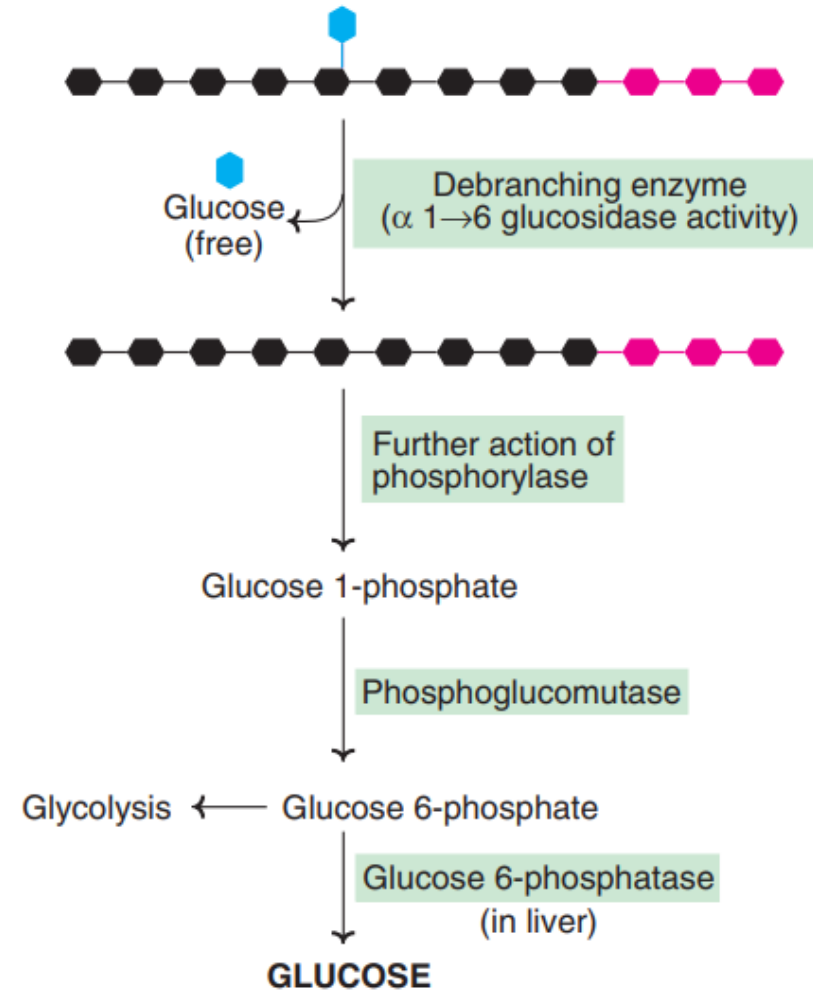
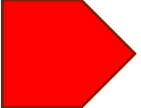


Fig. 13.15 : Glycogen degradation to glucose-glycogenolysis. (The ratio of glucose 1-phosphate to glucose is 8 : 1).



Regulation of Glycogen

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graph TD; A[Regulation of Glycogen] --> B[Allosteric regulation]; A --> C[Hormonal regulation]; A --> D[Influence of calcium];
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Allosteric
regulation

Hormonal
regulation

Influence of
calcium

Allosteric Regulation of Glycogen

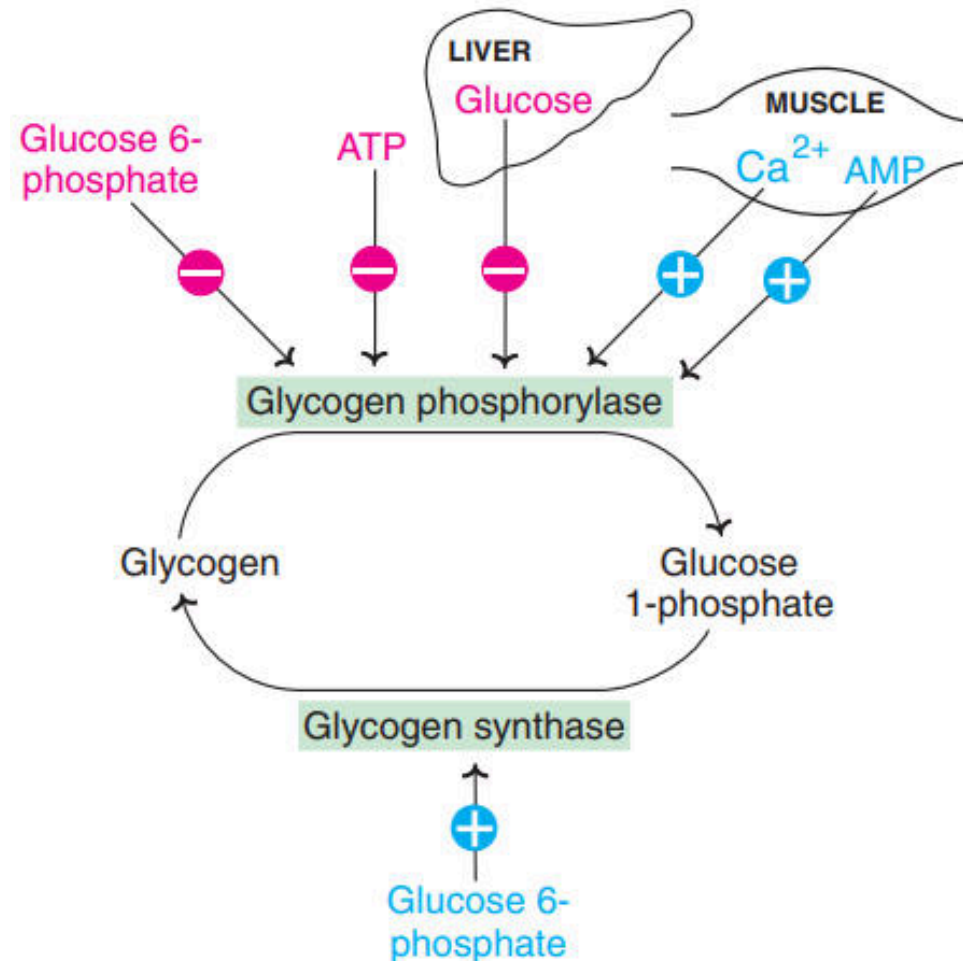
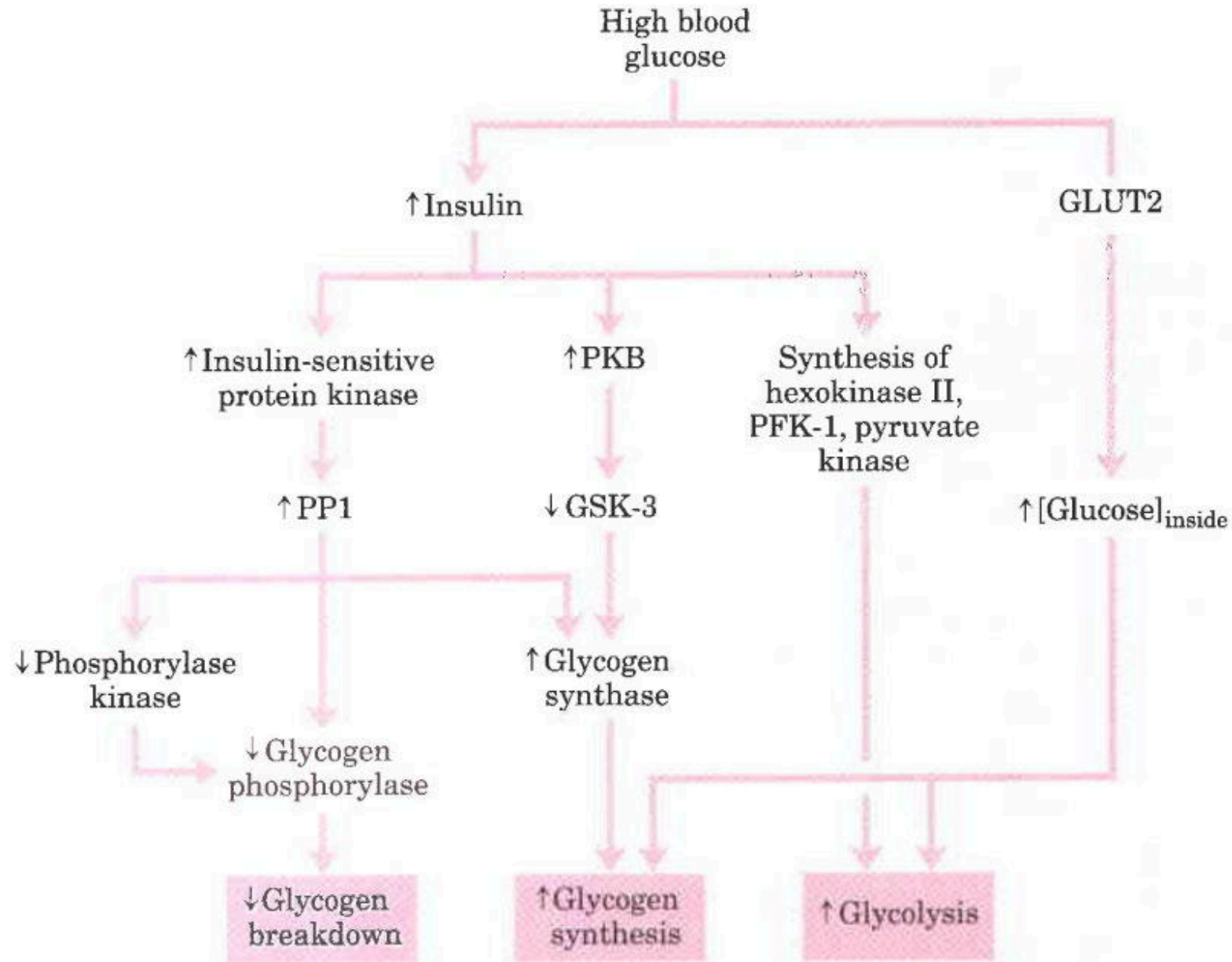
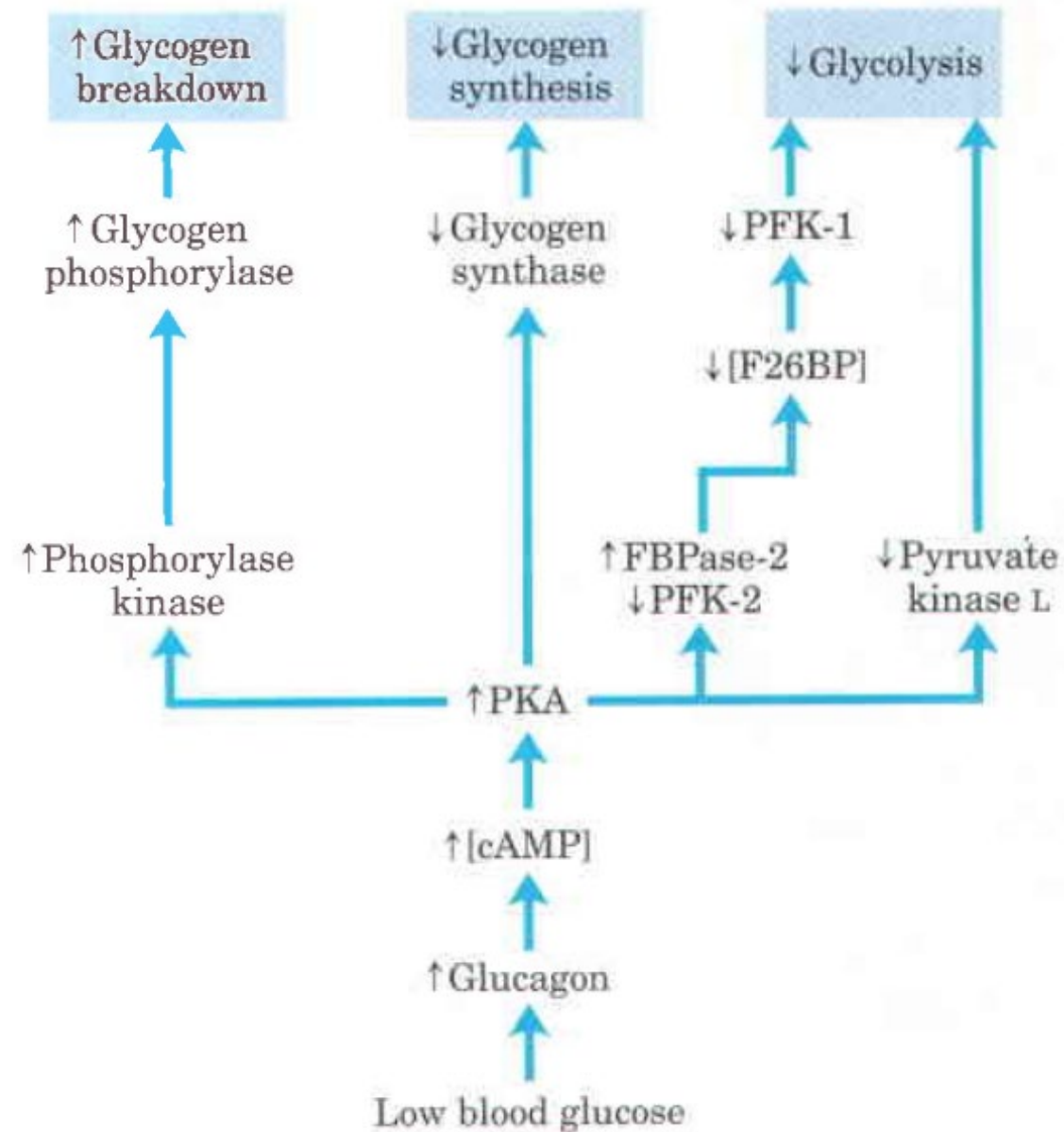


Fig. 13.16 : Allosteric regulation of glycogenolysis and glycogenesis ($\ominus$: Inhibition; $\oplus$: Activation).

Glycogen Regulation in Liver Hormonal



Glycogen regulation in liver (Hormonal)



Hormonal Regulation of Glycogen

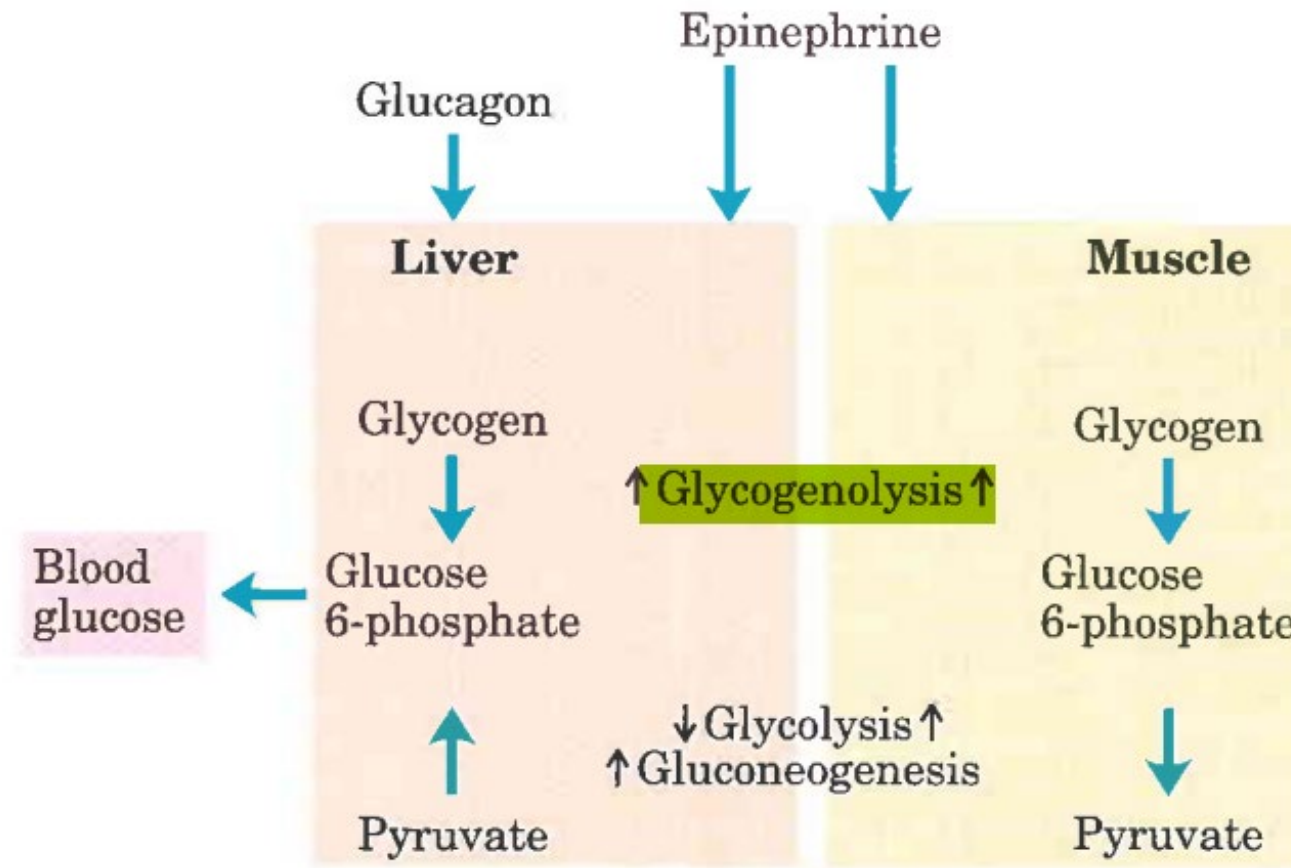
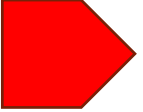


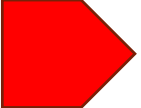
FIGURE 15-42 Difference in the regulation of carbohydrate metabolism in liver and muscle. In liver, either glucagon (indicating low



Fight & Flight Response & Glycogen regulation

- **Emotional or physical stress** releases the hormone **epinephrine** from the **adrenal medulla**.
- Epinephrine promotes glycogenolysis and inhibits glycogenesis.
- In emergency situations, when epinephrine is released in relatively large quantities, massive production of glucose provides the energy required to manage the situation.
- This effect is referred to as the **flight-or-fight response**.
- Epinephrine initiates the process by activating **adenylate cyclase** in liver and muscle cells
- Calcium ions and inositol trisphosphate are also believed to be involved in epinephrine's action





Reference

- Biochemistry by Satyanarayana and Chakrapani
- Lehninger Principles of Biochemistry, 5th Edition

Ketone bodies estimation in urine by dipstick method:

Ketone bodies:

Ketone is a name for a specific elemental structure in organic chemistry. A ketone consists of a single bond to two CH₃ or R groups with a double bond to an oxygen molecule.

Ketone bodies or simply ketones are substances produced by the liver during gluconeogenesis

The three main ketone bodies are acetoacetate (AcAc) and 3-beta-hydroxybutyrate (3HB), while acetone is the third, and least abundant, ketone body. The relative proportion of ketone bodies present in blood may vary from 78% (β-hydroxybutyric acid) to 20% (acetoacetic acid) and 2% (acetone).

Ketones build up when the body needs to break down fats and fatty acids to use as fuel. This is most likely to occur when the body does not get enough sugar or carbohydrates.

This may be due to diabetic ketoacidosis (DKA). DKA is a life-threatening problem that affects people with diabetes. It occurs when the body cannot use sugar (glucose) as a fuel source because there is no insulin or not enough insulin. Fat is used for fuel instead of carbohydrates.

TYPES of KETONE BODIES

There are three types of ketones produced when the body goes into ketosis:

ACETOACETATE



- Created from the breakdown of **fatty acids**.
- Either converted into **BHB** or turned into **acetone**.

BETA-HYDROXYBUTYRIC ACID



- Formed from acetoacetate.
- Not technically a ketone because of its structure, but we consider it as one within the keto diet.

ACETONE



- Created as a side product of acetoacetate.
- Breaks down **quickly**.
- **Is removed** from the body through the waste or the breath

Why the Test is performed?

- Ketone testing is most often done if you have type 1 diabetes and:
- Your blood sugar is higher than 240 milligrams per deciliter (mg/dl)
- Nausea or vomiting occur
- Pain in the abdomen
- Ketone testing may also be done if:
- You have an illness such as pneumonia, heart attack, or stroke
- Nausea or vomiting that does not go away.

Urine Reagent Strips can be used in general evaluation of health, and aids in the diagnosis and monitoring of metabolic or systemic diseases that affect kidney function, endocrine disorders and diseases or disorders of the urinary tract.

SPECIMEN COLLECTION AND PREPARATION:

A urine specimen must be collected in a clean and dry container and tested as soon as possible. Do not centrifuge. The use of urine preservatives is not recommended. If testing cannot be done within an hour after voiding, refrigerate the specimen immediately and let it return to room temperature before testing. Prolonged storage of unpreserved urine at room temperature may result in microbial proliferation with resultant changes in pH. A shift to alkaline pH may cause false positive results with the protein test area.

Urine containing glucose may decrease in pH as organisms metabolize the glucose.

Principle of urine dipstick:

This test is based on ketones reacting with nitroprusside and acetoacetic acid to produce a color change ranging from light pink or cream color for negative results to a darker pink or purple color for positive results.

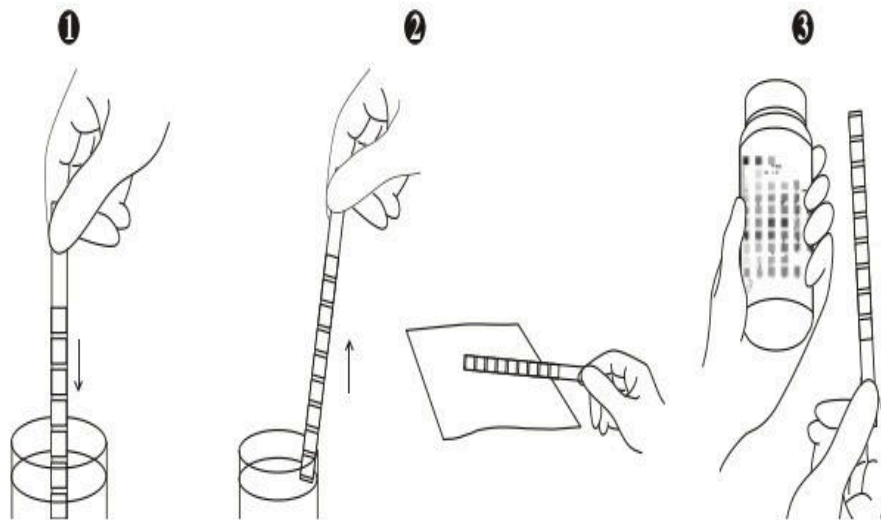
DIRECTION FOR USE:

1. Remove the strip from the closed canister and use it as soon as possible. Immediately close the canister tightly after removing the required number of strip(s).

Completely immerse the reagent areas of the strip in fresh, well-mixed urine and immediately remove the strip to avoid dissolving the reagents.

2. While removing the strip from the urine, run the edge of the strip against the rim of the urine container to remove excess urine.

3. Compare the reagent areas to the corresponding color blocks on the canister label at the specified times. Hold the strip close to the color blocks and match carefully.



Interpretation:

Results are obtained by direct comparison of the color blocks printed on the canister label.

The color blocks represent nominal values; actual values will vary close to the nominal values.

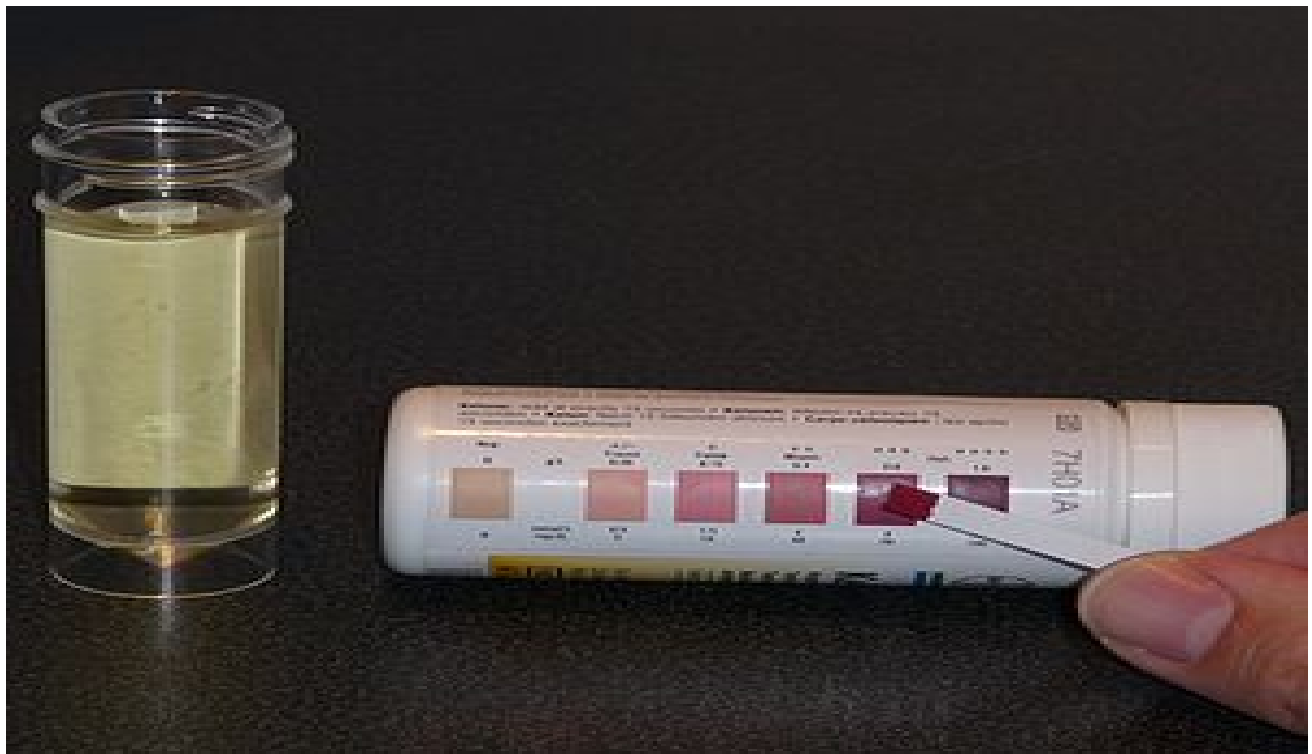
In the event of unexpected or questionable results, the following steps are recommended;

Confirm that the specimens have been tested within the expiration date printed on the canister label.

Compare results with known positive and negative controls and repeat the test using a new strip.

If the problem persists, discontinue using the strip immediately.

Test Result	Values
Normal/ Negative	Less than 0.6 millimoles per liter (mmol/L)
Low to Moderate	0.6 to 1.5 mmol/L
High	1.6 to 1.5 mmol/L
Very High	Greater than 3.0 mmol/L

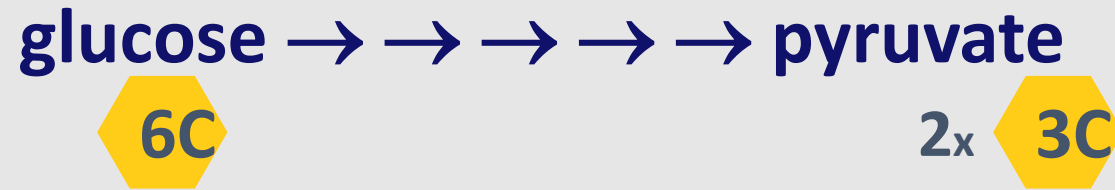


Test	Results					
Urine Urobilinogen	0.11(16) Normal 2(33)4(66)8(131) mg/dℓ (μ mol/L)					
Urine Glucose		neg.± 100(5.5)+250(14)++500(28)++1000(55)	mg/dℓ (mmol/L)			
Urine Bilirubin		neg.+ ++ +++				
Urine Ketone		neg.± 5(0.5)+ 15(1.5)++40(3.9)+++100(10)	mg/dℓ (mmol/L)			
Urine Specific Gravity		1.0001.0051.0101.0151.0201.0251.030				
Blood in Urine		neg.Hemolysis +10++50+++250Non Hemolysis+10++50	RBC/μ L			
Urine pH Level		566.5789				
Urine Protein		neg.trace+30(0.3)++100(1.0)+++300(3.0)++++1000(10)	mg/dℓ(g/L)			
Urine Nitrites		neg.tracepos.				
Urine Leukocytes		neg.+25++75++500	WBC/μ L			

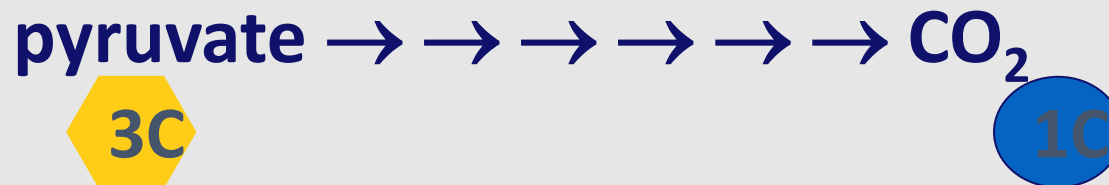
Krebs Cycle

Glycolysis is only the start

- Glycolysis



- Pyruvate has more energy to yield
 - 3 more C to strip off (to oxidize)
 - if O_2 is available, pyruvate enters mitochondria
 - enzymes of Krebs cycle complete the full oxidation of sugar to CO_2



Krebs cycle

1937 | 1953



Hans Krebs
1900-1981

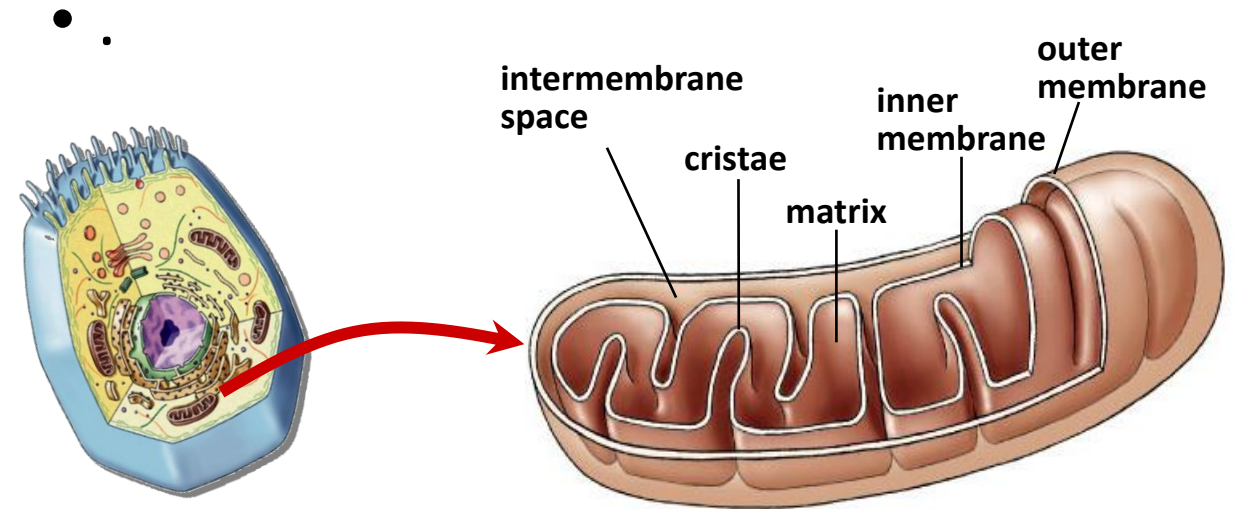
- aka Citric Acid Cycle
 - in mitochondrial matrix
 - 8 step pathway
 - each catalyzed by specific enzyme
 - step-wise catabolism of 6C citrate molecule
- Evolved later than glycolysis
 - does that make evolutionary sense?
 - bacteria → 3.5 billion years ago (glycolysis)
 - free O₂ → 2.7 billion years ago (photosynthesis)
 - eukaryotes → 1.5 billion years ago (aerobic respiration = organelles → mitochondria)

TCA /Citric Acid/KREBS Cycle

- The cycle is located in the mitochondria
- All cells have a mitochondria except RBCs
- This is the Final common pathway of **oxidative metabolism**
- Acetyl coenzyme A condenses with OAA to begin the cycle.
- Catabolism of **CHO, Fats and Proteins provide the acetyl CoA**
- The bulk of ATP used by many cells to maintain homeostasis is produced by the oxidation of pyruvate in the TCA cycle
- During this oxidation process, reduced **NADH** and reduced **FADH₂** are generated.
- The NADH and FADH₂ are principally used to drive the processes of oxidative phosphorylation which are responsible for converting the reducing potential of NADH and FADH₂ to the high energy phosphate in ATP

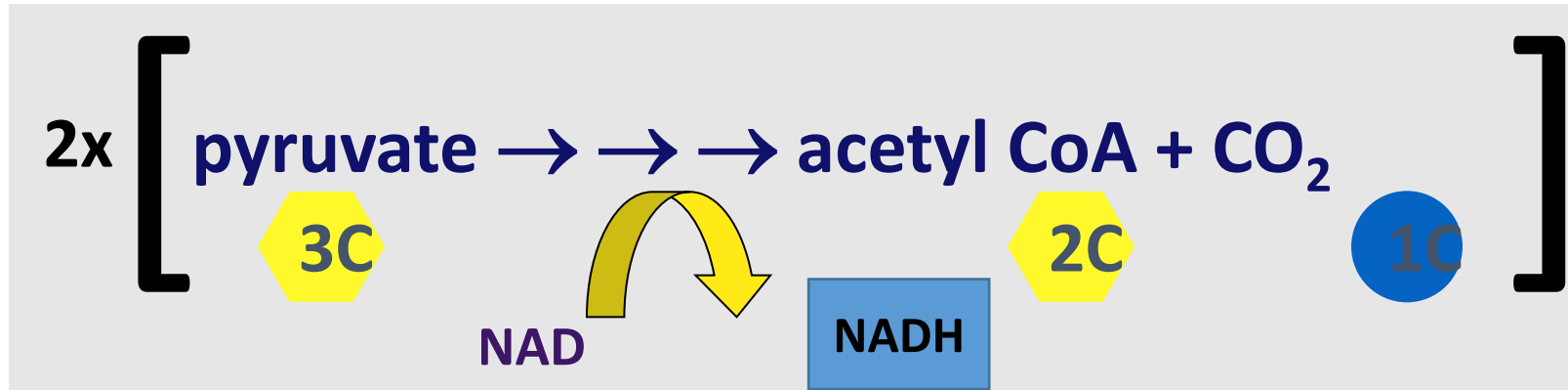
Mitochondria Structure

- Double membrane energy harvesting organelle
 - smooth outer membrane
 - highly folded inner membrane
 - cristae
 - intermembrane space
 - fluid-filled space between membranes
 - matrix
 - inner fluid-filled space
 - DNA, ribosomes
 - enzymes
 - free in matrix & membrane-bound



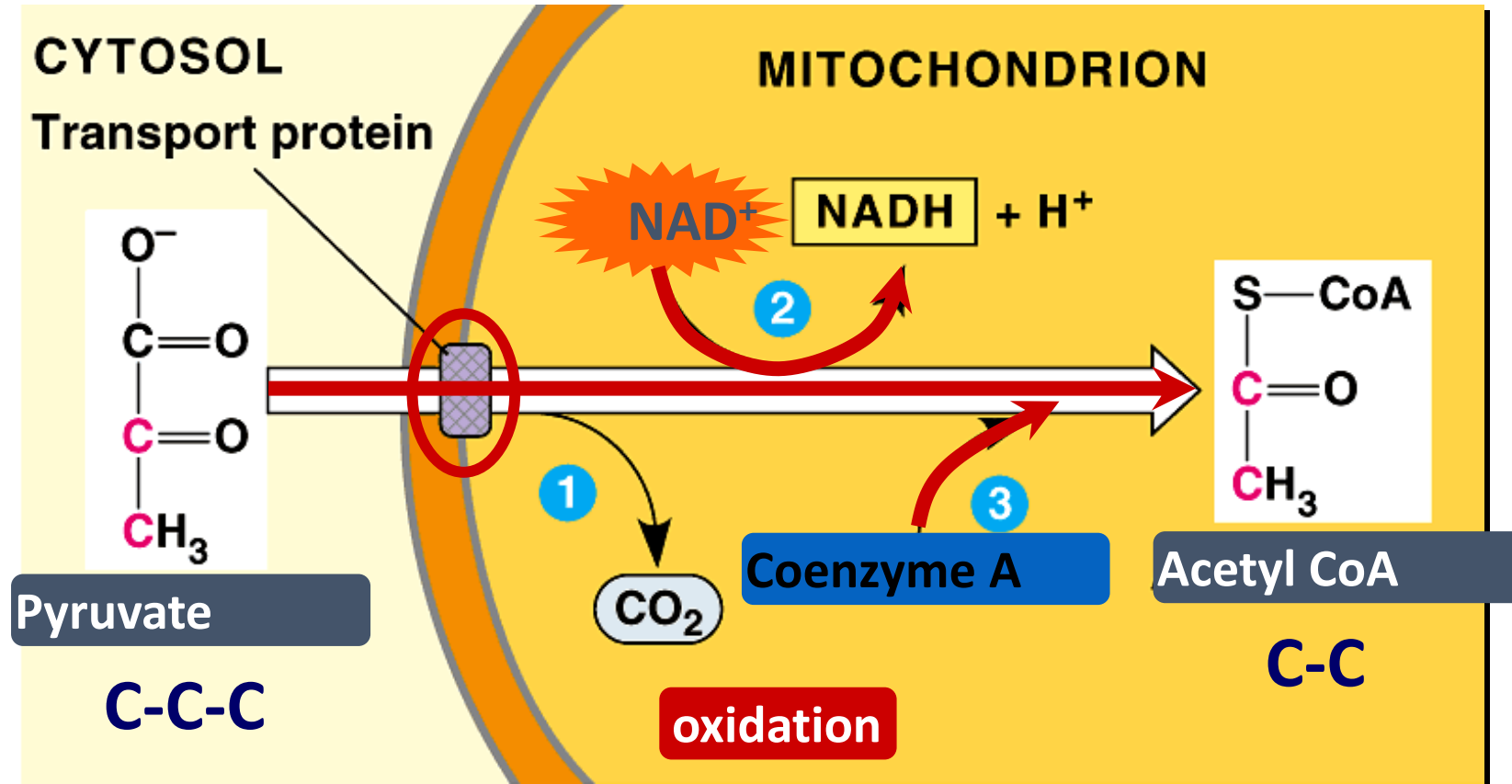
Oxidation of pyruvate

- Pyruvate enters mitochondrial matrix



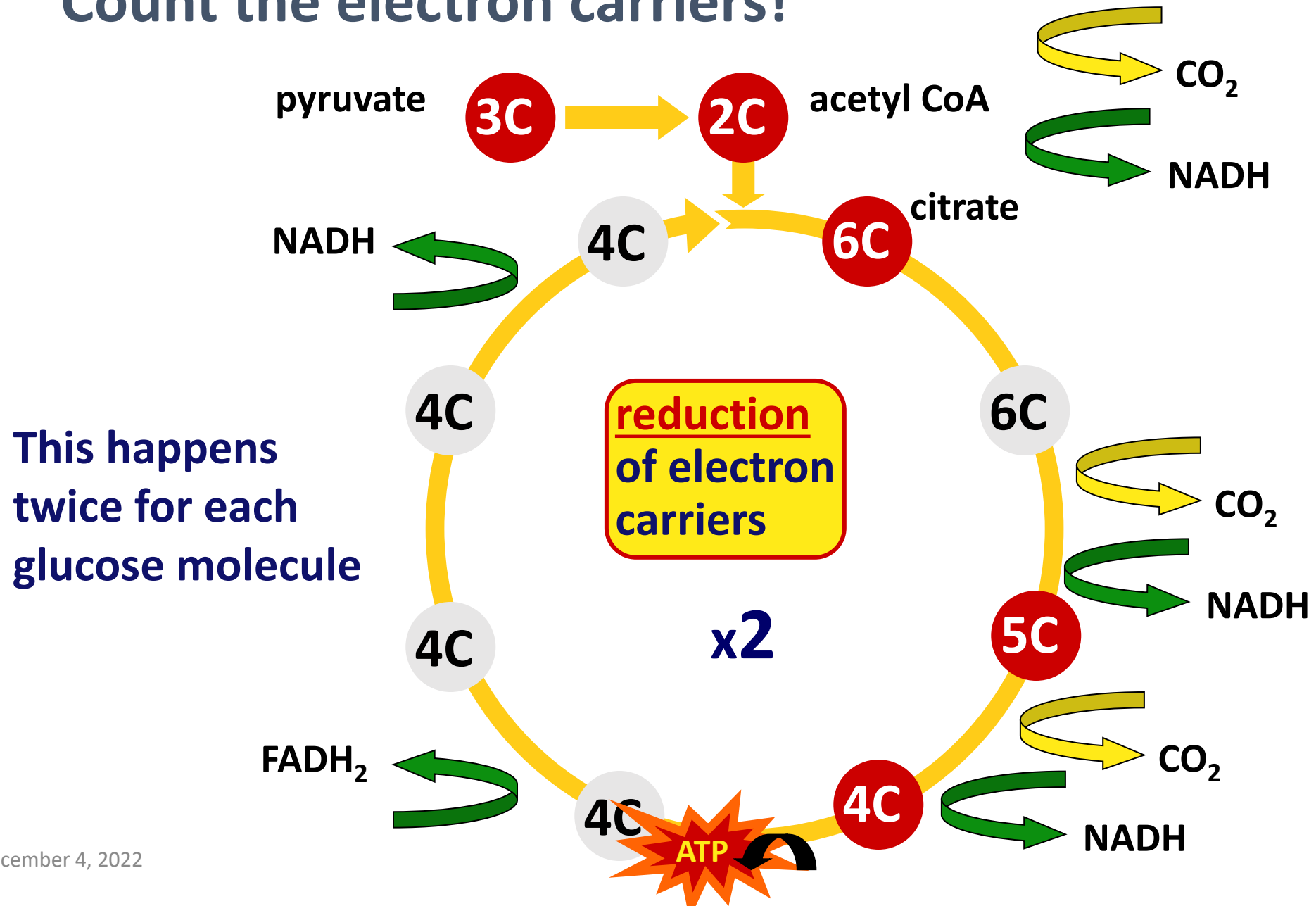
- 3 step oxidation process
- releases 2 CO₂ (count the carbons!)
- reduces 2 NAD → 2 NADH (moves e⁻)
- produces 2 acetyl CoA
- Acetyl CoA enters Krebs cycle

Pyruvate oxidized to Acetyl CoA

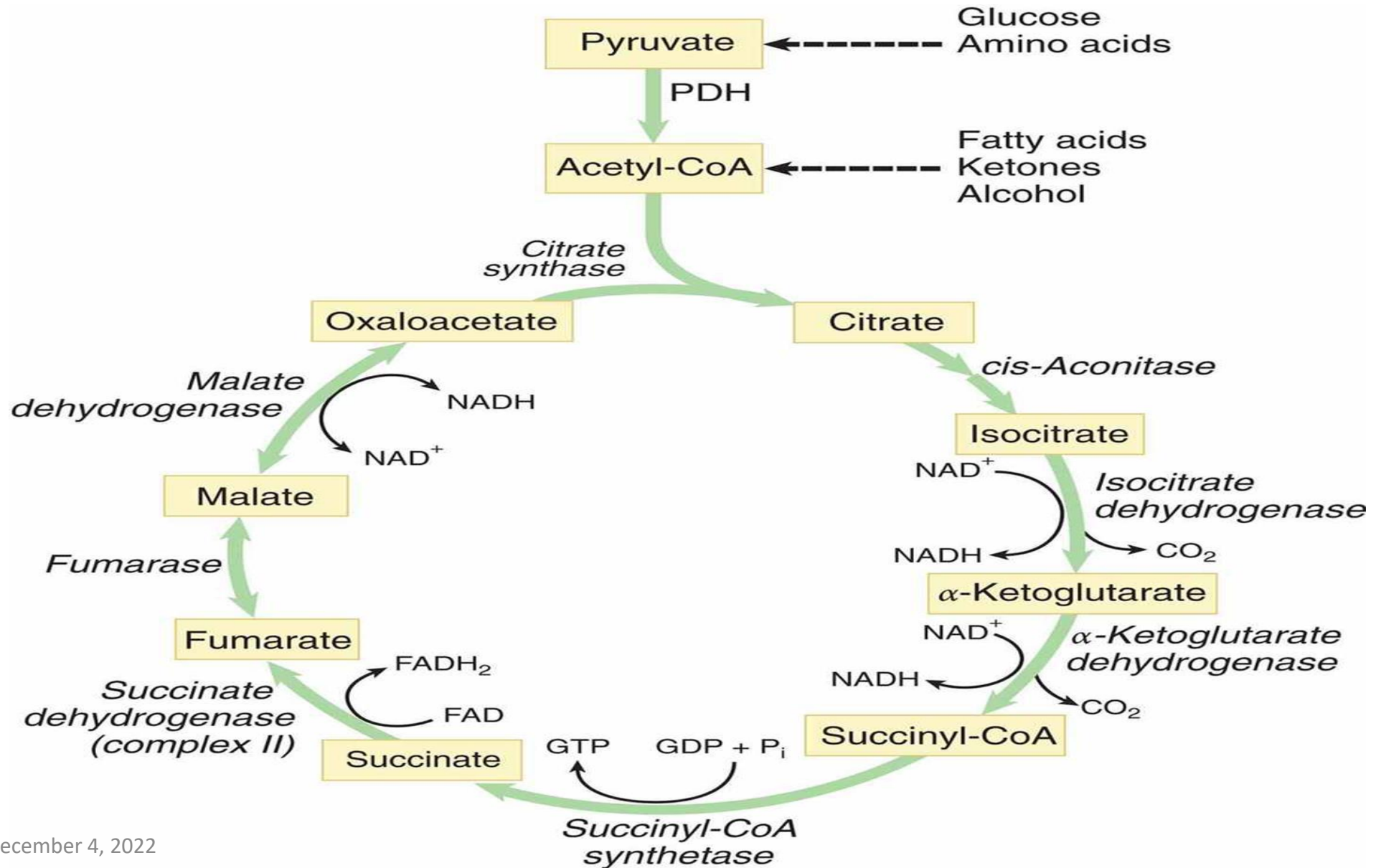


$$2 \times \left[\text{Yield} = 2\text{C sugar} + \text{NADH} + \text{CO}_2 \right]$$

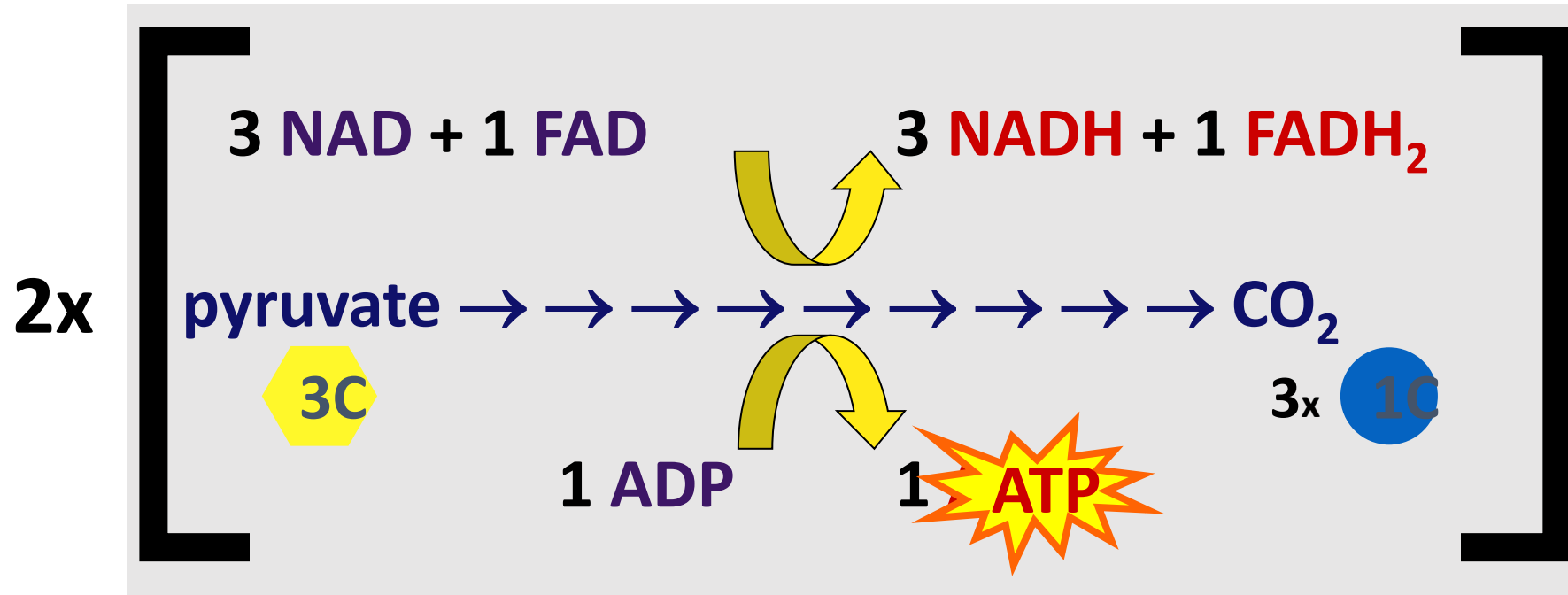
Count the electron carriers!



Krebs cycle



Energy accounting of Krebs cycle



Net gain = 2 ATP
= 8 NADH + 2 FADH₂

Regulation of TCA

- Regulation of the TCA cycle like that of glycolysis, occurs at both the level of entry of substrates into the cycle as well as at the key reactions of the cycle. Fuel enters the TCA cycle primarily as acetyl-CoA. The generation of acetyl-CoA from carbohydrates is a major control point of the cycle. This is the reaction catalyzed by the PDH complex
 - PDH complex is inhibited by acetyl-CoA, ATP, and NADH
 - PDH activated by non-acetylated CoA (CoASH) and NAD⁺.
 - The *pyruvate dehydrogenase* activities of the PDH complex are regulated by their state of phosphorylation. This modification is carried out by a specific kinase (*PDH kinase*) and the phosphates are removed by a specific phosphatase (*PDH phosphatase*).
 - The phosphorylation of PDH inhibits its activity which leads to decreased oxidation of pyruvate.
 - PDH kinase is activated by NADH and acetyl-CoA and inhibited by pyruvate, ADP, CoASH, Ca²⁺ and Mg²⁺. The *PDH phosphatase*, in contrast, is activated by Mg²⁺ and Ca²⁺

Citrate Synthase: inhibited by ATP and citrate

Isocitrate Dehydrogenase: Isocitrate, AMP, ADP activates, ATP and NADH inhibits

A-ketoglutarate dehydrogenase: succinoyl CoA and NADH inhibits



Basic Concept Of Metabolism

By: Abdur Rehman

Lecturer IPMS KMU

Subject: Biochemistry (2nd semester)



Metabolism: Basic Concepts and Design

- The concepts of conformation and dynamics especially those dealing with the **specificity and catalytic power of enzymes**, the **regulation of their catalytic activity**, and the **transport of molecules and ions across membranes** enable us to now ask questions fundamental to biochemistry:
 1. *How does a cell extract energy and reducing power from its environment?*
 2. *How does a cell synthesize the building blocks of its macromolecules and then the macromolecules themselves*



Cells Transform Different Types of Energy

- Living organisms require a continual input of free energy for three major purposes:
- The performance of mechanical work in muscle contraction and other cellular movements
- The active transport of molecules and ions
- The synthesis of macromolecules and other biomolecules from simple precursors.



Cont..

- **Photosynthetic organisms:** transform light energy into chemical energy. Indeed, this transformation is ultimately the primary source of chemical energy for the vast majority of organisms, **human beings included**.
- **Chemotrophs**, which include animals, obtain chemical energy through the oxidation of foodstuffs generated by phototrophs
- **Chemical energy** obtained from the oxidation of carbon compounds may be transformed into the unequal distribution of ions across a membrane, resulting in an **ion gradient**.



Cont..

- This gradient, in turn, is an **energy source** that can be used to **move molecules across membranes** that can convey **information in the form of nerve impulses**
- In addition, **chemical energy can be transduced into mechanical energy**. E.g contraction and movement
- Finally, **chemical energy powers the reactions** that result in the synthesis of biomolecules
- These transformations are referred to as **metabolism or intermediary metabolism**



Metabolism

- **Metabolism** is the sum of all the chemical transformations taking place in a cell or organism which occurs through a series of enzyme catalyzed reactions that constitute metabolic pathways.
- The term metabolite is applied to a substrate or an intermediate or a product in the metabolic reactions
- Metabolism is broadly divided into two categories (Fig.12.1)



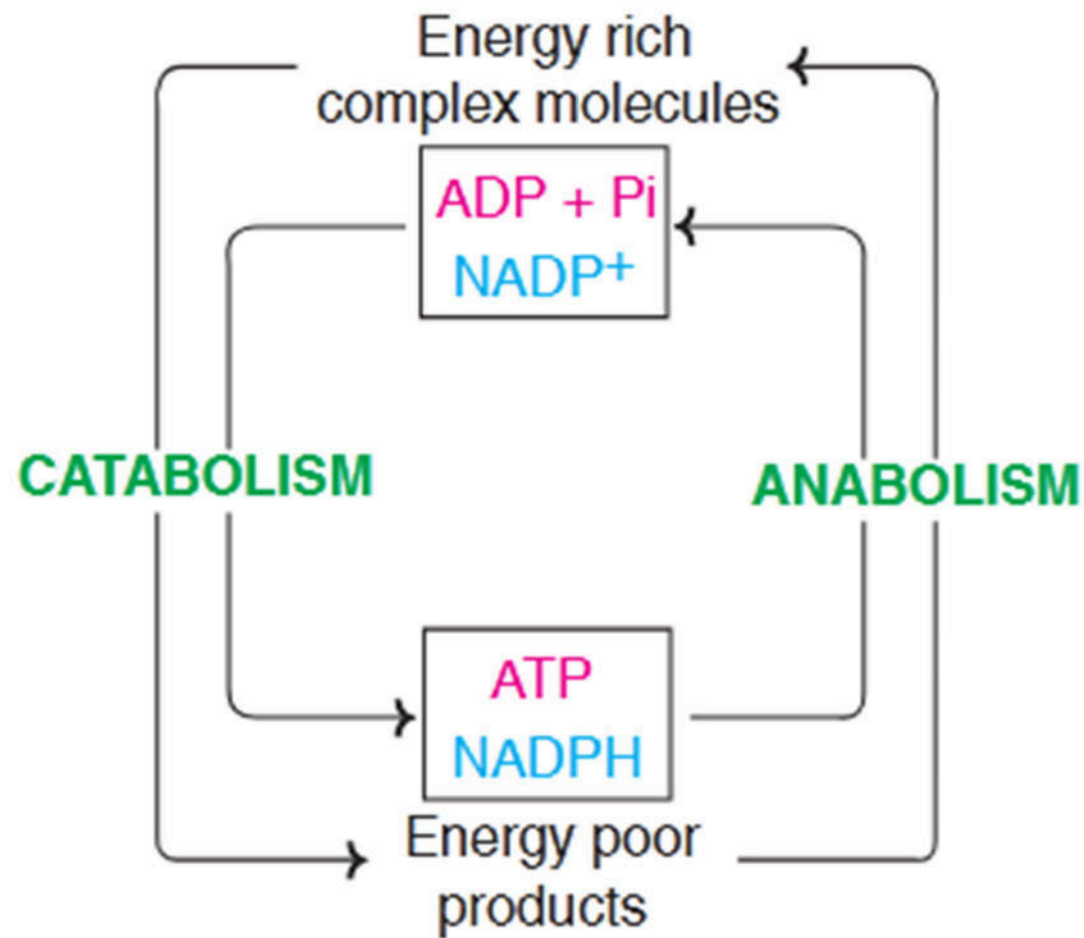


Fig. 12.1 : An outline of catabolism and anabolism.



Catabolism

- **Catabolism** is the **degradative phase of metabolism** in which organic nutrient molecules (carbohydrates, fats, and proteins) are converted into smaller, simpler end products (such as lactic acid, CO₂, and NH₃). **Catabolic pathways release energy, .**
- Catabolism occurs in three stages (Fig.12.2).
 1. Conversion of complex molecules into their building blocks
 2. Formation of simple intermediates
 3. Final oxidation of acetyl CoA



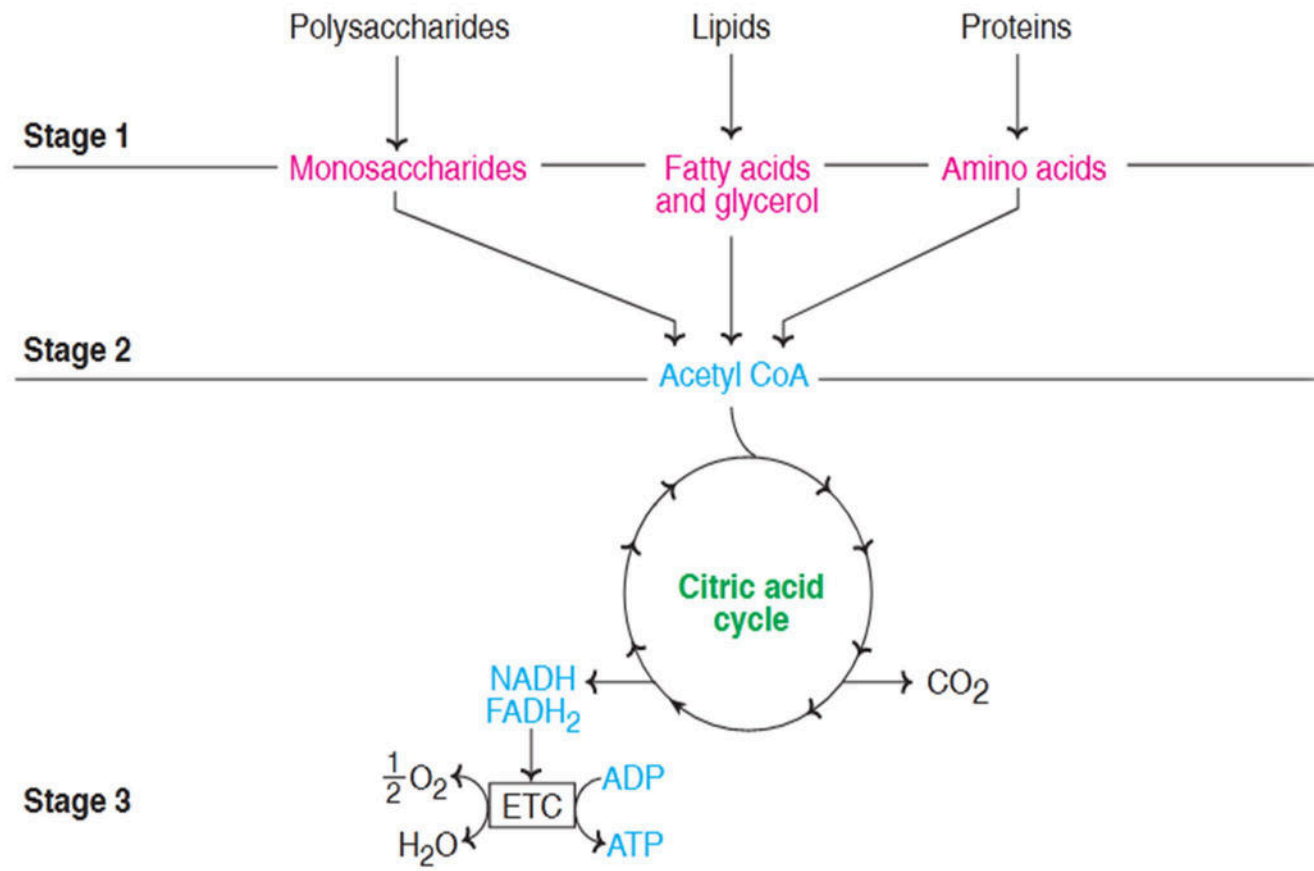


Fig. 12.2 : The three stages of catabolism (ETC–Electron transport chain).

Anabolism

- In **anabolism**, also called **biosynthesis**, small, simple precursors are built up into larger and more complex molecules, including lipids, polysaccharides, proteins, and nucleic acids.
- Anabolic reactions **require an input of energy**, generally in the form of the phosphoryl group transfer potential of ATP and the reducing power of NADH, NADPH, and FADH₂.



Types of metabolic reactions

- The biochemical reactions are mainly of four types
 1. Oxidation-reduction.
 2. Group transfer.
 3. Rearrangement and isomerization.
 4. Make and break of carbon-carbon bonds.
- These reactions are catalysed by specific enzymes—more than 2,000 known so far.



Method Employed to Study Metabolism

- The basic metabolic pathways in most organisms are essentially identical. For this reason, many organisms can be used to understand metabolisms
- The actual methods employed may be either in vivo (in the living system) or in vitro (in the test tube) or, more frequently, both
- Three methods are broadly used



1. Use of whole organism or its components :

- a) Whole organisms** : The ultimate aim of a biochemist is to know the metabolism in the organism as a whole. Glucose tolerance test (GTT) employed to measure the response of man (or other animals) towards carbohydrate metabolism
- b)** Isolated organs, tissue slices, whole cells, subcellular organelles, cell-free systems and recently purified components are frequently used to elucidate biochemical reactions and metabolic pathways.



2. Utility of metabolic probes

- inhibitors and mutations are metabolic probes commonly used to trace out biochemical pathways.
- Inhibitors of electron transport chain have been largely responsible to elucidate the sequence of electron carriers
- The inborn errors of metabolism in higher organisms and the genetic manipulations in the microorganisms have also contributed a lot to the understanding of metabolisms.



3. Application of isotopes :

- Isotopes are the atoms with the same number of protons but different neutrons. By use of isotopes, the molecules of the living system can be labeled without altering their chemical properties
- Application of isotopes in biochemistry has revolutionized the study of metabolisms.





Nucleotide Metabolism

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Nucleic acids- Introduction

- The final class of **biomolecules** to be considered, the **nucleotides plus molecules derived from them (i.e., nucleic acids)**, represent a clear case in which **last is not least**.
- “**Nucleotides** themselves participate in a plethora of **crucial supporting roles in cell metabolism**, and the **nucleic acids provide the script for everything that occurs in a cell**”
 - (Lehninger, Nelson and Cox, 1993).

Nucleic acids- Introduction

- **Nucleotides** are energy rich compounds that **drive metabolic processes** (esp., **biosynthetic**) in all cells.
- They also serve as **chemical signals**, key links in cellular systems that respond to hormones and other extracellular stimuli, and are **structural components of a number of enzyme cofactors** and metabolic intermediates.
- **The nucleic acids** (DNA and RNA) are the **molecular storehouses** for **genetic information** and are jointly referred to as the '**molecules of heredity**'.
- The structure of **every protein**, and ultimately of **every cell constituent**, is a **product of information programmed into the nucleotide sequence of a cell's nucleic acids**.

Historical resume

- The **nucleic acids** have been the subject of biochemical investigations since 1869 when **Friedrich Miescher**, a 25-year old Swiss chemist, isolated nuclei from **pus cells**- contained a hitherto **unknown phosphate rich substance**- named ***nuclein***.
- He continued his studies on **salmon sperm**, a **prime source of nuclein** and isolated it as a nucleoprotein complex in 1871.
- These nuclei were then analyzed and found to have different composition from any cellular component- was called ***nuclein***.
- When all the proteins was completely removed, it became clear that the new material was an **acid- nucleic acid**.

Definitions- Nucleic acids

- The **nucleic acids** are the hereditary determinants of living organisms.
- They are the **macromolecules** present in **most living cells** either in the free state or protein bound- **nucleoproteins**.
- Just as **amino acids** are the repeating units of proteins, **mononucleotide** as their repeating units of nucleic acids.
- Elemental composition- the **nucleic acids** contain **carbon, hydrogen, oxygen, nitrogen** and, **strangely enough, phosphorus** ; the percentage of the last two elements being about **15 and 10, respectively**.

Types- Nucleic acids

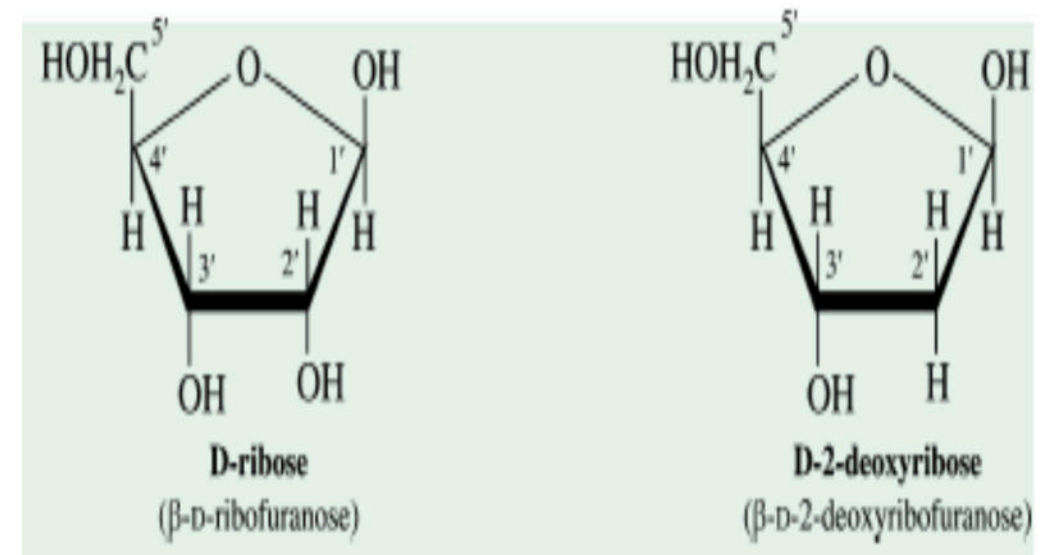
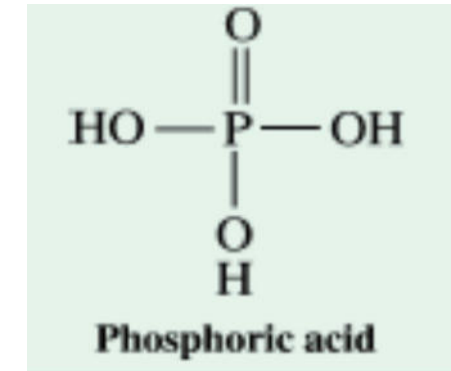
- Two kinds of nucleic acids- deoxyribonucleic acid (DNA) and ribonucleic acid (RNA).
- Both are present in **all plants and animals**. Viruses contain either **RNA or DNA, but not both**.
- DNA- in the **chromatin of the cell nucleus** whereas (90%) of the RNA- present in the cell **cytoplasm** and **(10%) in the nucleus**.
- Hydrolysis of nucleic acids yield **3 components** : phosphoric acid, a pentose sugar and nitrogenous bases.

Table 15-1. Structural components of RNA and DNA

<i>Components</i>	<i>Ribonucleic acid</i>	<i>Deoxyribonucleic acid</i>
Acid	Phosphoric acid	Phosphoric acid
Pentose sugar	D-ribose	D-2-deoxyribose
Nitrogenous bases		
Purines	Adenine	Adenine
	Guanine	Guanine
Pyrimidines	Cytosine	Cytosine
	Uracil	Thymine

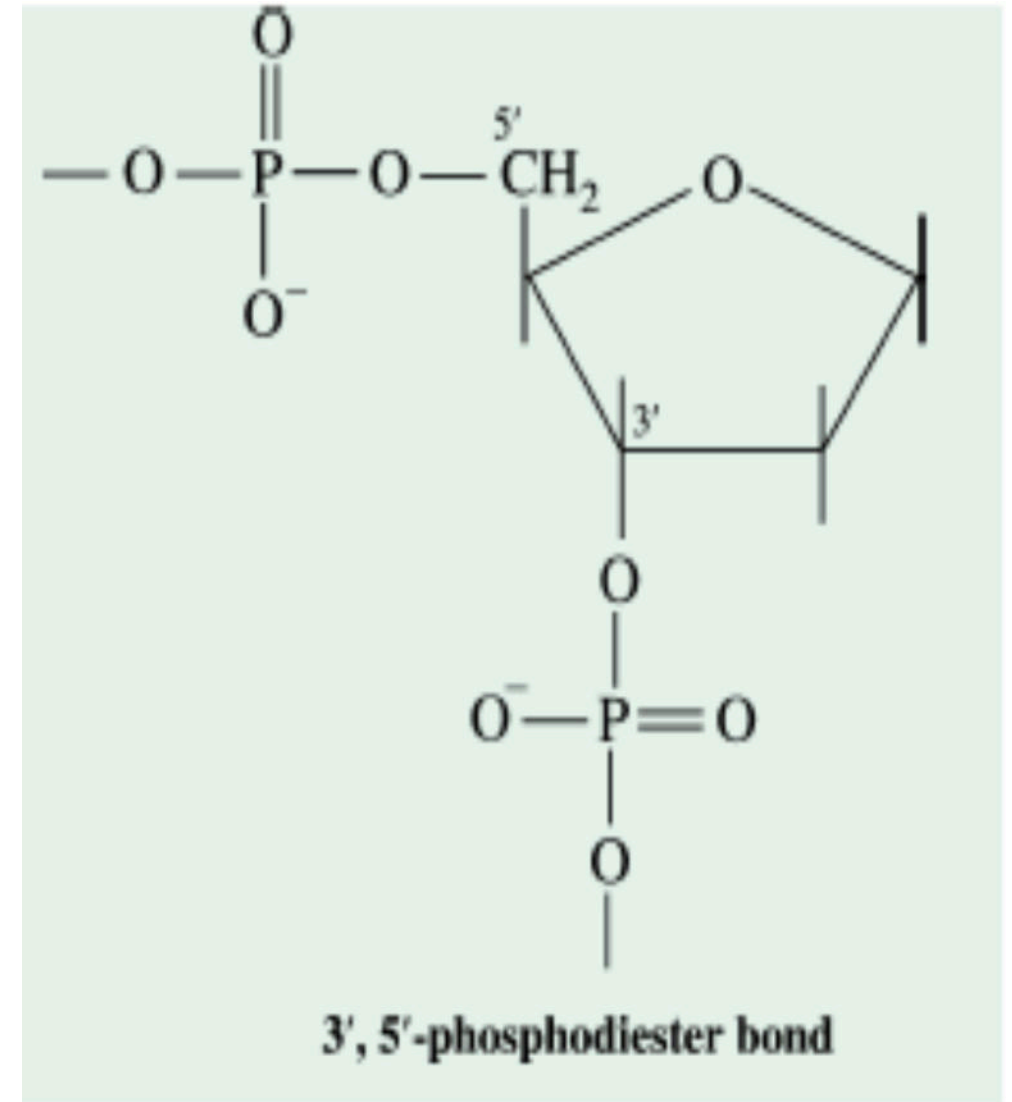
Structural components- Phosphoric Acid

- **Phosphoric Acid:** H_3PO_4 - contains 3 monovalent hydroxyl groups and a divalent oxygen atom, all linked to the pentavalent phosphorus atom
- **Pentose Sugar:** DNA- D-2-deoxyribose, hence the name **deoxyribonucleic acid**, while RNA- D-ribose, hence the name **ribonucleic acid**.
- Both sugars in nucleic acids are present in the **furanose** form and are of β



Structural components- Phosphoric Acid

- An important property of the **pentoses**- capacity to form **esters with phosphoric acid**.
- The **OH groups of the pentose at C3 and C5**, form a **3', 5'- phosphodiester bond** between adjacent pentose residues.
- This bond is an **integral part of the structure of nucleic acids**.



Structural components- Nitrogenous Bases

- **Nitrogenous Bases:** Two types- They are derivatives of **Pyrimidine and Purine**.
- **Pyrimidine Derivatives-** derived from a heterocyclic compound pyrimidine- a six-membered ring with two-nitrogen atoms and three double bonds.
- The common pyrimidine derivatives found in nucleic acids are **uracil, thymine and cytosine**.

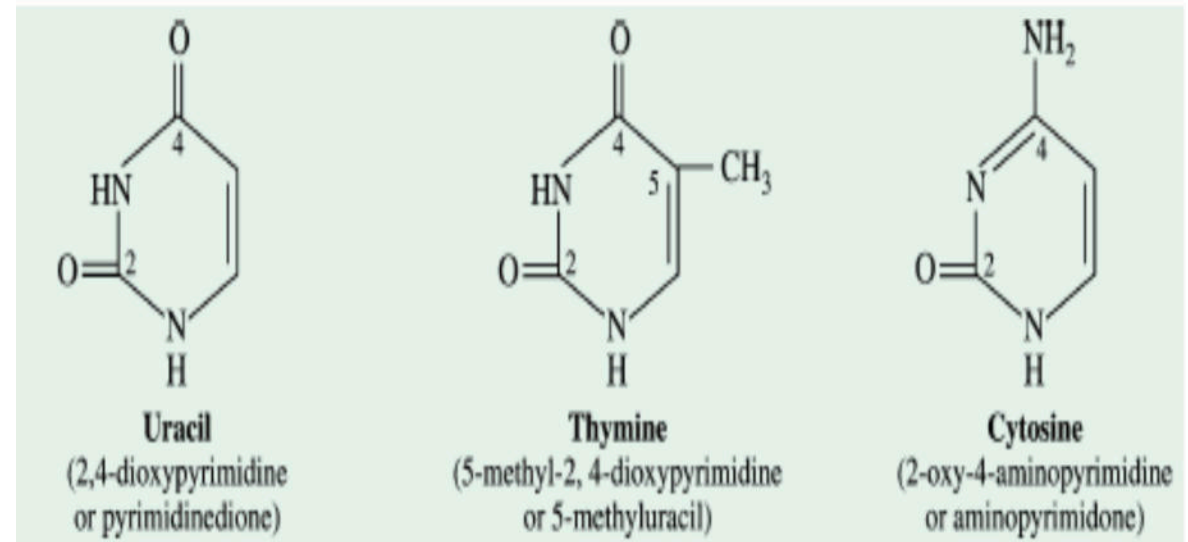
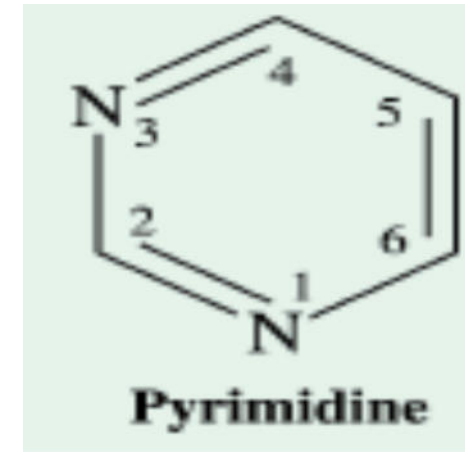


Fig. 15-2. Major pyrimidine derivatives

Structural components- Nitrogenous Bases

- **Purine Derivatives-** derived from purine- a six-membered pyrimidine ring fused to the five-membered imidazole ring and is related to uric acid.
- The **purine derivatives** found in nucleic acids are **adenine** and **guanine**

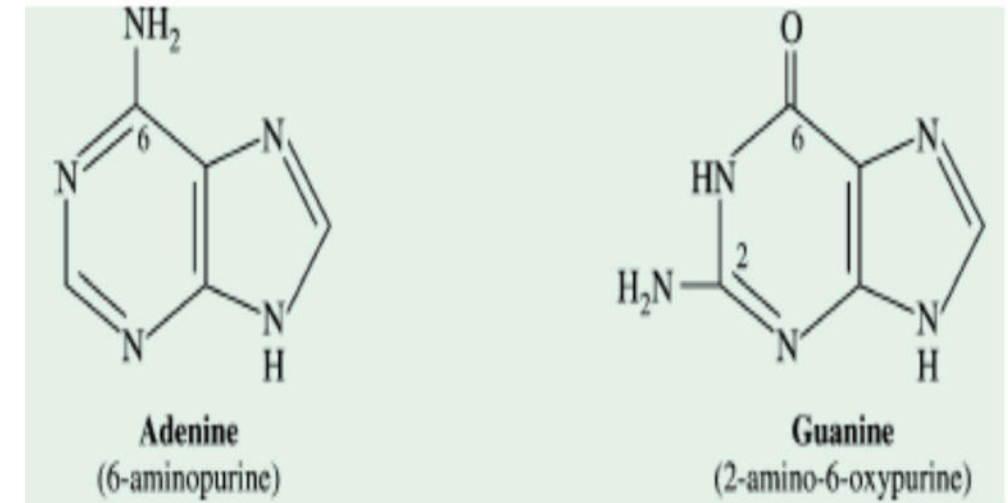
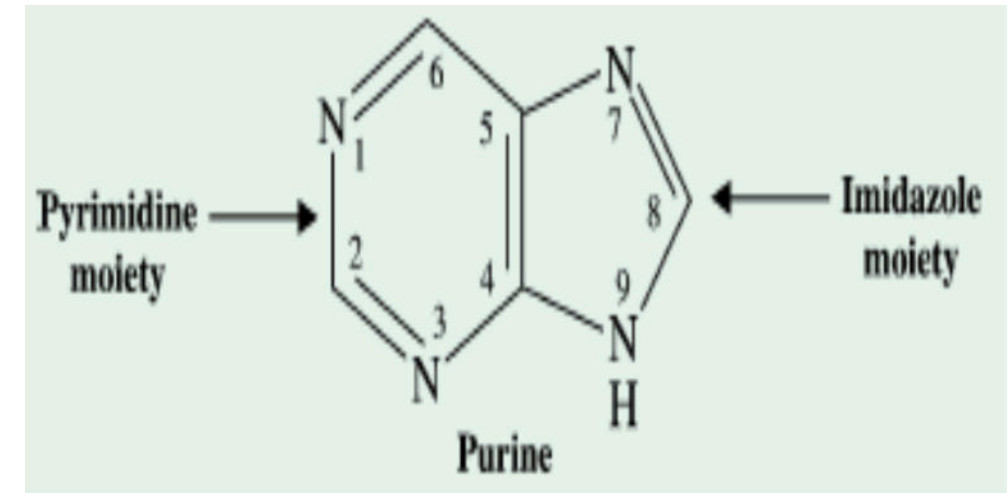


Fig. 15-3. Major purine derivatives

Nucleosides

- Result from linking one of the sugars with a **purine or pyrimidine base** through an **N-glycosidic linkage**
 - Purines bond to the C1' carbon of the sugar at their N9 atoms
 - Pyrimidines bond to the C1' carbon of the sugar at their N1 atoms
 - Therefore, the purine nucleosides are N-9 glycosides and the pyrimidine nucleosides are N-1 glycosides.

Table 15–2. The nucleosides

<i>Base</i>	<i>Sugar</i>	<i>Nucleoside</i>	<i>Trivial name*</i>	<i>Abbreviation</i>
Ribonucleosides				
Adenine	Ribose	Adenine ribonucleoside	Adenosine	AR
Guanine	Ribose	Guanine ribonucleoside	Guanosine	GR
Cytosine	Ribose	Cytosine ribonucleoside	Cytidine	CR
Thymine	Ribose	Thymine ribonucleoside	Thymidine	TR
Uracil	Ribose	Uracil ribonucleoside	Uridine	UR
Deoxyribonucleosides				
Adenine	Deoxyribose	Adenine deoxyribonucleoside	Deoxyadenosine	AdR
Guanine	Deoxyribose	Guanine deoxyribonucleoside	Deoxyguanosine	GdR
Cytosine	Deoxyribose	Cytosine deoxyribonucleoside	Deoxycytidine	CdR
Thymine	Deoxyribose	Thymine deoxyribonucleoside	Deoxythymidine**	TdR
Uracil	Deoxyribose	Uracil deoxyribonucleoside	Deoxyuridine	UdR

* Note that the trivial names for the pyrimidine nucleosides end with the suffix *-dine*, whereas those of purine nucleosides end with the suffix *-sine*.

Nucleotides

- Nucleotides are the **phosphoric acid esters of nucleosides**.
- Occur either in the **free form** or **as subunits in nucleic acids**.
- The **phosphate** is always **esterified to the sugar moiety**.

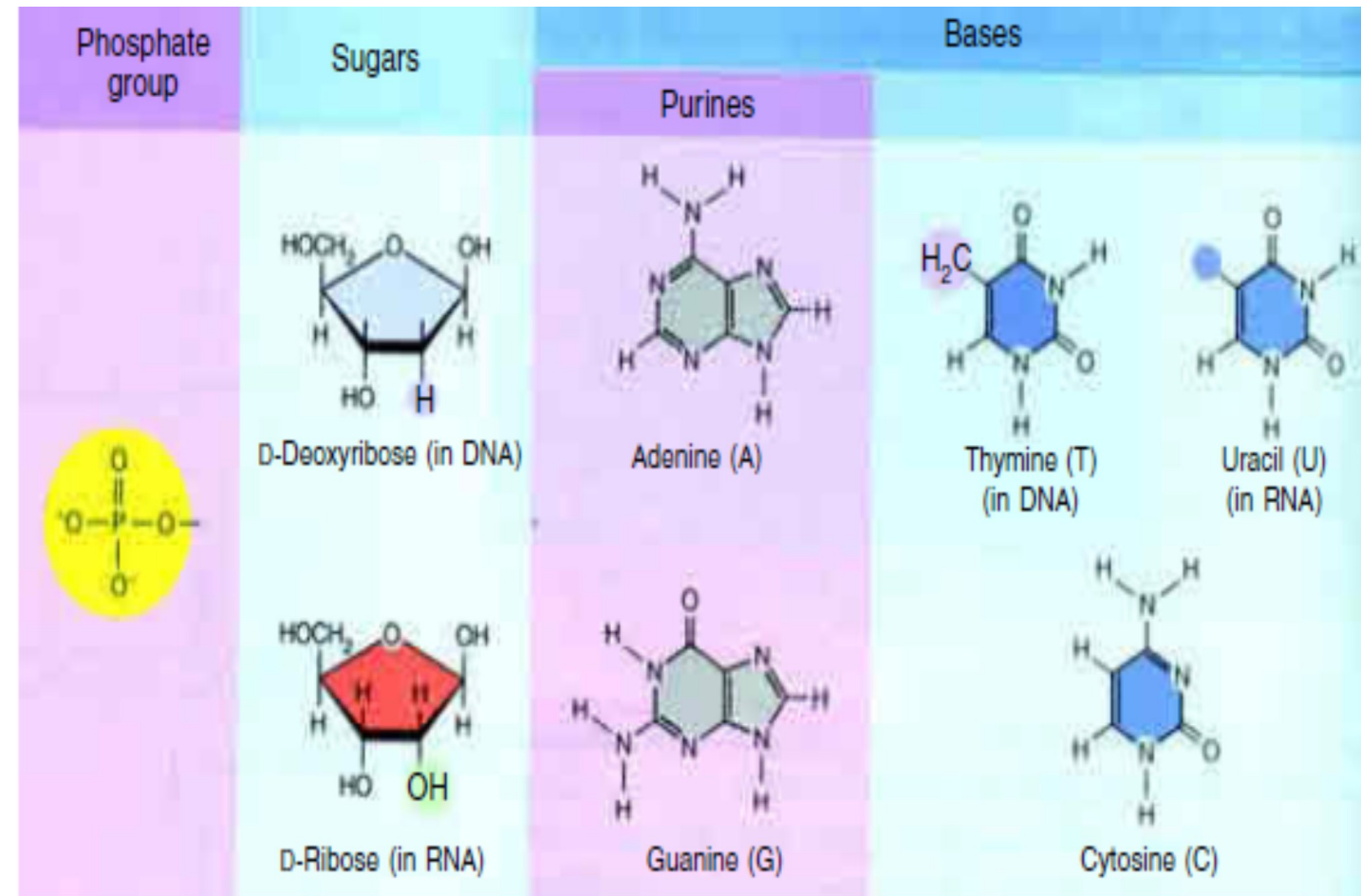


Fig. 15-10. The components of nucleotides

Nucleotides- Functions

- **Functions of Nucleotides:**

1. **As carriers of chemical energy.**

- Nucleotides may have **one, two or three phosphate groups** covalently linked at **5'-OH of ribose-** nucleoside mono-, di- and tri phosphates- NMPs, NDPs and NTPs, respectively.
- The **3 phosphate groups** called- α , β and γ , starting from the ribose.
- **NTPs are used as a source of chemical energy-** Adenosine triphosphate (**ATP**) is by far, the most widely used.
- The **hydrolysis of ATP** and other nucleoside triphosphates is an **exergonic reaction**.
- The bond between the **ribose and the α -phosphate** is an **ester linkage**. On hydrolysis yields about **14 kJ/mol**.
- The **α - β and β - γ linkages** are **phosphoric acid anhydrides**. On hydrolysis each anhydride bond yields about **30 kJ/mol**.

ATP

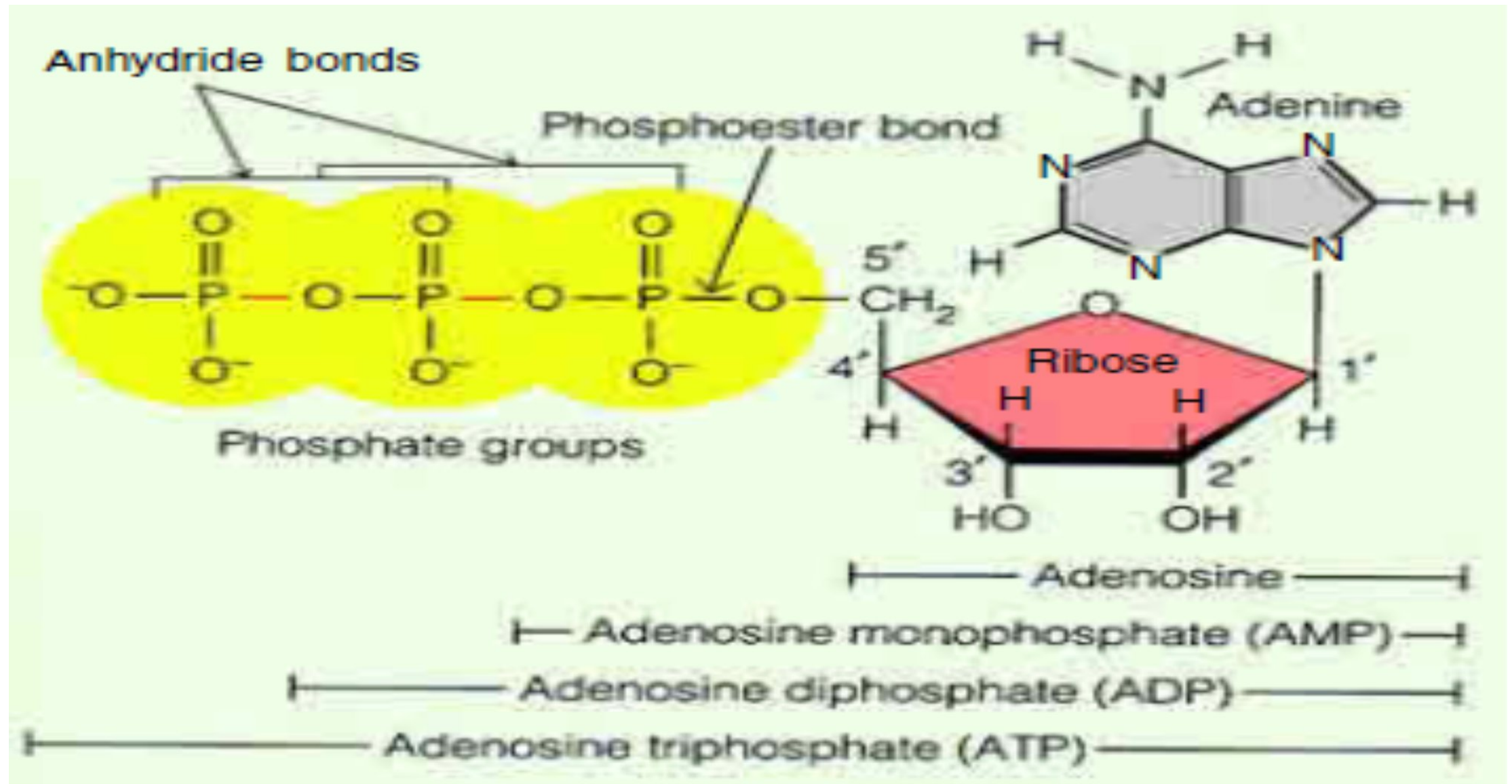


Fig. 15-12. The phosphate ester and phosphoric acid anhydride bonds of ATP

Nucleotides- Functions

2. As components of enzyme factors:

- Many **enzyme cofactors and coenzymes** (such as coenzyme A, NAD^+ and FAD) contain adenosine as part of their structure.

3. As chemical messengers:

- The cells **respond to their environment** by **chemical signals** in the surrounding medium- these chemical signals (**first messengers**) with **receptors on the cell surface often leads** to the formation of **second messengers** inside the cell- lead to adaptive changes inside the cell.
- Often, the **second messenger is a nucleotide**.
- One of the most common second messengers is the **nucleotide adenosine cyclic**

Synthesis of Purine Nucleotides

- Two pathways of nucleotide synthesis:
 1. **De novo synthesis:** The synthesis of nucleotides begins with their metabolic precursors: *amino acids, ribose-5-phosphate, CO₂, and one-carbon units.*
 2. **Salvage pathways:** The synthesis of nucleotide by **recycle the free bases or nucleosides released from nucleic acid breakdown.**

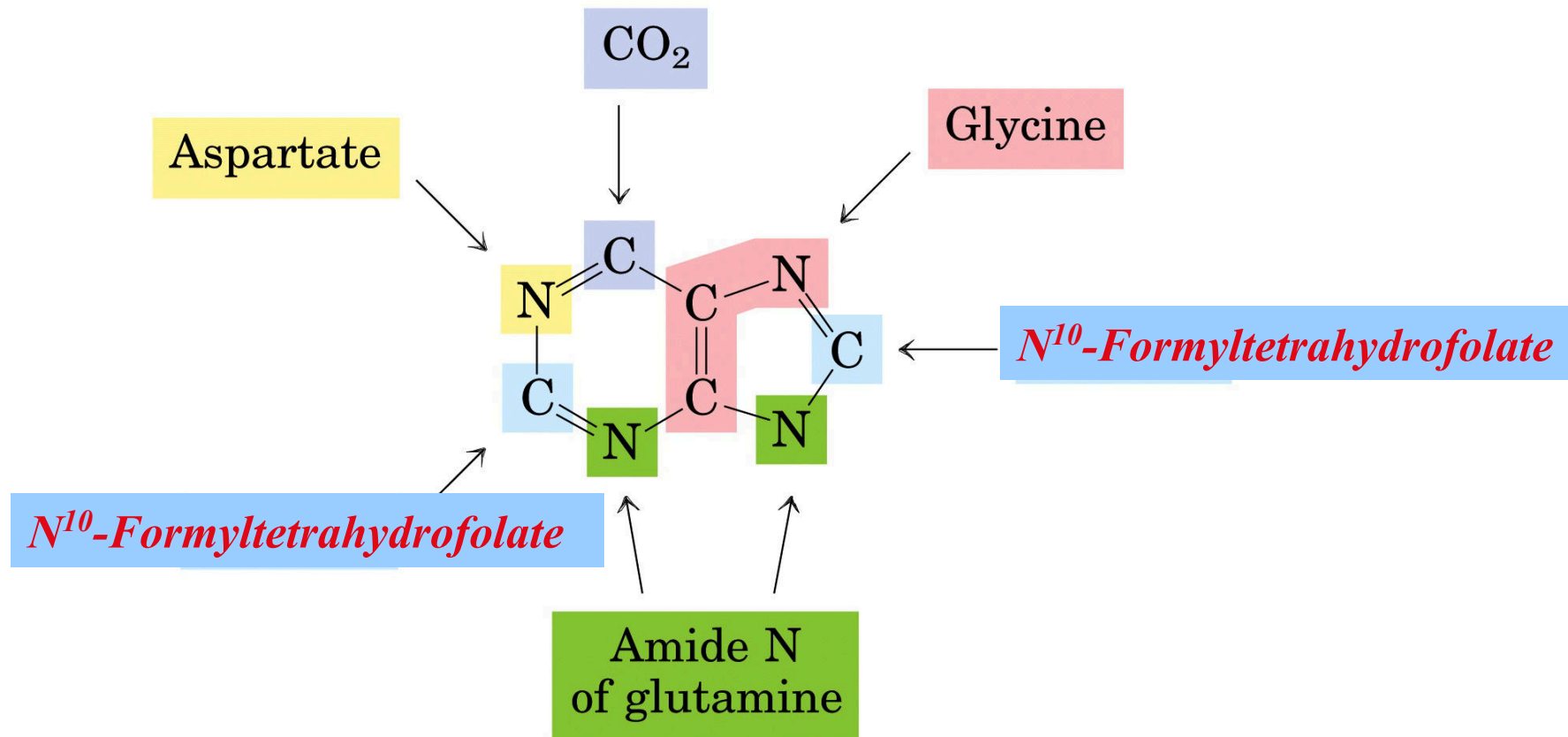
Synthesis of Purine Nucleotides

- **De Novo synthesis:**
- **Site:**
 - in cytosol of liver, small intestine and thymus
- **Characteristics:**
 - a. Purines are synthesized using **ribose-5-phosphate (R-5-P)** as the starting material step by step.
 - b. **PRPP(5-phosphoribosyl-1-pyrophosphate)** is active donor of R-5-P
 - c. AMP and GMP are synthesized from **IMP(Inosine-5'-Monophosphate)**

Synthesis of Inosine Monophosphate (IMP)

- Ribose phosphate is formed in pentose-phosphate pathway from glucose
- Basic pathway for biosynthesis of purine ribonucleotides starts from **ribose-5-phosphate (R-5-P)**
- Purine ring is synthesized on ribose-5-phosphate by the way of gradual adding of nitrogen and carbon atoms and cyclization.
- The pathway ends with the formation of a purine nucleotide called **Inosine monophosphate (IMP)** which is the precursor of **AMP** and **GMP** which then converted into ATP and GTP, respectively
- Requires **11 steps**.
- Occurs primarily in the liver

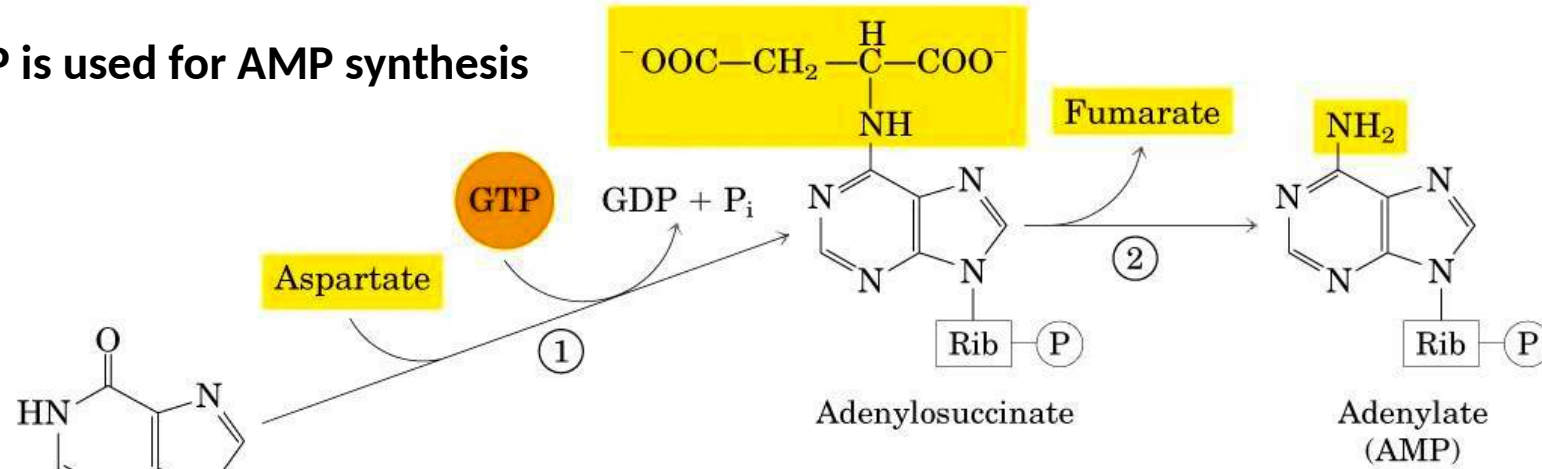
Element sources of purine bases



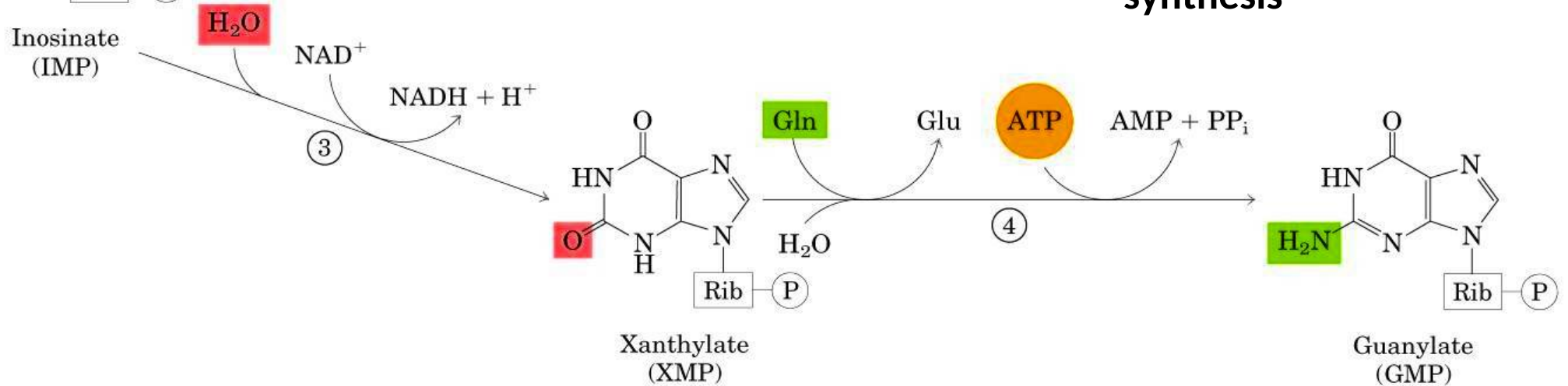
Conversion of IMP to AMP and GMP

IMP is the precursor for both AMP and GMP.

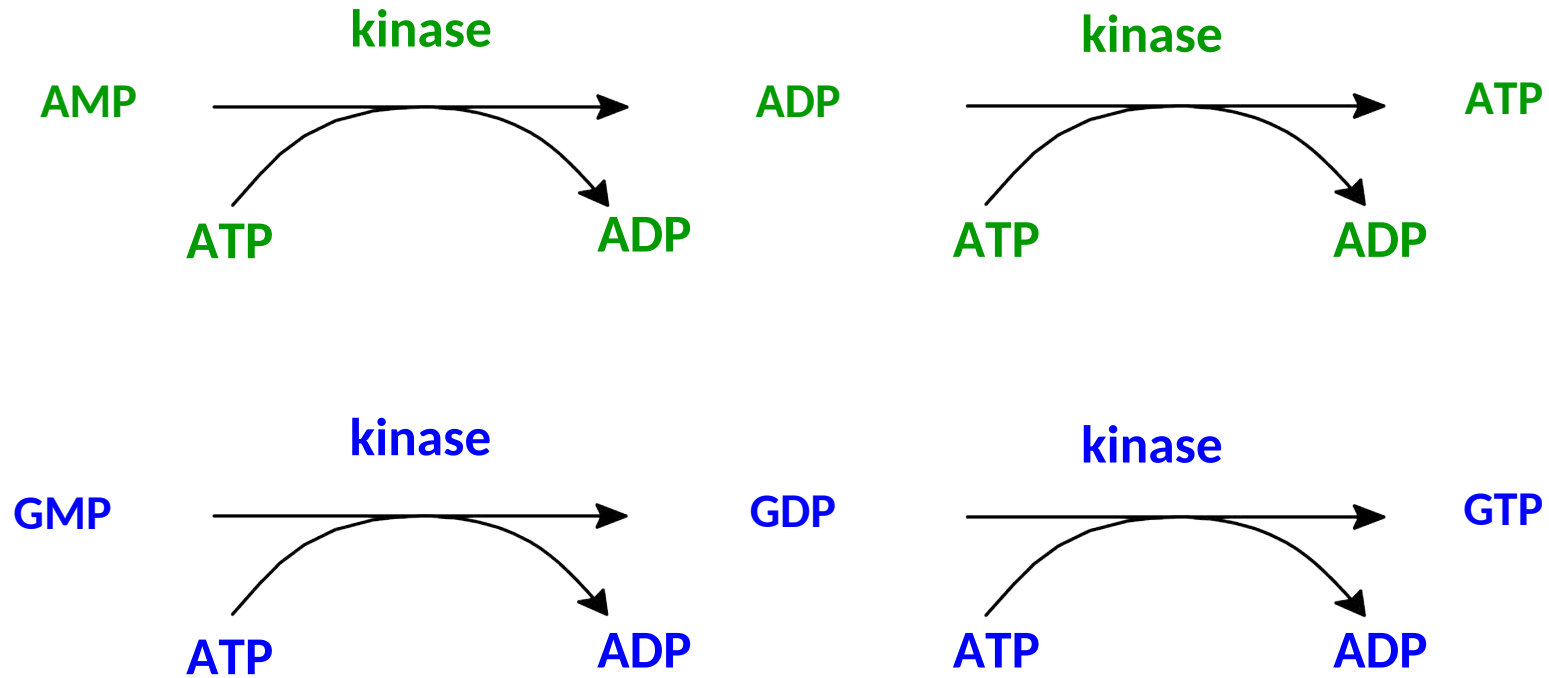
GTP is used for AMP synthesis



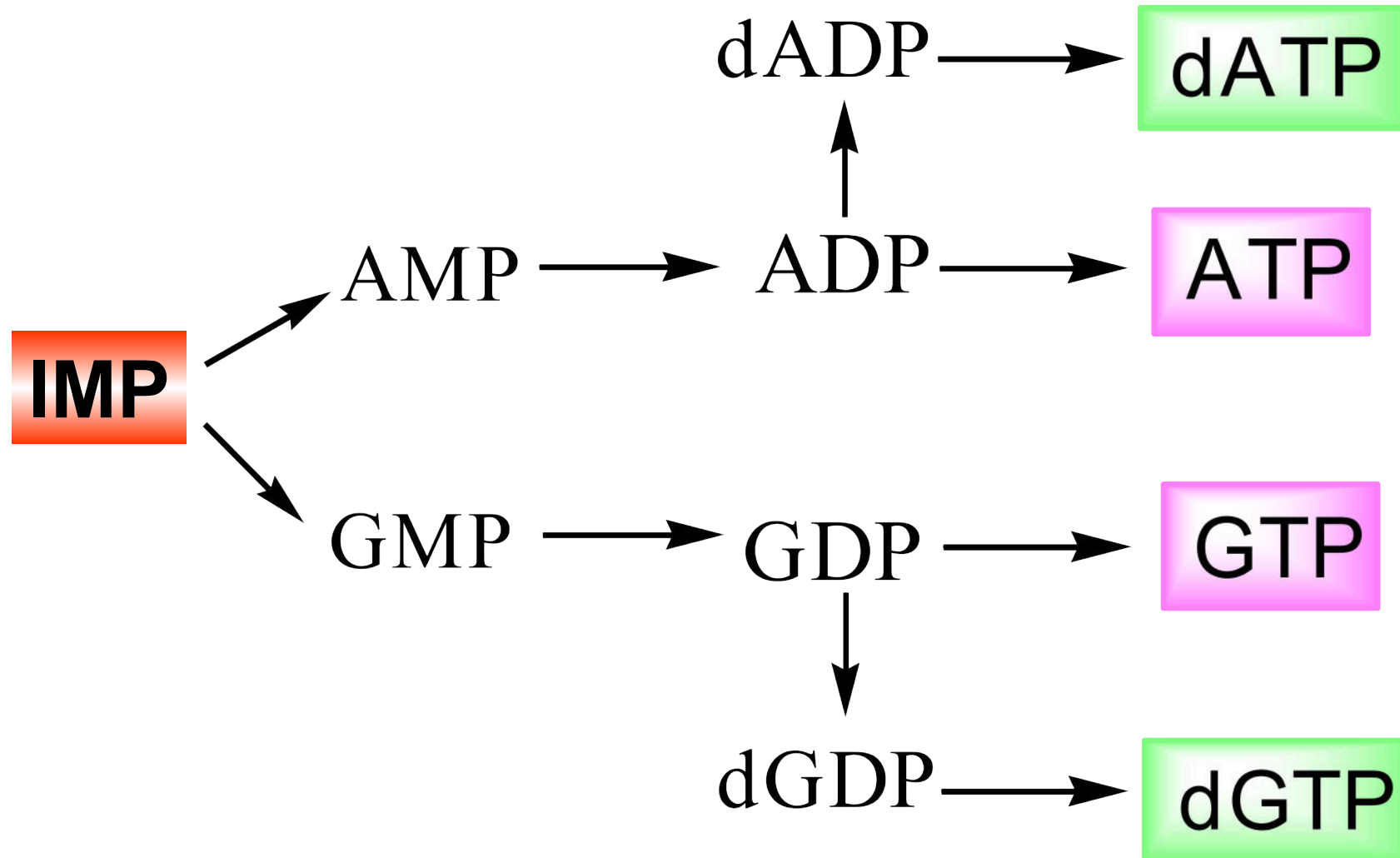
ATP is used for GMP synthesis



ADP, ATP, GDP and GTP biosynthesis



Summary of purine biosynthesis

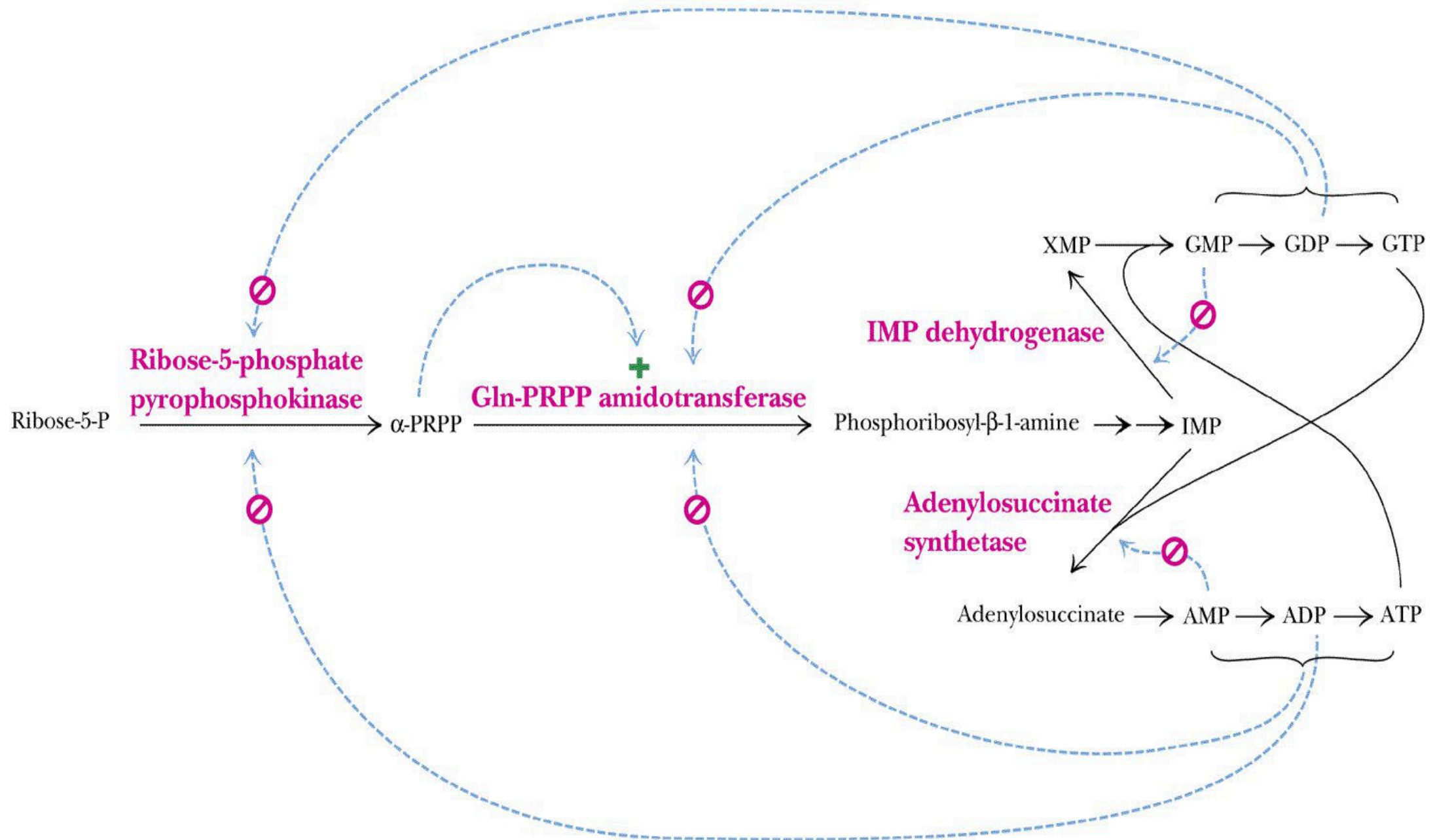


Regulation of de novo synthesis

- The significance of regulation:
 1. Meet the need of the body, without wasting.
 2. AMP and GMP control their respective synthesis from IMP by a feedback mechanism, $[GTP]=[ATP]$

Regulation of de novo synthesis

- **Amidophosphoribosyl transferase catalyses the rate limiting step of the pathway which is activated by PRPP.**
- **So PRPP is an activator of the pathway. Increased PRPP leads to overproduction of purine nucleotides.**
- **Inhibitors of the amidotransferase:**
- **The enzyme is inhibited by the final products of the pathway (IMP, AMP and GMP).**



Salvage pathway of purines- Resynthesis of purine nucleotides

- ***De novo*** biosynthesis **occurs in liver** due to the presence of enzymes.
- Other tissues can't do de novo synthesis.
- In these organs, free purine bases (**guanine, hypoxanthine and adenine**) reacts with PRPP again to resynthesize purine nucleotides.
- These free purine bases are obtained from **diet** or by **degradation of RNA or DNA**

Salvage pathway of purines- Resynthesis of purine nucleotides

- The significance of salvage pathway :
 - Save the fuel
 - Some tissues and organs such as brain and bone marrow are only capable of synthesizing nucleotides by salvage pathway.
- Two phosphoribosyl transferases are involved:
 - APRT (adenine phosphoribosyl transferase) for adenine.
 - HGPRT (hypoxanthine guanine phosphoribosyl transferase) for guanine or hypoxanthine.
 - Both enzymes use PRPP as the source of ribose-5-phosphate

Salvage pathway of purines- Resynthesis of purine nucleotides

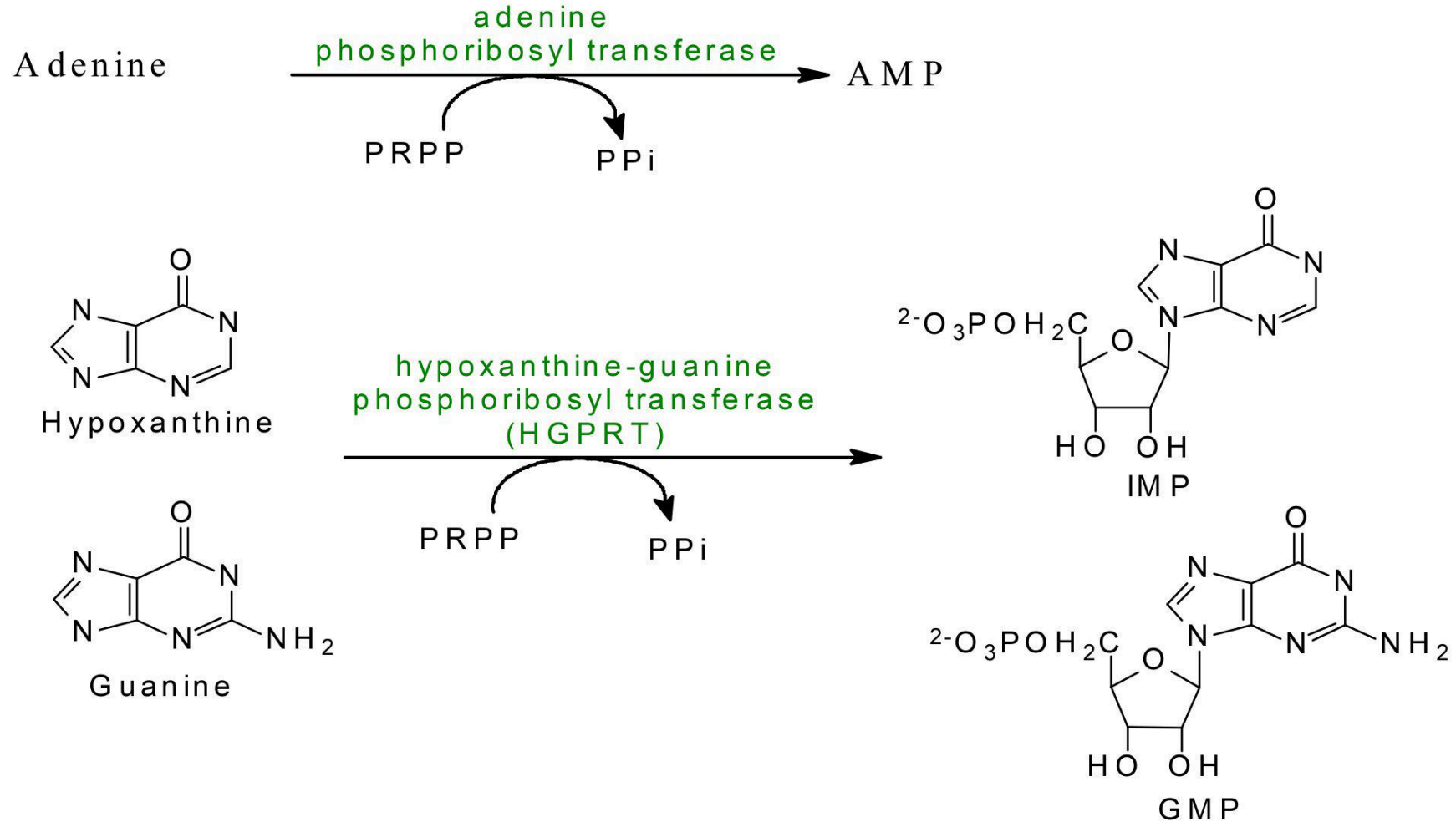
- The first enzyme catalyze the **transfer of ribose-5-P from PRPP to adenine to synthesize AMP**



- The second enzyme of salvage pathway catalyzes the **transfer of ribose-5-P into hypoxanthine or guanine to regenerate IMP or GMP**

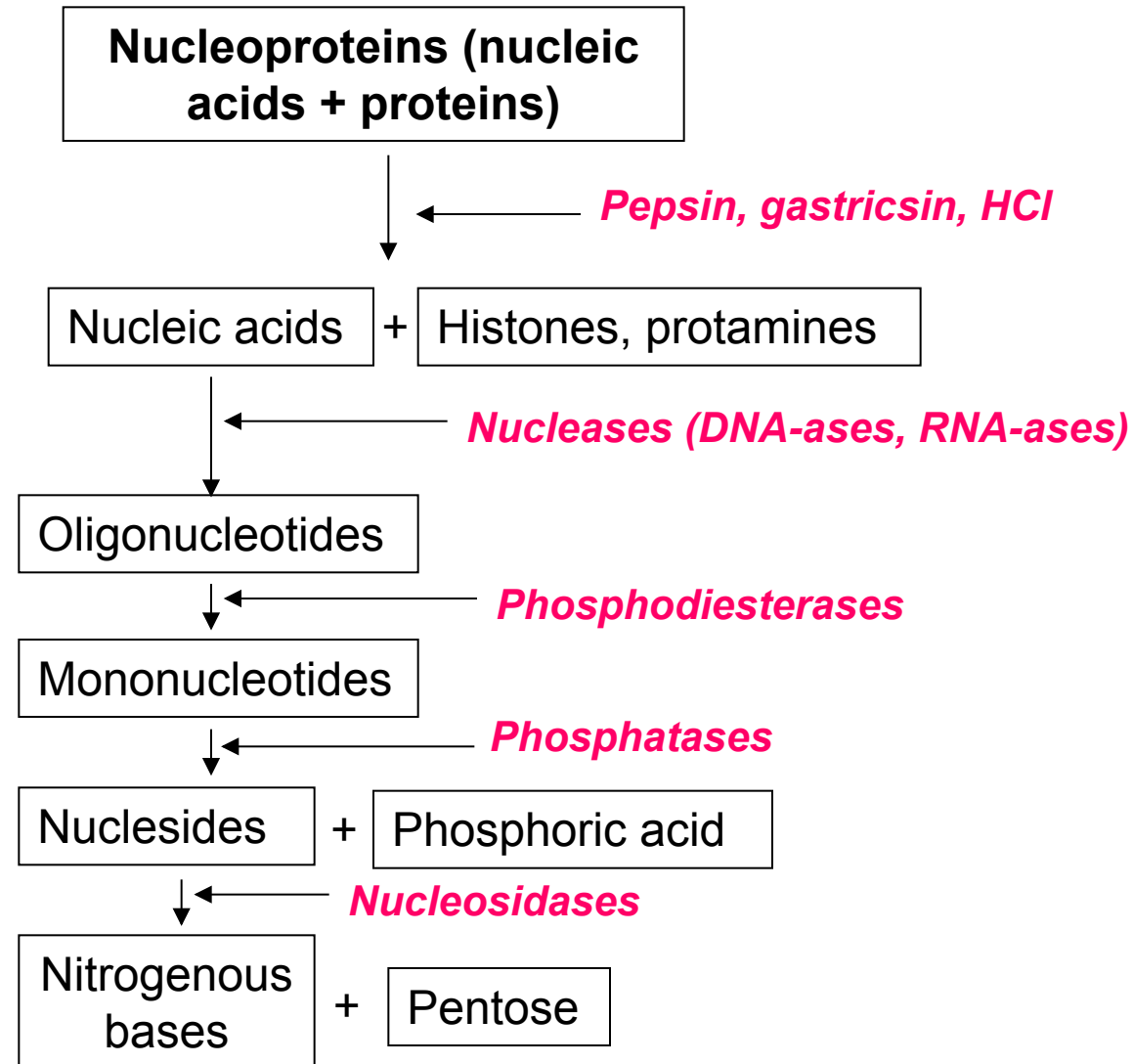


Purine Salvage Pathway



Absence of activity of HGPRT leads to Lesch-Nyhan syndrome.

Decomposition of nucleic acids in intestine and tissue



Destiny of nitrogenous bases, pentoses and phosphoric acids in the organism

**Nitrogenous
bases**



oxidation to the end products

Pentoses



oxidation with energy formation;
synthesis of nucleotides, hexoses,
coenzymes

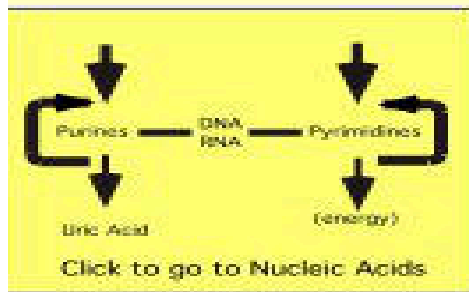
Phosphoric acid



phosphorylation;
ATP synthesis;
synthesis of phospholipids;
buffer systems;
constituent of bones, cartilages

Catabolism of purine nucleotides-Uric acid

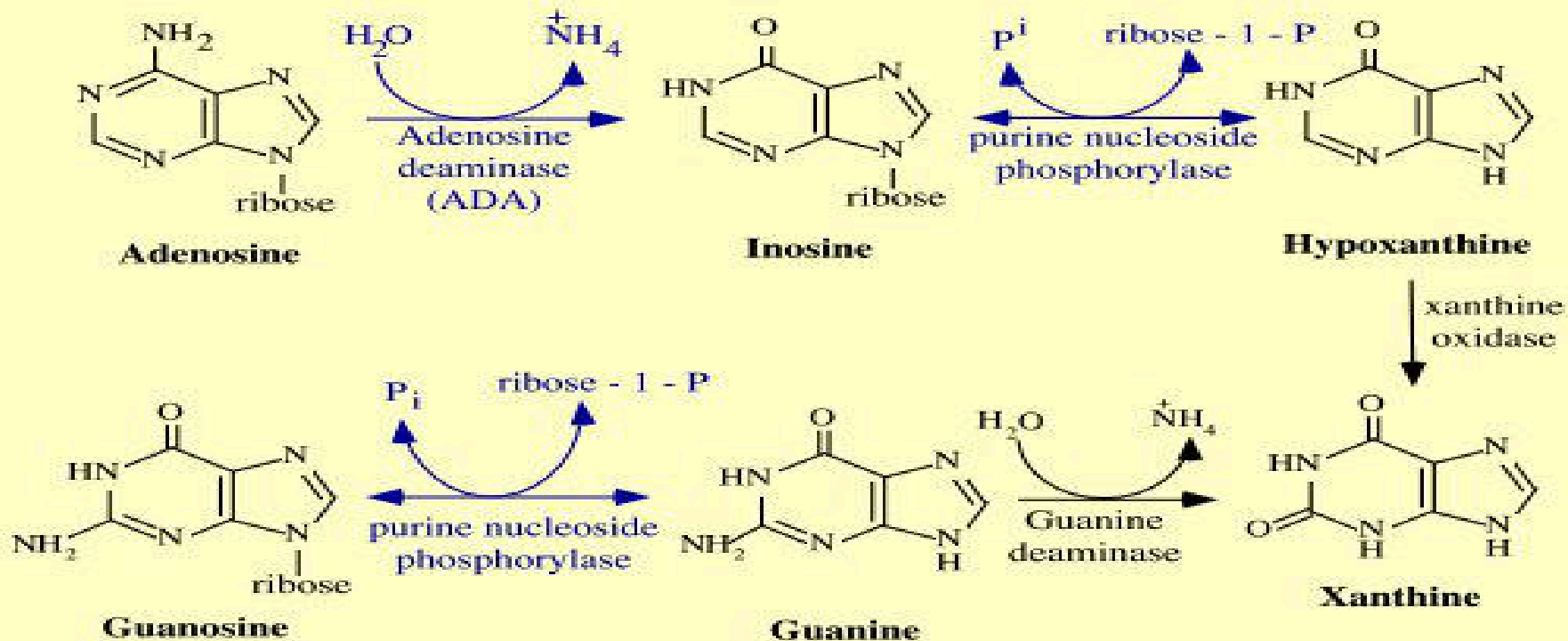
- **Uric acid** is the **end product** of **purine metabolism** in primates, birds and terrestrial reptiles
- **AMP or GMP** is metabolized to give **hypoxanthine** which is then converted into **xanthine** and finally into **uric acid**.
- Most of uric acid is excreted by the kidneys. The rate of uric acid excretion by the normal adult human is about **0.6 g/24 h**, arising in part from **ingested purines** and in part from the **turnover of the purine nucleotides of nucleic acids**.
- The normal serum uric acid concentration in adults is in the range of **3-7 mg/dl**.
- But under certain circumstances, the body produces **too much uric acid** or **removes too little**.
- In either case, concentrations of uric acid increase in the blood. This condition is known as **hyperuricemia**.



Degradation of Purine Nucleotides

Dietary and cellular nucleic acids are constantly being degraded. Many of the resultant purines are catabolized and excreted as Uric acid.

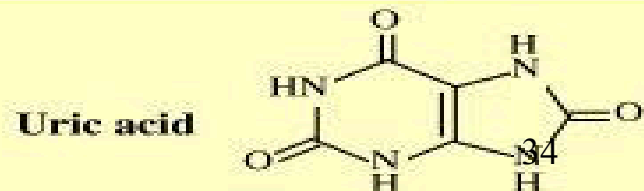
Note the first step in adenosine degradation is deamination. This is significant due to certain pathologies.

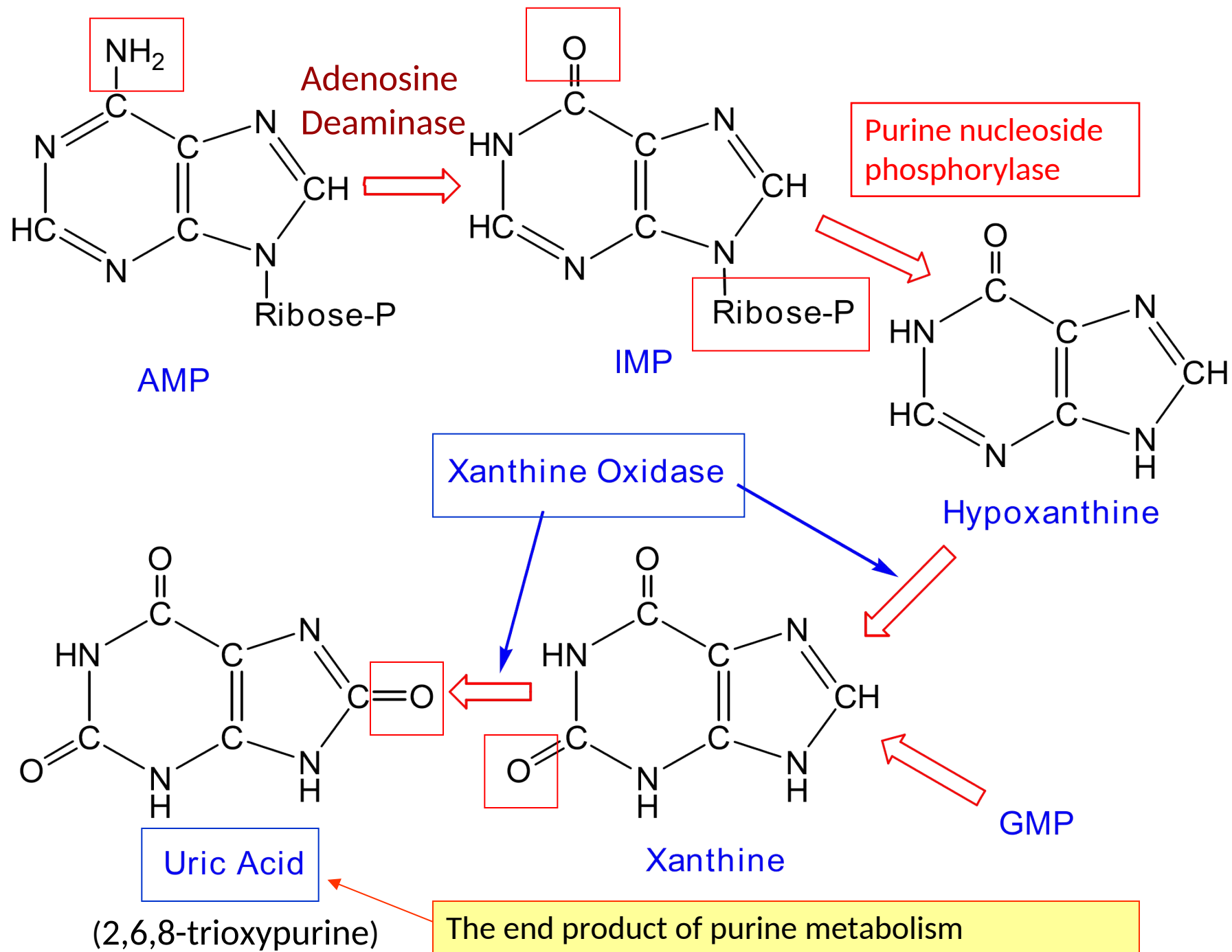


Unlike for adenosine, the first step in guanosine degradation is deribosylation.

The end product of purine degradation is Uric Acid, which is excreted in urine.

Uric Acid can build up and cause problems in some individuals. Click to find out more.





Disorders of purine nucleotides metabolism

- **A- Gout:** is a disorder characterized by **high levels of uric acid in blood (hyperuricemia)**, with deposition of **monosodium urate crystals** in **joints** and surrounding tissues.
- The **joints become inflamed, painful, and arthritic**, owing to the abnormal deposition of crystals of sodium urate.
- The **kidneys** are also affected, because excess uric acid is deposited in the kidney tubules.

Gout

- **Primary causes:**

1. **decreased excretion of uric acid by the kidney** due to renal disease-sometimes it is inherent.
2. **Defect in purine metabolism: increased synthesis of purine nucleotides which may be idiopathic or due to increased levels of PRPP that stimulate synthetic pathway of purine nucleotides.**

- **Secondary causes:** High intake of **purine rich diet** such as red meat, liver, xanthine beverages like tea, coffee, cola or due to medication in **radiation injury, blood diseases, tumors, toxemia, kidney diseases**

Gout

- **Symptoms:**

1. Hyperuricemia
2. Arthritis- inflammation of the joints due to deposition of urate crystals- hot red and swollen joints with severe pain.
3. Redness and swelling of big toe.
4. It may also present as tophi (masses of urate crystals deposited under skin) appears after several years.
5. It may lead to kidney stones,



Gout

- **Treatment:**

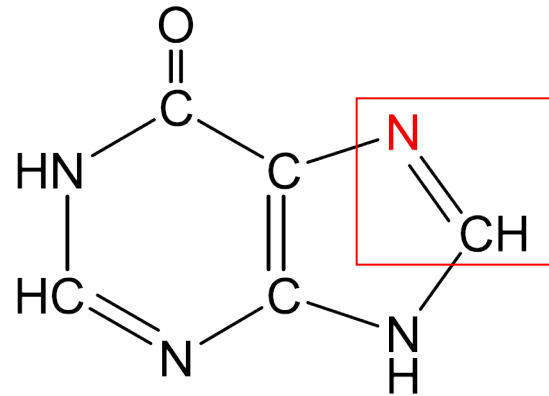
1. **Allopurinol**, analogue (structurally similar) of **hypoxanthine**.

- It **competitively inhibits xanthine oxidase**- prevents **conversion of hypoxanthine to xanthine and xanthine to uric acid**.

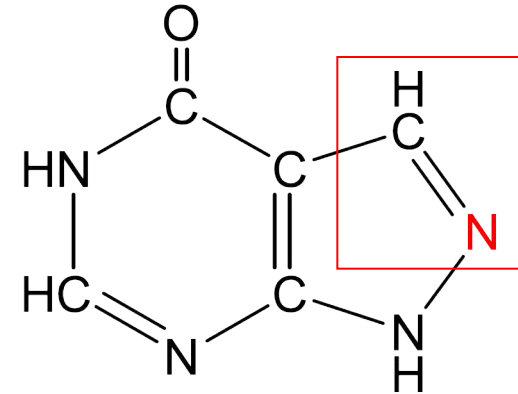
2. **Uricosuric agents**: drugs which increase excretion of uric acid by the kidneys such as **probenecid**.

3. **NSAIDS**- to decrease pain and inflammation

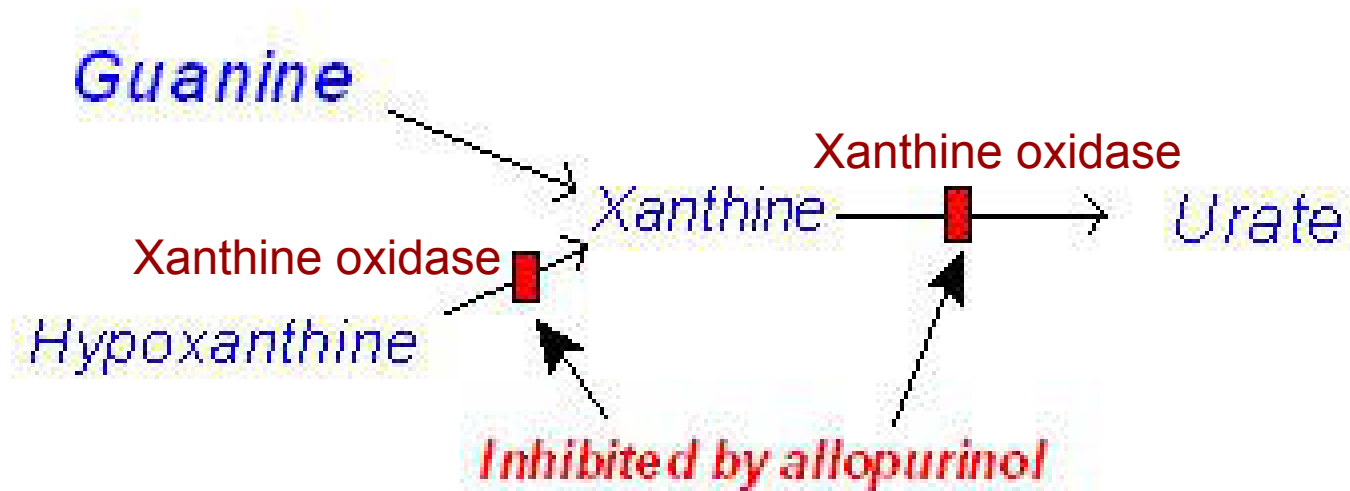
Allopurinol – a suicide inhibitor used to treat Gout



Hypoxanthine



Allopurinol



Lesch- Nyhan syndrome

- First described by [Michael Lesch](#) and [William L. Nyhan](#) in 1964
- There is a defect or lack in the [HGPRT](#) enzyme
- Sex-linked metabolic disorder: only males
- The rate of purine synthesis is increased about 200-fold
 - Loss of HGPRT → block (inhibit) salvage pathway of guanine and hypoxanthine → ↓ use of PRPP in salvage pathway and ↑ de novo purine synthesis leading to overproduction of purine nucleotides which by catabolism, will give increased levels of uric acid

Symptoms: appear at age 3-6 months. The first symptom is orange colored crystals in the diaper of the baby.

1- Hyperuricemia: in aggressive way than in gout.

2- urate kidney stones:

Some symptoms of unknown mechanism are:

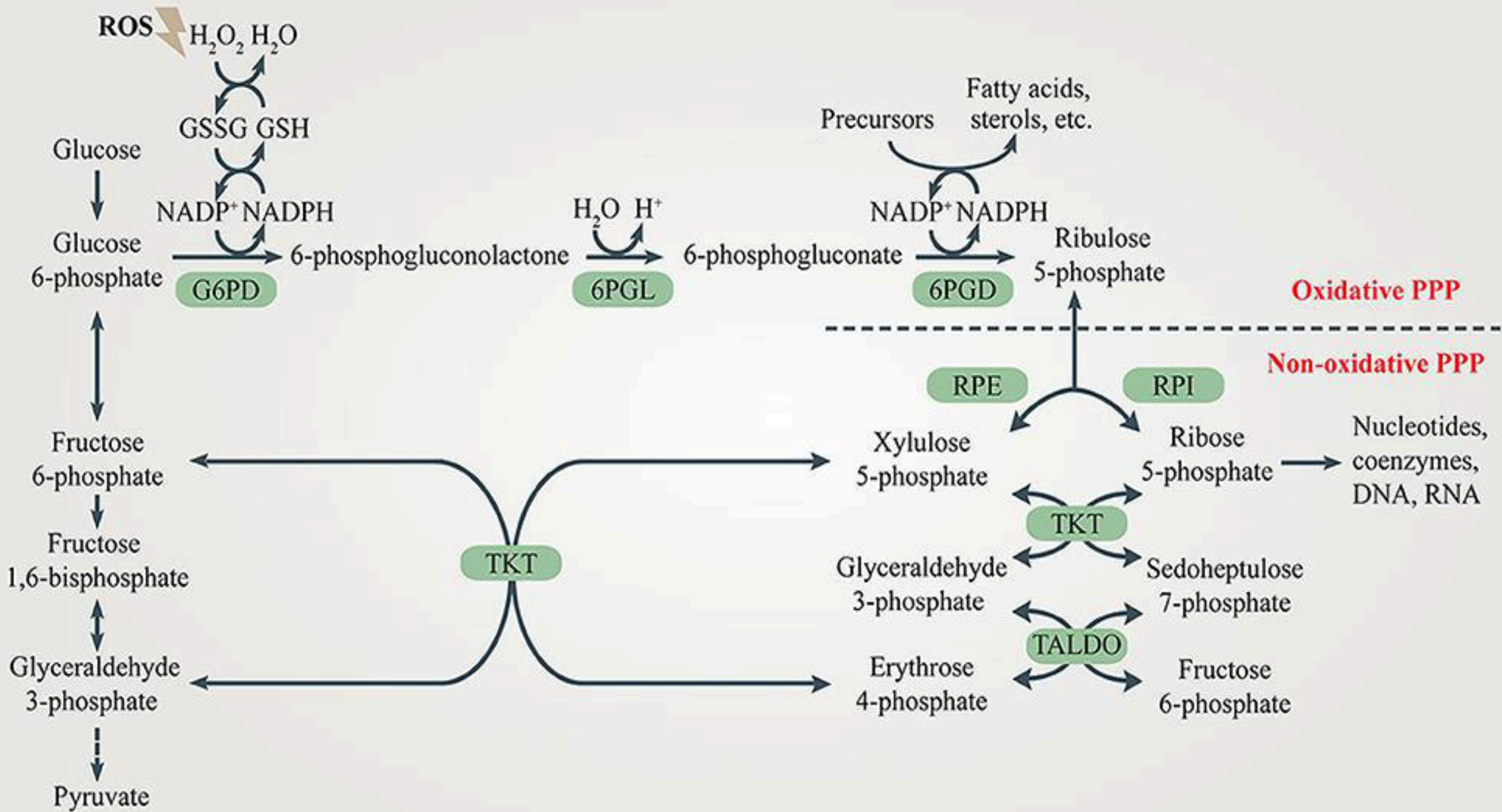
3- mental retardation

4- involuntary movements of legs and arms

5- lack of muscle coordination

6- **self mutilation** (biting of fingers and lips leading to lip lesions).

Pentose Phosphate Pathway

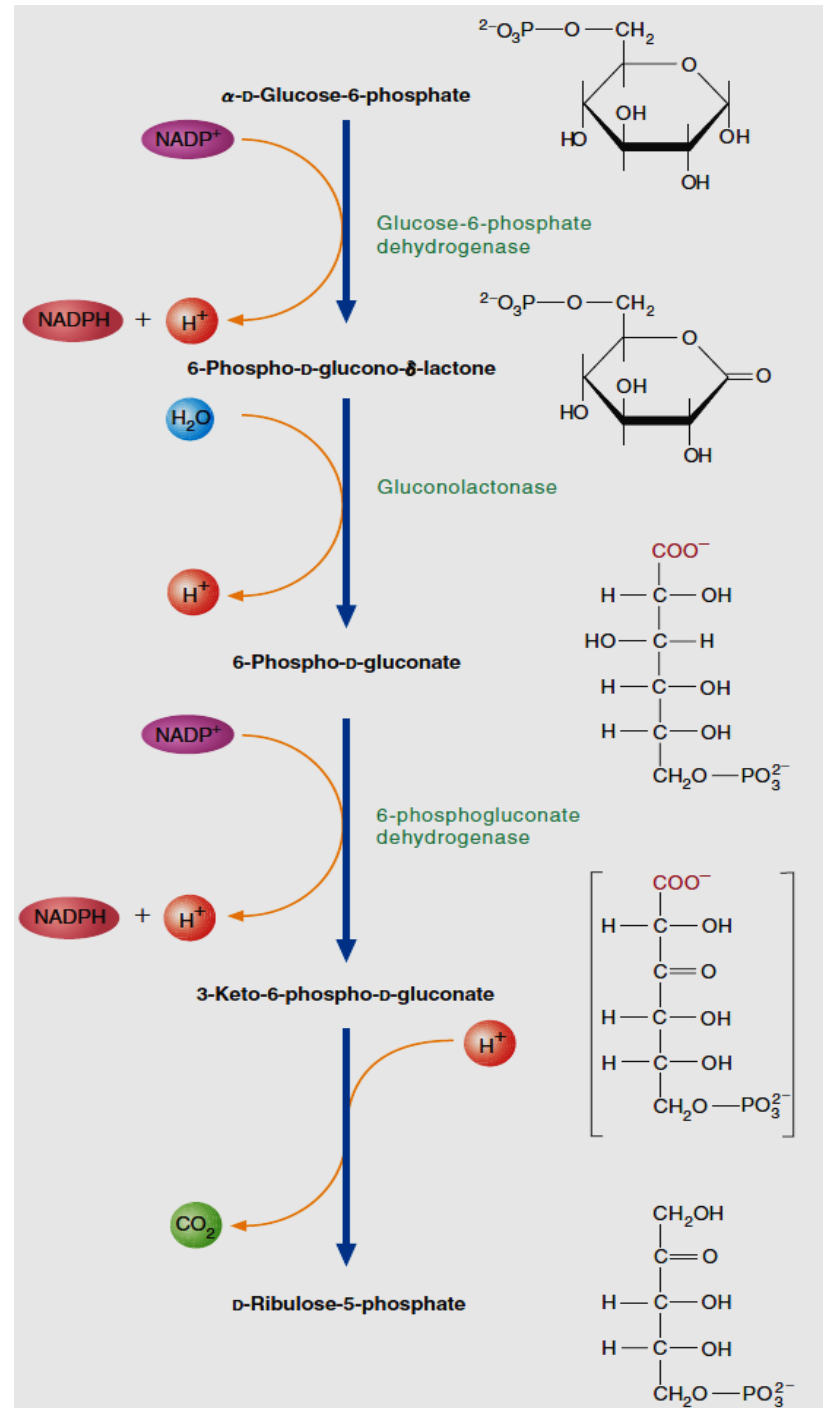


The pentose phosphate pathway (PPP)

- The **pentose phosphate pathway** is an alternative metabolic pathway for **glucose oxidation in which no ATP is generated**.
- Its principal products are **NADPH**, a reducing agent required in several anabolic processes, and **ribose-5-phosphate**, a structural component of nucleotides and nucleic acids.
- The pentose phosphate pathway occurs in the **cytoplasm** in two phases: **oxidative and non oxidative**.
- In the **oxidative phase**, **glucose-6-phosphate is converted to ribulose-5-phosphate and two molecules of NADPH** are produced.
- The non oxidative phase involves the **isomerization and condensation** of a number of different sugar molecules.
- Three intermediates in this process that are useful in other pathways are **ribose-5-phosphate, fructose-6-phosphate, and glyceraldehyde-3-phosphate**.

Oxidative phase

- The oxidative phase of the pentose phosphate pathway consists of three reactions.
- In the **first** reaction, **glucose-6-phosphate dehydrogenase (G-6-PD)** **catalyzes the oxidation of glucose-6-phosphate.**
- **6-Phosphogluconolactone** and **NADPH** are **products in this reaction.**
- **6-Phospho-D-glucono-lactone** is then hydrolyzed to produce 6 phospho-D-gluconate.
- A second molecule of NADPH is produced during the oxidative decarboxylation of 6-phosphogluconate, a **reaction that yields ribulose-5-phosphate.**



NADPH

- A substantial amount of the NADPH required for reductive processes (i.e., lipid biosynthesis) is supplied by these reactions.
- For this reason this pathway is most active in cells in which relatively **large amounts of lipids are synthesized, (e.g., adipose tissue, adrenal cortex, mammary glands, and the liver).**
- NADPH is also a **powerful antioxidant.** (substances that prevent the oxidation of other molecules).
- Consequently, the oxidative phase of the pentose phosphate pathway is also **quite active in cells that are at high risk for oxidative damage, such as red blood cells.**

TABLE 20.2 **Pathways requiring NADPH**

Synthesis

Fatty acid biosynthesis

Cholesterol biosynthesis

Neurotransmitter biosynthesis

Nucleotide biosynthesis

Detoxification

Reduction of oxidized glutathione

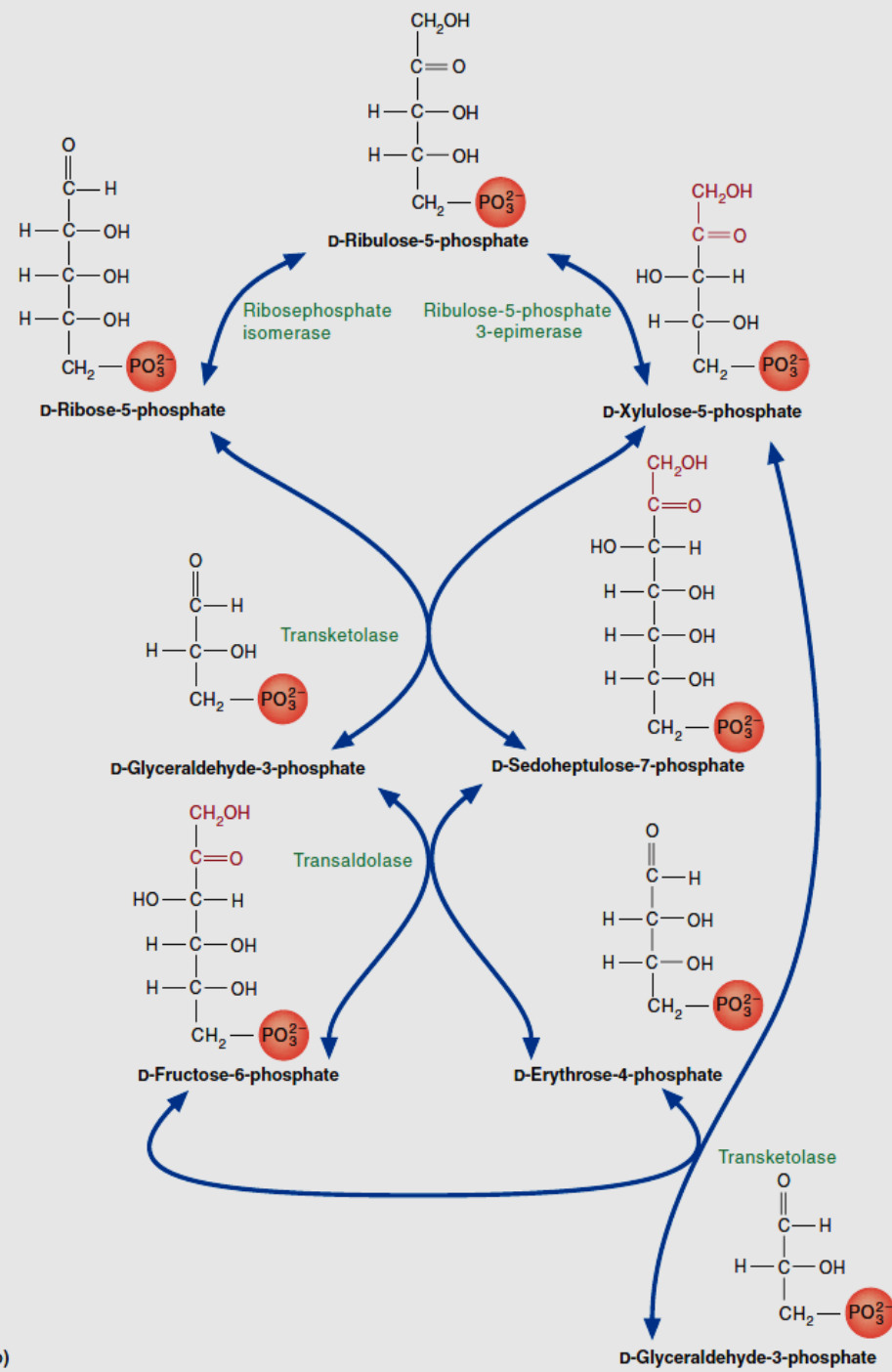
Cytochrome P450 monooxygenases

TABLE 20.4 Tissues with active pentose phosphate pathways

Tissue	Function
Adrenal gland	Steroid synthesis
Liver	Fatty acid and cholesterol synthesis
Testes	Steroid synthesis
Adipose tissue	Fatty acid synthesis
Ovary	Steroid synthesis
Mammary gland	Fatty acid synthesis
Red blood cells	Maintenance of reduced glutathione

Non oxidative phase

- The non oxidative phase of the pathway begins with the **conversion of ribulose- 5-phosphate to ribose-5-phosphate** by **ribulose-5-phosphate isomerase** or to xylulose-5-phosphate by ribulose-5-phosphate epimerase.
- During the remaining reactions of the pathway , **transketolase and transaldolase** catalyze the **interconversions of trioses, pentoses, and hexoses**.
- Transketolase is a TPP-requiring enzyme that transfers two-carbon units from a ketose to an aldose. (TPP, thiamine pyrophosphate, is the coenzyme form of thiamine, also known as vitamin B1.)



(b)

Regulation

- The pentose phosphate pathway is regulated to meet the cell's requirements for **NADPH and ribose-5 phosphate**.
- The **oxidative phase** is very active in cells such as **red blood cells or hepatocytes in which demand for NADPH is high** and is virtually absent in cells (e.g., muscle cells) that synthesize little or no lipid.
- **G-6-PD catalyzes** a key regulatory step in the pentose phosphate pathway.
- Its activity is **inhibited by NADPH and stimulated by GSSG, the oxidized form of glutathione**, an important cellular **anti oxidant** and glucose-6-phosphate.
- In addition, diets high in carbohydrate increase the synthesis of both G-6PD and phosphogluconate dehydrogenase.

Standard Operating Procedure for Collection of blood specimens by vein puncture:

This standard operating procedure (SOP) is applicable to the collection of blood specimens by venepuncture for the purpose of medical diagnostic testing, red cell cross-matching, ongoing monitoring and / or evaluation of patient responses to treatment. This SOP intends to standardize the venepuncture procedure for blood specimens collection executed by all related medical staffs involved in patient care.

1. An authorized prescriber's written order of blood specimens is required.
2. Vein puncture must be performed using aseptic technique.

3. In the case of a difficult vein puncture, it is recommended NOT MORE THAN TWO attempts of

Vein puncture on any one patient by a single phlebotomist at any one time. Rest the patient for a

While and get help from colleagues or superiors. (This is NOT applicable to emergency cases)

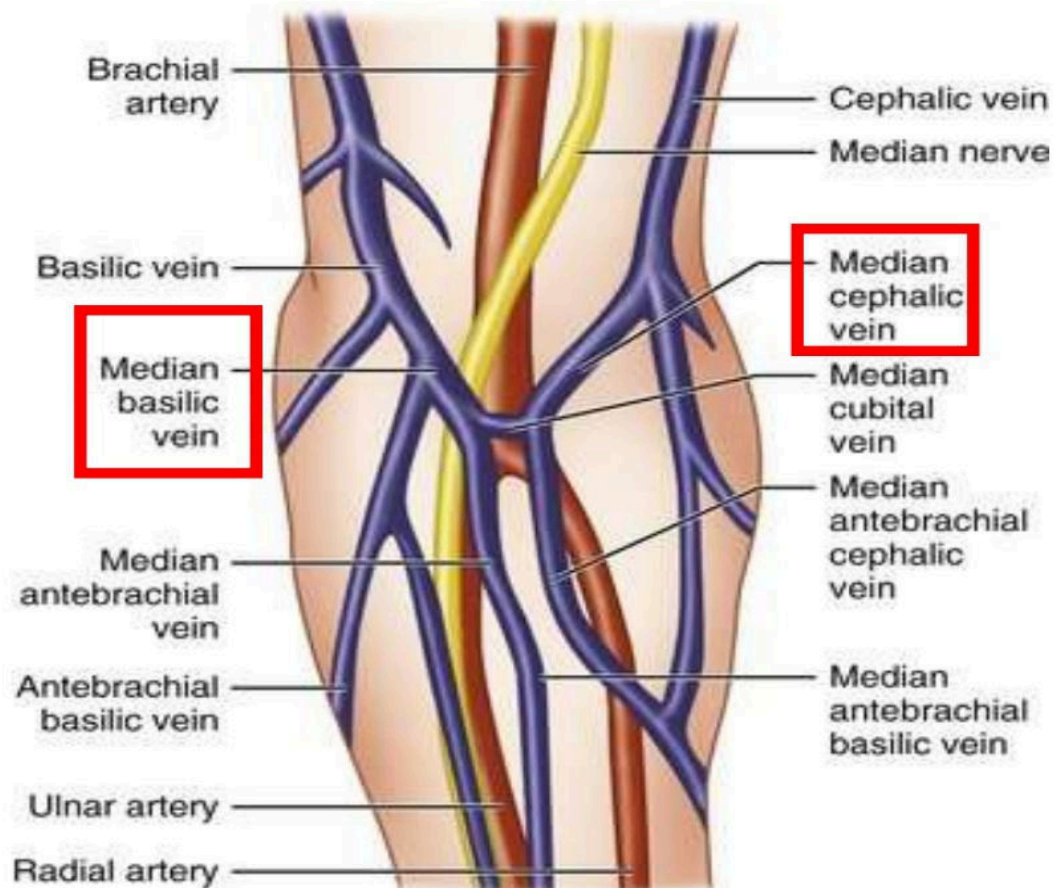
4. At the following below situations Vein puncture shall not be performed.

- a. on an extremity with an active shunt or fistula, or central vascular access device.
 - b. on the same side as axillary lymph node surgery as in the case of a mastectomy
 - c. from an area where a hematoma is present
 - d. from an artery or proximal to an active intravenous site
 - e. on the skin area with evidence of infection such as cellulitis or abscess.
5. The Blood Collection Order of Draw must be followed.

Procedure:

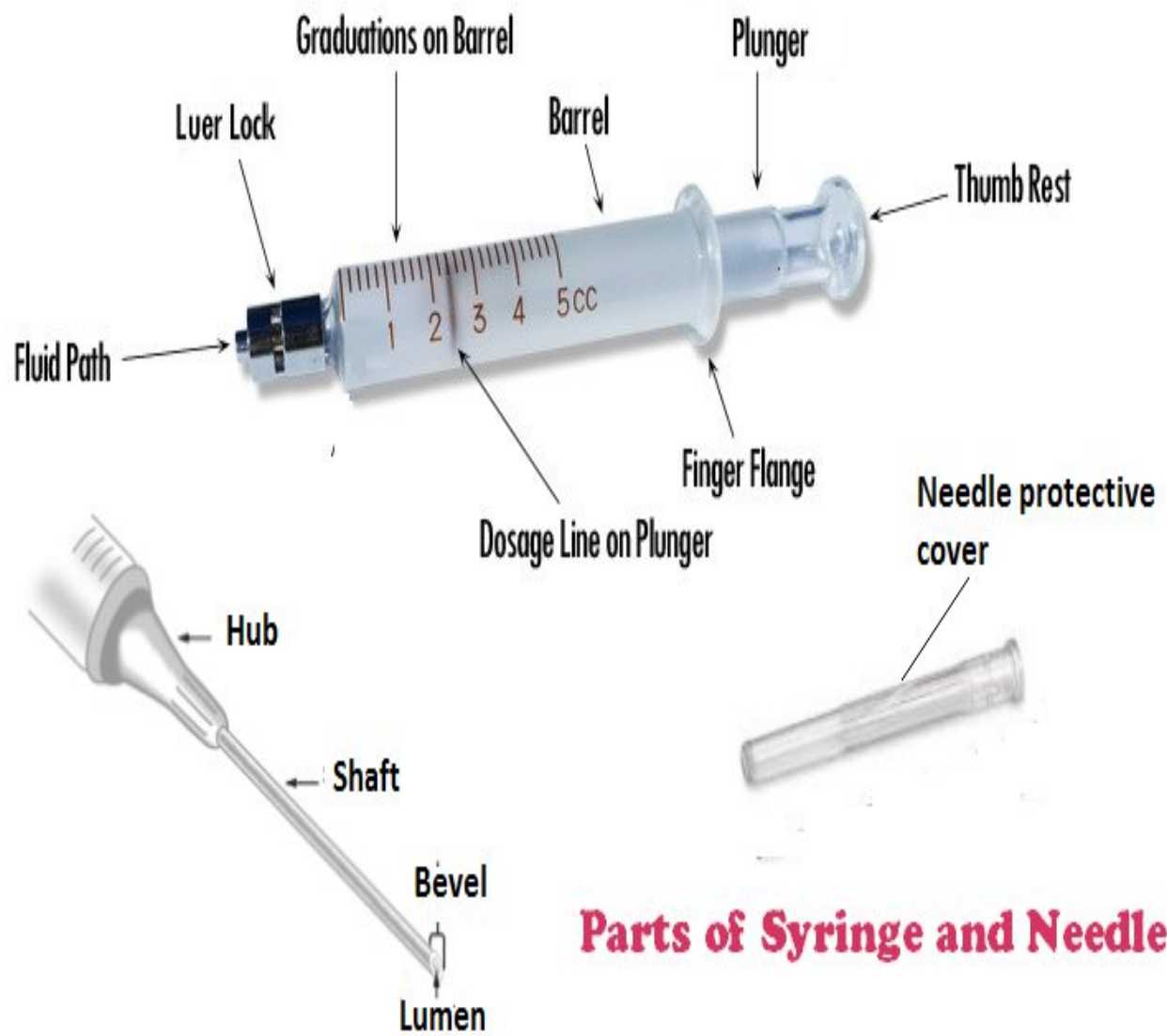
1. Check the authorized prescriber's order.
2. Assemble all equipment and supplies (Gloves, alcohol swab, vacutainer(s), needle, tourniquet, gauze and tape, and sharp container) and ensure all vacutainer tubes have not expired.
3. Wash hands thoroughly and put on gloves. When more than one patient is having vein puncture, gloves should be changed between patients, and use of hand sanitizer or hand washing should be done each time when gloves are changed.
4. Greet the patient, introduce yourself and indicate the procedure to be performed.
5. Confirm the identity of the patient. This confirmation involves ensuring the two unique patient identifiers either given by the patient verbally (i.e. to ask patient for their full name and also birthday or MRN or IC no etc.) or from the patient's wrist band or other recognized ID, that match with the information on the request order and the patient labels.
6. Check to ensure the patient's unique identifiers on the label are correct and label the correct specimen vacutainer(s) in front of the patient.
7. Once identification confirmed, do not leave the patient, otherwise the identification process must be repeated.

8. Position the patient so that he/she is seated or reclined comfortably with his/her arm extended to form a straight line from the shoulder to the wrist. In either situation, the patient's arm and elbow should be firmly supported with a pillow or towel and not bent.
9. Check both arms to identify a vein, preferable one which runs along to inner part of the forearm close to the surface of the skin. Use of the median cephalic vein or median basilica vein is preferable. Select the larger and fuller vein.

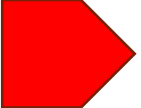


10. Palpate and trace the path of the vein several times with your index finger.
11. Tap the vein at the site of the draw with your index finger and second finger; this will cause the vein to dilate.
12. Apply tourniquet 8-10 cm above the desired site of puncture with enough pressure to compress the vein but not the artery.
13. Ask the patient to hold a fist tightly. They should avoid opening and closing the fist as studies show that this can increase blood potassium.
14. Clean the draw site with an alcohol swab (70% isopropyl alcohol) in a circular motion from the center of the area outwards and allow the alcohol to dry (30-60 seconds). DO NOT touch the vein puncture site again.

15. Using a sterile 21-23 gauge needle attached to a holder for vacutainer or a syringe, gently insert the needle into the vein at an angle roughly 15-30 degrees parallel to the vein making sure that the bevel of the needle is pointing up.
16. Push the vacutainer tube into the holder according to Blood Collection Order of Draw; or draw the correct amount of blood required into a syringe.
17. As the blood begins to flow into the specimen tubes, have the patient to open the hand.
18. Change the tubes for multiple specimen collection and gently invert tubes immediately upon filling following the instruction as outline in the Blood Collection Order of Draw.
19. Once the last vacutainer tube has been filled, remove the collection tube from the holder and remove the tourniquet; or once the adequate amount of blood has been collected, loosen the tourniquet.
20. Remove the needle at the same angle it was inserted.
21. Using gauze apply pressure to the site of the vein puncture for 2 minutes or until bleeding stops.
22. Dispose off the needle into the designated sharps container.
23. For the blood specimens taken by the syringe, filled the pre labeled specimen tubes according to the Blood Collection Order of Draw.
24. Apply tape and gauze to the vein puncture site.
25. Perform final patient identification with the patient either verbally or check on patient's wrist band and the label on the specimen tubes.
26. All blood specimens for blood grouping, Group screen and hold (GSH) and cross-match (GXM), the phlebotomist must write down his/her name and initial at the column "Blood taken by"and blood labeled by"on the Blood Transfusion request form and initial at the specimen label in the presence of the patient.
27. Advise the patient to consult the doctor if any complications develop at the site of vein puncture.
28. Remove the gloves and perform hand hygiene.
29. Ensure the specimens are transported to the laboratory as soon as possible.



Parts of Syringe and Needle

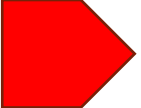


The Pentose Phosphate Pathway (PPP) or HMP Shunt

By; Abdur Rehman

Lecturer

Biochemistry

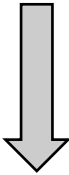


The Pentose Phosphate Pathway (PPP)

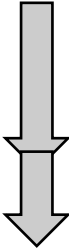
- An alternative metabolic pathway for glucose oxidation in which no ATP is generated
- **NADPH** is the important product of this pathway, which is utilized in
 - Several anabolic processes
 - **Ribose-5-phosphate**, a structural component of nucleotides and nucleic acids
- Alternative branch off glycolysis to produce the sugars that make up DNA and RNA

Glycolysis

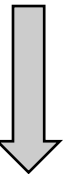
Glucose



Glucose 6,Phosphate



Kreb Cycle



ETC

Pentose phosphate
pathway



Phases of the pentose phosphate pathway operated in cytoplasm

Oxidative Phase

**Ribulose-5-
Phosphate**

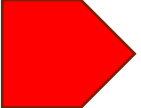
2NADPH

**Non-Oxidative
Phase**

**Ribose-5-
phosphate**

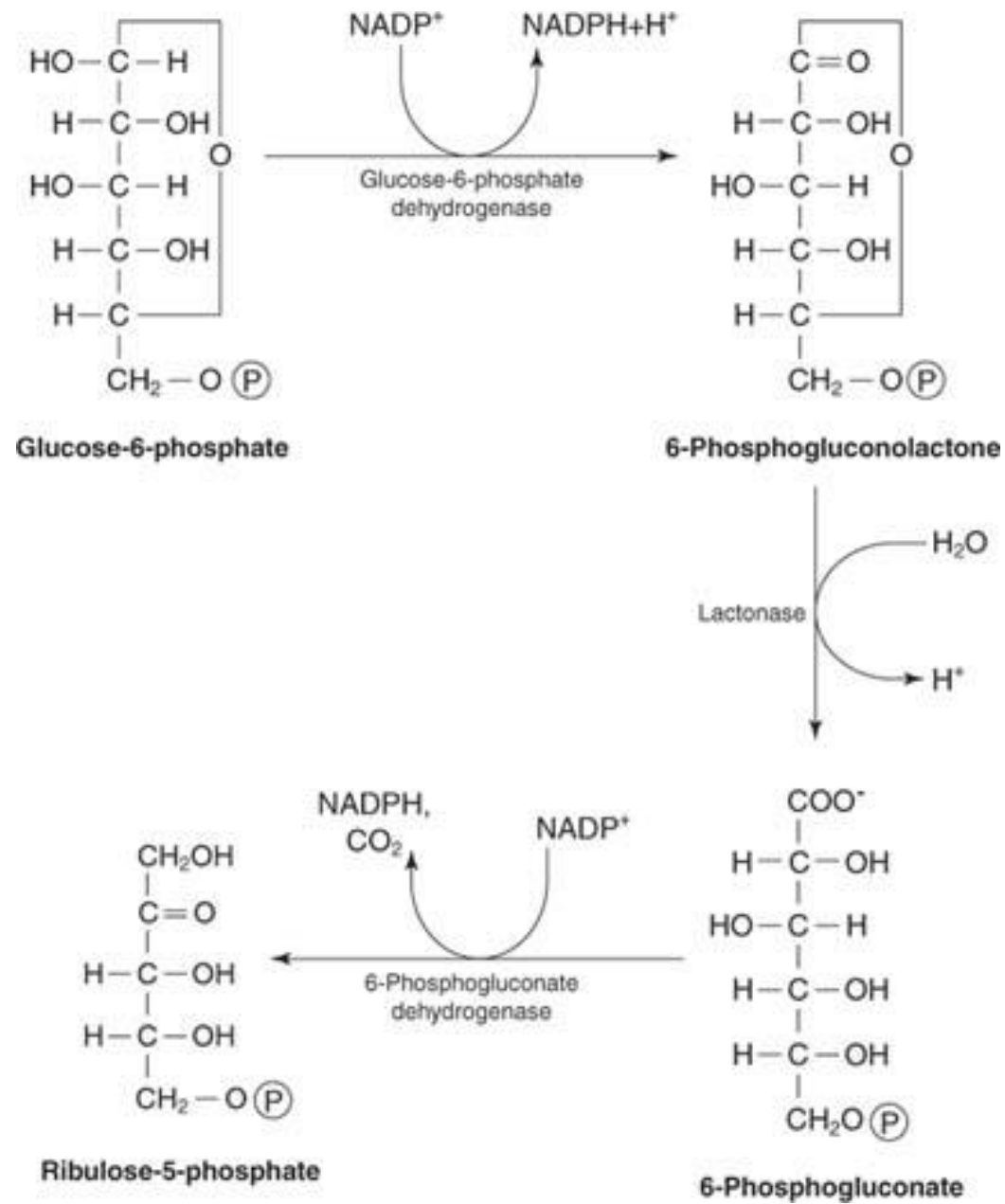
**Fructose-6-
phosphate**

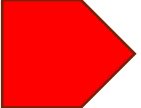
**Glyceraldehyde-
3-phosphate**



Oxidative Phase

- The oxidative phase of the pentose phosphate pathway consists of three reactions
 - i. glucose-6-phosphate dehydrogenase (G-6-PD) catalyzes the oxidation of glucose-6-phosphate give product of 6-Phosphogluconolactone and NADPH**
 - ii. 6-Phospho-D-glucono-lactone is then hydrolyzed to produce 6 phospho-D-gluconate**
 - iii. The oxidative decarboxylation of 6-phosphogluconate yield ribulose-5-phosphate with second molecule of NADPH**





Utilization of NADPH

- The oxidative phase of PPP kept the NADPH reserves high in the cells to
 - Utilize NADPH for reductive processes of lipid Biosynthesis
 - NADPH is also a **powerful antioxidant**(substances that prevent the oxidation of other molecules)
 - Important role in combating free radicals
 - NADPH Important for RBCs help iron in reduced form to increase affinity of Hemoglobin with oxygen
 - It also maintain glutathione in reduced state which is essentially required to preserve the integrity of RBC membrane
 - Phagocytosis by white blood cells requires the supply of NADPH
 - High concentration of NADPH in lens of eyes is necessary to preserve the transparency of the lenses

TABLE 20.2 **Pathways requiring
NADPH**

Synthesis

Fatty acid biosynthesis

Cholesterol biosynthesis

Neurotransmitter biosynthesis

Nucleotide biosynthesis

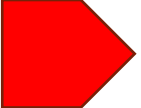
Detoxification

Reduction of oxidized glutathione

Cytochrome P450 monooxygenases

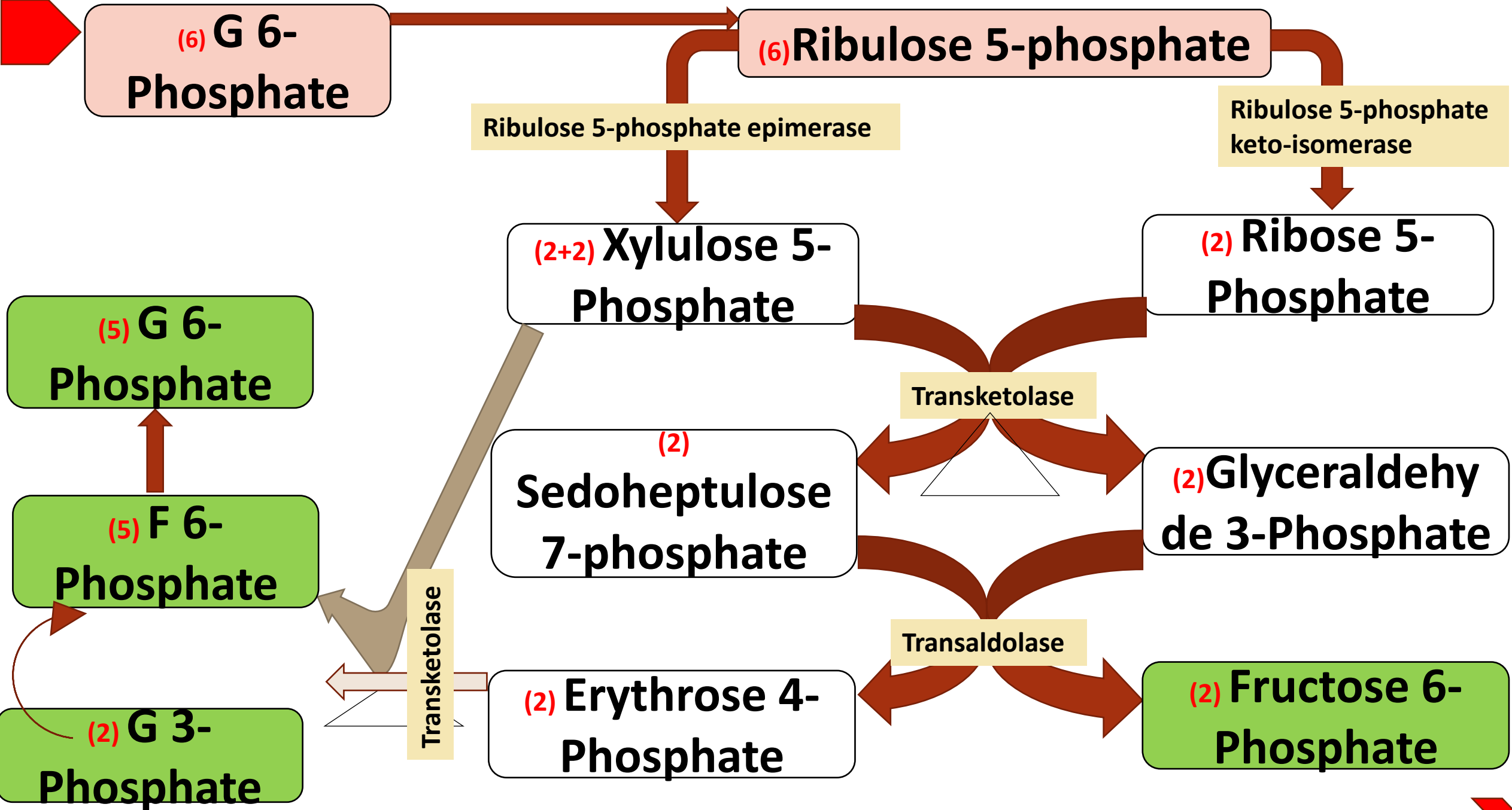
TABLE 20.4 **Tissues with active pentose phosphate pathways**

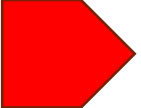
Tissue	Function
Adrenal gland	Steroid synthesis
Liver	Fatty acid and cholesterol synthesis
Testes	Steroid synthesis
Adipose tissue	Fatty acid synthesis
Ovary	Steroid synthesis
Mammary gland	Fatty acid synthesis
Red blood cells	Maintenance of reduced glutathione



Non-Oxidative Phase

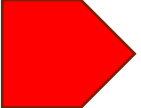
- Non oxidative pathway begin with conversion of
 - ribulose- 5-phosphate to ribose-5-phosphate by ribulose phosphate isomerase
- OR
- Xylulose-5-phosphate by ribulose-5-phosphate epimerase
- While the remaining reactions of pathway,
 - Enzymes transketolase and trans-aldolase catalyze the interconversions of trioses, pentoses, and hexoses
 - Transketolase is a TPP (thiamine pyrophosphate also known as Vitamin B1 act as coenzyme)-requiring enzyme that transfers two-carbon units from a ketose to an aldose





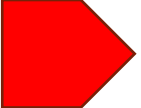
Glucose 6-phosphate dehydrogenase deficiency

- G6PD deficiency is an inherited sex-linked(X) trait
- Deficiency occurs in all the cells of affected individual but more severe in RBCs
 - HMP shunt is the only means of providing NADPH in the erythrocytes
 - Decreased activity of G6PD impairs the synthesis of NADPH in RBC
 - This results in the accumulation of methemoglobin and peroxides in erythrocytes leading to hemolysis
 - Severe Infection, oxidant drugs and ingestion of fava beans may result in hemolytic jaundice



G6PD deficiency and Resistance to Malaria

- The parasites that cause malaria are dependent on HMP shunt and reduced glutathione for their optimum growth in RBC
- G6PD deficiency more prevalent in African population hence more protected from malaria....**Natural selection?**
 - An adaptability of the people living in malaria-infected regions of the world



Reference

- Biochemistry by Satyanarayana and Chakrapani
- Lehninger Principles of Biochemistry, 5th Edition

Tubes , preservatives in tubes and their uses:

Most blood collection tubes contain an additive that either accelerates clotting of the blood (clot activator) or prevents the blood from clotting (anticoagulant).

Light Blue:

The blue bottle is used for hematology tests involving the clotting system, which require inactivated whole blood for analysis.

Additive: Sodium Citrate

Mechanism of action: Sodium Citrate Binds and remove calcium to prevent blood from clotting

Laboratory uses: Coagulation Profile.

PT (Prothrombin Time – evaluates the extrinsic system of the coagulation cascade.)

APTT/ PTT (Activated Partial Thromboplastin Time – evaluates the intrinsic system of the coagulation cascade.)

TT (Thrombin Time)



Green:

This less commonly used bottle is for biochemistry tests which require heparinised plasma or whole blood for analysis.

Additive: Heparin (Sodium/Lithium/Ammonium)

Mechanism of action: Heparin Inhibits thrombin formation to prevent clotting

Laboratory uses: For all biochemistry Testing (Plasma determinations in chemistry).



Lavender:

These bottles are generally used for haematology tests where whole blood is required for analysis.

Additive: EDTA (Ethylenediaminetetraacetic Acid)

Mechanism of action: EDTA removes calcium preventing clotting of blood.

Laboratory uses: Hematology testing (ESR, CBC, HgBA1c) blood film for abnormal cells or malaria parasites etc.



Grey:

Additive: Potassium oxalate and Sodium fluoride

Mechanism of action: Sodium fluoride acts as an antiglycolytic agent by inhibiting enolase enzyme to ensure that no further glucose breakdown occurs within the sample after it is taken. Potassium oxalate removes calcium and acts as an anticoagulant.

Laboratory uses: Chemistry testing, especially glucose(sugar) , Glucose tolerance test (GTT)



Gold or yellow:

Gel or clot activator tube also known as serum separating tube.

Additive: Clot activator and gel for serum separation.

Mechanism of action: Clot activator promotes blood clotting with glass or silica particles. Gel separates serum from cells.

Laboratory uses: Chemistry, serology, immunology



Tube cap color	Additive	Function of Additive	Common laboratory tests
Light-blue 	3.2% Sodium citrate	Prevents blood from clotting by binding calcium	Coagulation
Red or gold (mottled or "tiger" top used with some tubes is not shown) 	Serum tube with or without clot activator or gel	Clot activator promotes blood clotting with glass or silica particles. Gel separates serum from cells.	Chemistry, serology, immunology
Green 	Sodium or lithium heparin with or without gel	Prevents clotting by inhibiting thrombin and thromboplastin	Stat and routine chemistry
Lavender or pink 	Potassium EDTA	Prevents clotting by binding calcium	Hematology and blood bank
Gray 	Sodium fluoride, and sodium or potassium oxalate	Fluoride inhibits glycolysis, and oxalate prevents clotting by precipitating calcium.	Glucose (especially when testing will be delayed), blood alcohol, lactic acid